

LLM 后训练：深入探究大语言模型的推理能力

Komal Kumar*, Tajamul Ashraf*, Omark Thawakar, Rao Muhammad Anwer, Hisham Cholakkal, Mubarak Shah, Ming-Hsuan Yang, Phillip H.S. Torr, Fahad Shahbaz Khan, Salman Khan

摘要—大语言模型 (LLMs) 已经改变了自然语言处理领域的格局，并催生了多样化的应用。在海量网络规模数据上进行预训练使得这些模型奠定了基础，然而，研究界已将焦点转向训练后技术，以寻求进一步的突破。虽然预训练提供了强大的语言基础，但训练后方法使 LLMs 能够弥补其知识、改进推理、增强事实准确性，并更有效地向用户输出答案及伦理考量支持。微调、强化学习和奖励微调放大了优化 LLMs 性能、确保准确性以及解决在现实世界中存在适应性挑战的能力。本研究系统地探讨了训练后方法，分析了它们在预训练基础上精调 LLMs 的作用，并解决了灾难性遗忘、奖励策略攻击和推理时权衡等关键挑战。我们强调了模型对齐、可扩展适应和推理时推理等新兴方向，并概述了未来的研究方向。我们还提供了一个公共存储库，以持续跟踪这个快速发展领域的进展：<https://github.com/mzbui-ovr/Awesome-LLM-Post-training>。

Index Terms—Reasoning Models, Large Language Models, Reinforcement Learning, Reward Modeling, Test-time Scaling

1 Introduction

CONTEMPORARY 当代大型语言模型 (LLMs) 展现出跨越广泛任务领域的卓越能力, 不仅涵盖文本生成 [1, 2, 3] 和问答 [4, 5, 6, 7], 还包括复杂的多步推理 [8, 9, 10, 11]。它们驱动着自然语言理解 [12, 13, 14, 15, 16, 17]、内容生成 [18, 19, 20, 21, 22, 23, 24, 25]、自动推理 [26, 27, 28, 29] 以及多模态交互 [30, 31, 32, 33] 等应用。通过利用海量的自监督训练语料, 这些模型常常近似于人类的认知 [34, 35, 36, 37, 38], 在现实世界场景中展现出令人印象深刻的适应性。

尽管取得了这些令人瞩目的成就, LLMs 仍然存在关键的缺陷。它们可能生成误导性或事实错误的内容 (通常被称为“hallucinations”), 并且在长篇论述中难以保持逻辑一致性 [41, 42, 43, 44, 45, 46]。此外, LLMs 中的推理模型仍然是一个有争议的话题。虽然这些模型能够产生看似逻辑连贯的回应, 但其推理从根本上不同于人类的逻辑推断 [47, 34, 48, 49]。这种区别至关重要, 因为它有助于解释为何 LLMs 能够产生引人入胜的输出, 却仍然会在相对简单的逻辑任务上出错。与操作显式规则和事实的符号推理不同, LLMs 以隐式概率性的方式运作 [50, 42, 51]。在本工作范围内, LLMs 中的

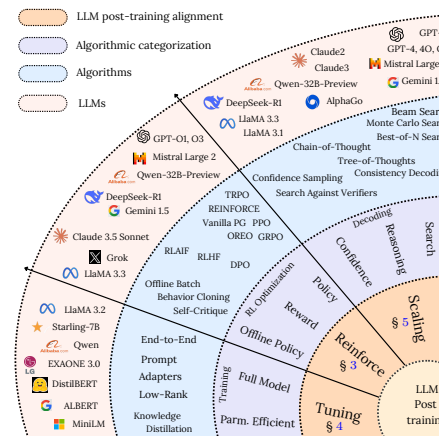


图 1: 大型语言模型 (LLMs) 的后训练方法分类, 分为微调、强化学习和测试时缩放方法。我们总结了近期 LLM 模型 (如 GPT-4 [39]、LLaMA 3.3 [13] 和 Deepseek R1 [40]) 中使用的关键技术。

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“reasoning”指的是它们基于数据中的统计模式生成逻辑连贯回应的能力，而非基于显式的逻辑推断或符号操作。此外，纯粹通过下一词预测训练的模型可能无法与用户期望或伦理标准对齐，尤其是在模糊或恶意场景下 [4, 52]。这些问题凸显了需要专门策略来解决 LLM 输出中的可靠性、偏见和上下文敏感性。

‘reasoning’ in LLMs refers to their ability to generate logically coherent responses based on statistical patterns in data rather than explicit logical inference or symbolic manipulation. Additionally, models trained purely via next-token prediction can fail to align with user expectations or ethical standards, especially in ambiguous or malicious scenarios [4, 52]. These issues underscore the need for specialized strategies that address reliability, bias, and context sensitivity in LLM outputs.

LLMs training can be broadly categorized into two stages: pre-training, which generally relies on a next-token prediction objective over large-scale corpora, and post-training, encompassing multiple rounds of fine-tuning and alignment. Post-training mechanisms aim to mitigate LLMs limitations by refining model behavior and aligning outputs with human intent, mitigating biases or inaccuracies [53].

Adapting LLMs to domain-specific tasks often involves techniques like **fine-tuning** [54, 55, 56], which enables task-specific learning but risks overfitting and incurs high computational costs. To address these challenges, approaches such as **Reinforcement Learning (RL)** [57, 58, 59] enhance adaptability by leveraging dynamic feedback and optimizing sequential decision-making. Additionally, advances in **scaling** techniques, including Low-Rank Adaptation (LoRA) [60], adapters [61, 62], and Retrieval-Augmented Generation (RAG) [63, 64, 65], improve both computational efficiency and factual accuracy. These strategies, coupled with distributed training frameworks, facilitate large-scale deployment and further boost the usability of LLMs across diverse applications (Figure 1). Through these targeted post-training interventions, LLMs become better aligned with human intent and ethical requirements, ultimately enhancing their real-world applicability. Below, we summarize key post-training stages.

a) Fine-Tuning in LLMs: Fine-tuning adapts pre-trained LLMs to specific tasks or domains by updating parameters on curated datasets [66, 67, 68, 54, 55, 69, 56]. While LLMs generalize well after large-scale pretraining, fine-tuning enhances performance in tasks like sentiment analysis [70, 71], question answering, and domain-specific applications such as medical diagnosis [72, 73, 74]. This process, typically supervised, aligns models with task requirements but poses challenges like overfitting, high computational costs, and sensitivity to data biases [56, 31, 16]. To this end, parameter-efficient techniques like LoRA [60] and adapters learn task-specific adaptation by updating explicit parameters, significantly reducing computational overhead. As models specialize, they may struggle with out-of-domain generalization, underscoring the trade-off between specificity and versatility.

💡 **Fine-tuning tailors LLMs for specific tasks, improving performance but risking overfitting, high compute costs, and reduced generalization.**

b) Reinforcement Learning in LLMs: In conventional RL, an agent interacts with a structured environment, taking discrete actions to transition between states while maximizing cumulative rewards [75]. RL domains—such as robotics, board games, and control systems—feature well-defined state-action spaces and clear objectives [76, 77]. RL in LLMs differs significantly. Instead of a finite action set, LLMs select tokens from a vast vocabulary, and their evolving state comprises an

ever-growing text sequence [16, 59, 78, 57]. This complicates planning and credit assignment, as the impact of token selection may only emerge later. Feedback in language-based RL is also sparse [79], subjective, and delayed, relying on heuristic evaluations and user preferences rather than clear performance metrics [80, 81, 58]. Additionally, LLMs must balance multiple, sometimes conflicting, objectives, unlike conventional RL, which typically optimizes for a single goal. Hybrid approaches combining process-based rewards (e.g., chain-of-thought reasoning) with outcome-based evaluations (e.g., response quality) help refine learning [8, 82, 83]. Thus, RL for LLMs requires specialized optimization techniques to handle high-dimensional outputs, non-stationary objectives, and complex reward structures, ensuring responses remain contextually relevant and aligned with user expectations.

💡 **Reinforcement in LLMs extends beyond conventional RL as it navigates vast action spaces, handles subjective and delayed rewards, and balances multiple objectives, necessitating specialized optimization techniques.**

c) Test Time Scaling in LLMs: Test Time Scaling is optimizing model performance and efficiency without altering the core architecture. It enables better generalization while minimizing computational overhead. It is crucial for enhancing the performance and efficiency of LLMs. It helps improve generalization across tasks but introduces significant computational challenges [84, 85]. Balancing performance and resource efficiency requires targeted strategies at inference. Techniques like CoT [8] reasoning and Tree-of-Thought (ToT) [86] frameworks enhance multi-step reasoning by breaking down complex problems into sequential or tree-structured steps. Additionally, search-based techniques [87, 88, 89, 90] enable iterative exploration of possible outputs, helping refine responses and ensure higher factual accuracy. These approaches, combined with methods like LoRA [60], adapters, and RAG [63, 64, 91], optimize the model’s ability to handle complex, domain-specific tasks at scale. RAG enhances factual accuracy by dynamically retrieving external knowledge, mitigating limitations of static training data [64, 24, 92]. Distributed training frameworks leverage parallel processing to manage the high computational demands of large-scale models. Test-time scaling optimizes inference by adjusting parameters dynamically based on task complexity [85, 93]. Modifying depth, width, or active layers balances computational efficiency and output quality, making it valuable in resource-limited or variable conditions. Despite advancements, scaling presents challenges such as diminishing returns, longer inference times, and environmental impact, especially when search techniques are performed at test time rather than during training [84]. Ensuring accessibility and feasibility is essential to maintain high-quality, efficient LLM deployment.

💡 **Test-time scaling enhances the adaptability of LLMs by dynamically adjusting computational resources during inference.**

大语言模型的训练可大致分为两个阶段：预训练，通常依赖于在大规模语料库上进行下一个词预测的目标；以及后训练，包含多轮微调与对齐。后训练机制旨在通过精炼模型行为、使输出与人类意图对齐来缓解大语言模型的局限性，减轻偏见或不准确性 [53]。

使大语言模型适应特定领域任务通常涉及诸如 **fine-tuning** [54, 55, 56] 等技术，这些技术能够实现任务特定学习，但存在过拟合风险并带来高昂的计算成本。为应对这些挑战，诸如 **Reinforcement Learning (RL)** [57, 58, 59] 等方法通过利用动态反馈和优化序列决策来增强适应性。此外，**scaling** 技术的进步，包括低秩适应 [60]、适配器 [61, 62] 以及检索增强生成 [63, 64, 65]，提高了计算效率和事实准确性。这些策略与分布式训练框架相结合，促进了大规模部署，并进一步提升了大语言模型在不同应用中的可用性（图 1）。通过这些有针对性的后训练干预，大语言模型能更好地与人类意图及伦理要求对齐，最终增强其现实世界的适用性。下文我们总结关键的后训练阶段。

a) Fine-Tuning in LLMs: 微调通过在精选数据集上更新参数，使预训练的大语言模型适应特定任务或领域 [66, 67, 68, 54, 55, 69, 56]。尽管大语言模型在大规模预训练后展现出良好的泛化能力，微调仍能提升其在情感分析 [70, 71]、问答以及诸如医疗诊断 [72, 73, 74] 等特定领域应用中的性能。这一过程通常是有监督的，使模型与任务要求对齐，但也带来了过拟合、高计算成本以及对数据偏差敏感 [56, 31, 16] 等挑战。为此，参数高效技术如低秩适应 [60] 和适配器通过更新显式参数来学习任务特定的适应，显著降低了计算开销。随着模型专业化，它们可能在领域外泛化方面遇到困难，这凸显了特定性与通用性之间的权衡。

💡 **微调使大语言模型适应特定任务，虽能提升性能，但存在过拟合风险、高昂计算成本以及泛化能力下降的问题。**

b) Reinforcement Learning in LLMs: 在传统的强化学习中，智能体与结构化环境交互，采取离散动作以在状态间转移，同时最大化累积奖励 [75]。强化学习领域——如机器人学、棋盘游戏和控制系统——具有定义明确的状态-动作空间和清晰的目标 $\langle \text{LEX}_{CMD_2} \rangle$ [16, 59, 78, 57][79] [80, 81, 58] [8, 82, 83]

💡 **大语言模型中的强化学习超越了传统强化学习，因为它需要驾驭庞大的动作空间、处理主观且延迟的奖励，并平衡多个目标，这必然要求采用专门的优化技术。**

[76, 77] **c) Test Time Scaling in LLMs:** 测试时缩放是在不改变核心架构的情况下优化模型性能和效率。它能在最小化计算开销的同时实现更好的泛化能力。这对于提升大型语

言模型的性能和效率至关重要。它有助于改善跨任务的泛化能力，但也带来了显著的计算挑战 [84, 85]。平衡性能与资源效率需要在推理阶段采取有针对性的策略。诸如 CoT [8] 推理和思维树 (ToT) [86] 框架等技术，通过将复杂问题分解为顺序或树状结构的步骤，增强了多步推理能力。此外，基于搜索的技术 [87, 88, 89, 90] 能够迭代探索可能的输出，有助于精炼回答并确保更高的事实准确性。这些方法，与诸如 LoRA [60]、适配器以及 RAG [63, 64, 91] 等方法相结合，优化了模型处理大规模、特定领域复杂任务的能力，缓解了静态训练数据的局限性 [64, 24, 92]。分布式训练框架利用并行处理来应对大规模模型的高计算需求。测试时缩放通过根据任务复杂性动态调整参数来优化推理过程 [85, 93]。修改深度、宽度或激活层可以在计算效率和输出质量之间取得平衡，使其在资源有限或条件多变的情况下具有重要价值。尽管取得了进展，缩放仍面临诸如收益递减、推理时间延长以及环境影响等挑战，尤其是在搜索技术在测试时而非训练时执行的情况下 [84]。确保可访问性和可行性对于维持高质量、高效的大型语言模型部署至关重要。

💡 **测试时缩放通过动态调整推理过程中的计算资源，增强了大型语言模型的适应性。**

1.1 Prior Surveys

近期关于强化学习与大语言模型的调查提供了宝贵的见解，但往往侧重于特定方面，导致关键的训练后组件未被充分探索 [51, 94, 95, 96]。许多研究探讨了诸如基于人类反馈的强化学习 [58]、基于人工智能反馈的强化学习 (RLAIF) [97] 以及直接偏好优化 [57] 等强化学习技术，然而它们忽视了微调、扩展以及对实际应用至关重要的基准测试。此外，这些研究尚未探索在各种框架（例如采用 R1 [59]）中，即使没有人工标注监督微调的情况下强化学习的潜力。其他调查探讨了大语言模型在传统强化学习任务（如多任务学习和决策制定）中的应用，但它们主要对大语言模型的功能进行分类，而非解决测试时扩展和集成的训练后策略问题 [98, 99]。同样，关于大语言模型推理的研究 [100, 101, 102, 55, 103, 104, 105, 106] 讨论了学习推理技术，但缺乏关于结合微调、强化学习和扩展的结构化指导。教程的缺失，以及对软件库和实现工具的综述不足，进一步限制了其实用性。相比之下，本调查通过系统地涵盖微调、强化学习和扩展作为相互关联的优化策略，提供了关于大语言模型训练后的全面视角，如图 1 所示。我们提供了实用资源——基准测试、数据集和教程——以助力大语言模型在实际应用中的精炼。

1.2 Contributions

本调查的主要贡献如下：

1.1 Prior Surveys

Recent surveys on RL and LLMs provide valuable insights but often focus on specific aspects, leaving key post-training components underexplored [51, 94, 95, 96]. Many works examine RL techniques like Reinforcement Learning from Human Feedback (RLHF) [58], Reinforcement Learning from AI Feedback (RLAIF) [97], and Direct Preference Optimization (DPO) [57], yet they overlook fine-tuning, scaling, and critical benchmarks essential for real-world applications. Furthermore, these studies have not explored the potential of RL even without human annotation supervised finetuning in various frameworks such as DeepSeek R1 with GRPO [59]. Other surveys explore LLMs in traditional RL tasks, such as multi-task learning and decision-making, but they primarily classify LLM functionalities rather than addressing test-time scaling and integrated post-training strategies [98, 99]. Similarly, studies on LLM reasoning [100, 101, 102, 55, 103, 104, 105, 106] discuss learning-to-reason techniques but lack structured guidance on combining fine-tuning, RL, and scaling. The absence of tutorials, along with reviews of software libraries and implementation tools, further limits their practicality. In contrast, this survey offers a comprehensive view of LLM post-training as shown in Figure 1 by systematically covering fine-tuning, RL, and scaling as interconnected optimization strategies. We offer practical resources—benchmarks, datasets, and tutorials—to aid LLM refinement for real-world applications.

1.2 Contributions

The key contributions of this survey are as follows:

- We provide a comprehensive and systematic review of post-training methodologies for LLMs, covering **fine-tuning**, **RL**, and **scaling** as integral components of model optimization.
- We offer a **structured taxonomy** of post-training techniques, clarifying their roles and interconnections, and present insights into open challenges and future research directions in optimizing LLMs for real-world deployment.
- Our survey provides practical guidance by introducing key **benchmarks**, **datasets**, and **evaluation metrics** essential for assessing post-training effectiveness, ensuring a structured framework for real-world applications.

2 Background

The LLMs have transformed reasoning by learning to predict the next token in a sequence based on vast amounts of text data [107, 4] using Maximum Likelihood Estimation (MLE) [108, 3, 109], which maximizes the probability of generating the correct sequence given an input. This is achieved by minimizing the negative log-likelihood:

$$\mathcal{L}_{\text{MLE}} = -\sum_{t=1}^T \log P_{\theta}(y_t | y_{<t}, X).$$

Here, X represents the input, such as a prompt or context. $Y = (y_1, y_2, \dots, y_T)$ is the corresponding target output sequence, and $P_{\theta}(y_t | y_{<t}, X)$ denotes the model's predicted probability for token y_t , given preceding tokens.

💡 **Token-wise training can ensure fluency but may cause cascading errors due to uncorrected mistakes in inference.**

As these models scale, they exhibit emergent reasoning abilities, particularly when trained on diverse data that include code and mathematical content [110, 8]. However, despite their impressive capabilities, LLMs struggle to maintain coherence and contextual relevance over long sequences. Addressing these limitations necessitates a structured approach to sequence generation, which naturally aligns with RL.

Since LLMs generate text autoregressively—where each token prediction depends on previously generated tokens—this process can be modeled as a sequential decision-making problem within a Markov Decision Process (MDP) [111]. In this setting, the state s_t represents the sequence of tokens generated so far, the action a_t is the next token, and a reward $R(s_t, a_t)$ evaluates the quality of the output. An LLM's policy π_{θ} is optimized to maximize the expected return:

$$J(\pi_{\theta}) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right],$$

where γ is the discount factor that determines how strongly future rewards influence current decisions. A higher γ places greater importance on long-term rewards. The primary objective in RL is to learn a policy that maximizes the expected cumulative reward, often referred to as the return. This requires balancing exploration—trying new actions to discover their effects—and exploitation—leveraging known actions that yield high rewards. While LLMs optimize a likelihood function using static data, RL instead optimizes the expected return through dynamic interactions. To ensure that LLMs generate responses that are not only statistically likely but also aligned with human preferences, it is essential to go beyond static optimization methods. While likelihood-based training captures patterns from vast corpora, it lacks the adaptability needed for refining decision-making in interactive settings. By leveraging structured approaches to maximizing long-term objectives, models can dynamically adjust their strategies, balancing exploration and exploitation to improve reasoning, coherence, and alignment [112, 113, 49, 48].

💡 **LLMs exhibit emergent abilities due to scale, while RL refines and aligns them for better reasoning and interaction.**

2.1 RL based Sequential Reasoning.

The chain-of-thought reasoning employed in modern LLMs is naturally framed as an RL problem. In this perspective, each intermediate reasoning step is treated as an action contributing to a final answer. The objective function $J(\pi_{\theta})$ represents the expected reward of the policy π_{θ} , capturing how well the model performs over multiple reasoning steps. The policy gradient update is given by:

$$\nabla_{\theta} J(\pi_{\theta}) = \mathbb{E}_{\tau} \left[\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(x_t | x_{1:t-1}) A(s_t, a_t) \right],$$

- 我们对大语言模型的训练后方法进行了全面而系统的综述，涵盖 **fine-tuning**、**RL** 和 **scaling** 作为模型优化的组成部分。
- 我们提供了训练后技术的 **structured taxonomy**，阐明了它们的作用与相互联系，并就优化大语言模型以进行实际部署所面临的开放挑战和未来研究方向提出了见解。
- 我们的调查通过引入评估训练后效果所必需的关键 **benchmarks**, **datasets**, and **evaluation metrics**，提供了实用指导，确保为实际应用构建一个结构化框架。

2 Background

大型语言模型通过学习基于海量文本数据预测序列中的下一个词元，改变了推理方式 [107, 4] 其采用最大似然估计 (MLE) [108, 3, 109]，该目标旨在最大化给定输入条件下生成正确序列的概率。这是通过最小化负对数似然实现的：

$$\mathcal{L}_{\text{MLE}} = -\sum_{t=1}^T \log P_{\theta}(y_t | y_{<t}, X).$$

此处， X 代表输入，例如提示或上下文。 $Y = (y_1, y_2, \dots, y_T)$ 是对应的目标输出序列，而 $P_{\theta}(y_t | y_{<t}, X)$ 表示模型在给定前面词元条件下，对词元 y_t 的预测概率。

💡 **基于词元的训练可以确保流畅性，但由于推理中未纠正的错误，可能导致级联错误。**

随着这些模型规模的扩大，它们展现出涌现的推理能力，尤其是在包含代码和数学内容的多样化数据上训练时 [110, 8]。然而，尽管其能力令人印象深刻，大型语言模型难以在长序列中保持连贯性和上下文相关性。解决这些局限性需要对序列生成采取结构化方法，这自然与强化学习对齐。

$$J(\pi_{\theta}) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right],$$

其中 γ 是折扣因子，决定了未来奖励对当前决策的影响强度。较高的 γ 赋予长期奖励更大的重要性。强化学习中的主要目标是学习一个能最大化期望累积奖励（通常称为回报）的策略。这需要在探索——尝试新动作以发现其效果——与利用——利用已知能产生高奖励的动作——之间取得平衡。虽然大语言模型使用静态数据优化似然函数，但强化学习则通过动态交互来优化期望回报。为确保大语言模型生成的回应不仅在统计上可能，而且符合人类偏好，必须超越静态优化方法。虽然基于似然的训练能从海量语料中捕捉模式，但它缺乏在交互式环境中精化决策所需的适应性。通过利用结构化方法来最大化长期目标，模型可以动态调整其策略，平衡探索与利用，从而提升推理能力、连贯性和对齐性 [112, 113, 49, 48]。

💡 **大语言模型因规模而展现出涌现能力，而强化学习则对其进行精炼和对齐，以实现更好的推理与交互。**

现代大语言模型中采用的思维链推理自然地构建为一个强化学习问题。在此视角下，每个中间推理步骤被视为一个贡献于最终答案的动作。目标函数 $J(\pi_{\theta})$ 表示策略 π_{θ} 的期望奖励，它捕捉了模型在多个推理步骤上的表现。策略梯度更新由下式给出：

$$\nabla_{\theta} J(\pi_{\theta}) = \mathbb{E}_{\tau} \left[\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(x_t | x_{1:t-1}) A(s_t, a_t) \right],$$

其中优势函数 $A(s_t, a_t)$ 将信用分配给各个步骤，确保整体推理过程通过即时和延迟奖励得到优化。此类表述，包括逐步奖励分解 [114, 115]，对于提升大语言模型在复杂推理任务上的可解释性和性能至关重要。在传统的强化学习表述中，智能体具有：

Value function: $V(s) = \mathbb{E}[\text{future return} | s]$,

Action-value (Q-) function: $Q(s, a) = \mathbb{E}[\text{future return} | s, a]$,

Advantage function: $A(s, a) = Q(s, a) - V(s)$.

换言之， $A(s, a)$ 衡量了在状态 s 下采取特定动作 a 相比智能体原本预期（其基线 $V(s)$ ）有多好。how much better or worse normally

2.1 Early RL Methods for Language Modeling.

在此，我们简要概述为将强化学习应用于语言生成任务奠定基础的开创性方法。这些早期工作通过直接调整决策模型（策略 $\langle p_{\theta} \rangle$ ）的参数以最大化奖励来训练该模型。以下解释几种策略梯度方法：

Policy Gradient (REINFORCE). REINFORCE 算法 [116, 117] 是一种通过根据其行动获得的奖励调整模型策略（策略）来改进决策的方法。该算法并非直接学习每种情况下的最佳行动，而是优化选择不同行动的可能性，从而逐步改善长期结果。在每一步，模型根据其过去决策的表现更新其参数 $\langle \theta \rangle$ ：

$$\theta \leftarrow \theta + \alpha \left(G - b \right) \sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t).$$

其中： G 表示模型在一个回合中累积的总奖励， b 是一个有助于减少方差、使学习更稳定的基线值， $\nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$ 衡量 θ 的微小变化如何影响在给定状态 s_t 下选择行动 a_t 的概率， α 是学习率，控制每一步策略更新的幅度。

💡 **基于长期奖励优化行动——即考虑一系列推理步骤的累积收益而非仅即时结果——是近期大语言模型的基础。这种方法使模型能更有效地探索多种推理路径。**

Curriculum Learning with MIXER. Ranzato 等人 [118] 提出了一种从最大似然估计逐渐过渡到强化学习的方案。总体损失是加权组合：

$$\mathcal{L} = \lambda(t) \mathcal{L}_{\text{MLE}} + (1 - \lambda(t)) \mathcal{L}_{\text{RL}}.$$

where the advantage function $A(s_t, a_t)$ distributes credit to individual steps, ensuring that the overall reasoning process is refined through both immediate and delayed rewards. Such formulations, including step-wise reward decomposition [114, 115], have been crucial for enhancing the interpretability and performance of LLMs on complex reasoning tasks. In traditional RL formulations, an agent has:

$$\text{Value function: } V(s) = \mathbb{E}[\text{future return} \mid s],$$

$$\text{Action-value (Q-) function: } Q(s, a) = \mathbb{E}[\text{future return} \mid s, a],$$

$$\text{Advantage function: } A(s, a) = Q(s, a) - V(s).$$

In words, $A(s, a)$ measures *how much better or worse* it is to take a specific action a in state s compared to what the agent would *normally* expect (its baseline $V(s)$).

2.2 Early RL Methods for Language Modeling.

Here, we briefly overview pioneering methods that laid the groundwork for applying RL to language generation tasks. These initial efforts train a decision-making model (policy (p_θ)) by directly adjusting its parameters to maximize rewards. Some policy gradient approaches are explained below: **Policy Gradient (REINFORCE)**. The REINFORCE algorithm [116, 117] is a method used to improve decision-making by adjusting the model's strategy (policy) based on rewards received from its actions. Instead of directly learning the best action for every situation, the algorithm refines how likely different actions are to be chosen, gradually improving outcomes over time. At each step, the model updates its parameters (θ) based on how well its past decisions performed:

$$\theta \leftarrow \theta + \alpha \left(G - b \right) \sum_{t=1}^T \nabla_\theta \log \pi_\theta(a_t \mid s_t).$$

Here: G represents the total reward the model accumulates over an episode, b is a baseline value that helps reduce variance, making learning more stable, $\nabla_\theta \log \pi_\theta(a_t \mid s_t)$ measures how much a small change in θ affects the probability of choosing action a_t given state s_t , α is the learning rate, controlling how much the policy updates at each step.

💡 **Optimizing actions based on long-term rewards, which account for the cumulative benefits of a sequence of reasoning steps rather than just immediate outcomes, is fundamental in recent LLMs. This approach allows models to explore multiple reasoning paths more effectively.**

Curriculum Learning with MIXER. Ranzato et al. [118] introduces a gradual transition from maximum likelihood estimation (MLE) to RL. The overall loss is a weighted combination:

$$\mathcal{L} = \lambda(t) \mathcal{L}_{\text{MLE}} + (1 - \lambda(t)) \mathcal{L}_{\text{RL}},$$

where $\lambda(t)$ decreases with training time. This curriculum helps the model ease into the RL objective and mitigate the mismatch between training and inference.

Self-Critical Sequence Training (SCST). SCST [119] refines the policy gradient method by comparing the model's sampled outputs against its own best (greedy) predictions. Instead of using an arbitrary baseline, SCST uses the model's

own highest-scoring output, ensuring that updates directly improve performance relative to what the model currently considers its best response. The gradient update follows:

$$\nabla_\theta J(\pi_\theta) \approx \left(r(y^*) - r(\hat{y}) \right) \nabla_\theta \log \pi_\theta(y^*),$$

where y^* is a sampled sequence, $\hat{y}$ is the greedy output, and $r(y)$ represents an evaluation metric such as BLEU [120] for translation or CIDEr [121] for image captioning. Since the learning signal is based on the difference $r(y^*) - r(\hat{y})$, the model is explicitly trained to generate outputs that score higher than its own baseline under the evaluation metric. If the sampled output outperforms the greedy output, the model reinforces it; otherwise, it discourages that sequence. This direct feedback loop ensures that training aligns with the desired evaluation criteria rather than just maximizing likelihood. By leveraging the model's own best predictions as a baseline, SCST effectively reduces variance and stabilizes training while optimizing real-world performance metrics.

Minimum Risk Training (MRT). MRT [153] directly minimizes the expected risk over the output distribution. Given a task-specific loss $\Delta(y, y^*)$ comparing the generated output y with the reference y^* , the MRT objective is defined as:

$$\mathcal{L}_{\text{MRT}}(\theta) = \sum_{y \in \mathcal{Y}} p_\theta(y \mid x) \Delta(y, y^*).$$

This formulation incorporates evaluation metrics (e.g., 1 - BLEU) directly into training, enabling fine-grained adjustments of the policy.

Advantage Actor-Critic (A2C/A3C). RL methods like REINFORCE [116] rely solely on policy gradients, which suffer from high variance, leading to unstable and inefficient learning. Since the reward signal fluctuates across different trajectories, updates may be noisy, causing slow or erratic convergence. To mitigate this, Actor-Critic methods [154, 155, 156, 157] combine two components as follows: an actor and a critic. The actor is a policy $\pi_\theta(a_t \mid s_t)$ that selects actions a_t at state s_t , while the critic is a value function $V_\phi(s_t)$ that evaluates the expected return of a state. The critic provides a more stable learning signal, reducing variance in policy updates and enabling efficient learning in continuous action spaces. Actor updates are guided by the policy gradient theorem, where the advantage function $A(s_t, a_t)$ defined in Sec. 2.1, determines how much better an action a_t is compared to the expected value of state s_t . The policy with the learning rate α is updated as:

$$\theta \leftarrow \theta + \alpha A(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t \mid s_t).$$

Meanwhile, the critic is updated using temporal difference learning, minimizing the squared error between its estimate and the actual return:

$$\phi \leftarrow \phi - \beta \nabla_\phi \left(V_\phi(s_t) - G_t \right)^2.$$

where β is a learning rate for critic. To enhance stability and efficiency, several improvements have been proposed. Eligibility traces allow learning from recent states, enabling faster convergence. Function approximation with neural networks ensures effective handling of high-dimensional inputs. Advanced variants such as Natural Gradient methods [158] adjust updates using the Fisher Information Matrix, improving convergence speed.

RL Enhanced LLMs	Developer	Source	# Params	RL Methods	Fine-Tuning	Architecture Type	Model	TTS
DeepSeek-V2 [146]	Deepseek	链接	236B-A21B	GRPO	DPO + GRPO	MoE	开源	✓
GPT 4.5 [122]	OpenAI	链接	-	RLHF, PPO, RBRM	SFT + RLHF	MoE	闭源	✓
Gemini [15]	Google	链接	-	RLHF	SFT + RLHF	MoE	闭源	✓
Claude 3.7 [123]	Anthropic	链接	-	RLAIF	SFT + RLAI	MoE	闭源	✓
Roka [124]	Roka	链接	7B, 21B	RLHF, PPO	SFT + RLHF	MoE	闭源	✓
DeepSeek1 [40]	Deepseek	链接	240B-A22B	GRPO	DPO + GRPO	MoE	闭源	✓
Nemotron-4 340B [125]	NVIDIA	链接	340B	DPO, RPO	DPO + RPO	MoE	闭源	✓
Falcon [29]	TIH	链接	40B	-	SFT	MoE	闭源	✓
GPT-4 [39]	OpenAI	链接	-	RLHF, PPO, RBRM	SFT + RLHF	MoE	闭源	✓
Llama 3 [13]	Meta	链接	8B, 70B, 405B	DPO	SFT + DPO	MoE	闭源	✓
Qwen2 [127]	Alibaba	链接	(0.5-72)B, 57B-A14B	DPO	SFT + DPO	MoE	闭源	✓
Gemma2 [14]	Google	链接	2B, 9B, 27B	RLHF	SFT + RLHF	MoE	闭源	✓
Starling-7B [26]	Berkeley	链接	7B	RLAIF, PPO	SFT + RLAI	MoE	闭源	✓
Moshi [128]	Kyutai	链接	7B	-	-	MoE	闭源	✓
Athene-70B [129]	Nexusflow	链接	70B	RLHF	SFT + RLHF	MoE	闭源	✓
GPT-3.5 [39]	OpenAI	链接	3.5B, 175B	RLHF, PPO	SFT + RLHF	MoE	闭源	✓
Hermes 3 [130]	Nous	链接	8B, 70B, 405B	DPO	SFT + DPO	MoE	闭源	✓
Zed [131]	Zed AI	链接	500B	RLHF	RLHF	MoE	闭源	✓
PalM 2 [132]	Google	链接	-	RLHF	-	MoE	闭源	✓
InternLM2 [133]	SAII	链接	1.8B, 7B, 20B	RLHF, PPO	SFT + RLHF	MoE	闭源	✓
Supernova [134]	Nova AI	链接	220B	RLHF	RLHF	MoE	闭源	✓
Grok3 [135]	Grok-3	链接	175B	-	-	MoE	闭源	✓
Pixtral [136]	Mistral AI	链接	12B, 123B	-	-	MoE	闭源	✓
Minimax [137]	Minimax	链接	456B	-	SFT	MoE	闭源	✓
Amazonova [138]	Amazon	链接	-	DPO, RLHF, RLAI	SFT	MoE	闭源	✓
Fugaku [139]	Fujitsu	链接	13B	-	-	MoE	闭源	✓
Nova [140]	Rubik's AI	链接	-	-	SFT	MoE	闭源	✓
G3 [141]	OpenAI	链接	-	基于链式思维的强化学习	基于链式思维的强化学习	MoE	闭源	✓
Durix [142]	Databricks	链接	136B	-	-	MoE	闭源	✓
Instruct-GPT [58]	OpenAI	链接	1.3B, 6B, 175B	RLHF, PPO	SFT + RLHF	MoE	闭源	✓
OpenAssistant [143]	LAION	链接	17B	-	SFT	MoE	闭源	✓
ChatGLM [144]	Zhipu AI	链接	6B, 9B	ChatGLM-RLHF	SFT + RLHF	MoE	闭源	✓
Zephyr [145]	Argilla	链接	141B-A39B	ORPO	DPO + ORPO	MoE	闭源	✓
phi-3 [17]	Microsoft	链接	3.8B, 7B, 14B	DPO	SFT + DPO	MoE	闭源	✓
Jurassic [146]	AI21 Labs	链接	-	-	SFT	MoE	闭源	✓
Kimi K1.5 [147]	Moonshot AI	链接	150B	-	RLHF	MoE	闭源	✓
Phi-4 [148]	Microsoft	链接	28B, 70B, 140B	DPO	SFT + DPO	MoE	闭源	✓
Chameleon [149]	Meta AI	链接	34B	-	SFT	MoE	闭源	✓
Cerebrasgpt [150]	Cerebras	链接	130 亿	-	SFT	MoE	闭源	✓
Bloomberggpt [151]	Bloomberg L.P.	链接	500 亿	-	SFT	MoE	闭源	✓
Chinchilla [152]	DeepMind	链接	700 亿	RLHF, PPO	SFT	MoE	闭源	✓

表 1: 强化学习增强型大型语言模型概览，其中“141B-A39B”表示一个专家混合（MoE）模型，总参数量为 1410 亿，推理时实际使用 390 亿参数。TTS 代表测试时缩放。

其中 $\lambda(t)$ 随训练时间递减。这种课程学习有助于模型平滑适应强化学习目标，并缓解训练与推理之间的不匹配。

Self-Critical Sequence Training (SCST). 自临界序列训练 (SCST) [119] 通过将模型的采样输出与其自身的最佳（贪婪）预测进行比较，改进了策略梯度方法。SCST 不使用任意基线，而是采用模型自身得分最高的输出作为基线，确保更新直接针对模型当前认为的最佳响应提升性能。梯度更新如下：

$$\nabla_\theta J(\pi_\theta) \approx \left(r(y^*) - r(\hat{y}) \right) \nabla_\theta \log \pi_\theta(y^*),$$

其中 y^* 是采样序列， $\hat{y}$ 是贪婪输出， $r(y)$ 代表评估指标，如用于翻译的 BLEU [120] 或用于图像描述的 CIDEr [121]。由于学习信号基于差值 $r(y^*) - r(\hat{y})$ ，模型被明确训练以生成在评估指标下得分高于其自身基线的输出。若采样输出优于贪婪输出，则模型强化该序列；反之，则抑制该序列。这种直接反馈确保训练与期望的评估标准对齐，而非仅仅最大化似然。通过利用模型自身的最佳预测作为基线，SCST 有效降低了方差、稳定了训练，同时优化了实际性能指标。 **Minimum Risk Training (MRT)**. MRT [153] 直接最小化输出分布上的期

望风险。给定一个任务特定的损失函数 $\Delta(y, y^*)$ ，用于比较生成的输出 y 与参考 y^* ，MRT 目标定义为：

$$\mathcal{L}_{\text{MRT}}(\theta) = \sum_{y \in \mathcal{Y}} p_\theta(y \mid x) \Delta(y, y^*).$$

该公式将评估指标（例如 1 - BLEU）直接纳入训练，从而实现对策略的细粒度调整。

Advantage Actor-Critic (A2C/A3C). 像 REINFORCE [116] 这样的 RL 方法仅依赖于策略梯度，其方差较高，导致学习不稳定且低效。由于奖励信号在不同轨迹间波动，更新可能带有噪声，造成收敛缓慢或不稳定。为缓解此问题，Actor-Critic 方法 [154, 155, 156, 157] 结合了两个组件：一个执行器 (actor) 和一个评判器 (critic)。执行器是一个策略 $\pi_\theta(a_t \mid s_t)$ ，在状态 s_t 下选择动作 a_t ；而评判器是一个价值函数 $V_\phi(s_t)$ ，用于评估状态的期望回报。评判器提供了更稳定的学习信号，减少了策略更新的方差，并使得在连续动作空间中进行高效学习成为可能。执行器的更新遵循策略梯度定理，其中优势函数 $A(s_t, a_t)$ （在第 ?? 节中定义）衡量了动作 a_t 相对于状

RL Enhanced LLMs	Developer	Source	# Params	RL Methods	Fine-Tuning	Architecture Type	Model	TTS
DeepSeek-V2 [16]	Deepseek	Link	236B-A21B	GRPO	DPO + GRPO	MoE	Open	✓
GPT 4.5 [122]	OpenAI	Link	-	RLHF, PPO, RBRM	SFT + RLHF	MoE	Closed	✓
Gemini [15]	Google	Link	-	RLHF	SFT + RLHF	Single Model	Closed	✗
Claude 3.7 [123]	Anthropic	Link	-	RLAIF	SFT + RLAIF	Single Model	Closed	✓
Rak1 [124]	Rak1	Link	7B, 21B	RLHF, PPO	SFT + RLHF	Single Model	Closed	✗
DeepSeekR1 [40]	Deepseek	Link	240B-A22B	GRPO	DPO + GRPO	MoE	Open	✓
Nemotron-4 340B [125]	NVIDIA	Link	340B	DPO, RPO	DPO + RPO	Single Model	Closed	✓
Falcon [126]	TI	Link	40B	-	SFT	Single Model	Open	✗
GPT-4 [39]	OpenAI	Link	-	RLHF, PPO, RBRM	SFT + RLHF	MoE	Closed	✓
Llama 3 [13]	Meta	Link	8B, 70B, 405B	DPO	SFT + DPO	Single Model	Open	✗
Qwen2 [127]	Alibaba	Link	(0.5-72)B, 57B-A14B	DPO	SFT + DPO	Single Model	Open	✓
Gemma2 [14]	Google	Link	2B, 9B, 27B	RLHF	SFT + RLHF	Single Model	Open	✗
Starling-7B [26]	Berkeley	Link	7B	RLAIF, PPO	SFT + RLAIF	Single Model	Open	✗
Moshi [128]	Kyutai	Link	7B	-	-	Multi-modal	Open	✓
Athene-70B [129]	Nexusflow	Link	70B	RLHF	SFT + RLHF	Single Model	Open	✗
GPT-3.5 [39]	OpenAI	Link	3.5B, 175B	RLHF, PPO	SFT + RLHF	MoE	Closed	✓
Hermes 3 [130]	Nous	Link	8B, 70B, 405B	DPO	SFT + DPO	Single Model	Open	✗
Zed [131]	Zed AI	Link	500B	RLHF	RLHF	Multi-modal	Open	✓
PaLM 2 [132]	Google	Link	-	RLHF	-	Single Model	Closed	✓
InternLM2 [133]	SAII	Link	1.8B, 7B, 20B	RLHF, PPO	SFT + RLHF	Single Model	Closed	✓
Supernova [134]	Nova AI	Link	220B	RLHF	RLHF	Multi-modal	Open	✓
Grok3 [135]	Grok-3	Link	175B	-	DPO	Dense	Open	✓
Pixtral [136]	Mistral AI	Link	12B, 123B	-	PEFT	Multimodal	Open	✓
MiniMaxT1 [137]	MiniMax	Link	456B	-	SFT	Single Model	Closed	✓
Amazonova [138]	Amazon	Link	-	DPO, RLHF, RLAIF	SFT	Single Model	Closed	✗
Fugaku1m [139]	Fujitsu	Link	13B	-	-	Single Model	Closed	✗
Nova [140]	Rubik's AI	Link	-	-	SFT	Proprietary	Closed	✗
O3 [141]	OpenAI	Link	-	RL through CoT	RL through CoT	Single Model	Closed	✓
Drx [142]	Databricks	Link	136B	-	-	Single Model	Open	✓
Instruct-GPT [58]	OpenAI	Link	1.3B, 6B, 175B	RLHF, PPO	SFT + RLHF	Single Model	Closed	✗
Openassistant [143]	LAION	Link	17B	-	SFT	Single Model	Open	✗
ChatGLM [144]	Zhipu AI	Link	6B, 9B	ChatGLM-RLHF	SFT + RLHF	Single Model	Open	✗
Zephyr [145]	Argilla	Link	141B-A39B	ORPO	DPO + ORPO	MoE	Open	✓
phi-3 [17]	Microsoft	Link	3.8B, 7B, 14B	DPO	SFT + DPO	Single Model	Closed	✗
Jurassic [146]	A121 Labs	Link	-	-	SFT	Proprietary	Closed	✗
Kimi K1.5 [147]	Moonshot AI	Link	150B	-	RLHF	Multi-modal	Open	✓
Phi-4 [148]	Microsoft	Link	28B, 70B, 140B	DPO	SFT + DPO	Single Model	Closed	✓
Chameleon [149]	Meta AI	Link	34B	-	SFT	Single Model	Open	✓
Cerebrasgpt [150]	Cerebras	Link	13B	-	SFT	Single Model	Open	✗
Bloomberggpt [151]	Bloomberg L.P.	Link	50B	-	SFT	Single Model	Closed	✗
Chinchilla [152]	DeepMind	Link	70B	RLHF, PPO	SFT	Single Model	Closed	✗

TABLE 1: An overview of reinforcement learning-enhanced LLMs, where '141B-A39B' denotes a Mixture of Experts (MoE) model with 141 billion total parameters, of which 39 billion are utilized during inference. TTS stands for Test-Time Scaling.

A notable early example is Barto's Actor-Critic model [159], where the critic uses a linear function $V_\phi(s_t)$ and the actor follows a linear policy. Modern methods like A2C (Advantage Actor-Critic) [156] and A3C (Asynchronous Advantage Actor-Critic) [157] extend this approach by parallelizing training across multiple environments, leading to faster and more stable learning. By leveraging the critic's value estimation, actor-critic methods stabilize learning, improve sample efficiency, and accelerate convergence, making them more effective for complex decision-making tasks.

Connection with Modern Methods. The aforementioned early RL methods—REINFORCE [116], MIXER [118], SeqGAN [160], SCST [119], MRT [153], and actor-critic algorithms established the mathematical foundations for sequential reasoning in LLMs. These methods provided initial solutions to challenges such as exposure bias and high variance. Modern techniques such as large-scale RL from Human Feedback (RLHF) using PPO [75] and advanced reward models, e.g., Group Relative Policy Optimization (GRPO) [161] build directly upon these ideas. By integrating sophisticated reward signals and leveraging efficient policy updates, contemporary LLMs achieve improved reasoning, safety, and alignment with human values and pave the way for robust multi-step reasoning and improved quality of generated text. Table 1 provides an overview of recent models, including their parameters, architecture types, and the distilled RL methods employed, along with links for easy access.

3 Reinforced LLMs

From a methodological perspective, the integration of RL into LLM reasoning typically follows three core steps:

- 1) **Supervised Fine-Tuning (SFT):** Commences with a pretrained language model that is subsequently refined on a supervised dataset of high-quality, human-crafted examples. This phase ensures the model acquires a baseline compliance with format and style guidelines.
- 2) **Reward Model (RM) Training:** Generated outputs from the fine-tuned model are collected and subjected to human preference labeling. The reward model is then trained to replicate these label-based scores or rankings, effectively learning a continuous reward function that maps response text to a scalar value.
- 3) **RL Fine-Tuning:** Finally, the main language model is optimized via a policy gradient algorithm most e.g. PPO to maximize the reward model's output. By iterating this loop, the LLM learns to produce responses that humans find preferable along key dimensions such as accuracy, helpfulness, and stylistic coherence.
- 4) **Reward Modeling and Alignment:** Sophisticated reward functions are developed—drawing from human preferences, adversarial feedback, or automated metrics—to guide the model toward outputs that are coherent, safe, and contextually appropriate. These rewards are critical for effective credit assignment across multi-step reasoning processes.
- 5) **Pairwise preference:** For two responses y_j and y_k to the same query x , an annotator indicates whether y_j is

state s_t 期望值的优越程度。具有学习率 α 的策略更新如下:

$$\theta \leftarrow \theta + \alpha A(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t | s_t).$$

同时，评判器使用时序差分学习进行更新，最小化其估计值与实际回报之间的平方误差:

$$\phi \leftarrow \phi - \beta \nabla_\phi (V_\phi(s_t) - G_t)^2.$$

其中 β 是评判器的学习率。为了增强稳定性和效率，已提出了若干改进方法。资格迹允许从近期状态中学习，从而实现更快的收敛。使用神经网络进行函数逼近确保了高维输入的有效处理。高级变体如自然梯度方法 [158] 利用费雪信息矩阵调整更新，提高了收敛速度。一个值得注意的早期例子是巴托的演员-评论家模型 [159]，其中评论家使用线性函数 $V_\phi(s_t)$ ，而演员遵循线性策略。现代方法如 A2C（优势演员-评论家）[156] 和 A3C（异步优势演员-评论家）[157] 通过并行化多个环境中的训练来扩展这一方法，从而实现更快、更稳定的学习。通过利用评论家的价值估计，演员-评论家方法稳定了学习过程，提高了样本效率，并加速了收敛，使其在处理复杂决策任务时更为有效。

Connection with Modern Methods. 上述早期的强化学习方法——REINFORCE [116]、MIXER [118]、SeqGAN [160]、SCST [119]、MRT [153] 以及演员-评论家算法，为大型语言模型中的序列推理奠定了数学基础。这些方法为诸如曝光偏差和高方差等挑战提供了初步解决方案。现代技术，例如使用近端策略优化（PPO）[75] 和先进奖励模型（例如，组相对策略优化 [GRPO] [161]）的大规模人类反馈强化学习（RLHF），直接建立在这些思想之上。通过整合复杂的奖励信号并利用高效的策略更新，当代大型语言模型实现了改进的推理能力、安全性以及与人类价值观的对齐，并为稳健的多步推理和生成文本质量的提升铺平了道路。表 1 概述了最近的模型，包括其参数、架构类型以及所采用的蒸馏强化学习方法，并附有便于访问的链接。

3 Reinforced LLMs

从方法论的角度来看，将强化学习融入大语言模型的推理过程通常遵循三个核心步骤:

- 1) **Supervised Fine-Tuning (SFT):** 从一个预训练的语言模型开始，随后在一个由高质量人工编写示例组成的监督数据集上进行微调。此阶段确保模型获得对格式和风格指南的基本遵从性。
- 2) **Reward Model (RM) 训练:** 收集来自微调后模型的生成输出，并进行人类偏好标注。然后训练奖励模型以复现这些基于标注的分数或排名，从而有效地学习一个将响应文本映射到标量值的连续奖励函数。
- 3) **RL Fine-Tuning:** 最后，通过策略梯度算法（最常见的是近端策略优化）来优化主语言模型，以最大化奖励模型的输出。通过迭代此循环，大语言模型学会生成在准

确性、帮助性和风格连贯性等关键维度上更受人类偏好的响应。

- 4) **Reward Modeling and Alignment:** 开发复杂的奖励函数——借鉴人类偏好、对抗性反馈或自动化指标——以引导模型产生连贯、安全且符合上下文的输出。这些奖励对于在多步推理过程中进行有效的信用分配至关重要。
- 5) **Pairwise preference:** 对于同一查询 x 的两个响应 y_j 和 y_k ，标注者需指明 y_j 是否优于 y_k 。
- 6) **Rankings:** 候选响应的部分或完全排序，例如 $y_{j_1} \succ y_{j_2} \succ \dots \succ y_{j_{m_x}}$ 。
- 7) 每个 *state* s 在此可解释为用于生成下一个词元的部分对话或部分生成过程（在语言建模中）。
- 8) 每个 *action* a 是要生成的下一个词元（或下一段文本）。
- 9) *policy* $\pi_\phi(a | s)$ 是下一个词元的条件分布，由 ϕ 参数化。
- 10) 我们将 $R_\phi(x, y)$ 解释为对生成响应 y 的 *immediate* 或 *terminal* 奖励。
- 11) 因此，策略的 *future returns* 考虑了后续词元被 R_ϕ 给予正面评分的可能性。
- 12) *advantage* 函数仍然捕捉特定生成步骤相较于基线性能 $V(s_t)$ 的优越程度。
- 13) **DeepSeek-R1-Zero** 采用纯粹的强化学习方法，不包含任何 SFT。
- 14) **DeepSeek-R1** 结合了 *cold-start* 数据，并应用了多阶段训练流程。
- 15) **Outcome Reward Models (ORM):** 仅评判最终答案（例如数学解的正确性）。
- 16) **Process Reward Models (PRM):** 评估推理步骤（例如思维链中的逻辑连贯性），提供细粒度反馈以剪除无效路径。

早期使大语言模型与人类偏好对齐的方法利用了经典的强化学习算法，例如近端策略优化 [75] 和信任区域策略优化 [162]。这些算法通过最大化期望累积奖励来优化策略，同时通过代理目标函数和 KL 散度正则化对策略更新施加约束 [163]。针对可扩展的基于偏好的优化方法，已出现改进的替代方案，例如直接偏好优化（DPO）[57, 164] 和组相对策略优化（GRPO）[161, 59, 16]，它们将对齐目标重新表述为基于排序的对比损失函数 [165]，作用于人工标注的偏好数据。与依赖显式奖励模型和评论家网络的 PPO 和 TRPO [162] 不同，DPO 和 GRPO 分别通过利用对数似然比和组间奖励比较直接优化策略，从而在保持偏好一致学习动态的同时，消除了对显式价值函数近似的需求。这一从经典的基于，引入了诸如对比排序损失、策略似然比正则化和分组优势估计等新形式，这些内容将在后续章节中详细说明。

3.1 Reward modeling

令 $\mathcal{X}$ 为可能的 *queries*（例如用户提示）的集合。对于每个查询 $x \in \mathcal{X}$ ，我们收集一个或多个 *candidate responses* $\{y_j\}_{j=1}^{m_x}$

preferred to y_k .

- 6) **Rankings:** A partial or total ordering of the candidate responses, e.g. $y_{j_1} \succ y_{j_2} \succ \dots \succ y_{j_{m_x}}$.
- 7) Each *state* s here can be interpreted as the partial dialogue or partial generation process for the next token (in language modeling).
- 8) Each *action* a is the next token (or next chunk of text) to be generated.
- 9) The *policy* $\pi_\theta(a | s)$ is a conditional distribution over the next token, parameterized by ϕ .
- 10) We interpret $R_\theta(x, y)$ as the *immediate* or *terminal* reward for the generated response y .
- 11) The policy's *future returns* thus factor in how likely subsequent tokens are to be positively scored by R_θ .
- 12) The *advantage* function still captures how much better a particular generation step is compared to the baseline performance $V(s_t)$.

Early approaches to aligning LLMs with human preferences leveraged classical RL algorithms, such as PPO [75] and Trust Region Policy Optimization (TRPO) [162], which optimize a policy by maximizing the expected cumulative reward while enforcing constraints on policy updates via a surrogate objective function and KL-divergence regularization [163]. Improved alternatives to these methods for scalable preference-based optimization have emerged, such as Direct Preference Optimization (DPO) [57, 164] and Group Relative Policy Optimization (GRPO) [161, 59, 16], which reformulate the alignment objective as a ranking-based contrastive loss function [165] over human-labeled preference data. Unlike PPO and TRPO [162], which rely on explicit reward models and critic networks, DPO and GRPO directly optimize the policy by leveraging log-likelihood ratios and group-wise reward comparisons, respectively, eliminating the need for explicit value function approximation while preserving preference-consistent learning dynamics. This transition from classical RL-based alignment to preference-based direct optimization introduces novel formulations such as contrastive ranking loss, policy likelihood ratio regularization, and grouped advantage estimation, which are explained in subsequent sections.

3.1 Reward modeling

Let $\mathcal{X}$ be the space of possible *queries* (e.g., user prompts). For each query $x \in \mathcal{X}$, we collect one or more *candidate responses* $\{y_j\}_{j=1}^{m_x}$ where m_x is the number of candidate responses for query x . Typically, these responses are generated by a language model or policy under different sampling or prompting conditions. Human annotators provide preference judgments for these responses. These can take various forms:

We denote such human preference data by $\{r_j\}$ for each response or pair, where r_j might be a label, a rank, or an index indicating preference level. The overall dataset $\mathcal{D}$ then consists of N annotated examples:

$$\mathcal{D} = \left\{ (x^i, \{y_j^i\}_{j=1}^{m_i}, \{\text{preferences}^i\}) \right\}_{i=1}^N.$$

In practice, a large number of queries x are sampled from real or simulated user requests. Candidate responses $\{y_j\}_{j=1}^{m_x}$ are generated by either sampling from a base language model or using beam search or other decoding strategies. Human annotators then provide pairwise or ranking feedback on

which responses are better (or worse) according to predefined criteria (e.g., quality, correctness, helpfulness, etc.). We train a parametric model Reward Model ($R_\theta(x, y)$), referred to as the *reward model*, to map each (query, response) pair (x, y) to a scalar score. The goal is for R_θ to reflect the *alignment* or *preference* level, such that:

$$R_\theta : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}.$$

Here $\mathcal{Y}$ is the space of all possible responses.

To train R_θ , we use the human preference labels in $\mathcal{D}$ to define a suitable *ranking-based* loss, as explained below.

I. Bradley–Terry Model (Pairwise). For pairwise preferences, Bradley–Terry model [166] is often used. Suppose the dataset indicates that, for a given query x , human annotators prefer y_j to y_k , we denote it as $y_j \succ y_k$. Under Bradley–Terry, the probability of y_j being preferred over y_k is given by:

$$P(y_j \succ y_k | x; \theta) = \frac{\exp(R_\theta(x, y_j))}{\exp(R_\theta(x, y_j)) + \exp(R_\theta(x, y_k))}.$$

We train R_θ by *maximizing* the likelihood of observed preferences (or equivalently *minimizing* the negative log-likelihood):

$$\mathcal{L}_{\text{BT}}(\theta) = - \sum_{(x, y_j \succ y_k) \in \mathcal{D}} \log P(y_j \succ y_k | x; \theta).$$

II. Plackett–Luce Model¹ (Rankings). When full or partial *rankings* of m responses are available, i.e.,

$$y_{j_1} \succ y_{j_2} \succ \dots \succ y_{j_m},$$

the Plackett–Luce model [167] factorizes the probability of this ranking as:

$$P(y_{j_1}, \dots, y_{j_m} | x; \theta) = \prod_{\ell=1}^m \frac{\exp(R_\theta(x, y_{j_\ell}))}{\sum_{k=\ell}^m \exp(R_\theta(x, y_{j_k}))}.$$

Its negative log-likelihood is:

$$\mathcal{L}_{\text{PL}}(\theta) = - \sum_{(x, \text{rank}) \in \mathcal{D}} \sum_{\ell=1}^m \log \left(\frac{\exp(R_\theta(x, y_{j_\ell}))}{\sum_{k=\ell}^m \exp(R_\theta(x, y_{j_k}))} \right).$$

In practice, one minimizes the sum (or average) of the chosen ranking-based loss over all preference data:

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(x, \{y_j\}, \text{prefs}) \in \mathcal{D}} \mathcal{L}_{\text{ranking}}(\theta; x, \{y_j\}, \text{prefs}),$$

where $\mathcal{L}_{\text{ranking}}$ could be either $\mathcal{L}_{\text{BT}}$ or $\mathcal{L}_{\text{PL}}$. While the reward model $R_\theta(x, y)$ provides a *scalar reward signal* reflecting human preferences, this connects to common RL concepts, especially the *advantage function*.

 **Reward modeling uses ranking-based losses to learn a function from human preferences for policy optimization.**

Reward modeling Types. Rewards can be categorized into explicit and implicit approaches.

1. <https://hturner.github.io/PlackettLuce/>

其中 m_x 是查询 x 的候选响应数量。通常，这些响应是由语言模型或策略在不同的采样或提示条件下生成的。人工标注者为这些响应提供偏好判断。这些判断可以采取多种形式：

我们将此类人类偏好数据记为 $\{r_j\}$ ，用于每个响应或响应对，其中 r_j 可能是一个标签、一个排名或一个表示偏好程度的索引。整体数据集 $\mathcal{D}$ 则由 N 个标注样本组成：

$$\mathcal{D} = \left\{ (x^i, \{y_j^i\}_{j=1}^{m_i}, \{\text{preferences}^i\}) \right\}_{i=1}^N.$$

在实践中，大量查询 x 是从真实或模拟的用户请求中采样的。候选响应 $\{y_j\}_{j=1}^{m_x}$ 通过从基础语言模型采样或使用束搜索等解码策略生成。然后，人工标注者根据预定义的标准（例如质量、正确性、帮助性等）提供关于哪些响应更好（或更差）的成对比较或排序反馈。我们训练一个参数化模型——奖励模型 ($R_\theta(x, y)$)，称为 *reward model*，将每个（查询，响应）对 (x, y) 映射到一个标量分数。目标是使 R_θ 反映 *alignment* 或 *preference* 水平，使得：

$$R_\theta : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}.$$

这里 $\mathcal{Y}$ 是所有可能响应的集合。

为了训练 R_θ ，我们使用 $\mathcal{D}$ 中的人类偏好标签来定义一个合适的 *ranking-based* 损失，如下所述。

I. Bradley–Terry Model (Pairwise). 对于成对偏好，通常使用 Bradley–Terry 模型 [166]。假设数据集表明，对于给定查询 x ，人工标注者偏好 y_j 胜于 y_k ，我们将其记为 $y_j \succ y_k$ 。在 Bradley–Terry 模型下， y_j 被偏好于 y_k 的概率由下式给出：

$$P(y_j \succ y_k | x; \theta) = \frac{\exp(R_\theta(x, y_j))}{\exp(R_\theta(x, y_j)) + \exp(R_\theta(x, y_k))}.$$

我们通过 *maximizing* 观测到的偏好的似然（或等价地 *minimizing* 负对数似然）来训练 R_θ ：

$$\mathcal{L}_{\text{BT}}(\theta) = - \sum_{(x, y_j \succ y_k) \in \mathcal{D}} \log P(y_j \succ y_k | x; \theta).$$

II. Plackett–Luce Model¹ (排名). 当可获得 m 响应的完整或部分 *rankings* 时，即

$$y_{j_1} \succ y_{j_2} \succ \dots \succ y_{j_m},$$

Plackett–Luce 模型 [167] 将此排名的概率分解为：

$$P(y_{j_1}, \dots, y_{j_m} | x; \theta) = \prod_{\ell=1}^m \frac{\exp(R_\theta(x, y_{j_\ell}))}{\sum_{k=\ell}^m \exp(R_\theta(x, y_{j_k}))}.$$

其负对数似然为：

$$\mathcal{L}_{\text{PL}}(\theta) = - \sum_{(x, \text{rank}) \in \mathcal{D}} \sum_{\ell=1}^m \log \left(\frac{\exp(R_\theta(x, y_{j_\ell}))}{\sum_{k=\ell}^m \exp(R_\theta(x, y_{j_k}))} \right).$$

在实践中，人们最小化所有偏好数据上所选基于排名的损失之和（或平均值）：

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(x, \{y_j\}, \text{prefs}) \in \mathcal{D}} \mathcal{L}_{\text{ranking}}(\theta; x, \{y_j\}, \text{prefs}),$$

1. <https://hturner.github.io/PlackettLuce/>

其中 $\mathcal{L}_{\text{ranking}}$ 可以是 $\mathcal{L}_{\text{BT}}$ 或 $\mathcal{L}_{\text{PL}}$ 。虽然奖励模型 $R_\theta(x, y)$ 提供了一个反映人类偏好的 *scalar reward signal*，但这与常见的强化学习概念相关联，尤其是 *advantage function*。

 **奖励建模使用基于排名的损失来学习一个从人类偏好到策略优化的函数。**

Reward modeling Types. 奖励可分为显式和隐式方法。

3.1.1 Explicit Reward Modeling

显式奖励建模直接基于预定义的规则、启发式方法或人工标注来定义奖励函数。这种奖励结构涉及来自人类的直接数值信号，或来自经过训练以近似人类判断（例如排序或成对比较）的专用人工智能模块。该方法能产生精确的奖励估计，但在大规模应用时可能耗时或成本高昂。示例用例包括专家评估有害输出严重性的“红队”演练，或需要由领域专家验证正确性的专业领域任务。

3.1.2 Implicit Reward Modeling

隐式奖励建模从观察到的行为、交互或偏好信号中间接推断奖励，通常利用机器学习技术来揭示潜在的奖励结构。其信号来源于用户交互指标，如点赞数、采纳率、点击率模式或会话参与时长。虽然这种方法能以最小开销积累海量数据集，但它存在助长利用参与度启发式而牺牲内容质量或真实性的行为的风险。

Reward Function. 为文本生成任务定义奖励函数是一个不适定问题 [168, 169]。现有大型语言模型中的强化学习方法要么关注生成过程的结果（结果奖励建模），要么关注过程（过程奖励建模），以塑造大型语言模型的行为。下文我们将解释这两种奖励建模范式。

3.1.3 Outcome Reward Modeling

衡量最终结果（例如，最终答案是否事实正确或解决了用户的查询）。该模型易于实现，但可能对结论如何达成的洞察有限。它普遍应用于短响应任务中，用户主要关心最终陈述的正确性或简洁性。对于长响应任务，基于结果的奖励可能导致 *credit assignment problem*，即哪些具体行动或状态导致了特定的奖励结果。

3.1.4 Process Reward Modeling

在中间推理步骤分配反馈，激励连贯、逻辑一致且结构良好的思维链。这种方法对于涉及数学推导、法律论证或代码调试的任务尤其有价值，在这些任务中，得出答案的路径与最终陈述同等重要。在此类问题中，分配给单个步骤的奖励鼓励透明度和稳健的逐步推理。然而，它需要更复杂的标注过程，例如需要“黄金”推理步骤或部分得分评分。过程奖励可以与结果奖励结合，以提供强大的多阶段训练信号。

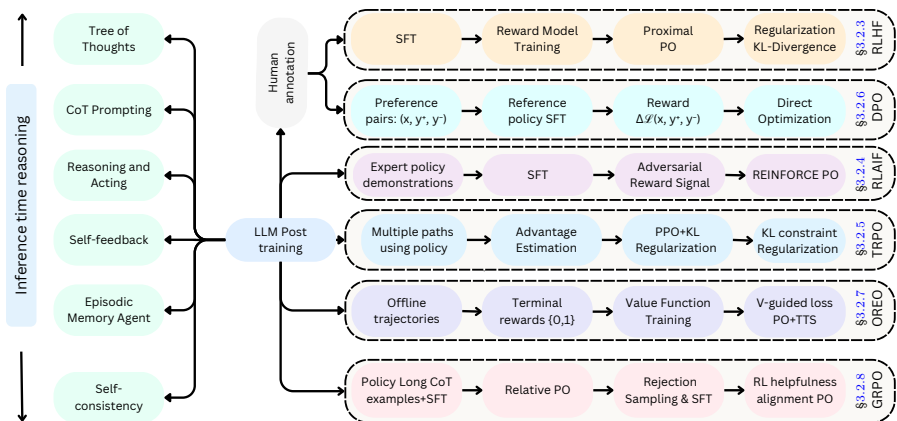


Fig. 2: Overview of Large Language Models (LLMs) reasoning methods, showcasing pathways for enhancing reasoning capabilities through approaches like Chain-of-Thought (CoT) prompting, self-feedback, and episodic memory. The diagram highlights multiple reinforcement learning-based optimization techniques, including GRPO, RLHF, DPO, and RLAIIF, for fine-tuning reasoning models with reward mechanisms and preference-based learning.

3.1.1 Explicit Reward Modeling

Explicit reward modeling defines reward functions directly based on predefined rules, heuristics, or human annotations. This reward structure involves direct, numeric signals from humans or from specialized AI modules trained to approximate human judgments (e.g., ranking or pairwise comparison). This method can produce precise reward estimates but may be time-consuming or costly at scale. Illustrative use cases include ‘red-teaming’ exercises where experts rate the severity of toxic outputs, or domain-specialist tasks in which correctness must be validated by a subject matter expert.

3.1.2 Implicit Reward Modeling

Implicit reward modeling infers rewards indirectly from observed behaviors, interactions, or preference signals, often leveraging machine learning techniques to uncover latent reward structures. It derives its signals from user interaction metrics such as upvotes, acceptance rates, click-through patterns, or session engagement times. While it can accumulate vast datasets with minimal overhead, this approach risks fostering behaviors that exploit engagement heuristics at the expense of content quality or veracity.

Reward Function. Defining a reward function for text generation tasks is an ill-posed problem [168, 169]. The existing RL methods in LLMs either focus on the generation process outcome (Outcome Reward Modeling) or the (Process Reward Modeling), to shape LLM behaviors. We explain these two reward modeling paradigms below.

3.1.3 Outcome Reward Modeling

Measures the end result (e.g., whether the final answer is factually correct or solves the user’s query). This model

is straightforward to implement but may offer limited insight into how the conclusion was reached. It is prevalent in short-response tasks, where the user’s primary concern is the correctness or succinctness of the final statement. For long-response tasks, outcome based reward can lead to *credit assignment problem*, i.e., which specific actions or states lead to a particular reward outcome.

3.1.4 Process Reward Modeling

Assigns feedback at intermediate reasoning steps, incentivizing coherent, logically consistent, and well-structured chains of thought. This approach is particularly valuable for tasks involving mathematical derivations, legal arguments, or code debugging, in which the path to the answer is as significant as the final statement. In such problems, the reward assigned in individual steps encourages transparency and robust step-by-step reasoning. However, it requires a more complex annotation process, e.g., requires ‘gold’ reasoning steps or partial credit scoring. Process rewards can be combined with outcome rewards for a strong multi-phase training signal.

💡 **Policy Reward Modeling (PRM) with last-step aggregation outperforms Outcome Reward Modeling (ORM) by leveraging final-step evaluations to optimize policy updates more effectively.**

3.1.5 Iterative RL with Adaptive Reward Models

Adaptive Reward Models is a training methodology designed to continuously improve the performance of LLMs by iteratively refining the reward models and the policy model. This approach addresses the challenges of reward hacking and reward model drift, which can occur when the reward model

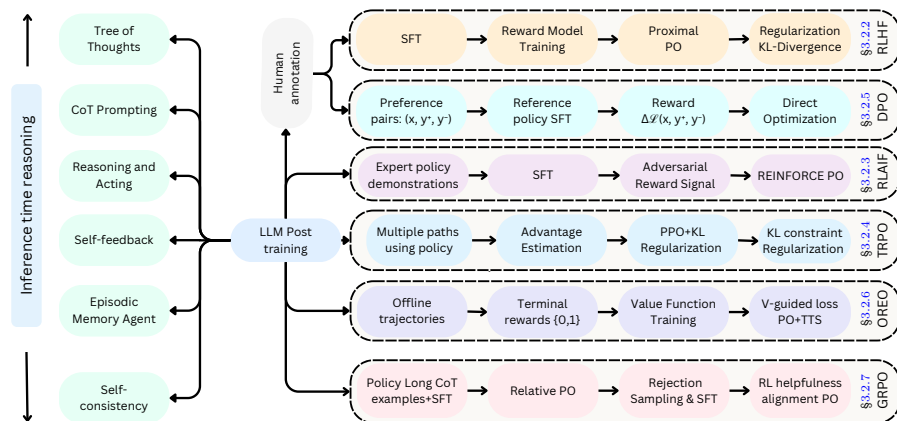


图 2: 大语言模型 (LLMs) 推理方法概述, 展示了通过思维链 (CoT) 提示、自我反馈和情景记忆等方法增强推理能力的途径。该图突出了多种基于强化学习的优化技术, 包括 GRPO、RLHF、DPO 和 RLAIIF, 用于通过奖励机制和基于偏好的学习对推理模型进行微调。

💡 **Policy Reward Modeling (PRM) 通过最后一步聚合, 利用最终步骤评估更有效地优化策略更新, 其性能优于 Outcome Reward Modeling (ORM)。**

3.1.5 Iterative RL with Adaptive Reward Models

自适应奖励模型是一种旨在通过迭代优化奖励模型与策略模型来持续提升大语言模型性能的训练方法。该方法解决了奖励破解与奖励模型漂移的挑战, 这些挑战可能在大规模强化学习训练过程中, 当奖励模型与期望目标失准时发生。强化学习过程被划分为多次迭代, 模型在循环中进行训练。每次迭代后, 奖励模型会根据最新的模型行为与人类反馈进行更新。奖励模型并非静态, 而是随时间演进, 以更好地对齐人类偏好与任务需求。这种适应机制确保随着模型改进, 奖励信号始终保持准确与相关。重复此迭代过程, 直至模型性能达到平台期或满足期望的基准。奖励模型与策略模型协同演化, 每一次迭代都使它们更接近最优对齐。

3.2 Policy Optimization

一旦我们有了一个经过训练的、能够捕捉人类偏好的奖励模型 $R_\theta(x, y)$, 我们就可以将其整合到强化学习框架中, 以 *optimize* 一个策略 π_ϕ 。本质上, 我们用 $R_\theta(x, y)$ 替换 (或增强) 环境固有的奖励信号, 从而使智能体专注于为给定的查询 x 生成人类更偏好的响应 y 。

在典型的强化学习表示法中:

我们寻求找到 ϕ , 以 *maximizes* 在 R_θ 下的期望奖励。具体来说, 令 x 为用户查询, 令 $y \sim \pi_\phi(\cdot | x)$ 为生成的响应。我们的目标是求解:

$$\max_{\phi} \mathbb{E}_{x \sim \mathcal{X}} \left[\mathbb{E}_{y \sim \pi_\phi(\cdot | x)} [R_\theta(x, y)] \right].$$

这意味着, 对于从策略 π_ϕ 中抽取的用户查询 x 和响应 y , 我们希望奖励模型的评分 $R_\theta(x, y)$ 尽可能高。

Policy Gradient and Advantage. 现代算法 (例如, PPO [75]、GRPO [59]、TRPO [162]) 依赖于 *policy gradients*。图 5 对这些主要的强化学习框架进行了结构化比较。每个框架都基于不同的策略学习、参考建模和奖励计算原则构建。回想一下, *advantage function* $A(s, a)$ 量化了一个动作 a 相对于基线期望回报 $V(s)$ 的优越程度。在高层次上, 我们沿着增加具有 *positive advantage* 的动作 a 的 $\pi_\phi(a | s)$ 的方向更新策略 π_ϕ , 并减少具有负优势的动作的优势。形式上, 时间 t 的优势函数 A_t 可以写为:

$$A_t = Q(s_t, a_t) - V(s_t),$$

其中 $Q(s_t, a_t)$ 是从 s_t 开始并采取动作 a_t 时的预期未来回报 (包括 R_θ 在内的未来奖励之和)。

当使用奖励模型 R_θ 时:

💡 **奖励模型学习的是相对偏好而非绝对分数。这避免了对校准的人类评分的需求, 并专注于成对比较。**

on average

becomes misaligned with the desired objectives during large-scale RL training. The RL process is divided into multiple iterations, where the model is trained in cycles. After each iteration, the reward model is updated based on the latest model behavior and human feedback. The reward model is not static but evolves over time to better align with human preferences and task requirements. This adaptation ensures that the reward signals remain accurate and relevant as the model improves. Repeat the iterative process until the model's performance plateaus or meets the desired benchmarks. The reward model and policy model co-evolve, with each iteration bringing them closer to optimal alignment.

3.2 Policy Optimization

Once we have a trained reward model $R_\theta(x, y)$ that captures human preferences, we can integrate it into a RL framework to *optimize* a policy π_ϕ . In essence, we replace (or augment) the environment's native reward signal with $R_\theta(x, y)$ so that the agent focuses on producing responses y that humans prefer for a given query x .

In typical RL notation:

We seek to find ϕ that *maximizes* the expected reward under R_θ . Concretely, let x be a user query, and let $y \sim \pi_\phi(\cdot | x)$ be the generated response. We aim to solve:

$$\max_{\phi} \mathbb{E}_{x \sim \mathcal{X}} \left[\mathbb{E}_{y \sim \pi_\phi(\cdot | x)} [R_\theta(x, y)] \right].$$

This means that *on average*, over user queries x and responses y drawn from the policy π_ϕ , we want the reward model's score $R_\theta(x, y)$ to be as high as possible.

Policy Gradient and Advantage. The modern algorithms (e.g., PPO [75], GRPO [59], TRPO [162]) rely on *policy gradients*. Figure 5 presents a structured comparison of these main RL frameworks. Each framework builds upon different principles for policy learning, reference modeling, and reward computation. Recall that the *advantage function* $A(s, a)$ quantifies how much better an action a is than the baseline expected return $V(s)$. At a high level, we update the policy π_ϕ in the direction that increases $\pi_\phi(a | s)$ for actions a with *positive advantage* and decreases it for negative-advantage actions. Formally, the advantage A_t at time t can be written as:

$$A_t = Q(s_t, a_t) - V(s_t),$$

where $Q(s_t, a_t)$ is the expected future return (sum of future rewards, including R_θ) starting from s_t when taking action a_t .

When using the reward model R_θ :

💡 **The reward model learns relative preferences rather than absolute scores. This avoids the need for calibrated human ratings and focuses on pairwise comparisons.**

3.2.1 Odds Ratio Preference Optimization (ORPO)

The simplest method is ORPO [170] which *directly* optimizing a policy from pairwise human preferences. Instead of first learning a separate reward model and then running standard RL, ORPO updates the policy to increase the likelihood of

preferred responses (according to human labels) relative to *dispreferred* ones. The key idea is to look at the *odds ratio*:

$$\frac{\pi_\phi(y_j | x)}{\pi_\phi(y_k | x)},$$

where y_j is the *preferred* response and y_k is the *less-preferred* response for a given query x .

Pairwise Preference Probability. In many direct preference approaches (e.g., Bradley-Terry style), one writes

$$P_\phi(y_j \succ y_k | x) = \sigma\left(\ln \frac{\pi_\phi(y_j | x)}{\pi_\phi(y_k | x)}\right) = \frac{1}{1 + \exp\left(\ln \frac{\pi_\phi(y_k | x)}{\pi_\phi(y_j | x)}\right)},$$

where $\sigma(\cdot)$ is the logistic (sigmoid) function. Intuitively, if the policy π_ϕ assigns higher probability to y_j than to y_k , the *odds* $\frac{\pi_\phi(y_j | x)}{\pi_\phi(y_k | x)}$ exceed 1, making y_j more likely to be the *preferred* outcome under the model.

In ORPO, one typically defines a *negative log-likelihood* loss for all pairs $\{(x, y_j \succ y_k)\}$ in the dataset:

$$\mathcal{L}_{\text{ORPO}}(\phi) = - \sum_{(x, y_j \succ y_k) \in \mathcal{D}} \log(P_\phi(y_j \succ y_k | x)).$$

Substituting the logistic form gives:

$$\mathcal{L}_{\text{ORPO}}(\phi) = - \sum_{(x, y_j \succ y_k) \in \mathcal{D}} \log\left(\frac{\pi_\phi(y_j | x)}{\pi_\phi(y_j | x) + \pi_\phi(y_k | x)}\right),$$

which can also be interpreted as maximizing the *log odds ratio* for the correct (preferred) label in each pairwise comparison.

Interpretation via Odds Ratios. By treating each preference label $(y_j \succ y_k)$ as a constraint on the *odds* $\frac{\pi_\phi(y_j | x)}{\pi_\phi(y_k | x)}$, ORPO pushes the policy to increase its probability mass on y_j while decreasing it on y_k . When viewed in logarithmic space:

$$\ln\left(\frac{\pi_\phi(y_j | x)}{\pi_\phi(y_k | x)}\right),$$

a higher value corresponds to a greater likelihood of selecting y_j over y_k . Hence, minimizing $\mathcal{L}_{\text{ORPO}}(\phi)$ aligns π_ϕ with the human-labeled preferences.

⚠️ **Odds Ratio Preference Optimization (ORPO) is potentially less flexible for combining multiple reward signals.**

3.2.2 Proximal Policy Optimization (PPO) in LLMs

A popular method for policy optimization is PPO [75], a strategy adapted to align LLMs with human feedback. Given a policy π_θ parameterized by θ and a reward function R , PPO updates the policy by optimizing a clipped objective that balances exploration and stability. Specifically, if $r_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{ref}}}(a_t | s_t)}$ denotes the probability ratio for an action a_t in state s_t , the clipped PPO objective is:

$$\mathcal{L}^{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta) A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t) \right],$$

where A_t is an estimator of the advantage function and ϵ is a hyperparameter controlling the allowable deviation from the previous policy. A_t is computed using Generalized Advantage Estimation (GAE) [171] based on rewards and a learned value function. The clipping objective of PPO restricts how drastically the updated policy distribution can diverge from the

3.2.1 Odds Ratio Preference Optimization (ORPO)

最简单的方法是 ORPO [170], 它 *directly* 通过对人类偏好来优化策略。与先学习一个独立的奖励模型再运行标准强化学习不同, ORPO 更新策略以提高 *preferred* 响应 (根据人类标注) 相对于 *dispreferred* 响应的可能性。其核心思想是考察 *odds ratio*:

$$\frac{\pi_\phi(y_j | x)}{\pi_\phi(y_k | x)},$$

其中 y_j 是给定查询 x 的 *preferred* 响应, 而 y_k 是 *less-preferred* 响应。

Pairwise Preference Probability. 在许多直接偏好方法中 (例如, Bradley-Terry 风格), 人们写作:

$$P_\phi(y_j \succ y_k | x) = \sigma\left(\ln \frac{\pi_\phi(y_j | x)}{\pi_\phi(y_k | x)}\right) = \frac{1}{1 + \exp\left(\ln \frac{\pi_\phi(y_k | x)}{\pi_\phi(y_j | x)}\right)},$$

其中 $\sigma(\cdot)$ 是逻辑 (Sigmoid) 函数。直观上, 如果策略 π_ϕ 分配给 y_j 的概率高于分配给 y_k 的概率, 则 *odds* $\frac{\pi_\phi(y_j | x)}{\pi_\phi(y_k | x)}$ 将超过 1, 使得 y_j 在模型下更可能成为 *preferred* 结果。

在 ORPO 中, 通常为数据集集中的所有配对 $\{(x, y_j \succ y_k)\}$ 定义一个 *negative log-likelihood* 损失:

$$\mathcal{L}_{\text{ORPO}}(\phi) = - \sum_{(x, y_j \succ y_k) \in \mathcal{D}} \log(P_\phi(y_j \succ y_k | x)).$$

代入逻辑函数形式得到:

$$\mathcal{L}_{\text{ORPO}}(\phi) = - \sum_{(x, y_j \succ y_k) \in \mathcal{D}} \log\left(\frac{\pi_\phi(y_j | x)}{\pi_\phi(y_j | x) + \pi_\phi(y_k | x)}\right),$$

这也可以解释为在每次成对比较中最大化正确 (偏好) 标签的 *log odds ratio*。

Interpretation via Odds Ratios. 通过将每个偏好标签 $(y_j \succ y_k)$ 视为对 *odds* $\frac{\pi_\phi(y_j | x)}{\pi_\phi(y_k | x)}$ 的约束, ORPO 推动策略增加其在 y_j 上的概率质量, 同时减少在 y_k 上的概率质量。在对数空间中观察:

$$\ln\left(\frac{\pi_\phi(y_j | x)}{\pi_\phi(y_k | x)}\right),$$

更高的值对应于选择 y_j 而非 y_k 的更大可能性。因此, 最小化 $\mathcal{L}_{\text{ORPO}}(\phi)$ 使 π_ϕ 与人类标注的偏好保持一致。

⚠️ **优势比偏好优化 (ORPO) 在结合多个奖励信号时可能灵活性较低。**

一种流行的策略优化方法是 PPO [75], 这是一种为 LLMs 与人类反馈对齐而调整的策略。给定一个由 θ 参数化的策略 π_θ 和一个奖励函数 R , PPO 通过优化一个平衡探索与稳定性的裁剪目标来更新策略。具体而言, 如果 $r_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{ref}}}(a_t | s_t)}$ 表示状态 s_t 下动作 a_t 的概率比, 则裁剪后的 PPO 目标为:

$$\mathcal{L}^{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta) A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t) \right],$$

其中 A_t 是优势函数的估计器, ϵ 是控制允许偏离先前策略程度的超参数。 A_t 是基于奖励和一个已学习的价值函数, 使用

广义优势估计 (GAE) [171] 计算得出的。PPO 的裁剪目标限制了更新后的策略分布与原始策略的偏离程度。这种调节避免了语言生成的灾难性转变, 并保持了训练稳定性。

Policy Optimization with KL Penalty. 在使用 PPO 进行 RL 微调期间, 策略 π 被优化以最大化奖励, 同时保持接近基础模型 ρ 。修改后的奖励函数包含一个 KL 散度惩罚项:

$$J(\pi) = \mathbb{E}_{(x, y) \sim \mathcal{D}} [r(x, y) - \beta \text{KL}(\pi(\cdot | x) \parallel \rho(\cdot | x))],$$

其中 β 控制惩罚强度。KL 项 $\text{KL}(\pi \parallel \rho)$ 防止对代理奖励 $r(x, y)$ 的过度优化 (即奖励黑客行为)。

💡 **KL 惩罚是一种正则化, 它确保策略保留基础模型的语言连贯性, 并避免产生退化的输出。**

3.2.2 Reinforcement Learning from Human Feedback (RLHF)

RLHF [58] 通过直接的人类偏好信号来精炼 LLMs, 使其更符合人类期望。该过程包含三个主要步骤。首先, 使用高质量标注数据对预训练模型执行 SFT, 以建立强大的语言和事实能力。其次, 使用人类标注的生成响应排名来训练奖励函数 R , 使其能够预测偏好并提供标量奖励信号。第三, 在 RLHF [58] 流程中采用 PPO, 利用人类提供的偏好分数 (或排名) 来塑造 R , 从而指导策略更新。这确保了模型优先输出符合人类偏好行为的结果。在噪声或部分奖励信号条件下, 其稳健的性能使得 PPO 非常适合文本生成任务, 因为这类任务通常具有巨大的动作空间和细微的奖励定义。

3.2.3 Reinforcement Learning from AI Feedback (RLAIF)

RLAIF [97] 是 RLHF 的一种替代方案, 它用人工智能生成的反馈取代了人类标注。RLAIF 不依赖人类标注的偏好, 而是利用一个次要的、能力强大的语言模型来生成偏好标签, 然后使用这些标签来训练奖励模型。该奖励模型指导对目标模型进行基于强化学习的微调。RLAIF 通过消除对人类标注员的需求, 降低了数据收集所需的成本和时间。它使得大规模模型对齐成为可能, 无需大量人工干预, 同时保持高性能和对齐度。实证研究表明, RLAIF [97, 172] 是 RLHF 的一个可扩展且高效的替代方案, 使其成为强化学习驱动的语言模型优化的一个有前景的方向。

💡 **裁剪机制将策略更新限制在安全的信任区域内, 这在处理复杂、高维动作空间时至关重要。**

3.2.4 Trust Region Policy Optimization (TRPO)

TRPO [162] 是另一种广泛使用的策略优化方法, 它先于 PPO 出现, 并共享其基本目标: 提高强化学习更新的稳定性。TRPO 在优化策略更新的同时, 确保其保持在由 KL 散度量度的约束信任区域内。

original policy. This moderation averts catastrophic shifts in language generation and preserves training stability.

Policy Optimization with KL Penalty. During RL fine-tuning with PPO, the policy π is optimized to maximize reward while staying close to the base model ρ . The modified reward function includes a KL divergence penalty:

$$J(\pi) = \mathbb{E}_{(x,y) \sim \mathcal{D}} [r(x,y) - \beta \text{KL}(\pi(\cdot|x) \parallel \rho(\cdot|x))],$$

where β controls the penalty strength. The KL term $\text{KL}(\pi \parallel \rho)$ prevents over-optimization to the proxy reward $r(x,y)$ (i.e., reward hacking).

💡 **The KL penalty is a regularization, which ensure policy retains the base model's linguistic coherence and avoids degenerate outputs.**

3.2.3 Reinforcement Learning from Human Feedback (RLHF)

RLHF [58] refines LLMs through direct human preference signals, making them more aligned with human expectations. The process involves three main steps. First, SFT is performed on a pretrained model using high-quality labeled data to establish strong linguistic and factual capabilities. Second, a reward function R is trained using human-annotated rankings of generated responses, allowing it to predict preferences and provide a scalar reward signal. Third, PPO is employed in the RLHF [58] pipeline by using human-provided preference scores (or rankings) to shape R and thereby guide the policy updates. This ensures that the model prioritizes outputs aligned with human-preferred behavior. The robust performance under conditions of noisy or partial reward signals makes PPO well-suited for text generation tasks, where large action spaces and nuanced reward definitions are common.

3.2.4 Reinforcement Learning from AI Feedback (RLAIF)

RLAIF [97] is an alternative to RLHF that replaces human annotation with AI-generated feedback. Instead of relying on human-labeled preferences, RLAIF employs a secondary, highly capable language model to generate preference labels, which are then used to train a reward model. This reward model guides reinforcement learning-based fine-tuning of the target model. RLAIF reduces the cost and time required for data collection by eliminating the need for human annotators. It enables large-scale model alignment without requiring extensive human intervention while maintaining high performance and alignment. Empirical studies indicate that RLAIF [97, 172] is a scalable and efficient alternative to RLHF, making it a promising direction for reinforcement learning-driven language model optimization.

💡 **The clipping mechanism constrains policy updates to remain within a safe trust region, which is crucial when dealing with complex, high-dimensional action spaces.**

3.2.5 Trust Region Policy Optimization (TRPO)

TRPO [162] is another widely used policy optimization method, preceding PPO and sharing its fundamental goal: improving stability in reinforcement learning updates. TRPO optimizes

policy updates while ensuring they remain within a constrained trust region, measured by KL divergence.

Instead of using a clipped objective like PPO, TRPO enforces a hard constraint on policy updates by solving the following optimization problem:

$$\max_{\theta} \mathbb{E}_t \left[\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} A_t \right]$$

subject to the constraint:

$$\mathbb{E}_t [D_{KL}(\pi_{\theta_{\text{old}}}(\cdot | s_t) \parallel \pi_{\theta}(\cdot | s_t))] \leq \delta.$$

where δ is a hyperparameter that controls how much the new policy can diverge from the old one.

Unlike PPO, which approximates this constraint using clipping, TRPO directly solves a constrained optimization problem, ensuring each update does not move too far in policy space. However, solving this constrained problem requires computationally expensive second-order optimization techniques like conjugate gradient methods, making TRPO less efficient for large-scale models like LLMs. In practice, PPO is preferred over TRPO due to its simplicity, ease of implementation, and comparable performance in large-scale applications like RLHF. However, TRPO remains an important theoretical foundation for stable policy optimization in deep reinforcement learning.

3.2.6 Direct Preference Optimization (DPO)

DPO [164] is a recently proposed method for training LLMs from human preference data without resorting to the traditional RL loop (as in RLHF with PPO). Instead of learning a separate reward function and then running policy-gradient updates, DPO directly integrates human preference signals into the model's training objective. So instead of the above PPO objective, DPO instead constructs an objective that directly pushes up the probability of a chosen (preferred) response (y^+) while pushing down the probability of a less-preferred response (y^-), all within a single log-likelihood framework. Rather than bounding policy changes with clip, the DPO loss uses the difference between log probabilities of 'winning' vs. 'losing' responses. This explicitly encodes the user's preference in the updated parameters.

Here, π_{θ} is the learnable policy, π_{ref} is a reference policy (often the SFT-trained model), $\sigma(\cdot)$ is the sigmoid function, β is a scaling parameter, and $\mathcal{D}_{\text{train}}$ is a dataset of triplets (x, y^+, y^-) where y^+ is the preferred output over y^- .

$$\mathcal{L}^{\text{DPO}}(\theta) = \mathbb{E}_{(x, y^+, y^-) \sim \mathcal{D}_{\text{train}}} \left[\sigma \left(\beta \log \frac{\pi_{\theta}(y^+ | x)}{\pi_{\text{ref}}(y^+ | x)} - \beta \log \frac{\pi_{\theta}(y^- | x)}{\pi_{\text{ref}}(y^- | x)} \right) \right].$$

The key insight is that an LLM can be treated as a "hidden reward model": we can reparameterize preference data so that the model's own log probabilities reflect how preferable one response is over another. By directly adjusting the log-likelihood of more-preferred responses relative to less-preferred ones, DPO sidesteps many complexities of RL-based methods (e.g., advantage functions or explicit clipping).

与使用裁剪目标的 PPO 不同, TRPO 通过解决以下优化问题来强制执行策略更新的硬约束:

$$\max_{\theta} \mathbb{E}_t \left[\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} A_t \right]$$

其约束条件为:

$$\mathbb{E}_t [D_{KL}(\pi_{\theta_{\text{old}}}(\cdot | s_t) \parallel \pi_{\theta}(\cdot | s_t))] \leq \delta.$$

其中 δ 是一个超参数, 用于控制新策略与旧策略的偏离程度。

与使用裁剪来近似此约束的 PPO 不同, TRPO 直接求解一个约束优化问题, 确保每次更新在策略空间中不会移动太远。然而, 求解这个约束问题需要计算成本高昂的二阶优化技术 (如共轭梯度法), 这使得 TRPO 对于像 LLMs 这样的大规模模型效率较低。在实践中, 由于 PPO 的简单性、易于实现性以及其在 RLHF 等大规模应用中可比的性能, 它比 TRPO 更受青睐。尽管如此, TRPO 仍然是深度强化学习中稳定策略优化的重要理论基础。

3.2.5 Direct Preference Optimization (DPO)

DPO [164] 是一种最近提出的方法, 用于根据人类偏好数据训练 LLMs, 而无需诉诸传统的 RL 循环 (如在采用 PPO 的 RLHF 中)。与学习一个单独的奖励函数然后运行策略梯度更新不同, DPO 直接将人类偏好信号整合到模型的训练目标中。因此, 不同于上述 PPO 目标, DPO 构建了一个目标, 该目标直接提升被选中 (偏好) 的响应 (y^+) 的概率, 同时降低低劣响应 (y^-) 的概率, 所有这些都在一个单一的对数似然框架内完成。DPO 损失不使用裁剪来限制策略变化, 而是利用 "获胜" 响应与 "失败" 响应的对数概率之差。这明确地将用户的偏好编码到更新后的参数中。

此处, π_{θ} 是可学习的策略, π_{ref} 是参考策略 (通常是经过 SFT 训练的模型), $\sigma(\cdot)$ 是 sigmoid 函数, β 是一个缩放参数, 而 $\mathcal{D}_{\text{train}}$ 是一个三元组 (x, y^+, y^-) 的数据集, 其中 y^+ 是优于 y^- 的偏好输出。

$$\mathcal{L}^{\text{DPO}}(\theta) = \mathbb{E}_{(x, y^+, y^-) \sim \mathcal{D}_{\text{train}}} \left[\sigma \left(\beta \log \frac{\pi_{\theta}(y^+ | x)}{\pi_{\text{ref}}(y^+ | x)} - \beta \log \frac{\pi_{\theta}(y^- | x)}{\pi_{\text{ref}}(y^- | x)} \right) \right].$$

关键洞见在于, LLM 可以被视为一个 "隐藏的奖励模型": 我们可以对偏好数据进行重新参数化, 使得模型自身的对数概率反映一个响应相对于另一个响应的偏好程度。通过直接调整更偏好响应相对于劣响应的对数似然, DPO 绕过了基于 RL 的方法的许多复杂性 (例如, 优势函数或显式裁剪)。

💡 **优势函数 $A_{\phi} = V_{\phi}(s_{t+1}) - V_{\phi}(s_t)$ 量化了每一步的贡献, 对于识别关键推理错误至关重要。这种粒度在 DPO 中丢失了, 因为 DPO 统一处理整个轨迹。**

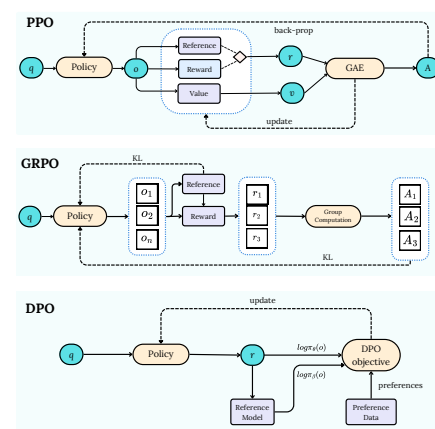


图 3: PPO [75]、GRPO [59] 与 DPO [164] 的对比。我们以相应的损失函数突出显示了策略模型、参考模型、奖励及优化流程。

Perplexity Filtering for Out-of-Distribution Data. 为确保 DPO 训练数据处于分布内 (与 ρ 对齐), 响应使用困惑度进行过滤。响应 $y = (y_1, y_2, \dots, y_T)$ 的困惑度定义为:

$$\text{PP}(y) = \exp \left(-\frac{1}{T} \sum_{i=1}^T \log P_{\rho}(y_i | y_{<i}) \right),$$

其中 y_i 是第 i 个词元。仅保留困惑度低于某个阈值 (例如, 由 ρ 生成的响应的第 95 百分位数) 的响应。

💡 **优势函数仍然是一个核心概念, 用于确定在每一步哪些动作 (词元选择) 优于基线。**

3.2.6 Offline Reasoning Optimization (OREO)

OREO [173] 是一种离线强化学习方法, 旨在通过优化软贝尔曼方程 [111] 来增强大型语言模型的多步推理能力。与依赖成对偏好数据的 DPO 不同, OREO 使用基于最终结果 (例如, 推理链的正确性) 的稀疏奖励, 并联合训练策略模型 π_{θ} 和价值函数 V_{ϕ} , 以实现细粒度的信用分配。其核心目标是最小化软贝尔曼方程中的不一致性:

$$V_{\phi}(s_t) - V_{\phi}(s_{t+1}) = r(s_t, a_t) - \beta \log \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\text{ref}}(a_t | s_t)},$$

其中 $s_{t+1} = f(s_t, a_t)$ 是下一个状态, r 是稀疏奖励, β 控制 KL 正则化。策略损失和价值损失分别为:

$$\mathcal{L}_V(\phi) = \frac{1}{T} \sum_{t=0}^{T-1} \left(V_{\phi}(s_t) - R_t + \beta \sum_{i=t}^{T-1} \log \frac{\pi_{\theta}(a_i | s_i)}{\pi_{\text{ref}}(a_i | s_i)} \right)^2,$$

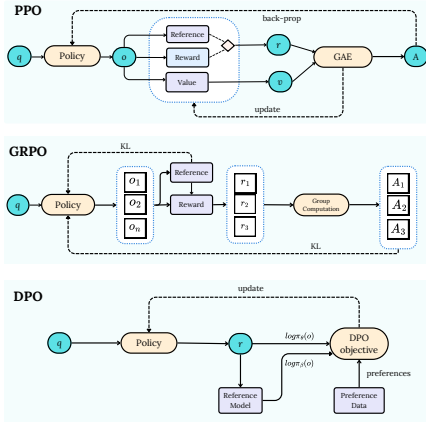


Fig. 3: Comparison of PPO [75], GRPO [59], and DPO [164]. We highlight policy models, reference models, rewards and optimization flows with corresponding loss functions.

💡 The advantage function $A_\phi = V_\phi(s_{t+1}) - V_\phi(s_t)$ quantifies per-step contributions, critical for identifying key reasoning errors. This granularity is lost in DPO, which treats entire trajectories uniformly.

Perplexity Filtering for Out-of-Distribution Data. To ensure DPO training data is on-distribution (aligned with ρ), responses are filtered using perplexity. The perplexity of a response $y = (y_1, y_2, \dots, y_T)$ is defined as:

$$PP(y) = \exp \left(-\frac{1}{T} \sum_{i=1}^T \log P_\rho(y_i | y_{<i}) \right),$$

where y_i is the i -th token. Only responses with perplexity below a threshold (e.g., the 95th percentile of ρ -generated responses) are retained.

💡 The advantage function remains a core concept to determine which actions (token choices) are better than the baseline at each step.

3.2.7 Offline Reasoning Optimization (OREO)

OREO [173] is an offline reinforcement learning method designed to enhance LLMs' multi-step reasoning by optimizing the soft Bellman equation [111]. Unlike DPO, which relies on paired preference data, OREO uses sparse rewards based on final outcomes (e.g., correctness of reasoning chains) and jointly trains a policy model π_θ and a value function V_ϕ for

fine-grained credit assignment. The core objective minimizes the inconsistency in the soft Bellman equation:

$$V_\phi(s_t) - V_\phi(s_{t+1}) = r(s_t, \mathbf{a}_t) - \beta \log \frac{\pi_\theta(\mathbf{a}_t | s_t)}{\pi_{\text{ref}}(\mathbf{a}_t | s_t)},$$

where $s_{t+1} = f(s_t, \mathbf{a}_t)$ is the next state, r is the sparse reward, and β controls KL regularization. The policy and value losses are:

$$\mathcal{L}_V(\phi) = \frac{1}{T} \sum_{t=0}^{T-1} \left(V_\phi(s_t) - R_t + \beta \sum_{i \geq t} \log \frac{\pi_\theta(\mathbf{a}_i | s_i)}{\pi_{\text{ref}}(\mathbf{a}_i | s_i)} \right)^2,$$

$$\mathcal{L}_\pi(\theta) = \frac{1}{T} \sum_{t=0}^{T-1} \left(V_\phi(s_t) - R_t + \beta \log \frac{\pi_\theta(\mathbf{a}_t | s_t)}{\pi_{\text{ref}}(\mathbf{a}_t | s_t)} \right)^2 + \alpha \mathcal{L}_{\text{reg}},$$

where $\mathcal{L}_{\text{reg}}$ penalizes deviations from π_{ref} , and α balances regularization.

💡 OREO's explicit value function enables test-time beam search (e.g., selecting high-value reasoning steps) and iterative training, where failed trajectories refine the policy. This contrasts with DPO implicit value function, which lacks stepwise credit assignment.

⚠️ OREO's computational cost scales with trajectory length and value-model training. While effective for math/agent tasks, its generalization to broader domains (e.g., coding) requires validation. Iterative training also demands careful data curation to avoid overfitting to failure modes.

3.2.8 Group Relative Policy Optimization (GRPO)

GRPO [59] simplifies the PPO framework by eliminating the need for a separate value function. Instead, GRPO estimates the baseline from the average reward of multiple sampled outputs for the same question. The primary contribution in GRPO is that it removes the need for a separate value model (critic model) and instead estimates the baseline reward from a group of sampled LLM outputs. This significantly reduces memory usage and stabilizes policy learning. The approach also aligns well with how reward models are trained, i.e., by comparing different LLM-generated outputs rather than predicting an absolute value.

For each question q , GRPO samples a group of outputs $\{o_1, o_2, \dots, o_G\}$ from the old policy π_θ^{old} . A reward model is used to score each output in the group, yielding rewards $\{r_1, r_2, \dots, r_G\}$. The rewards are normalized by subtracting the group average and dividing by the standard deviation:

$$\bar{r}_i = \frac{r_i - \text{mean}(r)}{\text{std}(r)}.$$

The advantage $\hat{A}_{i,t}$ for each token in the output is set as the normalized reward $\bar{r}_i$.

$$\mathcal{L}_\pi(\theta) = \frac{1}{T} \sum_{t=0}^{T-1} \left(V_\phi(s_t) - R_t + \beta \log \frac{\pi_\theta(\mathbf{a}_t | s_t)}{\pi_{\text{ref}}(\mathbf{a}_t | s_t)} \right)^2 + \alpha \mathcal{L}_{\text{reg}},$$

其中 $\mathcal{L}_{\text{reg}}$ 惩罚与 π_{ref} 的偏离，而 α 用于平衡正则化。

💡 OREO's 显式的价值函数支持测试时的束搜索（例如，选择高价值的推理步骤）和迭代训练，其中失败的轨迹可用于优化策略。这与 DPO 隐含的价值函数形成对比，后者缺乏逐步的信用分配能力。

⚠️ OREO's 其计算成本随轨迹长度和价值模型训练而增加。虽然该方法在数学/智能体任务上有效，但在更广泛领域（例如，代码生成）的泛化能力仍需验证。迭代训练也需要仔细的数据管理，以避免对失败模式过拟合。

3.2.7 Group Relative Policy Optimization (GRPO)

GRPO [59] 通过消除对独立价值函数的需求，简化了 PPO 框架。相反，GRPO 从同一问题的多个采样输出的平均奖励中估计基线。GRPO 的主要贡献在于，它移除了对独立价值模型（评判模型）的需求，转而从一组采样的 LLM 输出中估计基线奖励。这显著降低了内存使用并稳定了策略学习。该方法也与奖励模型的训练方式高度契合，即通过比较不同的 LLM 生成输出，而非预测一个绝对值。

对于每个问题 q ，GRPO 从旧策略 π_θ^{old} 中采样一组输出 $\{o_1, o_2, \dots, o_G\}$ 。使用奖励模型对组中的每个输出进行评分，得到奖励 $\{r_1, r_2, \dots, r_G\}$ 。通过减去组平均值并除以标准差对奖励进行归一化：

$$\bar{r}_i = \frac{r_i - \text{mean}(r)}{\text{std}(r)}.$$

输出中每个词元 i 的优势 $\hat{A}_{i,t}$ 被设定为归一化后的奖励 $\bar{r}_i$ 。

GRPO 首先采样一个问题 $q \sim P(Q)$ ，然后从 $\pi_\theta^{\text{old}}(O | q)$ 中采样 G 个输出 $\{o_i\}_{i=1}^G$ 。将每个输出的目标定义为

$$J(o_i, \theta, q) = \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left(\min \left\{ r_{\text{ratio},i,t} \hat{A}_{i,t}, \right. \right. \\ \left. \left. \text{clip} \left(r_{\text{ratio},i,t}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_{i,t} \right\} - \beta D_{\text{KL}} \left[\pi_\theta \| \pi_{\text{ref}} \right] \right).$$

那么，GRPO 目标变为

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{q \sim P(Q)} \left[\frac{1}{G} \sum_{i=1}^G J(o_i, \theta, q) \right],$$

其中概率比定义为

$$r_{\text{ratio},i,t} \triangleq \frac{\pi_\theta(o_{i,t} | q, o_{i,<t})}{\pi_\theta^{\text{old}}(o_{i,t} | q, o_{i,<t})}.$$

其中 ϵ 是一个类似于 PPO 的裁剪超参数，而 β 调整 KL 散度惩罚，以鼓励新策略 π_θ 不过度偏离参考策略 π_{ref} ，该参考策

略通常是初始的有监督微调（SFT）模型 [174, 175]。GRPO 可以在两种模式下应用：结果监督和过程监督。

Outcome Supervision: 仅在每个输出结束时提供一次奖励。输出中所有词元 i 的优势 $\hat{A}_{i,t}$ 被设定为归一化后的奖励 $\bar{r}_i$ 。

$$\bar{r}_i = \frac{r_i - \text{mean}(r)}{\text{std}(r)}.$$

Process Supervision: 在每个推理步骤结束时提供奖励。每个标记的优势值 $\hat{A}_{i,t}$ 计算为后续步骤归一化奖励的总和：

$$\hat{A}_{i,t} = \sum_{\text{index}(j) \geq t} \bar{r}_{i, \text{index}(j)},$$

其中 $\text{index}(j)$ 是第 j 个步骤的结束标记索引。

总体而言，GRPO 通过利用组级优势，成为 DeepSeekR1 [40] 中经典演员-评论员框架的高效替代方案，从而在不牺牲区分候选回答之间细微差异能力的前提下降低训练成本。

💡 细粒度的每步奖励使模型能够有效识别并强化高质量回答，提升在复杂多步推理任务中的整体性能。

与仅依赖单步比较不同，多样本比较优化 [176] 方法同时比较多个响应，以促进多样性并减轻偏差。具体而言，给定查询 $\{y_1, y_2, \dots, y_n\}$ 的一组响应 x ，观察到排序 $y_1 > y_2 > \dots > y_n$ 的概率由乘积决定：

$$P(y_1 > y_2 > \dots > y_n) = \prod_i \frac{e^{R(x, y_i)}}{\sum_j e^{R(x, y_j)}}.$$

在此公式中，每个响应 y_i 都在所有其他响应的上下文中被联合评估，确保比较不是孤立的成对事件，而是更广泛排序框架的一部分，这有助于捕捉更细微的偏好并减少潜在的偏差。

3.3 Pure RL Based LLM Refinement

Guo 等人 (2025) 的工作 [40] 引入了两个主要模型：DeepSeek-R1-Zero 和 DeepSeek-R1。

该方法涵盖多个步骤（主要步骤参见 GRPO 中的图2）：收集冷启动数据、执行强化学习训练、进行 SFT、使用蒸馏将知识迁移到较小模型，以及解决语言混合和可读性等特定挑战。这一多阶段流程确保了鲁棒性以及与人类偏好的对齐，同时蒸馏使得较小模型能够高效部署而不会造成显著的性能损失。

3.3.1 Cold-Start RL Phase

该过程始于一个 cold-start 强化学习阶段，其中收集少量精心策划的数据以微调一个初始的或 base 模型。在此初步微调之后，进行强化学习训练——通常通过诸如 GRPO 等算法直至收敛。冷启动阶段对于在全面强化学习训练之前稳定模型至关重要，可防止纯粹由强化学习驱动的更新所引起的不稳定性。cold-start data preparation 专注于捕捉人类可读的推理模式，以防止纯粹强化学习驱动更新带来的不稳定性。此步骤

GRPO first samples a question $q \sim P(Q)$ and then samples G outputs $\{o_i\}_{i=1}^G$ from $\pi_\theta^{old}(O \mid q)$. Define the per-output objective as

$$J(o_i, \theta, q) = \frac{1}{|o_i|} \sum_{i=1}^{|o_i|} \left(\min \left\{ r_{\text{ratio}, i, t} \hat{A}_{i, t}, \right. \right. \\ \left. \left. \text{clip} \left(r_{\text{ratio}, i, t}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_{i, t} \right\} \right. \\ \left. - \beta D_{KL} \left[\pi_\theta \parallel \pi_{\text{ref}} \right] \right).$$

Then, the GRPO objective becomes

$$J_{GRPO}(\theta) = \mathbb{E}_{q \sim P(Q)} \left[\frac{1}{G} \sum_{i=1}^G J(o_i, \theta, q) \right],$$

where the probability ratio is defined as

$$r_{\text{ratio}, i, t} \triangleq \frac{\pi_\theta(o_{i, t} \mid q, o_{i, < t})}{\pi_\theta^{old}(o_{i, t} \mid q, o_{i, < t})}.$$

where ϵ is a clipping hyperparameter akin to PP0, and β adjusts the KL-divergence penalty encouraging the new policy π_θ not to deviate excessively from a reference policy π_{ref} , which is typically the initial supervised fine-tuned (SFT) model [174, 175]. GRPO can be applied in two modes: outcome supervision and process supervision.

Outcome Supervision: Provides a reward only at the end of each output. The advantage $\hat{A}_{i, t}$ for all tokens in the output is set as the normalized reward $\bar{r}_i$.

$$\bar{r}_i = \frac{r_i - \text{mean}(r)}{\text{std}(r)}.$$

Process Supervision: Provides a reward at the end of each reasoning step. The advantage $\hat{A}_{i, t}$ for each token is calculated as the sum of the normalized rewards from the following steps:

$$\hat{A}_{i, t} = \sum_{\text{index}(j) \geq t} \bar{r}_{i, \text{index}(j)},$$

where $\text{index}(j)$ is the end token index of the j -th step.

Overall, GRPO serves as an efficient alternative to classic actor-critic frameworks in DeepSeekR1 [40] by leveraging group-level advantages, thereby reducing training costs without sacrificing the capacity to distinguish fine-grained differences among candidate responses.

💡 **Fine-grained per-step rewards enable the model to effectively identify and reinforce high-quality responses, boosting overall performance in complex, multi-step reasoning tasks.**

3.2.9 Multi-Sample Comparison Optimization

Instead of relying solely on single-pair comparisons, multi-sample comparison optimization [176] approach compares multiple responses simultaneously to promote diversity and mitigate bias. Specifically, given a set of responses $\{y_1, y_2, \dots, y_n\}$ for a query x , the probability of observing the ranking $y_1 > y_2 > \dots > y_n$ is determined by the product

$$P(y_1 > y_2 > \dots > y_n) = \prod_i \frac{e^{R(x, y_i)}}{\sum_j e^{R(x, y_j)}}.$$

In this formulation, each response y_i is jointly evaluated in the context of all other responses, ensuring that comparisons are not isolated pairwise events but rather part of a broader ranking framework that helps capture more nuanced preferences and reduces potential biases.

3.3 Pure RL Based LLM Refinement

The work from Guo et al. (2025) [40] introduces two main models: *DeepSeek-R1-Zero* and *DeepSeek-R1*.

- **DeepSeek-R1-Zero** operates with a purely Reinforcement Learning approach, excluding any SFT.
- **DeepSeek-R1** incorporates *cold-start* data and applies a multi-stage training pipeline.

The methodology encompasses several steps (See Figure 2 in GRPO for main steps): collecting cold-start data, performing RL training, carrying out SFT, using distillation to transfer knowledge to smaller models, and addressing specific challenges such as language mixing and readability. This multi-stage pipeline ensures robustness and alignment with human preferences, while distillation enables efficient deployment of smaller models without significant performance loss.

3.3.1 Cold-Start RL Phase

The process begins with a *cold-start* RL phase, where a small amount of curated data is gathered to fine-tune an initial, or *base*, model. Following this preliminary fine-tuning, RL is conducted—often via algorithms like GRPO until convergence. The cold-start phase is critical for stabilizing the model before full RL training, preventing instability that can arise from purely RL-driven updates. The *cold-start data preparation* focuses on capturing human-readable reasoning patterns to prevent instability from purely RL-driven updates. This step generates CoT-style examples with consistent `< reasoning_process >` and `< summary >` fields, usually involving thousands of carefully curated samples. Structured CoT formats and consistent fields ensure clarity and robustness in the model’s reasoning outputs, reducing errors and improving interpretability [8, 177, 178, 179].

💡 **Providing CoT reasoning traces before RL training establishes a stronger foundation for reasoning tasks, enhancing both robustness and interpretability of outputs.**

3.3.2 Rejection Sampling and Fine-tuning

This concept is also used in WebGPT [83]. Once RL stabilizes, a *rejection sampling* mechanism is employed to generate high-quality responses that are subsequently filtered for correctness, clarity, and other quality metrics. These filtered responses are then blended with additional datasets to produce a new, larger corpus for Supervised Fine-Tuning. Rejection sampling ensures that only high-quality outputs are used for further training, enhancing the model’s overall performance and reliability. After RL converges for high-stakes reasoning tasks, *rejection sampling* is used to filter a large number of generated outputs, expanding the training set. These newly generated reasoning examples (potentially up to hundreds of thousands in quantity) are mixed with existing SFT data to create a combined dataset of substantial size (often around

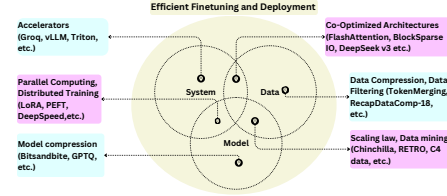


图 4: 此维恩图说明了系统、数据和模型之间为实现高效微调 and 部署的相互作用。它涵盖了诸如加速器 (Grok, vLLM)、适配 (LoRA, PEFT)、协同优化架构 (FlashAttention)、数据压缩 (TokenMerging)、缩放定律 (Chinchilla) 和模型压缩 (GPTQ) 等策略, 以提升性能和可扩展性。

生成具有一致 `< reasoning_process >` 和 `< summary >` 字段的思维链风格示例, 通常涉及数千个精心策划的样本。结构化的 CoT 格式和一致的字段确保了模型推理输出的清晰度和鲁棒性, 减少了错误并提高了可解释性 [8, 177, 178, 179]。

💡 在强化学习训练之前提供 CoT 推理轨迹, 为推理任务建立了更坚实的基础, 增强了输出的鲁棒性和可解释性。

3.3.2 Rejection Sampling and Fine-tuning

这一概念也用于 WebGPT [83]。一旦强化学习稳定下来, 便采用 *rejection sampling* 机制来生成高质量响应, 随后根据正确性、清晰度和其他质量指标进行筛选。这些经过筛选的响应随后与其他数据集混合, 以产生一个用于监督微调的新的、更大的语料库。拒绝采样确保只有高质量的输出被用于进一步训练, 从而提升模型的整体性能和可靠性。在强化学习针对高风险推理任务收敛后, 使用 *rejection sampling* 来筛选大量生成的输出, 从而扩展训练集。这些新生成的推理示例 (数量可能高达数十万) 与现有的 SFT 数据混合, 创建一个规模可观 (通常约 80 万样本) 的组合数据集。拒绝采样和数据集扩展显著增强了模型对通用任务的覆盖范围, 同时保持了其推理能力。

3.3.3 Reasoning-Oriented RL

reasoning-oriented RL 利用了 GRPO [59], 它从当前策略中采样一组输出, 并为每个输出计算奖励和优势。奖励可以通过基于规则的检查来计算, 例如确保数学或代码任务中的解决方案正确, 强制使用结构化的 CoT 标签, 以及对不期望的语言混合进行惩罚。GRPO 基于组群的采样和奖励计算确保模型优先考虑高质量、结构化的输出, 从而增强其推理能力。

3.3.4 Second RL Stage for Human Alignment

second RL stage 通过引入额外的奖励信号和提示分布, 进一步将模型与更广泛的人类偏好 (有益性、无害性、创造性等)

对齐。第二个强化学习阶段确保模型与人类价值观保持一致, 使其更具通用性和情境感知能力。在此组合数据集上对基础模型进行再训练后, 可以进行第二轮强化学习, 以使模型更紧密地符合人类偏好 (例如, 针对有益性和无害性)。此强化学习阶段微调模型, 以更好地与人类价值观对齐, 确保输出不仅准确, 而且在情境上恰当。

3.3.5 Distillation for Smaller Models

最后, *distillation* 技术被用于将主模型精炼后的能力迁移到更小的架构中, 从而在不大幅牺牲性能的情况下实现更高效的部署。它使得较小模型能够继承先进的推理能力, 使其在具有挑战性的基准测试中具备竞争力, 而无需承担完整规模强化学习训练的计算成本。最后, *distillation* 扮演着关键角色: 性能最佳的模型 *DeepSeek-R1* [40] 作为教师, 指导更小的架构 (例如, 参数规模从 15 亿到 700 亿不等的 Qwen 或 Llama 系列模型)。这种迁移使得较小模型能够继承先进的推理能力, 使其在具有挑战性的基准测试中具备竞争力, 而无需承担完整规模强化学习训练的计算成本。

💡 蒸馏技术民主化了先进的推理能力, 使得较小模型能够以更低的计算开销实现有竞争力的性能。

4 Supervised Finetuning in LLMs

如图 2 所示, 微调构成了大语言模型后训练方案的一个基本组成部分。在本节中, 我们总结不同类型的大语言模型微调机制。

4.1 Instruction finetuning

在指令微调中, 模型在精心构建的指令 (提示) 和响应 (补全) 配对数据上进行训练。其主要目标是引导大语言模型准确且有帮助地遵循用户提供的指令, 而不受任务领域的限制。这通常涉及编译大规模、多样化的指令-响应数据集, 涵盖多种任务类型 (例如, 摘要、问答、分类、创意写作)。诸如 T0 [180]、FLAN [181]、Alpaca [182]、Vicuna [183] 和 Dolly [184] 等模型展示了指令微调后的大语言模型如何凭借其增强的指令遵循能力, 在零样本或少样本任务上超越基础模型。

4.2 Dialogue (Multi-turn) Finetuning

一些大语言模型经过对话式微调, 以更好地处理多轮对话。这与上述描述的指令微调不同, 此处的数据形式是连续的对话 (多轮对话), 而非单一的提示-响应对。在这种方法中, 训练数据由包含多个用户查询和系统响应的聊天记录组成, 确保模型学会在多个轮次中保持上下文并生成连贯的回复。像 LaMDA [185] 和 ChatGPT [39] 这样的模型突显了经过对话微调的大语言模型如何能显得更具交互性和上下文感知能力。虽然对话微调可能与指令微调有所重叠 (因为许多指令以聊

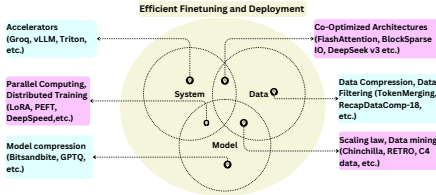


Fig. 4: This Venn diagram illustrates the interplay between System, Data, and Model for efficient finetuning and deployment. It covers strategies like accelerators (GROQ, vLLM), adaptation (LoRA, PEFT), co-optimized architectures (FlashAttention), data compression (TokenMerging), scaling laws (Chinchilla), and model compression (GPTQ) to boost performance and scalability.

800k samples). Rejection sampling and dataset expansion significantly enhance the model's coverage of general tasks while preserving its reasoning proficiency.

3.3.3 Reasoning-Oriented RL

The *reasoning-oriented RL* leverages GRPO [59], which samples a group of outputs from the current policy and computes rewards and advantages for each output. Rewards may be computed via rule-based checks, e.g., ensuring correct solutions in math or code tasks, enforcing structured CoT tags, and penalizing undesired language mixing. GRPO group-based sampling and reward computation ensure that the model prioritizes high-quality, structured outputs, enhancing its reasoning capabilities.

3.3.4 Second RL Stage for Human Alignment

A *second RL stage* further aligns the model with broader human preferences (helpfulness, harmlessness, creativity, etc.) by introducing additional reward signals and prompt distributions. The second RL stage ensures the model aligns with human values, making it more versatile and contextually aware. After re-training the base model on this combined dataset, a second round of RL can be conducted to align the model more closely with human preferences (e.g., for helpfulness and harmlessness). This RL stage fine-tunes the model to better align with human values, ensuring outputs are not only accurate but also contextually appropriate.

3.3.5 Distillation for Smaller Models

Finally, *distillation* techniques are used to transfer the refined capabilities of the main model to smaller architectures, enabling more efficient deployments without sacrificing much performance. It allows smaller models to inherit advanced reasoning capabilities, making them competitive on challenging benchmarks without the computational costs of full-scale RL training. Finally, *distillation* plays a pivotal role: the top-performing model, *DeepSeek-R1* [40], serves as a teacher to smaller architectures (e.g., Qwen or Llama families, ranging from 1.5B to 70B parameters). This transfer allows the smaller models to inherit advanced reasoning capabilities, making them competitive on challenging benchmarks without incurring the computational costs of full-scale RL training.

💡 **Distillation democratizes advanced reasoning capabilities, enabling smaller models to achieve competitive performance with reduced computational overhead.**

4 Supervised Finetuning in LLMs

As shown in Figure 2, finetuning forms a basic component of LLM post-training recipes. In this section, we summarize the different types of LLM fine-tuning mechanisms.

4.1 Instruction finetuning

In instruction finetuning, a model is trained on curated pairs of instruction (prompt) and response (completion). The main goal is to guide the LLM to follow a user-provided instruction accurately and helpfully, regardless of the task domain. This usually involves compiling large, diverse instruction-response datasets covering many task types (e.g., summarization, QA, classification, creative writing). Models such as T0 [180], FLAN [181], Alpaca [182], Vicuna [183] and Dolly [184] demonstrate how instruction-finetuned LLMs can outperform base models on zero-shot or few-shot tasks by virtue of their enhanced instruction-following abilities.

4.2 Dialogue (Multi-turn) Finetuning

Some LLMs undergo dialogue-style finetuning to better handle multi-turn conversations. Different from instruction tuning described above, here the data takes the form of a continuous dialogue (multi-turn conversations) instead of a single prompt-response pair. In this approach, training data consists of chat transcripts with multiple user queries and system responses, ensuring the model learns to maintain context across turns and produce coherent replies. Models like LaMDA [185] and ChatGPT [39] highlight how dialogue-tuned LLMs can feel more interactive and context-aware. While dialogue finetuning can overlap with instruction finetuning (because many instructions come in a chat format), specialized conversation data often yields more natural, multi-turn user experiences.

4.3 CoT Reasoning finetuning

Chain-of-Thought (CoT) reasoning finetuning teaches models to produce step-by-step reasoning traces instead of just final answers. By exposing intermediate rationales or *thoughts*, CoT finetuning can improve both interpretability and accuracy on complex tasks (e.g., math word problems, multi-hop QA). In practice, CoT finetuning uses supervised reasoning annotations (often handcrafted by experts) to show how a solution unfolds. Notable early work includes Chain-of-Thought Prompting [8] and Self-Consistency [186], which initially applied the idea to prompting; subsequent efforts (e.g., Chain-of-Thought Distillation [187]) adapt it to a full finetuning or student-teacher paradigm. These efforts have also been extended to the multimodal domain, e.g., LLaVA-CoT [188] and LlamaV-o1 [189] where image, QA and CoT reasoning steps are used in LLM finetuning.

天形式出现), 但专门的对话数据通常能带来更自然、多轮次的用户体验。

4.3 CoT Reasoning finetuning

思维链 (CoT) 推理微调教导模型生成逐步推理轨迹, 而不仅仅是最终答案。通过展示中间原理或 *thoughts*, CoT 微调可以提升复杂任务 (例如, 数学文字问题、多跳问答) 的可解释性和准确性。在实践中, CoT 微调使用监督式推理标注 (通常由专家手工制作) 来展示解决方案如何展开。著名的早期工作包括思维链提示 [8] 和自洽性 [186], 它们最初将该思想应用于提示; 后续工作 (例如, 思维链蒸馏 [187]) 将其适配到完整的微调或师生范式。这些努力也已扩展到多模态领域, 例如 LLaVA-CoT [188] 和 LlamaV-o1 [189], 其中图像、问答和思维链推理步骤被用于。

4.4 DomainSpecific (Specialized) Finetuning

当 (例如, 生物医学、金融或法律) 表现出色时, 会使用领域特定微调。这里, 采用一个精心策划的领域相关文本和标注示例语料库来微调 LLM。例如, BioGPT [73] 和 BiMediX [218] 专攻生物医学文献, FinBERT [219] 用于金融文本, ClimatGPT [220, 221] 用于气候与可持续性, 以及 CodeT5 [222] 用于代码理解。这些领域的监督微调通常包括使用领域特定数据进行分类、检索或问答任务, 确保模型的参数适应该领域的专业语言和概念。领域特定微调也扩展到视觉-语言模型, 例如, [223] 在遥感影像上微调, [224] 在医学成像模态上微调, [225, 226, 227] 在时空视频输入上微调, 以及 [228] 为图表理解而适配。

4.5 DistillationBased Finetuning

大型“教师”模型有时被用来生成标注数据或原理, 供较小的“学生”模型进行微调, 这通常被称为知识蒸馏 [229, 230]。在, CoT 蒸馏 [187] 是一个例子, 其中强大的教师, 而学生。逐步蒸馏 [231] 生成描述性原理以及最终答案, 通过蒸馏使用较小数据集来训练较小的模型。这种方法可以产生更轻量、更快的模型, 即使在零样本或少样本任务 [232] 中, 也能保留教师模型的大部分性能。

4.6 Preference and Alignment SFT

虽然 RLHF 并非纯粹的监督学习, 但它始于一个监督式 *preference* 或 *alignment* 微调阶段。该阶段使用人工标注或人工排序的示例来教导模型区分期望与不期望的输出 (例如, 安全与有害)。通过在明确的偏好上进行训练, 模型变得更加符合用户价值观, 减少有害或离题的补全。像 InstructGPT [58] 这样的工作说明了监督式偏好数据在奖励模型训练和。

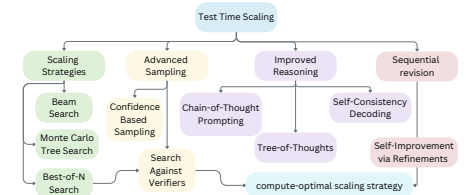


图 5: 测试时缩放方法概述: 并行缩放、顺序缩放和基于搜索的方法。它还展示了它们如何融入计算最优策略。

4.7 Efficient Finetuning

完全微调一个 LLM 可能在计算和内存方面非常密集, 尤其是当模型规模增长到数百亿或数千亿参数时。为了应对这些挑战, 参数高效微调 (PEFT) 技术引入了一小部分可训练参数或可学习的提示, 同时冻结模型的大部分权重。诸如 LoRA [60]、前缀微调 [233] 和适配器 [234] 等方法通过向特定层注入轻量级模块 (或提示) 来体现这一策略, 从而显著减少内存占用。

图 4 说明了这些技术如何融入一个更广泛的生态系统, 该系统涉及 LLMs 的系统级优化、数据管理和评估策略。特别是, PEFT 方法可以与量化和剪枝方法 [192, 190] 结合, 以进一步最小化内存使用和计算开销, 从而在较小的 GPU 甚至消费级硬件上进行微调。例如, QLoRA 将 4 位量化与低秩适应统一起来, 而 BitsAndBytes 提供了 8 位优化器, 使得在受限环境中进行 LLM 训练更加实用 (表 2)。

此外, 这些 PEFT 方法仍然需要监督数据来指导适应过程, 但可训练参数数量的减少使得使用领域内或特定任务的数据集更加可行。这对于专业领域 (例如医疗或软件开发) 尤其有价值, 因为这些领域的数据可能有限或标注成本高昂。如表 2 所示, PEFT (HF) 将其中几种方法 (LoRA、前缀微调等) 集成到一个单一的库中, 简化了研究和生产环境中的部署。

💡 **将高效的微调设计 (如 LoRA 和 QLoRA) 与系统和数据优化 (图 4) 相结合, 能够为特定领域文本生成等任务实现成本效益高的 LLM 适应, 而无需昂贵的完全微调。**

5 Test-time Scaling Methods

虽然 RL 微调模型的策略, 但测试时缩放 (TTS) 通常在无需更新模型的情况下增强推理能力。图 5 展示了 TTS 方法的分类法, 根据其底层技术对它们进行了分类。

5.1 Beam Search

束搜索最初是在语音识别领域被引入的 [235]。它作为一种序列模型的解码策略而受到重视, 后来被神经机器翻译和语音

Model	Category	Source	Description
1. Parameter-Efficient Fine-Tuning & Model Compression			
LoRA [60]	Low-Rank Adaptation	Link	Injects trainable low-rank adapters for efficient fine-tuning.
QLoRA [190]	Quantized Adaptation	Link	Combines 4-bit quantization with LoRA to enable fine-tuning on consumer GPUs
GPTQ [191]	Post-Training Quantization	Link	Optimal 4-bit quantization method for GPT-style models with minimal loss
SparseGPT [192]	Pruning	Link	One-shot pruning that preserves model quality with compensation.
PEFT (HF) [193]	Unified Fine-Tuning	Link	Library integrating LoRA, prefix tuning, and other parameter-efficient methods
BitsAndBytes [194]	Low-Precision Training	Link	Enables 8-bit optimizers and 4-bit quantization for memory-efficient training
AdaLoRA [195]	Adaptive Adaptation	Link	Dynamically allocates parameter budget between layers during fine-tuning
P-Tuning v2 [196]	Prompt Optimization	Link	Learns continuous prompt embeddings through deep prompt tuning
2. Data Management & Preprocessing			
HF Datasets [197]	Data Processing	Link	Unified API for 30k+ datasets with streaming, versioning, and preprocessing
WebDataset [198]	Data Streaming	Link	Efficient tar-based sharding format for petascale distributed training
DVC [199]	Data Versioning	Link	Git-like version control for datasets and machine learning pipelines
Apache Arrow [200]	Memory Format	Link	Language-agnostic columnar memory format for zero-copy data access
Zstandard [201]	Compression	Link	High-speed compression algorithm for training data storage/transfer
Cleanlab [202]	Data Quality	Link	Automatic detection of label errors and outliers in training datasets
3. Distributed Training & Optimization			
DeepSpeed [203]	Training Optimization	Link	ZeRO parallelism, 3D parallelism, and memory optimizations for giant models
Megatron-LM [204]	Model Parallelism	Link	NVIDIA's optimized framework for large transformer model training
Colossal-AI [205]	Heterogeneous Training	Link	Unified system supporting multiple parallelization strategies
Horovod [206]	Distributed Training	Link	MPI-inspired framework for multi-GPU/multi-node synchronization
Ray [207]	Distributed Computing	Link	Universal framework for distributed Python applications at scale
4. Efficient Inference & Deployment			
vLLM [208]	Serving Optimization	Link	Paged attention implementation for high-throughput LLM serving
TensorRT [209]	GPU Optimization	Link	NVIDIA's inference optimizer with kernel fusion and quantization support
Triton [210]	Serving Framework	Link	Production-grade serving with concurrent model execution support
ONNX [211]	Cross-Platform	Link	Unified inference engine with hardware-specific optimizations
OpenVINO [212]	Intel Optimization	Link	Runtime for Intel CPUs/iGPUs with pruning/quantization support
XNNPACK [213]	Mobile Inference	Link	Highly optimized floating-point kernels for ARM CPUs
Groq [214]	AI Accelerator	Link	Deterministic low-latency inference via custom tensor streaming processor
5. Integrated Development Ecosystems			
HF Ecosystem [215]	Full Stack	Link	Transformers + Datasets + Accelerate + Inference Endpoints
DeepSpeed [203]	Training/Inference	Link	Microsoft's end-to-end solution for billion-parameter models
PyTorch [216]	Unified Framework	Link	Native LLM support via torch.compile and scaled dot-product attention
LLM Reasoners [217]	Advanced Reasoning	Link	Enhances LLM reasoning capabilities using advanced search algorithms.

TABLE 2: Comprehensive Overview of Methods and Frameworks employed in Modern LLMs

4.4 Domain-Specific (Specialized) Finetuning

When an LLM needs to excel in a specific domain (e.g., biomedicine, finance, or legal), domain-specific finetuning is used. Here, a curated corpus of domain-relevant text and labeled examples is employed to finetune the LLM. For instance, BioGPT [73] and BiMediX [218] specialize in biomedical literature, FinBERT [219] for financial texts, ClimatGPT [220, 221] for climate and sustainability and CodeT5 [222] for code understanding. Supervised finetuning in these domains often includes classification, retrieval, or QA tasks with domain-specific data, ensuring the model's parameters adapt to the specialized language and concepts of the field. Domain-specific finetuning is also extended to vision-language models such as, [223] finetuned on remote sensing imagery, [224] on medical imaging modalities, [225, 226, 227] on spatiotemporal video inputs, and [228] adapted for chart understanding.

4.5 Distillation-Based Finetuning

Large 'teacher' models are sometimes used to produce labeled data or rationales, which a smaller 'student' model finetunes on, this is generally called knowledge distillation [229, 230]. In the context of LLMs, CoT Distillation [187] is one example where a powerful teacher LLM generates intermediate reasoning steps, and the student LLM is finetuned to reproduce both the final answer and the reasoning chain. Step-by-step distillation [231] generates descriptive rationales alongside

final answers to train smaller models through distillation with smaller datasets. This approach can yield lighter, faster models that retain much of the teacher's performance, even in zero-shot or few-shot tasks [232].

4.6 Preference and Alignment SFT

While RLHF is not purely supervised, it starts with a supervised *preference* or *alignment* finetuning stage. This stage uses human-labeled or human-ranked examples to teach the model about desirable vs. undesirable outputs (e.g., safe vs. toxic). By training on these explicit preferences, the model becomes more aligned with user values, reducing harmful or off-topic completions. Works like InstructGPT [58] illustrate how supervised preference data is critical before reward model training and RL updates begin.

4.7 Efficient Finetuning

Fully finetuning a LLM can be computationally and memory-intensive, particularly as model sizes grow into the tens or hundreds of billions of parameters. To address these challenges, parameter-efficient finetuning (PEFT) techniques introduce a small set of trainable parameters or learnable prompts while leaving most of the model weights frozen. Approaches such as LoRA [60], Prefix Tuning [233], and Adapters [234] exemplify this strategy by injecting lightweight modules (or

Model	Category	Source	Description
1. Parameter-Efficient Fine-Tuning & Model Compression			
LoRA [60]	低秩自适应	链接	注入可训练的低秩适配器以实现高效微调。
QLoRA [190]	量化自适应	链接	将 4 位量化与 LoRA 结合，实现在消费级 GPU 上进行微调
GPTQ [191]	训练后量化	链接	针对 GPT 风格模型的最优 4 位量化方法，损失极小
SparseGPT [192]	剪枝	链接	一次性剪枝，通过补偿保持模型质量。
PEFT (HF) [193]	统一微调	链接	集成 LoRA、前缀调优等参数高效方法的库
BitsAndBytes [194]	低精度训练	链接	支持 8 位优化器和 4 位量化，实现内存高效的训练
AdaLoRA [195]	自适应适配	链接	在微调期间动态分配各层间的参数预算
P-Tuning v2 [196]	提示优化	链接	通过深度提示调优学习连续的提示嵌入
2. Data Management & Preprocessing			
HF Datasets [197]	数据处理	链接	为 3 万多个数据集提供统一 API，支持流式处理、版本控制和预处理
WebDataset [198]	数据流式处理	链接	基于 tar 的高效分片格式，适用于 PB 级分布式训练
DVC [199]	数据版本控制	链接	为数据集和机器学习流水线提供类似 Git 的版本控制
Apache Arrow [200]	内存格式	链接	语言无关的列式内存格式，实现零拷贝数据访问
Zstandard [201]	压缩	链接	用于训练数据存储/传输的高速压缩算法
Cleanlab [202]	数据质量	链接	自动检测训练数据集中的标签错误和异常值
3. Distributed Training & Optimization			
DeepSpeed [203]	训练优化	链接	针对巨型模型的 ZeRO 并行、3D 并行和内存优化
Megatron-LM [204]	模型并行	链接	用于大型 Transformer 模型训练的 NVIDIA 优化框架
Colossal-AI [205]	异构训练	链接	支持多种并行化策略的统一系统
Horovod [206]	分布式训练	链接	受 MPI 启发的多 GPU/多节点同步框架
Ray [207]	分布式计算	链接	用于大规模分布式 Python 应用的通用框架
4. Efficient Inference & Deployment			
vLLM [208]	服务优化	链接	用于高吞吐量 LLM 服务的分页注意力实现
TensorRT [209]	GPU 优化	链接	NVIDIA 的推理优化器，支持内核融合和量化
Triton [210]	服务框架	链接	支持并发模型执行的生产级服务框架
ONNX [211]	跨平台	链接	具有硬件特定优化功能的统一推理引擎
OpenVINO [212]	Intel 优化	链接	适用于 Intel CPU/iGPU 的运行库，支持剪枝/量化
XNNPACK [213]	移动端推理	链接	针对 ARM CPU 高度优化的浮点内核
Groq [214]	AI 加速器	链接	通过定制张量流处理器实现确定性低延迟推理
5. Integrated Development Ecosystems			
HF Ecosystem [215]	全栈	链接	Transformers + Datasets + Accelerate + Inference Endpoints
DeepSpeed [203]	训练/推理	链接	微软针对数十亿参数模型的端到端解决方案
PyTorch [216]	统一框架	链接	通过 torch.compile 和缩放点积注意力提供原生 LLM 支持
LLM Reasoners [217]	高级推理	链接	使用高级搜索算法增强 LLM 的推理能力。

表 2: 现代大语言模型所采用方法与框架的全面概览

系统所采用 [236]。随着大语言模型的普及，该算法已被用于许多文本生成任务中的近似搜索。

束搜索的概念类似于剪枝的 *breadth-first search*，即在每一步保留 N 个概率最高的部分序列（即“束”），舍弃概率较低的路径。通过限制束宽（ N ），它在管理指数级搜索空间的同时，旨在找到一个接近最优的序列。这些束在每个解码步骤中被扩展，以找到多条可能的路径。在推理大语言模型中，此类路径使我们能够系统地并行探索多条推理链，并聚焦于最有希望的那些。这确保了高可能性的推理步骤被纳入考虑，与贪心解码相比，可以提高找到正确且连贯解决方案的机会。它传统上被用于翻译、摘要和代码生成等任务，这些任务的目标是获得一个高概率的正确序列 [95]。

尽管现代大语言模型通常倾向于随机采样（例如温度采样）以促进生成文本的多样性，但束搜索对于结构化推理问题仍然是一种有价值的技术。例如，思维树框架 [86] 允许嵌入不同的搜索算法来探索可能的“*thoughts*”或推理步骤树；通常使用束搜索（束宽为 b ）来在每个推理步骤中保持 b 个最有希望的状态。在这里，束搜索被用于系统地探索数学谜题和规划问题等任务的解决步骤，剪枝不太有希望的推理分支，从而提高模型解决问题的准确性。当希望模型在其学习到的分布下输出单条最可能的推理路径或答案时，束搜索仍然是测试时推理的一个强大基线。最佳 N 选一（BoN）[237] 搜索首先生成 N 个候选输出（通常通过采样），然后根据选定标准（例如奖励模型或模型自身的似然）选取最佳的一个 [238, 239, 240]。

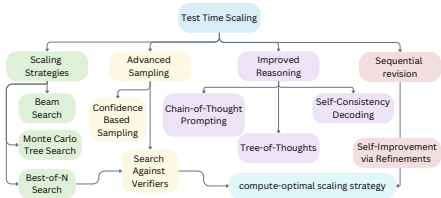


Fig. 5: An overview of Test-time Scaling methods: parallel scaling, sequential scaling, and search-based methods. It also shows how they integrate into a compute-optimal strategy.

prompts) in specific layers, thus significantly reducing the memory footprint.

Figure 4 illustrates how these techniques fit into a broader ecosystem that involves system-level optimizations, data management, and evaluation strategies for LLMs. In particular, PEFT approaches can be combined with quantization and pruning methods [192, 190] to further minimize memory usage and compute overhead, enabling finetuning on smaller GPUs or even consumer-grade hardware. For instance, QLoRA unifies 4-bit quantization with low-rank adaptation, while BitsAndBytes provides 8-bit optimizers to make LLM training more practical in constrained environments (Table 2).

Moreover, these PEFT methods still require supervised data to guide the adaptation process, but the reduction in the number of trainable parameters makes it more feasible to use in-domain or task-specific datasets. This is especially valuable for specialized domains (e.g., medical or software development), where data might be limited or expensive to annotate. As shown in Table 2, PEFT (HF) integrates several of these approaches (LoRA, prefix tuning, and more) into a single library, streamlining deployment in both research and production settings.

💡 Combining efficient tuning designs like LoRA and QLoRA with system and data optimizations (Figure 4) enables cost-effective LLM adaptation for tasks like domain-specific text generation, without expensive full fine-tuning.

5 Test-time Scaling Methods

While RL fine-tunes the model’s policy, test-time scaling (TTS) enhances reasoning during inference typically without model updates. Figure 5 presents a taxonomy of TTS methods, categorizing them based on their underlying techniques.

5.1 Beam Search

Beam search was first introduced in the context of speech recognition [235]. It gained prominence as a decoding strategy for sequence models and was later adopted in neural machine translation and speech systems [236]. With the popularity of LLMs, this algorithm has been used for approximate search in many text generation tasks.

The concept of Beam search is similar to pruned *breadth-first search*, where top N highest-probability partial sequences (the ‘beam’) are kept at each step, discarding lower-probability paths. By limiting the beam width (N), it manages the exponential search space while aiming to find a near-optimal sequence. These beams are expanded at each decoding step to find multiple probable paths. In reasoning LLMs, such paths allow us to systematically explore multiple reasoning chains in parallel, focusing on the most promising ones. This ensures that high-likelihood reasoning steps are considered, which can improve the chances of finding a correct and coherent solution compared to greedy decoding. It has traditionally been used in tasks such as translation, summarization, and code generation, where the goal is a highly probable correct sequence [95].

While modern LLMs often favor stochastic sampling (e.g., temperature sampling) to promote diversity in generated text, beam search is still a valuable technique for structured reasoning problems. For example, the Tree-of-Thoughts framework [86] allows plugging in different search algorithms to explore a tree of possible ‘thoughts’ or reasoning steps; usually a beam search (with beam width b) is used to maintain the b most promising states at each reasoning step. Here, beam search is used to systematically explore solution steps for tasks like mathematical puzzles and planning problems, pruning less promising reasoning branches and thus improving the model’s problem-solving accuracy. Beam search remains a strong baseline for test-time reasoning when one wants the model to output the single most likely reasoning path or answer under the model’s learned distribution.

5.2 Best-of-N Search (Rejection Sampling)

Best-of-N (BoN) [237] search generates N candidate outputs (usually via sampling) and then picks the best one according to a chosen criterion (e.g., a reward model or the model’s own likelihood) [238, 239, 240]. Conceptually, this is an application of rejection sampling: one draws multiple samples and rejects all but the top-rated result. Unlike Beam Search [235, 236], which incrementally expands and prunes partial hypotheses, BoN simply samples full solutions independently, allowing for greater diversity but at a higher computational cost. Beam Search systematically aims for the most probable sequence, while BoN may capture high-quality but lower-probability solutions through brute-force sampling.

💡 Beam search (effective for harder questions) outperforms best-of-N sampling at low compute budgets, while best-of-N scales better for easier tasks.

During LLM inference, BoN is used to enhance correctness or alignment without retraining the model. By sampling multiple answers and selecting the top candidate (e.g., via a reward model or a checker), BoN effectively boosts accuracy on tasks like QA or code generation. BoN is easy to understand and implement and is almost hyper-parameter-free, with N being the only parameter that can be adjusted at inference. In reinforcement learning contexts, BoN sampling can serve as a baseline exploration mechanism i.e., to generate many roll-outs, pick the best outcome according to the learned reward,

从概念上讲，这是拒绝采样的一种应用：抽取多个样本，然后拒绝除评分最高的结果之外的所有样本。与束搜索 [235, 236] 不同（后者逐步扩展并剪枝部分假设），BoN 只是独立地采样完整的解决方案，这允许更大的多样性，但计算成本也更高。束搜索系统地以最可能的序列为目标，而 BoN 则可以通过暴力采样捕获高质量但概率较低的解决方案。

💡 对于较难的问题，束搜索（在低计算预算下）优于最佳 N 选一采样，而对于较简单的任务，最佳 N 选一的可扩展性更好。

在，BoN 被用于在不重新训练模型的情况下增强正确性或对齐性。通过采样多个答案并选择最佳候选（例如通过奖励模型或检查器），BoN 有效地提高了问答或代码生成等任务的准确性。BoN 易于理解和实现，并且几乎无需超参数调整， N 是推理时唯一可调整的参数。在强化学习背景下，BoN 采样可以作为一种基线探索机制，即生成多次推演，根据学习到的奖励选择最佳结果，然后继续，尽管这会增加计算开销。OpenAI’s WebGPT 使用 BoN 通过奖励模型选择最佳响应，从而获得了强大的问答性能 [83]。BoN 也被用作一种简单的对齐方法，与其他训练后技术（例如 RLHF [58] 和 DPO [80]）相比具有高度竞争力。研究表明，当由足够鲁棒的奖励模型引导时，BoN 可以接近或匹配 [84, 241]。诸如推测性拒绝 [242] 等替代方法基于此思想，并利用更好的奖励模型来提高效率。研究还强调了如果用于 BoN 的（代理）奖励函数不完善 [243]，则可能出现 *reward hacking* 问题，或者如果 N 参数变得非常大，则可能出现不稳定性问题。

💡 选择使用束搜索的过程奖励模型还是最佳 N 选一，取决于任务难度和计算预算。

5.2 Compute-Optimal Scaling

计算最优缩放策略 (COS) [85] 是一种动态方法，旨在大型语言模型推理过程中高效分配计算资源，在优化准确性的同时避免不必要的开销。该方法并非对所有输入应用统一的采样策略，而是将提示词分为五个难度级别——从易到难——分类依据可以是预言机难度（真实成功率）或模型预测难度（例如，来自偏好排序模型的验证器分数）。一旦完成分类，该策略会调整计算分配：较简单的提示词进行顺序优化，即模型迭代优化其输出以提高正确性；而较难的提示词则触发并行采样或束搜索，通过探索多个响应变体来增加找到正确解决方案的可能性。这种双重方法平衡了探索（针对挑战性输入）和优化（针对接近正确的响应），确保了每单位计算努力下的最优性能。值得注意的是，该方法在保持同等性能的同时，计算使用量比传统的 N 选一最佳采样降低了四倍。其关键洞见在于，通过使计算策略与问题难度相匹配，它避免了对简单

案例的资源浪费，同时确保了复杂任务有足够的采样多样性。本质上，它充当了大型语言模型推理的“智能恒温器”，根据输入复杂度动态调整计算努力，从而实现大规模语言模型更高效、更具成本效益的部署。

💡 COS 通过最优平衡顺序/并行计算，相比 N 选一基线实现了 4 倍效率提升。在简单/中等问题上，束搜索 + 修订优于更大的模型。

5.3 Chain-of-thought prompting

CoT 提示引导大型语言模型产生中间推理步骤，而不是直接跳转到最终答案。通过将问题分解为逻辑子步骤，CoT 挖掘了模型执行多步推理的潜在能力，显著提升了在数学应用题、逻辑谜题和多跳问答等任务上的性能。

Wei 等人 [8] 证明了 CoT 在算术和逻辑任务上的有效性，显示出相比直接提示的巨大提升。Kojima 等人 [244] 引入了零样本 CoT，揭示了即使添加像“让我们一步步思考”这样简单的短语，也能在足够大的模型中触发连贯的推理。后续工作（例如，Wang 等人，2022 [186]）将 CoT 与基于采样的策略（自洽性）结合，实现了更高的准确性。如第 5.3 节所述，CoT 格式数据也被用于 SFT，并显示有助于重塑模型响应，使其更具逐步性。

💡 对模型进行微调以顺序修订答案，使其能够在先前尝试的基础上进行构建，从而随时间提高准确性。这种方法对于较简单的问题特别有效，而并行采样（探索）则被证明对较难的问题更有益。

5.4 Self-Consistency Decoding

自洽性是一种由 Wang 等人提出的解码策略 [245]。它被提出作为思维链提示中简单贪婪解码的替代方案。它建立在为问题采样多种不同推理路径的想法之上，并且首次表明对这些路径进行边缘化可以显著提高算术和推理问题的准确性。换言之，它允许模型以多种方式进行思考，然后信任共识，这在许多推理场景中提高了正确性。

自洽方法的工作原理是：从模型中采样一组多样化的推理链（通过提示工程鼓励不同的 CoTs，并使用温度采样），然后让模型为每条链输出一个最终答案。该方法不信任单一链条，而是选择在这些多重推理路径中最一致的答案，本质上是在边缘化潜在推理后的一种 *majority vote* 或 *highest probability answer*。其直觉是，如果一个复杂问题有唯一正确答案，不同的有效推理路径应该收敛于该答案。通过汇集多条链的结果，模型可以“决定”哪个答案得到最多支持。在实际应用中，例如，可以为一个数学问题采样 20 条 CoTs，观察哪个最终答案出现频率最高；该答案随后被作为模型的输出。这种方法将一次性 CoT 过程转变为模型交叉验证其答案的集成方法。它

and proceed, although at increased computational overhead. *OpenAI's* WebGPT used BoN to pick the best response via a reward model, yielding strong QA performance [83]. BoN is also used as a simple alignment method that is highly competitive with other post-training techniques e.g., RLHF [58] and DPO [80]. Studies have shown BoN can approach or match RLHF results when guided by a sufficiently robust reward model [84, 241]. Alternatives such as speculative rejection [242] build on this idea and utilize a better reward model to improve efficiency. The studies also highlight issues of *reward hacking* if the (proxy) reward function used for BoN is imperfect [243] or instability issues if the N parameter gets very large.

💡 Choice of either process reward models with beam search vs best-of-N depends on the difficulty and compute budget.

5.3 Compute-Optimal Scaling

The Compute-Optimal Scaling Strategy (COS) [85] is a dynamic method designed to allocate computational resources efficiently during inference in LLMs, optimizing accuracy without unnecessary expense. Instead of applying a uniform sampling strategy across all inputs, this approach categorizes prompts into five difficulty levels—ranging from easy to hard—either by leveraging oracle difficulty (ground-truth success rates) or model-predicted difficulty (e.g., verifier scores from Preference Ranking Models). Once categorized, the strategy adapts compute allocation: easier prompts undergo sequential refinement, where the model iteratively refines its output to improve correctness, while harder prompts trigger parallel sampling or beam search, which explores multiple response variations to increase the likelihood of finding a correct solution. This dual approach balances exploration (for challenging inputs) and refinement (for near-correct responses), ensuring optimal performance per unit of computational effort. Remarkably, this method achieves four times lower compute usage than traditional best-of-N sampling while maintaining equivalent performance. The key insight is that by matching computational strategy to problem difficulty, it avoids wasted resources on trivial cases while ensuring sufficient sampling diversity for complex tasks. In essence, it functions as a “smart thermostat” for LLM inference, dynamically adjusting computational effort in response to input complexity, leading to a more efficient and cost-effective deployment of large-scale language models.

💡 COS achieves 4× efficiency gains over best-of-N baselines by optimally balancing sequential/parallel compute. Beam search + revisions outperform larger models on easy/intermediate questions.

5.4 Chain-of-thought prompting

CoT prompting induces LLMs to produce intermediate reasoning steps rather than jumping directly to the final answer. By breaking down problems into logical sub-steps, CoT taps into a model's latent ability to perform multi-step inferences,

significantly improving performance on tasks like math word problems, logical puzzles, and multi-hop QA.

Wei et al. [8] demonstrated CoT's effectiveness on arithmetic and logic tasks, showing large gains over direct prompting. Kojima et al. [244] introduced Zero-Shot CoT, revealing that even adding a simple phrase like “Let's think step by step” can trigger coherent reasoning in sufficiently large models. Subsequent works (e.g., Wang et al., 2022 [186]) combined CoT with sampling-based strategies (Self-Consistency) for even higher accuracy. As described in Sec. 5.4, CoT format data have also been used for SFT and are shown to help reshape the model responses to be more step-by-step.

💡 Fine-tuning models to revise answers sequentially allows them to build on previous attempts, improving accuracy over time. This approach is particularly effective for easier questions, while parallel sampling (exploration) proves more beneficial for harder ones.

5.5 Self-Consistency Decoding

Self-Consistency is a decoding strategy introduced by Wang et al. [245]. It was proposed as an alternative to simple greedy decoding for chain-of-thought prompts. It built upon the idea of sampling multiple distinct reasoning paths for a question and was the first to show that marginalizing over those paths can significantly improve accuracy on arithmetic and reasoning problems. In other words, it allows the model to think in many ways and then trust the consensus, which improves correctness in many reasoning scenarios.

The self-consistency method works by sampling a diverse set of reasoning chains from the model (via prompt engineering to encourage different CoTs, and using temperature sampling) and then letting the model output a final answer for each chain. Instead of trusting a single chain, the method selects the answer that is most consistent across these multiple reasoning paths, effectively a *majority vote* or *highest probability answer* after marginalizing out the latent reasoning. The intuition is that if a complex problem has a unique correct answer, different valid reasoning paths should converge to that same answer. By pooling the outcomes of many chains, the model can “decide” which answer is most supported. In application, one might sample, e.g., 20 CoTs for a math problem and see what final answer appears most frequently; that answer is then taken as the model's output. This approach turns the one-shot CoT process into an ensemble where the model cross-verifies its answers. It is especially useful for arithmetic and commonsense reasoning tasks where reasoning diversity helps.

💡 Smaller models with test-time compute can outperform much larger models in certain scenarios.

Self-consistency is often combined with other methods: e.g., sampling multiple chains and then applying a verifier to the most common answer. Its strength lies in requiring no new training, only extra sampling, making it a popular test-time scaling strategy to obtain more reliable answers from

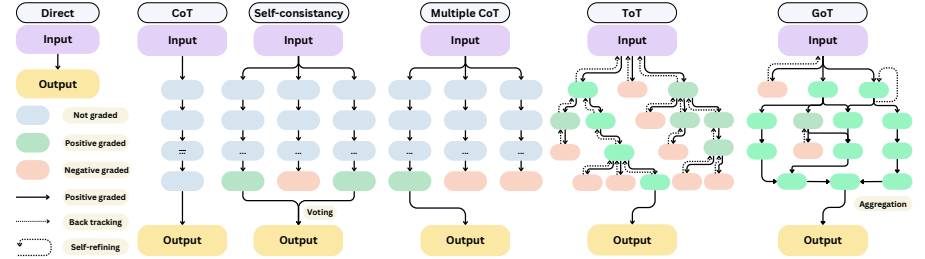


图 6: 本图比较了, 从直接将输入映射到输出而无推理的直接提示, 到更结构化的方法。思维链 (CoT) 引入了逐步推理, 而自洽性 (CoT-SC) 则生成多个 CoT 路径并选择最频繁的答案。多重 CoTs 独立探索多样化的推理路径。思维树 (ToT) 将推理结构化, 支持回溯和精炼, 而思维图 (GoT) 则通过动态聚合和连接思想对此进行了推广。图例解读了关键机制, 如评分、回溯和自我精炼, 这些对于优化推理效率至关重要。

对于算术和常识推理任务特别有用, 在这些任务中推理多样性有益。

💡 在特定场景下, 具有测试时计算的小模型可以超越多得多的模型。

自洽性常与其他方法结合: 例如, 采样多条链, 然后对最常见的答案应用验证器。其优势在于无需新的训练, 只需额外采样, 使其成为一种流行的测试时扩展策略, 以从。它也启发了其他变体, 例如, 通用自洽性 [246] 将原始思想 (仅适用于对单一最终答案进行多数投票) 扩展到更一般的生成任务, 如摘要和开放式问答。

5.5 Tree-of-thoughts

ToT 框架 [86] 通过允许模型分支到多个可能的思维序列, 而非遵循单一的线性链, 从而推广了思维链方法。因此, 它将语言模型推理问题表述为树搜索, 借鉴了受人类问题解决启发的经典人工智能搜索方法 [247, 37]。思维树将中间推理步骤视为搜索树中的“节点”, 并使用语言模型从给定状态扩展可能的后续步骤 (思想)。模型并非采样一条长的推理路径, 而是探索一个分支思想的树, 并能执行前瞻和回溯。在每一步, 而启发式或价值函数评估每个部分解状态。然后, 搜索算法 (例如深度优先、广度优先、束搜索) 导航此树, 决定进一步探索哪些分支。这种方法允许系统地探索不同的推理策略: 如果一条路径导致死胡同, 模型可以返回到较早的状态并尝试不同的分支 (这与标准的 CoT 不同, 后者只遵循一条推理线)。实际上, ToT 是一种迭代提示过程, 模型在其中生成思想、评估它们并精炼其方法, 模仿人类如何在脑海中规划解决问题的各种途径。

ToT 对于需要多步骤和策略性探索的复杂问题 (如谜题、规划任务或游戏) 尤其有用, 并且通过系统地搜索解空间, 其

性能优于更简单的 CoT 方法。它提供了一个灵活的框架——可以根据任务插入各种生成策略 (例如采样与提示) 和搜索算法 (BFS, DFS, A*, MCTS)。尽管计算量更大, 但 ToT 表明, 分配额外的“思考时间” (计算资源) 来探索替代方案可以显著提高推理和规划性能。它催生了旨在改进或利用它以获得更好推理效果的后续研究, 例如, 多智能体系统已与 ToT 结合: 不同的 LLM “智能体” 并行生成思想, 验证器智能体修剪不正确的分支, 从而在单智能体 ToT [248] 的基础上提高了准确性。

💡 对于, 推理时的计算可以超越扩展模型参数, 尤其是在数学问题等具有挑战性的推理任务上。

5.6 Graph of Thoughts

思维图 (GoT) [249] 框架通过基于图的结构而非严格的层次树, 实现了更灵活高效的推理过程, 从而扩展了 ToT。两种方法的思维表示存在差异: 在 ToT 中, 推理的每一步都被构建为树中的一个节点, 具有固定的父子关系; 而 GoT 则将思维表示为图中的节点, 从而实现了更具适应性的依赖关系和相互连接。

在思维扩展策略方面, ToT 遵循传统方法, 即在每一步生成多个思维候选, 使用基于树的搜索策略进行探索, 并根据启发式方法进行剪枝, 然后选择最优路径。相比之下, GoT 采用了基于图的思维扩展, 允许思维动态地相互连接。这实现了三种关键转换: 聚合 (将多个解决方案合并为统一答案)、精炼 (随时间迭代改进思维) 和生成 (产生多样化的候选)。GoT 不再通过严格的层次结构进行导航, 而是使用体积量对思维进行优先级排序, 并优化探索路径, 从而减少了不必要的计算。

ToT 的一个关键限制是其回溯能力受限——一旦某个分支被丢弃, 便不再被重新考虑。GoT 通过允许迭代精炼克服了

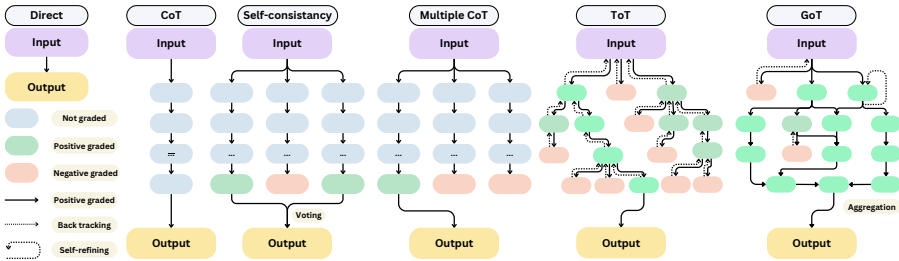


Fig. 6: This figure compares reasoning strategies in LLMs, evolving from Direct Prompting, which maps input to output without reasoning, to more structured approaches. Chain-of-Thought (CoT) introduces step-by-step reasoning, while Self-Consistency (CoT-SC) generates multiple CoT paths and selects the most frequent answer. Multiple CoTs explores diverse reasoning paths independently. Tree-of-Thoughts (ToT) structures reasoning as a tree, enabling backtracking and refinement, whereas Graph-of-Thoughts (GoT) generalizes this by dynamically aggregating and connecting thoughts. The legend deciphers key mechanisms like grading, backtracking, and self-refinement, crucial for optimizing reasoning efficiency.

LLMs. It has also inspired other variants, e.g., Universal Self-Consistency [246] extend the original idea (which worked only with majority vote on single final answer) to more general generation tasks such as summarization and open-ended QA.

5.6 Tree-of-thoughts

ToT framework [86] generalizes the chain-of-thought approach by allowing the model to branch out into multiple possible thought sequences instead of following a single linear chain. It thus formulates the problem of language-model reasoning as a tree search, drawing on classic AI search methods inspired by human problem-solving [247, 37]. Tree of Thoughts treats intermediate reasoning steps as “nodes” in a search tree and uses the language model to expand possible next steps (thoughts) from a given state. Rather than sampling one long reasoning path, the model explores a tree of branching thoughts and can perform lookahead and backtracking. At each step, the LLM might generate several candidate next thoughts, and a heuristic or value function evaluates each partial solution state. Then a search algorithm (e.g., depth-first, breadth-first, beam search) navigates this tree, deciding which branches to explore further. This approach allows systematic exploration of different reasoning strategies: if one path leads to a dead-end, the model can return to an earlier state and try a different branch (unlike standard CoT which commits to one line of reasoning). In effect, ToT is an iterative prompting procedure where the model generates thoughts, evaluates them, and refines its approach, mimicking how a human might mentally map out various ways to solve a problem.

ToT is especially useful for complex problems like puzzles, planning tasks, or games where multiple steps and strategic exploration are needed and outperforms simpler CoT methods by systematically searching through the solution space. It provides a flexible framework – one can plug in various generation strategies (e.g. sampling vs. prompting) and search algorithms (BFS, DFS, A*, MCTS) depending on the task. Although more computationally heavy, ToT shows that allocating extra “thinking time” (compute) to explore alternatives can yield significantly better reasoning and planning performance. It

has spawned follow-up research aiming to improve or utilize it for better reasoning e.g., multi-agent systems have been combined with ToT: different LLM “agents” generate thoughts in parallel and a validator agent prunes incorrect branches, leading to improved accuracy over the single-agent ToT [248].

💡 Inference-time computation for LLMs can outperform scaling model parameters, especially for challenging reasoning tasks like math problems.

5.7 Graph of Thoughts

The Graph of Thoughts (GoT) [249] framework extends the ToT by allowing more flexible and efficient reasoning processes through graph-based structures rather than strict hierarchical trees. Thought representation differs between the two approaches: in ToT, each step in reasoning is structured as a node in a tree with fixed parent-child relationships, whereas GoT represents thoughts as nodes in a graph, enabling more adaptable dependencies and interconnections.

In terms of thought expansion strategies, ToT follows a traditional approach where multiple thought candidates are generated at each step, explored using tree-based search strategies, and pruned based on heuristics before selecting the most optimal path. In contrast, GoT incorporates graph-based thought expansion, allowing thoughts to interconnect dynamically. This enables three key transformations: aggregation (merging multiple solutions into a unified answer), refinement (iteratively improving thoughts over time), and generation (producing diverse candidates). Instead of navigating through a rigid hierarchy, GoT prioritizes thoughts using a volume metric and explores paths optimally, reducing unnecessary computations.

A critical limitation of ToT is its restricted backtracking—once a branch is discarded, it is not reconsidered. GoT overcomes this by allowing iterative refinement, where previous thoughts can be revisited, modified, and improved upon. This iterative nature is particularly useful in complex reasoning tasks where initial thoughts may require adjustments.

这一限制，即可以重新访问、修改和改进先前的思维。这种迭代特性在复杂的推理任务中尤其有用，因为初始思维可能需要调整。此外，GoT 通过合并部分解决方案减少了冗余计算，从而显著提高了计算效率。

💡 GoT 提高了问题解决的效率和适应性，使其在需要复杂推理的任务中优于 ToT。

5.7 Confidence-based Sampling

在基于置信度的采样中，语言模型生成多个候选解决方案或推理路径，然后根据模型对每个结果的自身置信度进行优先级排序或选择 [250]。这可以通过两种方式实现：(a) 选择：生成 N 个输出，并选择对数概率最高的那个（即模型最确信的输出）。这本质上是基于概率的 N 选一最佳策略——模型选择其认为最可能正确的答案。(b) 引导式探索：在探索推理树或多步骤解决方案时，利用模型的标记概率来决定扩展哪个分支（置信度较高的分支优先探索）。换言之，模型的概率估计充当了一种启发式方法，引导在解空间中的搜索 [251]。与纯随机采样相比，基于置信度的方法使过程偏向于模型认为正确的方向，从而可能减少在低可能性（且通常错误）路径上的无效探索。

基于置信度的策略已在推理时被采用，例如，用于 [250] 为每个可能的完成项（叶节点）分配一个置信度分数。算法根据这些置信度分数按比例采样叶节点，以决定扩展哪些路径 [252]。类似地，一些推理方法利用模型对答案的估计似然来决定何时停止或是否提出后续问题——本质上，如果模型的置信度较低，可能会触发进一步的推理（一种自我反思的形式）。基于置信度的选择也用于集成设置中：例如，一个，而一个辅助模型评估每个答案正确的置信度，并选择置信度最高的答案。这在医疗问答 & 等任务中进行了探索，其中，只有高置信度的答案才被信任或返回 [253]。

5.8 Search Against Verifiers

这种验证方法 [254] 于大型语言模型中，通过生成多个候选回答并使用自动化验证系统选择最佳答案来提高回答质量。该方法将重点从增加预训练计算量转向优化测试时计算量，允许模型在推理过程中通过结构化推理步骤或迭代优化进行“更长时间的思考”。该方法包含两个主要步骤：

Generation: 模型（或“提议者”）生成多个答案或推理路径，通常使用高温采样或多样化解码等方法。

Verification: 验证器（例如奖励模型）根据预定标准（如正确性、连贯性或期望过程的契合度）评估这些候选答案。验证器根据其评估重点进行分类：

有几种技术属于此范式，增强了基于验证的优化。N 选一最佳采样涉及生成多个答案并通过验证器（ORM/PRM）进行排序，选择得分最高的答案，这是一种简单而有效的提高答案正确

性的方法。带 PRM 的束搜索跟踪得分最高的推理路径（束）并及时修剪低质量步骤，类似于思维树方法，在推理路径探索中平衡广度与深度。蒙特卡洛树搜索通过扩展有前景的推理分支、模拟推演和反向传播分数来平衡探索与利用，在搜索深度和验证置信度之间提供最优权衡。多数投票（自治性）聚合多个样本的答案并选择最频繁出现的答案，避免使用显式验证器，这在多个回答的一致性表明正确性的场景中效果良好。

💡 ORM 适用于正确性为二元（对/错）且易于评估的任务。

💡 PRM 在多数推理中很有用，可确保中间步骤遵循逻辑进展。

5.9 Self-Improvement via Refinements

这种方法指的是大语言模型通过自我评估和迭代修订来增强其输出的能力。这一过程使模型能够在推理过程中动态优化其回答，而非仅仅依赖预训练的权重。一种值得注意的方法是 **Self-Refinement** [255]，即大语言模型先生成一个初始回答，对其进行批判，然后基于自我生成的反馈来优化输出。这一迭代过程持续进行，直到模型获得满意的结果。此类技术已被证明能提升在多种任务上的表现，包括数学推理和代码生成。该过程遵循以下关键步骤：a) **Initial Generation:** 模型生成一个答案或推理路径。b) **Self-Critique:** 模型审查自身的回答，识别错误、不一致之处或可改进的方面。c) **Refinement:** 模型基于批判调整其回答，并生成一个改进版本。d) **Iteration:** 该过程重复进行，直到输出达到预定的质量阈值或不再有改进。

另一种方法称为 **Self-Polish** [256]，即模型逐步优化给定的问题，使其更易于理解和解决。通过重新表述或重构问题，模型增强其理解并提供更准确的解决方案。自我打磨涉及对问题陈述进行渐进式优化，以使其更易于理解和解决。模型首先重新表述或重构问题以提高清晰度，然后将复杂查询分解为更简单的子问题，并优化模糊输入以确保精确理解。通过在解决问题前重构问题，模型提高了其理解能力并生成更准确的解决方案。

💡 自我改进方法论代表了大语言模型优化范式的转变，强调主动推理和内部反馈，而非静态的预训练。通过对自身回答进行迭代，模型在广泛的应用中实现了更高的一致性和准确性。

5.10 Monte Carlo Tree Search

MCTS [257] 基于将蒙特卡洛模拟应用于博弈树搜索。它因在游戏领域的成功而声名鹊起，特别是在 2016 年，它通过策略网

Moreover, computational efficiency in GoT is significantly improved by reducing redundant calculations through the merging of partial solutions.

💡 GoT enhances problem-solving efficiency and adaptability, making it superior to ToT for tasks requiring complex reasoning.

5.8 Confidence-based Sampling

In confidence-based sampling, the language model generates multiple candidate solutions or reasoning paths and then prioritizes or selects among them based on the model's own confidence in each outcome [250]. This can happen in two ways: (a) Selection: Generate N outputs and pick the one with the highest log probability (i.e., the model's most confident output). This is essentially best-of-N by probability – the model chooses the answer it thinks is most likely correct. (b) Guided exploration: When exploring a reasoning tree or multi-step solution, use the model's token probabilities to decide which branch to expand (higher confidence branches are explored first). In other words, the model's probability estimates act as a heuristic guiding the search through solution space [251]. Compared to pure random sampling, confidence-based methods bias the process toward what the model believes is right, potentially reducing wasted exploration on low-likelihood (and often incorrect) paths.

Confidence-based strategies have been incorporated at inference time e.g., a tree-based search for LLM generation [250] assigns each possible completion (leaf) a confidence score. The algorithm samples leaves in proportion to these confidence scores to decide which paths to extend [252]. Similarly, some reasoning approaches use the model's estimated likelihood of an answer to decide when to halt or whether to ask a follow-up question – essentially if the model's confidence is low, it might trigger further reasoning (a form of self-reflection). Confidence-based selection is also used in ensemble settings: e.g., an LLM may generate multiple answers and a secondary model evaluates the confidence of each answer being correct, picking the answer with the highest confidence. This was explored in tasks like medical Q&A, where an LLM gave an answer and a confidence score, and only high confidence answers were trusted or returned [253].

5.9 Search Against Verifiers

This verification approach [254] in LLMs enhances answer quality by generating multiple candidate responses and selecting the best one using automated verification systems. This approach shifts focus from increasing pre-training compute to optimizing test-time compute, allowing models to “think longer” during inference through structured reasoning steps or iterative refinement. The method involves two main steps: **Generation**: The model (or “proposer”) produces multiple answers or reasoning paths, often using methods like high-temperature sampling or diverse decoding.

Verification: A verifier (e.g., a reward model) evaluates these candidates based on predefined criteria, such as correctness, coherence, or alignment with desired processes. Verifiers are categorized based on their evaluation focus:

- 1) **Outcome Reward Models (ORM)**: Judge only the final answer (e.g., correctness of a math solution).
- 2) **Process Reward Models (PRM)**: Evaluate the reasoning steps (e.g., logical coherence in a thought chain), providing granular feedback to prune invalid paths.

Several techniques fall under this paradigm, enhancing verification-based optimization. Best-of-N Sampling involves generating multiple answers and ranking them via a verifier (ORM/PRM), selecting the highest-scoring one, making it a simple yet effective approach for improving answer correctness. Beam Search with PRM tracks top-scoring reasoning paths (beams) and prunes low-quality steps early, similar to Tree of Thought approaches, balancing breadth and depth in reasoning path exploration. Monte Carlo Tree Search balances exploration and exploitation by expanding promising reasoning branches, simulating rollouts, and backpropagating scores, providing an optimal trade-off between search depth and verification confidence. Majority Voting (Self-Consistency) aggregates answers from multiple samples and selects the most frequent one, avoiding explicit verifiers, which works well in settings where consistency across multiple responses indicates correctness.

💡 ORM is suitable for tasks where correctness is binary (right/wrong) and can be easily assessed.

💡 PRM is useful in multi-step reasoning, ensuring intermediate steps follows logical progression.

5.10 Self-Improvement via Refinements

This approach refers to the ability of LLMs to enhance their outputs through self-evaluation and revision iteratively. This process enables models to refine their responses dynamically during inference rather than relying solely on pre-trained weights. One notable method is **Self-Refinement** [255], where an LLM generates an initial response, critiques it, and then refines the output based on its self-generated feedback. This iterative process continues until the model achieves a satisfactory result. Such techniques have been shown to improve performance on various tasks, including mathematical reasoning and code generation. This process follows these key steps: a) **Initial Generation**: The model produces an answer or reasoning path. b) **Self-Critique**: The model reviews its own response and identifies errors, inconsistencies, or areas for improvement. c) **Refinement**: The model adjusts its response based on the critique and generates an improved version. d) **Iteration**: The process repeats until the output meets a predefined quality threshold or stops improving.

Another approach is called **Self-Polish** [256], where the model progressively refines given problems to make them more comprehensible and solvable. By rephrasing or restructuring problems, the model enhances its understanding and provides more accurate solutions. Self-Polish involves progressive refinement of problem statements to make them more comprehensible and solvable. The model first rephrases or restructures the problem for better clarity, then breaks down complex queries into simpler sub-problems and refines ambiguous inputs to ensure precise understanding. By restructuring

和价值网络引导搜索可能的走法，驱动了 AlphaGo [258]。这一点，以及在其他棋盘游戏和视频游戏中的应用，展示了 MCTS 在不确定性下进行序列决策的强大能力。MCTS 是一种随机搜索算法，通过执行大量随机模拟来构建决策树。它最为人知的应用是在游戏状态中寻找最佳走法，但也可应用于任何能够模拟结果的问题。该算法迭代地执行以下步骤：(a) 根据某种启发式方法（例如 UCT [259]，它选择具有高置信上限的节点）从根节点选择一条路径；(b) 从该路径末端扩展一个新节点（一个先前未访问过的状态）；(c) 从该新状态开始进行一次随机模拟推演以获取结果（例如，游戏中的胜负或某种奖励）；(d) 将结果反向传播回树中，更新节点值并为未来的选择提供信息。重复这些模拟数千次，可以将搜索集中在树中最有希望的分支上。本质上，MCTS 使用随机抽样来评估不同动作序列的潜力，逐渐使搜索偏向那些具有更好平均结果的方向。在 LLM 推理中，我们可以将文本生成视为一个决策过程，并使用 来探索不同的续写路径。例如，对于一个给定的问题（根节点），每个可能的下一个推理步骤或答案都是一个动作；一次模拟可能意味着让 LLM 继续生成直至得到一个最终答案（可能带有一定的随机性），而奖励可以是答案是否正确。通过反复执行此过程，MCTS 可以识别出哪些思维链或答案具有最高的经验成功率。MCTS 对于推理的吸引力在于，它能够通过智能采样而非穷举搜索来处理巨大的搜索空间，并且自然地融入了不确定性和探索性。

💡 训练验证器来评估中间步骤（通过蒙特卡洛推演），而不仅仅是最终答案。

最近的研究已将 MCTS 与 LLMs 结合，以应对复杂的推理和决策任务。一个例子是将 MCTS 用于查询规划：蒙特卡洛思维搜索 [260]，其中引导 LLM 提出一系列子问题以寻找答案。Jay 等人 [260] 使用了一种基于 MCTS 的算法，称为“蒙特卡洛推理器”，它将 LLM 视为一个环境：每个节点是一个提示（状态），每条边是一个动作（例如，要问的特定问题或要采取的步骤），并使用随机推演来评估结果。这种方法使得系统能够高效地探索可能的推理路径空间，并选择一条高奖励的答案路径，在一个科学 Q&A 任务中表现优于简单采样。类似地，MCTS 已被应用于 LLMs 的代码生成 [261]——该算法探索不同的代码路径（使用模型提出代码补全并测试它们）以找到正确的解决方案。另一项工作将多个 LLMs 与 MCTS 集成，将每个模型的输出视为一个分支，并使用奖励模型来模拟结果 [262]。早期结果表明，基于 MCTS 的推理可以解决单次推理或贪婪方法常常错过的问题，尽管需要更多的计算 [76]。缺点是 MCTS 可能比直接采样或束搜索慢得多，最近的研究正在通过提高效率（例如，通过状态合并 [89]）来解决这个问题。总的来说，MCTS 将规划算法的优势带入了 LLM 推理，使 LLM 能够通过模拟推演进行“前瞻”，并做出更明智的推理选择，就像它在游戏 AI 中所做的那样。

💡 测试时计算并非预训练的 1 对 1 替代品，但在许多情况下提供了一种可行的替代方案。

5.11 Chain-of-Action-Thought reasoning

大型语言模型在推理任务中表现出色，但严重依赖外部指导（例如验证器）或推理时的大量采样。现有方法如 CoT [8] 缺乏自我纠正和自适应探索的机制，限制了其自主性和泛化能力。Satori [263] 引入了一种两阶段训练范式，其工作原理是首先调整模型的输出格式，然后通过自我改进来增强其推理能力。在第一阶段（格式调优）中，模型接触到一个由生成器、批评者和奖励模型组成的多智能体框架生成的大量 10K 合成轨迹集合。这种监督式微调帮助模型使用元动作标记生成特定推理格式的输出，尽管它可能仍然难以在这些示例之外进行泛化。在第二阶段（通过）中，模型采用带有重启与探索策略的 PPO [263]，这使其能够从中间步骤（无论正确与否）重新开始，以完善其推理过程。模型获得的奖励基于基于规则的正确性、反思奖励以及 § 5.8 中解释的基于偏好的结果奖励模型反馈的组合，从而激励将更多计算资源分配给更困难的问题，并在测试复杂任务时实现扩展推理。

多智能体框架和先进的微调策略正被越来越多地探索，以增强大型语言模型的推理能力。多智能体大型语言模型训练 (MALT) [264] 引入了一种结构化方法，其中生成、验证和细化步骤分布在专门的智能体之间，允许进行迭代式自我纠正和改进推理链。同样，优化偏好对齐仍然是确保大型语言模型安全性和有用性的关键挑战 [265]。像双因素偏好优化 (BFPO) [266] 这样的方法将，减少了人为干预，同时保持了稳健的对齐。除了基于文本的推理之外，像多模态思维可视化 (Mvot) [267] 这样的多模态方法通过融入视觉表示扩展了 CoT 提示，显著提升了空间推理任务的性能。这些进展凸显了日益增长的需求，即需要结构化的多智能体协作、安全感知的优化以及多模态推理，以解决大型语言模型推理中的根本局限性 [268, 269, 270]。

5.12 Pretraining vs. Test-Time Scaling

预训练与 TTS 是提升大语言模型性能的两种不同策略，各自在计算成本和效果上存在权衡。预训练通过扩展模型参数或增加训练数据来增强能力，需要大量的前期计算投入 [3]。相比之下，TTS 优化推理时计算（如迭代优化、基于搜索的解码或自适应采样），允许在不修改基础模型的情况下提升性能。

从性能与成本的角度看，TTS 在简单到中等难度任务（例如 MATH 基准测试）上取得了与规模大 14 倍的模型相当的结果，同时在计算密集型场景中推理成本降低了 4 倍 FLOPs [271]。然而，对于最困难的任务或当推理计算约束较高时，预训练仍然更优，因为更大的预训练模型本质上编码了更深层的推理能力。

turing problems before solving them, the model improves its comprehension and generates more accurate solutions.

💡 **Self-improvement methodologies represent a paradigm shift in LLM optimization, emphasizing active reasoning and internal feedback over static pre-training. By iterating on their own responses, models achieve greater consistency and accuracy across a wide range of applications.**

5.11 Monte Carlo Tree Search

MCTS [257] is based on the application of Monte Carlo simulations to game-tree search. It rose to prominence with successes in games, notably, it powered AlphaGo [258] in 2016 by searching possible moves guided by policy and value networks. This, as well as the application to other board and video games, demonstrates the power of MCTS for sequential decision-making under uncertainty.

MCTS is a stochastic search algorithm that builds a decision tree by performing many random simulations. It is best known for finding good moves in game states, but it can be applied to any problem where we can simulate outcomes. The algorithm iteratively: (a) Selects a path from the root according to a heuristic (like UCT [259], which picks nodes with a high upper-confidence bound), (b) Expands a new node (a previously unvisited state) from the end of that path, (c) Simulates a random rollout from that new state to get an outcome (e.g., win or loss in a game, or some reward), and (d) Backpropagates the result up the tree to update the values of nodes and inform future selections. Repeating these simulations thousands of times concentrates the search on the most promising branches of the tree. In essence, MCTS uses random sampling to evaluate the potential of different action sequences, gradually biasing the search towards those with better average outcomes. In LLM reasoning, we can treat the generation of text as a decision process and use to explore different continuations. For example, at a given question (root), each possible next reasoning step or answer is an action; a simulation could mean letting the LLM continue to a final answer (perhaps with some randomness), and a reward could be whether the answer is correct. By doing this repeatedly, MCTS can identify which chain of thoughts or answers has the highest empirical success rate. The appeal of MCTS for reasoning is that it can handle large search spaces by sampling intelligently rather than exhaustively, and it naturally incorporates uncertainty and exploration.

💡 **Train verifiers to score intermediate steps (via Monte Carlo rollouts) instead of just final answers.**

Recent efforts have integrated MCTS with LLMs to tackle complex reasoning and decision-making tasks. One example is using MCTS for query planning: Monte Carlo Thought Search [260], where an LLM is guided to ask a series of sub-questions to find an answer. Jay et al. [260] used an MCTS-based algorithm called ‘Monte Carlo Reasoner’ that treats the LLM as an environment: each node is a prompt (state) and each edge

is an action (e.g., a particular question to ask or step to take), and random rollouts are used to evaluate outcomes. This approach allowed the system to efficiently explore a space of possible reasoning paths and pick a high-reward answer path, outperforming naive sampling in a scientific Q&A task. Similarly, MCTS has been applied to code generation with LLMs [261] – the algorithm explores different code paths (using the model to propose code completions and test them) to find a correct solution. Another line of work ensembles multiple LLMs with MCTS, treating each model’s output as a branch and using a reward model to simulate outcomes [262]. Early results show that MCTS-based reasoning can solve problems that single-pass or greedy methods often miss, although with more compute [76]. The downside is that MCTS can be significantly slower than straightforward sampling or beam search, which recent research is addressing by improving efficiency (e.g., by state merging [89]). In general, MCTS brings the strength of planning algorithms to LLM inference and enables an LLM to ‘look ahead’ through simulated rollouts and make more informed reasoning choices, much like it has done for AI in gameplay.

💡 **Test-time compute is not a 1-to-1 replacement for pretraining but, offers a viable alternative in many cases.**

5.12 Chain-of-Action-Thought reasoning

LLMs excel in reasoning tasks but rely heavily on external guidance (e.g., verifiers) or extensive sampling at inference time. Existing methods like CoT [8] lack mechanisms for self-correction and adaptive exploration, limiting their autonomy and generalization. Satori [263] introduced a two-stage training paradigm, which works by initially tuning the model’s output format and then enhancing its reasoning capabilities through self-improvement. In Stage 1 (Format Tuning), the model is exposed to a large set of 10K synthetic trajectories generated by a multi-agent framework comprising a generator, a critic, and a reward model. This supervised fine-tuning helps the model to produce outputs in specific reasoning format using meta-action tokens, although it may still have difficulty generalizing beyond these examples. In Stage 2 (Self-Improvement via RL), the model employs PPO with a Restart and Explore strategy [263], which allows it to restart from intermediate steps, whether they were correct or not, to refine its reasoning process. The model receives rewards based on a combination of rule-based correctness, reflection bonuses, and preference-based Outcome Reward Model feedback explained in § 5.9, thereby incentivizing the allocation of more computational resources to tougher problems and enabling extended reasoning during testing for complex tasks.

Multi-agent frameworks and advanced fine-tuning strategies are increasingly being explored to enhance reasoning in LLMs. Multi-Agent LLM Training (MALT) [264] introduces a structured approach where generation, verification, and refinement steps are distributed across specialized agents, allowing for iterative self-correction and improved reasoning chains. Similarly, optimizing preference alignment remains a crucial challenge in ensuring both safety and helpfulness in LLMs [265]. Approaches like Bi-Factorial Preference Optimization (BFPO) [266] reframe RLHF objectives into a single

💡 **在推理令牌 (Y) 受限的情况下 (例如自改进设置中), 一个配备测试时计算的小模型可以在简单/中等难度问题上超越一个规模大 14 倍的模型。**

就使用场景而言, TTS 适用于推理预算灵活或基础模型已在任务中表现出合理能力的场景。相反, 预训练对于需要根本性新能力 (例如在新领域进行推理) 的任务至关重要, 仅靠推理时优化可能不足。

这两种方法之间存在显著的权衡。TTS 降低了前期训练成本, 使其对灵活的、即时的优化具有吸引力, 但需要在推理时动态分配计算资源。另一方面, 预训练虽然初始成本高, 却能保证一致的性能而无需额外的运行时开销, 因此非常适合大规模 API 部署或对延迟敏感的应用。总体而言, TTS 与预训练本质上是互补的。未来的大语言模型系统可能采用混合方法, 即较小基础模型经过预训练获得核心知识, 而 TTS 通过自适应的、按需计算动态增强响应。这种协同作用使得大规模模型的部署更具成本效益和效率。

💡 **选择预训练以建立基础能力, 选择测试时扩展以实现精确的上下文感知优化。**

6 Benchmarks for LLM Post-training Evaluation

为评估大语言模型后训练阶段的效果, 一系列涵盖多个领域的多样化基准测试被提出, 包括推理任务、对齐、多语言能力、通用理解以及对话与搜索任务。一个结构良好的评估框架能够确保全面理解大语言模型在不同任务中的优势与局限。这些基准测试在大语言模型的后处理阶段 (包括微调、校准、对齐和优化) 中发挥着关键作用, 旨在提升模型响应的准确性、鲁棒性和伦理合规性。接下来, 我们将解释主要的基准测试组别。表 3 概述了按这些基准组别分类的关键数据集。

Reasoning Benchmarks. 这些基准测试评估大语言模型执行逻辑、数学和科学推理的能力。数学推理数据集如 MATH [272]、GSM8K [273] 和 MetaMathQA [274] 测试模型在问题解决、多步算术以及基于定理的问题表述上的表现。科学及多模态推理基准测试, 如 WorldTree V2 [252] 和 MMMU [276], 评估模型在物理、化学和多模态理解方面的知识, 这对于大语言模型生成响应中的事实核查与验证过程至关重要。此外, 像 PangeaBench [275] 这样的数据集将推理任务扩展到多语言和文化领域, 使模型能够优化跨语言推理能力。这些基准测试有助于确定模型处理结构化知识和应用逻辑推理的能力。

RL Alignment Benchmarks. 强化学习对齐基准测试是大语言模型对齐和后训练优化的核心。它们通过人类反馈强化学习来优化响应生成、伦理约束和用户对齐的输出。诸如 HelpSteer [282] 和 UltraFeedback [283] 等数据集基于多属性评分和与用户指令的对齐程度来评估模型。Anthropic 的

表 3: 推理、强化学习对齐与多语言数据集的综合概览。此处, 逐点与成对指的是在不同任务中评估模型性能的不同方法。

Datasets	Domain	Type	#Samples	Evaluation Criteria
Reasoning Benchmarks				
MATH [272]	数学推理	逐点	7,500	分步解答
GSM8K [273]	数学推理	逐点	8.5K	多步推理
MetaMathQA [274]	数学推理	逐点	40K+	自我验证, FOBAR
WorldTree V2 [252]	科学问答	逐点	1,680	多跳解释
PangeaBench [275]	多模态推理	成对	47 种语言	文化理解
MMMU [276]	科学/数学	逐点	大学水平	物理、化学、双语
TruthfulQA [277]	问答/推理	逐点	不适用	真实性
MathInstruct [278]	数学推理	逐点	202K	正确性
MMLU [279, 280]	多任务推理	逐点	57 项任务	广泛知识评估
MMLU-Fairness [279]	公平性/推理	逐点	不适用	偏见/公平性分析
DROP [281]	阅读/推理	逐点	96K	段落级推理
BBH [177]	困难推理	成对	不适用	复杂逻辑问题解决
VRC-Bench [189]	多模态推理	成对	不适用	视觉推理与分类
RL Alignment Benchmarks				
HelpSteer [282]	强化学习对齐	成对	37K+	多属性评分
Anthropic HH-RLHF [123]	强化学习对齐	成对	42.5K	无毒性对齐
UltraFeedback [283]	强化学习对齐	成对	64K	指令遵循、真实性
D4RL [284]	强化学习/控制	逐点	不适用	跨领域离线强化学习
Meta-World [285]	强化学习/控制	逐点	不适用	多任务机器人强化学习
MineRL [286]	强化学习/游戏	成对	不适用	模仿学习、奖励
Multilingual Evaluation				
CulturaX [287]	多语言	逐点评分	6.3T	去重、质量
PangeaIns [288]	多语言	逐点评分	6M	多语言指令
TydiQA [289]	多语言	逐点评分	N/A	跨语言问答
XGLUE [290]	多语言	逐点评分	N/A	跨语言语言任务
MM-Eval [291]	多语言	成对比较	4,981	面向任务的多语言问答
ALM-Bench [291]	多语言问答	逐点评分	N/A	多语言评估
Dialogue and Search Benchmarks				
BigBench [292]	通用理解	逐点评分	200+ 任务	广泛多领域评估
Chatbot Arena [293]	理解	成对比较	33K	用户偏好
MTBench [293]	理解	成对比较	3K	多轮对话
RewardBench [169]	理解	成对比较	2,998	用户偏好
General Comprehension Benchmarks				
ConvAI2 [294]	对话	逐点评分	N/A	吸引力、一致性
MultiWOZ [295]	对话	逐点评分	N/A	任务成功率、连贯性
Tree DL1&L2 [296, 297]	搜索	逐点评分	1.549/2.673	相关性评分
BEIR [298]	搜索	逐点评分	18 个数据集	信息检索
Story & Recommendation Benchmarks				
HANNA [299]	故事	逐点评分	1,056	相关性、连贯性、复杂性
StoryER [300]	故事	成对比较	100K	基于用户偏好的排序
PKU-SafeRLHF [301]	价值观	成对比较	83.4K	有益性、无毒性
Cvalue [302]	价值观	成对比较	145K	安全性、责任性
NaturalInst. [303, 304]	指令微调	逐点评分	1,600+	指令遵循评估

HH-RLHF [123] 探索模型如何通过人类反馈强化学习来优化人类偏好。D4RL [284] 和 Meta-World [285] 专注于机器人控制和离线强化学习, 这对自主模型决策具有启示意义。MineRL [286] 将强化学习测试扩展到复杂环境 (如基于《我的世界》的交互), 有助于在大语言模型中训练自适应决策能力。

Multilingual Evaluation. 多语言基准测试对于大语言模型在跨语言泛化、翻译适应以及低资源语言微调方面的后处理至关重要。CulturaX [287] 和 PangeaIns [288] 评估超过 150 种语言中的分词、翻译和指令遵循能力, 确保模型输出的公平性和多样性。TydiQA [289] 和 MM-Eval [291] 针对双语和任务导向的多语言评估, 促进大语言模型微调的改进。这些数据集确保大语言模型不仅以英语为中心, 还能优化多语言适应能力。**General Comprehension Benchmarks.** 通用理解

supervised learning task, reducing human intervention while maintaining robust alignment. Beyond text-based reasoning, multimodal approaches like Multimodal Visualization-of-Thought (MvOT) [267] extend CoT prompting by incorporating visual representations, significantly enhancing performance in spatial reasoning tasks. These advancements highlight the growing need for structured multi-agent collaboration, safety-aware optimization, and multimodal reasoning to address fundamental limitations in LLM reasoning [268, 269, 270].

5.13 Pretraining vs. Test-Time Scaling

Pretraining and TTS are two distinct strategies for improving LLM performance, each with different tradeoffs in computational cost and effectiveness. Pretraining involves scaling model parameters or increasing training data to enhance capabilities, requiring substantial upfront computational investment [3]. In contrast, TTS optimizes inference-time compute (such as iterative refinements, search-based decoding, or adaptive sampling), allowing performance improvements without modifying the base model.

From a performance vs. cost perspective, TTS achieves results comparable to a model $14\times$ larger on easy to intermediate tasks (e.g., MATH benchmarks), while reducing inference costs by $4\times$ fewer FLOPs in compute-intensive scenarios [271]. However, pretraining remains superior for the hardest tasks or when inference compute constraints are high, as larger pretrained models inherently encode deeper reasoning capabilities.



A smaller model with test-time compute can outperform a $14\times$ larger model on easy/intermediate questions, when inference tokens (Y) are limited (e.g., self-improvement settings).

In terms of use cases, TTS is useful for scenarios with flexible inference budget or when base models already exhibit reasonable competence in the task. Conversely, pretraining is essential for tasks requiring fundamentally new capabilities (e.g., reasoning on novel domains) where inference-time optimizations alone may not suffice.

There are notable tradeoffs between the two approaches. TTS reduces upfront training costs, making it attractive for flexible, on-the-go optimization, but requires dynamic compute allocation at inference. Pretraining, on the other hand, incurs high initial costs but guarantees consistent performance without additional runtime overhead, making it ideal for large-scale API deployments or latency-sensitive applications. Overall, TTS and pretraining are complementary in nature. Future LLM systems may adopt a hybrid approach, where smaller base models are pretrained with essential knowledge, while TTS dynamically enhances responses through adaptive, on-demand computation. This synergy enables more cost-effective and efficient large-scale model deployment.



Choose pretraining for foundational capabilities and test-time scaling for accurate context-aware refinement.

TABLE 3: Comprehensive Overview of Reasoning, RL Alignment, and Multilingual Datasets. Here, pointwise and pairwise refer to different methods of evaluating model performance across various tasks.

Datasets	Domain	Type	#Samples	Evaluation Criteria
Reasoning Benchmarks				
MATH [272]	Math Reasoning	Pointwise	7,500	Step-by-step solutions
GSM8K [273]	Math Reasoning	Pointwise	8.5K	Multi-step reasoning
MetaMathQA [274]	Math Reasoning	Pointwise	40K+	Self-verification, FoRAIR
WorldTree V2 [252]	Science QA	Pointwise	1,680	Multi-hop explanations
PangeaBench [275]	Multimodal Reasoning	Pairwise	7 Langs.	Cultural understanding
MMMU [276]	Science/Math	Pointwise	College-Level	Physics, Chemistry, Bilingual
TruthfulQA [277]	QA/Reasoning	Pointwise	N/A	Truthfulness
MetaInstruct [278]	Math Reasoning	Pointwise	262K	Correctness
MMLU [279, 280]	Multitask Reasoning	Pointwise	57 Tasks	Broad knowledge evaluation
UltraFeedback [279]	Fairness/Reasoning	Pointwise	N/A	Bias/Equity Analysis
DROP [281]	Reading/Reasoning	Pointwise	46K	Discrete reasoning over paragraphs
BBH [177]	Hard Reasoning	Pairwise	N/A	Complex logical problem-solving
VRCC-Bench [188]	Multimodal Reasoning	Pairwise	N/A	Visual Reasoning and Classification
RL Alignment Benchmarks				
HelpSteer [282]	RL Alignment	Pairwise	37K+	Multi-attribute scoring
Anthropic HH-RLHF [123]	RL Alignment	Pairwise	42.5K	Harmlessness alignment
UltraFeedback [283]	RL Alignment	Pairwise	64K	Instruction-following, Truthfulness
D4RL [284]	RL/Control	Pointwise	N/A	Officer RL across domains
Meta-World [285]	RL/Control	Pointwise	N/A	Multi-task robotic RL
MisoRL [286]	RL/Games	Pairwise	N/A	Imitation learning, rewards
Multilingual Evaluation				
CulturaX [287]	Multilingual	Pointwise	6.3T	Desliplication, Quality
PangeaBench [275]	Multilingual	Pointwise	6M	Multilingual instructions
TyDiQA [288]	Multilingual	Pointwise	N/A	Cross-lingual QA
GLUE [289]	Multilingual	Pointwise	N/A	Cross-lingual language tasks
MM-Eval [291]	Multilingual	Pairwise	4,981	Task-oriented multilingual QA
ALM-Bench [291]	Multilingual QA	Pointwise	N/A	Multilingual Evaluation
Dialogue and Search Benchmarks				
BigBench [292]	General Comprehension	Pointwise	200+ Tasks	Broad multi-domain evaluation
Chatbot Arena [293]	Comprehension	Pairwise	33K	User preference
MTBench [295]	Comprehension	Pairwise	3K	Multi-turn conversations
ResearchBench [169]	Comprehension	Pairwise	2,998	User preference
General Comprehension Benchmarks				
ConvAI2 [294]	Dialogue	Pointwise	N/A	Engagingness, Consistency
MultiWOZ [295]	Dialogue	Pointwise	N/A	Task success, Coherence
StoryIE [296]	Search	Pointwise	1,140/1,673	Relevance scoring
D4RL [284]	Search	Pointwise	18 Datasets	Information retrieval
BEIR [298]	Search	Pointwise	18 Datasets	Information retrieval
Story & Recommendation Benchmarks				
HANNA [299]	Story	Pointwise	1,056	Relevance, Coherence, Complexity
MultiWOZ [295]	Story	Pointwise	100K	User preference-based ranking
PKU-SafeRLHF [301]	Values	Pairwise	83.4K	Helpfulness, Harmlessness
CySaaS [302]	Values	Pairwise	145K	Safety, Responsibility
NaturalInstruct [303, 304]	Instruction Tuning	Pointwise	1,600+	Instruction-following evaluation

6 Benchmarks for LLM Post-training Evaluation

To evaluate the success of LLM post-training phases, a diverse set of benchmarks have been proposed covering multiple domains: reasoning tasks, alignment, multilinguality, general comprehension, and dialogue and search tasks. A well-structured evaluation framework ensures a comprehensive understanding of an LLM strengths, and limitations across various tasks. These benchmarks play a crucial role in LLM post-processing stages, where models undergo fine-tuning, calibration, alignment, and optimization to improve response accuracy, robustness, and ethical compliance. Next, we explain the main benchmark groups. Table 3 provides an overview of key datasets categorized under these benchmark groups.

Reasoning Benchmarks. These benchmarks assess LLMs on their ability to perform logical, mathematical, and scientific reasoning. Mathematical reasoning datasets like MATH [272], GSM8K [273], and MetaMathQA [274] test models on problem-solving, multi-step arithmetic, and theorem-based problem formulations. Scientific and multimodal reasoning benchmarks such as WorldTree V2 [252] and MMMU [276] evaluate knowledge in physics, chemistry, and multimodal understanding, which are crucial for fact-checking and verification processes in LLM-generated responses. Additionally, datasets like PangeaBench [275] extend reasoning tasks into multilingual and cultural domains, enabling models to refine cross-lingual reasoning. These benchmarks help determine how well models can process structured knowledge and apply logical deductions.

RL Alignment Benchmarks. RL alignment benchmarks are central to LLM alignment and post-training optimiza-

tion. These benchmarks assess LLMs on their ability to perform logical, mathematical, and scientific reasoning. Mathematical reasoning datasets like MATH [272], GSM8K [273], and MetaMathQA [274] test models on problem-solving, multi-step arithmetic, and theorem-based problem formulations. Scientific and multimodal reasoning benchmarks such as WorldTree V2 [252] and MMMU [276] evaluate knowledge in physics, chemistry, and multimodal understanding, which are crucial for fact-checking and verification processes in LLM-generated responses. Additionally, datasets like PangeaBench [275] extend reasoning tasks into multilingual and cultural domains, enabling models to refine cross-lingual reasoning. These benchmarks help determine how well models can process structured knowledge and apply logical deductions.

Dialogue and Search Benchmarks. Dialogue with search benchmarks in optimization of large language models based on retrieval, multi-turn dialogue coherence, and information retrieval accuracy. For example, ConvAI2 [294] and MultiWOZ [295] are datasets for evaluating multi-turn dialogue models. For example, Chatbot Arena [293] and RewardBench [169] are datasets for evaluating dialogue models. For example, Chatbot Arena [293] and RewardBench [169] are datasets for evaluating dialogue models. For example, Chatbot Arena [293] and RewardBench [169] are datasets for evaluating dialogue models.

7 Future Directions

We collected all research on large language models (LLMs) after training methods. We analyzed its trends, as shown in Figure 7. Since 2020, the application of reinforcement learning (RL) technology [305, 57, 40] to refine large language models has become increasingly prominent (Figure 7a), highlighting the demand for **interactive approaches**, such as humans in the loop [35, 306] to enhance scalability [113, 84, 307]. Simultaneously, **reward modeling** [308, 168, 169] (Figure 7b) has shown a steady upward trend, benefiting from the emergence of self-rewarding language models. However, this field still faces the **reward hacking** [309, 310] challenge, as well as the growing demand for **reward attacks** [312] and the need for robustness [311] in reward functions. **Decoding and search** (Figure 7c) methods include思维树 [86] and蒙特卡洛 [313, 260] strategies, aiming to iteratively self-critique [314, 306, 29] to enhance model reasoning capabilities, but these technologies also require reliable [315, 113] uncertainty estimators to prevent excessive calculation. Safety [301, 316, 317], robustness [318], and interpretability [319, 320, 321] have also become key focus points (Figure 7d), driving the need for uncertainty-aware [325] models, bias perception [322, 323], and uncertainty-aware [324] reinforcement learning methods. To maintain user trust and prevent adversarial attacks, another key challenge involves **personalization** [326, 327] and **adaptation** [195] (Figure 7e), where domain-specific large language model tuning must balance privacy and privacy risks [328]—especially in scenarios involving enterprise data or sensitive personal information. Simultaneously, **process** [329, 330] and **outcome reward optimization** [331] (Figure 7f) represent a delicate balance between unresolved issues: while process-based rewards aid in guiding iterative improvement, outcome-centric metrics are simpler but may fail to capture key intermediate decision steps. Beyond reward structures, **fine-tuning** of large language models for new tasks still encounters challenges like **catastrophic forgetting** [332] and potential data leakage [333, 334] issues, highlighting the need for efficient parameter methods [60] and privacy protection strategies (such as differential privacy [335] and federated learning [336]) requirements. Human feedback remains the core, but its high cost and limited scope pose challenges; for example, **Constitutional AI** [53] and **RLAIF** [97] methods

attempt to automate part of the supervision process, but they also bring concerns about bias calibration [337] and model self-consistency [186] as new challenges. Finally, expanding [113] and **dynamic reasoning** [338] frameworks bring further challenges: models must learn when to allocate more computational resources for complex queries, how to efficiently adjust verification modules [339], and how to maintain robustness when faced with adversarial inputs. These research directions—covering reward modeling, decoding strategies, interpretability, personalization, and safety-aware fine-tuning—highlight the challenges of reinforcement learning in large language models across multiple dimensions, and jointly shaped the development of large language model development in the future. In the following, we will explore some of these directions in more detail.

Fine-tuning challenges. Fine-tuning remains the most direct method for adapting large language models to specific tasks or domains, but it faces numerous challenges. A fundamental issue is **catastrophic forgetting**—when new data is added to a pre-trained model, it may lose previously learned information. Even advanced PEFT methods, such as LoRA [60] (which significantly reduces the number of trainable parameters), cannot fully solve this problem [332]. Future work can explore better continuous learning strategies and regularization techniques to enable models to acquire new skills without forgetting old knowledge. For example, new fine-tuning algorithms (such as CURLoRA [332]) aim to stabilize training and retain new tasks while preserving prior knowledge. Research directions with potential include task-specific fine-tuning [340] (incremental or in-context learning within known contexts), and combining retrieval with external knowledge bases for hybrid training. For example, by adjusting the weights of different knowledge bases, retrieval can be used for large language model fine-tuning, enabling the model to handle queries beyond the original training distribution. When querying, the model can retrieve knowledge from the knowledge base or use tools (such as database queries or calculations) [341, 342]. Such retrieval-augmented fine-tuning [343] can help the model in reasoning by integrating new information, reducing the need for internalizing knowledge. Another method is to train models to explicitly represent uncertainty in new knowledge, thereby enabling them to handle queries involving unseen content during pre-training by saying “I don’t know” or seeking help from external sources. By updating weights with external knowledge integration, future fine-tuning of large language models will maintain higher factual accuracy and lower hallucination rates. **Safe Fine-tuning.** From the perspective of ethics and safety, fine-tuning poses important open research questions. Fine-tuning data often contains sensitive or proprietary information [328], and if the model memorizes and reproduces this data, it may lead to privacy risks. A recent comprehensive survey [344] emphasizes the fragility of the fine-tuning process, such as membership inference attacks (detecting whether a record is in the fine-tuning dataset) and data leakage (retrieving partial fine-tuning data). Mitigating these risks is an open question: how to balance privacy and privacy risks [328]—the relationship, especially in scenarios involving enterprise data or sensitive personal information. Simultaneously, **process** [329, 330] and **outcome reward optimization** [331] (Figure 7f) represent a delicate balance between unresolved issues: while process-based rewards aid in guiding iterative improvement, outcome-centric metrics are simpler but may fail to capture key intermediate decision steps. Beyond reward structures, **fine-tuning** of large language models for new tasks still encounters challenges like **catastrophic forgetting** [332] and potential data leakage [333, 334] issues, highlighting the need for efficient parameter methods [60] and privacy protection strategies (such as differential privacy [335] and federated learning [336]) requirements. Human feedback remains the core, but its high cost and limited scope pose challenges; for example, **Constitutional AI** [53] and **RLAIF** [97] methods

Limitations of Human Feedback. Human feedback is highly costly and subjective. Addressing the limitations of human feedback is a promising avenue for using artificial intelligence feedback and automation to assist or replace human evaluators. Anthropic’s constitutional AI [53] is a notable example: it does not rely on feedback for every harmful or beneficial behavior from a large number of humans, but rather through a set of principles (a “constitution”) to guide the model, and the training model uses another AI model as a reviewer to critique and improve its own responses [345].

tion. They refine response generation, ethical constraints, and user-aligned outputs through RLHF. Datasets such as Help-Steer [282] and UltraFeedback [283] evaluate models based on multi-attribute scoring and alignment with user instructions. Anthropic’s HH-RLHF [123] explores how well models learn human preference optimization through reinforcement learning with human feedback. D4RL [284] and Meta-World [285] focus on robotic control and offline RL, which have implications for autonomous model decision-making. MineRL [286] extends RL testing into complex environments such as Minecraft-based interactions, useful for training LLMs in adaptive decision-making settings.

Multilingual Evaluation. Multilingual benchmarks are essential for LLM post-processing in cross-lingual generalization, translation adaptation, and fine-tuning for low-resource languages. CulturaX [287] and PangeaIns [288] evaluate tokenization, translation, and instruction-following in over 150 languages, ensuring fairness and diversity in model outputs. TydiQA [289] and MM-Eval [291] target bilingual and task-oriented multilingual evaluation, enabling improvements in LLM fine-tuning. These datasets ensure that LLMs are not just English-centric but optimized for multilingual adaptability.

General Comprehension Benchmarks. General comprehension benchmarks contribute to model fine-tuning, response coherence, and preference optimization. Datasets such as Chatbot Arena [293], MTBench [293], and RewardBench [169] test user preference modeling and conversational fluency, crucial for LLM response ranking and re-ranking methods. BigBench [292] evaluates broad multi-domain comprehension, while MMLU [279, 280] measures correctness and informativeness. These datasets help in refining LLM fluency, factual correctness, and open-ended response generation.

Dialogue and Search Benchmarks. Dialogue and search benchmarks play a key role in optimizing LLM retrieval-based responses, multi-turn coherence, and information retrieval accuracy. Datasets such as ConvAI2 [294] and MultiWOZ [295] evaluate multi-turn conversational models, essential for dialogue history tracking and adaptive response fine-tuning. For search relevance assessment, BEIR [298] provides large-scale human-annotated judgments for retrieval fine-tuning, ensuring LLMs generate and rank responses effectively. TREC DL21/22 [296, 297] contributes to document relevance ranking and fact retrieval.

7 Future Directions

We gathered all papers related to post-training methods in LLMs and analyzed their trends, as shown in Figure 7. Application of RL techniques [305, 57, 40] for refining the LLMs have a noticeable increase in prominence since 2020 (Figure 7a), emphasizing the demand for **interactive approaches** such as human-in-the-loop [35, 306] reinforcement and scalability [113, 84, 307]. At the same time, **reward modeling** [308, 168, 169] (Figure 7b) has seen a steady rise in interest due to the emergence of self-rewarding language models, yet the field still struggles with **reward hacking** [309, 310] and the design of robust [311], failure-aware reward functions beyond reward hacking [312]. **Decoding and search** (Figure 7c) methods include tree-of-thoughts [86] and Monte Carlo [313, 260] strategies aiming to enhance model reasoning through iterative self-critique [314, 306, 29], but

these techniques also demand reliable uncertainty estimators to prevent excessive computational overhead [315, 113]. Safety [301, 316, 317], robustness [318], and interpretability [319, 320, 321] have likewise become central concerns (Figure 7d), motivating the development of bias-aware [322, 323] and uncertainty-aware [324] RL methods beyond correlation with human uncertainty [325] that safeguard user trust and prevent adversarial attacks. Another crucial area involves **personalization** [326, 327] and **adaptation** [195] (Figure 7e), where efforts to tailor LLMs for specific domains must be balanced against risks to privacy [328], particularly when enterprise data or sensitive personal information is involved.

In parallel, **process** [329, 330] vs. **outcome reward optimization** [331] (Figure 7f) remains an open question: while process-based rewards help guide incremental improvements, outcome-focused metrics are simpler but may not capture crucial intermediate decision-making steps. Beyond reward structure, **fine-tuning** LLMs on new tasks still encounter issues like **catastrophic forgetting** [332] and potential data leakage [333, 334], underscoring the need for parameter-efficient methods [60] and privacy-preserving strategies such as differential privacy [335] and federated learning [336]. Human feedback, while central to alignment, is inherently costly and limited in scope; methods like **Constitutional AI** [53] and **RLAIF** [97] seek to automate parts of this oversight, though they introduce fresh concerns about bias calibration [337] and model self-consistency [186]. Finally, test-time scaling [113] and **dynamic reasoning** [338] frameworks pose further challenges: models must learn when to allocate more computation for complex queries, how to adapt verification modules [339] efficiently, and how to maintain robust performance even when facing adversarial inputs. These converging research directions—spanning reward modeling, decoding strategies, interpretability, personalization, and safe fine-tuning—highlight the multifaceted role of RL in LLMs and collectively shape the future trajectory of large-scale language model development. Below, we delve into some of these directions in greater detail.

Fine-tuning challenges. Fine-tuning remains one of the most direct post-training methods to adapt LLMs to specific tasks or domains, yet it faces several open challenges. One fundamental issue is catastrophic forgetting – when updating an LLM on new data causes it to lose or degrade previously learned capabilities. Even advanced PEFT methods like LoRA [60], which greatly reduce the number of trainable weights, do not fully solve this problem [332]. Future work can explore better continual learning strategies and regularization techniques so that models can acquire new skills without erasing old ones. For example, new fine-tuning algorithms (e.g. CURLoRA [332]) explicitly aim to stabilize training and preserve prior knowledge while adding new tasks. Promising research directions include curriculum-based fine-tuning [340] (introducing new facts gradually or in context with known facts) and hybrid training that combines retrieval or external knowledge bases. For instance, rather than solely adjusting the model’s weights, one could fine-tune LLMs to consult a knowledge repository or perform tool use (such as database queries or computations) when faced with queries outside their original training distribution [341, 342]. This retrieval-augmented fine-tuning [343] could let models incorporate fresh information at inference time, reducing the need to overwrite their internal weights with new facts. Another approach

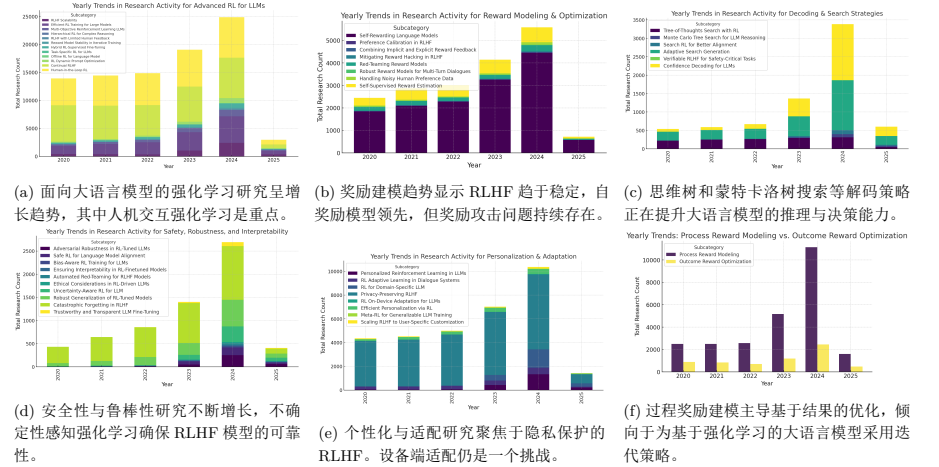


图 7: 大语言模型特定后训练方法中强化学习的年度趋势及新兴研究方向。

这一领域的新兴方向包括 **RLAIF** [97] 以及其他半自动化反馈技术 [346]: 使用强大的模型来评估或指导较弱的模型，甚至让多个 AI 智能体就一个问题进行辩论，并将它们的一致意见作为奖励信号 [347, 348]。这种人工智能辅助的反馈可以极大地扩展调优过程，并有助于克服人类专家时间有限的瓶颈。然而，它也引发了新的理论问题：我们如何确保 AI 评判者本身是对齐且正确的？如果自动化偏好存在缺陷，则存在反馈循环或偏见回音室的风险。一个开放的空白是创建稳健的 AI 反馈系统，这些系统需根据人类价值观进行校准（或许通过周期性的人类监督或一套多样化的宪法原则来“接地”）。在分层方案中融合人类和 AI 反馈，可以为大型语言模型提供一个可扩展且可靠的强化学习范式。

Test-time scaling challenges. TTS 中的开放挑战主要围绕如何高效、可靠地编排推理时过程。一个关键问题是：对于给定的查询，多少计算量是足够的？以及如何动态地确定这一点？使用过少资源可能导致错误，但使用过多则效率低下，并可能引入不一致性。Snell 等人 [85] 的最新研究通过提出一个包含“提议者”和“验证者”的统一框架来系统地探索和评估答案，从而应对了这一挑战。在他们的框架中，提议者（通常是基础 LLM）生成多个候选解决方案，而验证者（另一个模型或启发式方法）则进行判断并选择最佳方案。最优策略可能因问题难度而异：对于较简单的查询，并行生成多个答案并选取最优可能就足够了；而对于较难的问题，采用带有每一步验证的、顺序的、逐步推理则效果更好。一个重要的未来方向是构建自适应系统，其中 LLM 能基于对问题复杂度的估计动态分配计算资源。这一想法与人工智能中的元认知 [316] 相关联，使模型能够感知它

们不知道什么或哪些问题值得更多思考。为 LLMs 开发可靠的信度度量或难度预测器是一个开放的研究领域，但此处的进展将使 TTS 变得实用得多，即模型仅在必要时才“放慢速度并思考”，就像人类在难题上花费额外时间一样。此外，通过将推理时扩展重新定义为概率推理问题并采用基于粒子的蒙特卡洛方法 [349]，小型模型仅用 32 次推演就达到了 o1 级别的准确度，在各种数学推理任务中，扩展效率提升了 4-16 倍。最近的研究 [350] 表明，将测试时计算提炼为合成训练数据能产生协同的预训练效益，这一点也值得进一步探索。

Reward Modeling and Credit Assignment. 当前的 RL 方法受到奖励错误泛化的困扰，模型过度优化表面的代理指标，而非真正的推理质量。多步任务中终端奖励的稀疏性加剧了信用分配的挑战，尤其是在长视野推理场景中。传统方法如 DPO 需要低效的成对偏好数据，且未能有效利用失败轨迹。可以通过整合过程监督与基于结果的奖励，并采用对比式逐步评估 [351] 来研究混合奖励模型。这种方法能够对中间决策步骤进行更细粒度的评估，同时与长期目标保持一致。近期工作 [173] 表明，步骤级策略优化可以在保持安全约束的同时提高价值函数的准确性。可以通过为 Transformer 适配的时序差分学习 [352, 353] 来探索动态信用分配机制。此类适配可能增强模型捕捉长期依赖关系的能力，并优化在长序列上的奖励传播。可以通过对抗性数据增强 [354] 将负面示例纳入 RL 循环，从而开发失败感知训练策略。这可以通过系统性地将模型暴露于具有挑战性的场景并鼓励更具韧性的策略学习，来提高模型的鲁棒性。

Efficient RL Training and Distillation. 当前用于大语言模型的强化学习方法需要极高

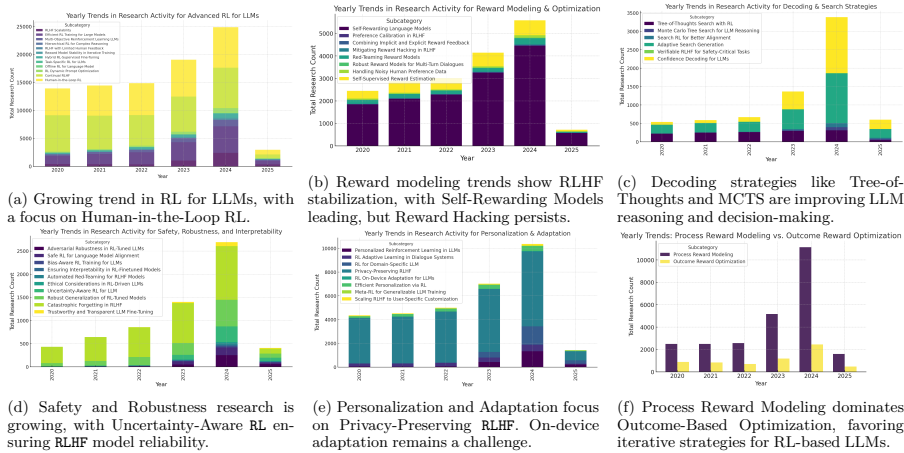


Fig. 7: Yearly Trends in RL specific post-training methods for LLMs and emerging research directions.

is training models to explicitly represent uncertainty about new knowledge, thereby enabling them to say ‘I don’t know’ or defer to an external source if a query concerns content not seen in pre-training. By blending weight updates with external knowledge integration, future fine-tuned LLMs will maintain higher factual accuracy and lower hallucination rates on emerging information.

Safe Fine-tuning. From an ethical and safety perspective, fine-tuning raises important open research questions. Fine-tuning data often contains sensitive or proprietary information [328], which can lead to privacy risks if the model memorizes and later regurgitates that data. A recent comprehensive survey [344] highlights vulnerabilities in the fine-tuning stage, such as membership inference attacks (detecting if a specific record was in the fine-tuning set) and data extraction (recovering parts of the fine-tuning data from the model’s outputs). Mitigating these risks is an open problem: methods like differential privacy fine-tuning [335] (adding noise to the weight updates) and federated fine-tuning (where data never leaves user devices and only aggregated updates are sent to the model) are being actively explored. However, these methods often come at the cost of model utility or require careful calibration to avoid degrading performance.

Limitations of Human Feedback. Human feedback is costly and subjective. One promising avenue to address the limitations of human feedback is using AI feedback and automation to assist or replace human evaluators. Constitutional AI [53], introduced by Anthropic, is a notable example: instead of relying on extensive human feedback for every harmful or helpful behavior, the model is guided by a set of written principles (a ‘constitution’) and is trained to critique and refine its own responses using another AI model as the judge [345]. Emerging directions here include RLATF [97] and other semi-automated feedback techniques [346]: using strong models to evaluate or guide weaker models, or even having

multiple AI agents debate a question and using their agreement as a reward signal [347, 348]. Such AI-aided feedback could vastly scale the tuning process and help overcome the bottleneck of limited human expert time. However, it raises new theoretical questions: how do we ensure the AI judge is itself aligned and correct? There is a risk of feedback loops or an echo chamber of biases if the automated preferences are flawed. An open gap is the creation of robust AI feedback systems that are calibrated to human values (perhaps periodically ‘grounded’ by human oversight or by a diverse set of constitutional principles). The blending of human and AI feedback in a hierarchical scheme could provide a scalable yet reliable RL paradigm for LLMs.

Test-time scaling challenges. Open challenges in TTS revolve around how to orchestrate the inference-time processes efficiently and reliably. A key question is how much computing is enough for a given query, and how to determine this on the fly? Using less resources can result in mistakes, but using too much is inefficient and could introduce inconsistencies. Recent research by Snell et al. [85] tackled it by proposing a unified framework with a ‘Proposer’ and a ‘Verifier’ to systematically explore and evaluate answers. In their framework, the Proposer (usually the base LLM) generates multiple candidate solutions, and the Verifier (another model or a heuristic) judges and selects the best. The optimal strategy can vary by problem difficulty: for easier queries, generating many answers in parallel and picking the top might be sufficient, whereas for harder problems, sequential, step-by-step reasoning with verification at each step works better. An important future direction is building adaptive systems where the LLM dynamically allocates computation based on an estimate of the question’s complexity. This idea connects to meta-cognition in AI [316], enabling models to have a sense of what they don’t know or what deserves more thought. Developing reliable

的计算资源 [355], 而其性能却常常不及知识蒸馏技术 [95]。这种低效性限制了可扩展性和实际部署, 因为蒸馏模型尽管所需的训练开销更少, 却经常超越经过强化学习训练模型。此外, 纯粹的强化学习方法难以平衡语言质量与推理能力的提升 [99, 95], 从而形成了性能瓶颈。

开发混合框架是一个有趣的方向, 这类框架利用从大模型蒸馏出的知识来初始化强化学习策略, 从而结合强化学习的探索优势与监督学习的稳定性。同样, 课程采样策略逐步增加任务复杂度, 同时利用蒸馏来保持语言连贯性, 也能提供帮助。PEFT 方法 [60] 可以在强化学习更新过程中被利用, 以在增强推理能力的同时保持基础能力。

整合: 将 PRM 引导的树搜索与在线蒸馏相结合, 相比基线方法实现了 4 倍的效率提升, 同时在 MATH 数据集上保持了 94% 的解题准确率。

Privacy-Preserving Personalization. 为企业及个人用例定制模型, 会因记忆化而增加暴露私有训练数据的风险, 这使得保护隐私的 [328] 适配变得至关重要。有前景的解决方案包括: 同态指令微调 [356], 它在推理过程中处理加密的用户查询并保持端到端加密; 通过奖励噪声化实现差分隐私 [357], 该方法在对齐过程中向 RLHF 偏好排名引入数学上有界的噪声; 以及联邦蒸馏, 它聚合来自分散的用户特定模型的知识, 而无需共享原始数据。随着单模型 [358, 359, 360] 扩展逼近物理极限, 诸如多智能体大语言模型协作 [361, 362, 180] 等替代范式变得必要。研究人员正在探索涌现的通信协议, 这些协议训练模型发展用于模型间知识传输的有损压缩“语言”, 例如 GenAInet[363]; 构建鲁棒集成系统, 其中压力测试诱导的个性化驱动基于失败分析的问题空间自动划分 [364]; 以及通过进化策略进行无梯度协同学习, 旨在发现互补的模型组合而无需依赖反向传播 [365]。

Multimodal RL Integration. 多模态强化学习 [366, 49, 367] 面临着组合状态爆炸的障碍, 尤其是在上下文超过 128k 令牌的场景中。克服这一障碍的开拓性方法包括: 采用具有跨注意力门控的模态特定策略的分层注意力框架 [61]; 在压缩上下文的同时保留关键推理片段的自适应截断策略 [368]; 以及利用自监督 [369, 370, 371] 复杂度预测来促进渐进式多模态集成的闪存课程方法。高效的强化学习训练范式仍然是一个关键的研究前沿, 因为现有方法表现出显著的样本效率低下和计算开销。要解决诸如过度思考 [372, 373] 现象 (其中过长的推理链浪费了宝贵的计算 [147]) 等问题, 需要采用诸如部分推演策略 [374]、采用学习型压缩变换器的自适应长度惩罚机制, 以及将蒙特卡洛树搜索与先进强化学习优化器相结合的混合架构等方法。这些创新对于将强化学习扩展到长上下文任务, 同时最大限度地减少计算资源浪费至关重要。

过度思考现象: 分析表明, 有 22% 的计算浪费在超过最优推理长度的推理链中。

强化学习方法表现出样本效率低下和计算开销大的问题, 尤其是在扩展到超过 128k 令牌上下文时。‘过度思考’现象——即模型生成长度过长的推理链——进一步降低了令牌效率并增加了部署成本 [375]。研究用于长上下文处理的、结合闪存注意力机制的部分推演策略。开发使用学习型压缩变换器的长度惩罚机制, 用于迭代式的长到短蒸馏。将 MCTS [76] 与分组相对策略优化 [59] 相结合的混合架构可能实现更好的探索-利用权衡。谢等人 [76] 的并行工作通过自适应树搜索剪枝展示了有希望的结果。该领域仍存在若干开放性挑战。不确定性传播仍然存在问题, 因为当前的置信度估计器增加了约 18% 的延迟开销, 而灾难性遗忘导致在强化学习微调 [376] 期间基础能力下降 29%。此外, 基准测试饱和也是一个问题, MMLU 分数与实际性能 [377] 的相关性很差 ($r = 0.34$)。

对抗性漏洞: 压力测试揭示了基于梯度的提示注入攻击具有高成功率。

8 Conclusion

本综述与教程系统回顾了大语言模型的后训练方法, 重点关注微调、强化学习和规模化。我们分析了关键技术, 以及提升效率与对齐人类偏好的策略。此外, 我们探讨了强化学习在通过推理、规划与多任务泛化增强大语言模型中的作用, 并在智能体-环境范式中对其功能进行分类。强化学习与测试时规模化的最新进展显著提升了大语言模型的推理能力, 使其能够应对日益复杂的任务。通过整合最新研究并指出开放挑战, 我们旨在为优化大语言模型以应用于现实世界指引未来方向。

参考文献

- [1] A. Vaswani, “Attention is all you need,” *Advances in Neural Information Processing Systems*, 2017. 1
- [2] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al., “Language models are few-shot learners,” *Advances in neural information processing systems*, vol. 33, pp. 1877–1901, 2020. 1
- [3] Z. Yang, “Xlnet: Generalized autoregressive pretraining for language understanding,” *arXiv preprint arXiv:1906.08237*, 2019. 1, 3, 17
- [4] J. Devlin, “Bert: Pre-training of deep bidirectional transformers for language understanding,” *arXiv preprint arXiv:1810.04805*, 2018. 1, 3
- [5] Z. Lan, “Albert: A lite bert for self-supervised learning of language representations,” *arXiv preprint arXiv:1909.11942*, 2019. 1
- [6] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, “Exploring the limits

confidence metrics or difficulty predictors for LLMs is an open research area, but progress here would make TTS far more practical i.e., the model would only ‘slow down and think’ when necessary, much like a human spending extra time on a hard problem. Additionally, By reframing inference-time scaling as a probabilistic inference problem and employing particle-based Monte Carlo methods [349], the small models achieved 0.1 level accuracy in only 32 rollouts, a 4–16x improvement in scaling efficiency across various mathematical reasoning tasks. Recent study [350] shows distilling test-time computations into synthetic training data creates synergistic pretraining benefits which can also be further explored.

Reward Modeling and Credit Assignment. Current RL approaches suffer from reward misgeneralization, where models over-optimize superficial proxy metrics rather than genuine reasoning quality. The sparse nature of terminal rewards in multi-step tasks increases credit assignment challenges, particularly in long-horizon reasoning scenarios. Traditional methods like DPO require inefficient pairwise preference data and fail to utilize failure trajectories effectively. Hybrid reward models can be investigated by integrating process supervision with outcome-based rewards using contrastive stepwise evaluation [351]. This approach enables a more granular assessment of intermediate decision-making steps while aligning with long-term objectives. Recent work [173] suggests step-level policy optimization could improve value function accuracy while maintaining safety constraints. Dynamic credit assignment mechanisms can be explored through temporal difference learning adapted for transformers [352, 353]. Such adaptations may enhance the model’s ability to capture long-range dependencies and optimize reward propagation over extended sequences. Failure-aware training strategies can be developed by incorporating negative examples into the RL loop via adversarial data augmentation [354]. This can improve model robustness by systematically exposing it to challenging scenarios and encouraging more resilient policy learning.

Efficient RL Training and Distillation. Current RL methods for LLMs require prohibitive computational resources [355] while often underperforming knowledge distillation techniques [95]. This inefficiency limits scalability and practical deployment, as distilled models frequently surpass RL-trained counterparts despite requiring less training overhead. Additionally, pure RL approaches struggle to balance language quality with reasoning improvement [99, 95], creating a performance ceiling.

The development of hybrid frameworks that initialize RL policies with distilled knowledge from large models, combining the exploratory benefits of RL with the stability of supervised learning is an interesting direction. Similarly, curriculum sampling strategies that progressively increase task complexity while using distillation to preserve linguistic coherence can also help. PEFT methods [60] can be leveraged during RL updates to maintain base capabilities while enhancing reasoning.



Integration: Combining PRM-guided tree search with online distillation achieves 4× efficiency gains over baseline methods, while maintaining 94% solution accuracy on MATH dataset.

Privacy-Preserving Personalization. Customizing mod-

els for enterprise and individual use cases raises the risk of exposing private training data through memorization, making privacy-preserving [328] adaptation essential. Promising solutions include homomorphic instruction tuning [356], which processes encrypted user queries while maintaining end-to-end encryption during inference; differential privacy via reward noising [357], which introduces mathematically bounded noise into RLHF preference rankings during alignment; and federated distillation, which aggregates knowledge from decentralized user-specific models without sharing raw data.

Collaborative Multi-Model Systems. As single-model [358, 359, 360] scaling approaches physical limits, alternative paradigms such as multi-agent LLM collaboration [361, 362, 180] become necessary. Researchers are investigating emergent communication protocols that train models to develop lossy compression “languages” for inter-model knowledge transfer such as GenAInet [363], robust ensembles where stress-test induced specialization drives automatic division of problem spaces based on failure analysis [364], and gradient-free synergy learning through evolutionary strategies designed to discover complementary model combinations without relying on backpropagation [365].

Multimodal RL Integration. Multimodal reinforcement learning [366, 49, 367] faces the obstacle of a combinatorial state explosion, especially in contexts exceeding 128k tokens. Pioneering methods to overcome this include hierarchical attention frameworks that employ modality-specific policies with cross-attention gating [61], adaptive truncation strategies that compress context while preserving critical reasoning segments [368], and flash curriculum approaches that leverage self-supervised [369, 370, 371] complexity prediction to facilitate progressive multimodal integration.

Efficient RL Training. Efficient RL training paradigms continue to be a critical research frontier as current methods exhibit significant sample inefficiency and computational overhead. Addressing issues like the overthinking [372, 373] phenomenon, where excessive reasoning chains waste valuable computation [147], requires approaches such as partial rollout strategies [374], adaptive length penalty mechanisms employing learned compression transformers, and hybrid architectures that combine MCTS with advanced RL optimizers. These innovations are essential for scaling RL to long-context tasks while minimizing wasted computational resources.

⚠ Overthinking Phenomenon: Analysis reveals 22% wasted computation is in reasoning chains exceeding optimal reasoning length.

RL methods exhibit sample inefficiency and computational overhead, particularly when scaling to contexts exceeding 128k tokens. The ‘overthinking’ phenomenon, where models generate excessively long reasoning chains, further reduces token efficiency and increases deployment costs [375]. Investigate partial rollout strategies with flash attention mechanisms for long-context processing. Develop length penalty mechanisms using learned compression transformers for iterative long2short distillation. Hybrid architectures combining MCTS [76] with GRPO [59] could enable better exploration-exploitation tradeoffs. Parallel work by Xie et al. [76] demonstrates promising results through adaptive tree search prun-

ing of transfer learning with a unified text-to-text transformer,” *Journal of machine learning research*, vol. 21, no. 140, pp. 1–67, 2020. 1

- [7] P. Verga, S. Hofstatter, S. Althammer, Y. Su, A. Piktus, A. Arkhangorodsky, M. Xu, N. White, and P. Lewis, “Replacing judges with juries: Evaluating llm generations with a panel of diverse models,” *arXiv preprint arXiv:2404.18796*, 2024. 1
- [8] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q. V. Le, D. Zhou, et al., “Chain-of-thought prompting elicits reasoning in large language models,” *Advances in neural information processing systems*, vol. 35, pp. 24824–24837, 2022. 1, 2, 3, 11, 12, 14, 17
- [9] C. Wang, Y. Deng, Z. Lyu, L. Zeng, J. He, S. Yan, and B. An, “Q*: Improving multi-step reasoning for llms with deliberative planning,” *arXiv preprint arXiv:2406.14283*, 2024. 1
- [10] Y. Wu, X. Han, W. Song, M. Cheng, and F. Li, “Mindmap: Constructing evidence chains for multi-step reasoning in large language models,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, pp. 19270–19278, 2024. 1
- [11] Y. Ding, Z. Wang, W. Ahmad, H. Ding, M. Tan, N. Jain, M. K. Ramanathan, R. Nallapati, P. Bhatia, D. Roth, et al., “Cross-codeeval: A diverse and multilingual benchmark for cross-file code completion,” *Advances in Neural Information Processing Systems*, vol. 36, 2024. 1
- [12] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altschmidt, S. Altmann, S. Anadkat, et al., “Gpt-4 technical report,” *arXiv preprint arXiv:2303.08774*, 2023. 1
- [13] A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang, A. Fan, et al., “The llama 3 herd of models,” *arXiv preprint arXiv:2407.21783*, 2024. 1, 4
- [14] G. Team, M. Riviere, S. Pathak, P. G. Sessa, C. Hardin, S. Bhupatiraju, L. Hussonot, T. Mesnard, B. Shahriari, A. Ramé, et al., “Gemini 2: Improving open language models at a practical size,” *arXiv preprint arXiv:2408.00118*, 2024. 1, 4
- [15] G. Team, R. Anil, S. Borgeaud, J.-B. Alayrac, J. Yu, R. Soricut, J. Schalkwyk, A. M. Dai, A. Hauth, K. Millican, et al., “Gemini: a family of highly capable multimodal models,” *arXiv preprint arXiv:2312.11805*, 2023. 1, 4
- [16] A. Liu, B. Feng, B. Wang, B. Wang, B. Liu, C. Zhao, C. Deng, C. Ruan, D. Dai, D. Guo, et al., “Deepseek-v2: A strong, economical, and efficient mixture-of-experts language model,” *arXiv preprint arXiv:2405.04434*, 2024. 1, 2, 4, 5
- [17] M. Abdin, J. Aneja, H. Awadalla, A. Awadallah, A. A. Awan, N. Bach, A. Bahree, A. Bakhtiari, J. Bao, H. Behl, et al., “Phi-3 technical report: A highly capable language model locally on your phone,” *arXiv preprint arXiv:2404.14219*, 2024. 1, 4
- [18] A. Fan, M. Lewis, and Y. Dauphin, “Hierarchical neural story generation,” *arXiv preprint arXiv:1805.04833*, 2018. 1
- [19] C. Chhun, P. Colombo, C. Clavel, and F. M. Suchanek, “Of human criteria and automatic metrics: A benchmark of the evaluation of story generation,” *arXiv preprint arXiv:2208.11646*, 2022. 1
- [20] S. Arif, S. Farid, A. H. Azeemi, A. Athar, and A. A. Raza, “The fellowship of the llms: Multi-agent workflows for synthetic preference optimization dataset generation,” *arXiv preprint arXiv:2408.08688*, 2024. 1
- [21] S. Ye, Y. Jo, D. Kim, S. Kim, H. Hwang, and M. Seo, “Selfee: Iterative self-revising llm empowered by self-feedback genera-
- [22] M. Musolesi, “Creative beam search: Llm-as-a-judge for improving response generation,” *ICCC*, 2024. 1
- [23] J. Ren, Y. Zhao, T. Vu, P. J. Liu, and B. Lakshminarayanan, “Self-evaluation improves selective generation in large language models,” in *Proceedings on*, pp. 49–64, PMLR, 2023. 1
- [24] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, et al., “Retrieval-augmented generation for knowledge-intensive nlp tasks,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020. 1, 2
- [25] Y. Lai, C. Li, Y. Wang, T. Zhang, R. Zhong, L. Zettlemoyer, W.-t. Yih, D. Fried, S. Wang, and T. Yu, “Ds-1000: A natural and reliable benchmark for data science code generation,” in *International Conference on Machine Learning*, pp. 18319–18345, PMLR, 2023. 1
- [26] B. Zhu, E. Frick, T. Wu, H. Zhu, K. Ganesan, W.-L. Chiang, J. Zhang, and J. Jiao, “Starling-7b: Improving helpfulness and harmlessness with rlai,” in *First Conference on Language Modeling*, 2024. 1, 4
- [27] D. Paul, M. Ismayilzada, M. Peyrard, B. Borges, A. Bosselut, R. West, and B. Faltings, “Refiner: Reasoning feedback on intermediate representations,” *arXiv preprint arXiv:2304.01904*, 2023. 1
- [28] Y. Xie, K. Kawaguchi, Y. Zhao, J. X. Zhao, M.-Y. Kan, J. He, and M. Xie, “Self-evaluation guided beam search for reasoning,” *Advances in Neural Information Processing Systems*, vol. 36, 2024. 1
- [29] H. Ye and H. T. Ng, “Self-judge: Selective instruction following with alignment self-evaluation,” *arXiv preprint arXiv:2409.00935*, 2024. 1, 19
- [30] Z. Luo, H. Wu, D. Li, J. Ma, M. Kankanhalli, and J. Li, “Videoautoarena: An automated arena for evaluating large multimodal models in video analysis through user simulation,” *arXiv preprint arXiv:2411.13281*, 2024. 1
- [31] S. Deng, W. Zhao, Y.-J. Li, K. Wan, D. Miranda, A. Kale, and Y. Tian, “Efficient self-improvement in multimodal large language models: A model-level judge-free approach,” *arXiv preprint arXiv:2411.17760*, 2024. 1, 2
- [32] T. Ashraf and J. Bashir, “Fate: Focal-modulated attention encoder for temperature prediction,” 2024. 1
- [33] D. Chen, R. Chen, S. Zhang, Y. Liu, Y. Wang, H. Zhou, Q. Zhang, Y. Wan, P. Zhou, and L. Sun, “Mllm-as-a-judge: Assessing multimodal llm-as-a-judge with vision-language benchmark,” *arXiv preprint arXiv:2402.04788*, 2024. 1
- [34] T. Hagendorff, S. Fabi, and M. Kosinski, “Human-like intuitive behavior and reasoning biases emerged in large language models but disappeared in chatgpt,” *Nature Computational Science*, vol. 3, no. 10, pp. 833–838, 2023. 1
- [35] Q. Pan, Z. Ashtorab, M. Desmond, M. S. Cooper, J. Johnson, R. Nair, E. Daly, and W. Geyer, “Human-centered design recommendations for llm-as-a-judge,” *arXiv preprint arXiv:2407.03479*, 2024. 1, 19
- [36] G. H. Chen, S. Chen, Z. Liu, F. Jiang, and B. Wang, “Humans or llms as the judge? a study on judgement biases,” *arXiv preprint arXiv:2402.10669*, 2024. 1
- [37] A. Newell, “Human problem solving,” *Upper Saddle River/Prentice Hall*, 1972. 1, 15
- [38] X. Wang, H. Kim, S. Rahman, K. Mitra, and Z. Miao, “Human-llm collaborative annotation through effective verification of

ing. Several open challenges persist in the field. Uncertainty propagation remains problematic as current confidence estimators add approximately 18% latency overhead, while catastrophic forgetting results in a degradation of 29% of base capabilities during RL fine-tuning [376]. Moreover, benchmark saturation is an issue, with MMLU scores correlating poorly ($r = 0.34$) with real-world performance [377].

⚠️ **Adversarial Vulnerabilities: Stress tests reveal a high success rate on gradient-based prompt injections.**

8 Conclusion

This survey and tutorial provides a systematic review of post-training methodologies for LLMs, focusing on fine-tuning, reinforcement learning, and scaling. We analyze key techniques, along with strategies for improving efficiency and alignment with human preferences. Additionally, we explore the role of RL in enhancing LLMs through reasoning, planning, and multi-task generalization, categorizing their functionalities within the agent-environment paradigm. Recent advancements in reinforcement learning and test-time scaling have significantly improved LLMs reasoning capabilities, enabling them to tackle increasingly complex tasks. By consolidating the latest research and identifying open challenges, we aim to guide future efforts in optimizing LLMs for real-world applications.

References

- [1] A. Vaswani, "Attention is all you need," *Advances in Neural Information Processing Systems*, 2017. **1**
- [2] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, *et al.*, "Language models are few-shot learners," *Advances in neural information processing systems*, vol. 33, pp. 1877–1901, 2020. **1**
- [3] Z. Yang, "Xlnet: Generalized autoregressive pretraining for language understanding," *arXiv preprint arXiv:1906.08237*, 2019. **1, 3, 19**
- [4] J. Devlin, "Bert: Pre-training of deep bidirectional transformers for language understanding," *arXiv preprint arXiv:1810.04805*, 2018. **1, 2, 3**
- [5] Z. Lan, "Albert: A lite bert for self-supervised learning of language representations," *arXiv preprint arXiv:1909.11942*, 2019. **1**
- [6] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, "Exploring the limits of transfer learning with a unified text-to-text transformer," *Journal of machine learning research*, vol. 21, no. 140, pp. 1–67, 2020. **1**
- [7] P. Verga, S. Hofstätter, S. Althammer, Y. Su, A. Piktus, A. Arkhangorodsky, M. Xu, N. White, and P. Lewis, "Replacing judges with juries: Evaluating llm generations with a panel of diverse models," *arXiv preprint arXiv:2404.18796*, 2024. **1**
- [8] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q. V. Le, D. Zhou, *et al.*, "Chain-of-thought prompting elicits reasoning in large language models," *Advances in neural information processing systems*, vol. 35, pp. 24824–24837, 2022. **1, 2, 3, 11, 12, 15, 18**
- [9] C. Wang, Y. Deng, Z. Lyu, L. Zeng, J. He, S. Yan, and B. An, "Q": Improving multi-step reasoning for llms with deliberative planning," *arXiv preprint arXiv:2406.14283*, 2024. **1**
- [10] Y. Wu, X. Han, W. Song, M. Cheng, and F. Li, "Mindmap: Constructing evidence chains for multi-step reasoning in large language models," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, pp. 19270–19278, 2024. **1**
- [11] Y. Ding, Z. Wang, W. Ahmad, H. Ding, M. Tan, N. Jain, M. K. Ramanathan, R. Nallapati, P. Bhatia, D. Roth, *et al.*, "Cross-codeval: A diverse and multilingual benchmark for cross-file code completion," *Advances in Neural Information Processing Systems*, vol. 36, 2024. **1**
- [12] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, *et al.*, "Gpt-4 technical report," *arXiv preprint arXiv:2303.08774*, 2023. **1**
- [13] A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang, A. Fan, *et al.*, "The llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024. **1, 5**
- [14] G. Team, M. Riviere, S. Pathak, P. G. Sessa, C. Hardin, S. Bhupatiraju, L. Hussenot, T. Mesnard, B. Shahriri, A. Ramé, *et al.*, "Gemma 2: Improving open language models at a practical size," *arXiv preprint arXiv:2408.00118*, 2024. **1, 5**
- [15] G. Team, R. Anil, S. Borgeand, J.-B. Alayrac, J. Yu, R. Soricut, J. Schalkwyk, A. M. Dai, A. Hauth, K. Millican, *et al.*, "Gemini: a family of highly capable multimodal models," *arXiv preprint arXiv:2312.11805*, 2023. **1, 5**
- [16] A. Liu, B. Feng, B. Wang, B. Wang, B. Liu, C. Zhao, C. Deng, C. Ruan, D. Dai, D. Guo, *et al.*, "Deepseek-v2: A strong, economical, and efficient mixture-of-experts language model," *arXiv preprint arXiv:2405.04434*, 2024. **1, 2, 5, 6**
- [17] M. Abdin, J. Anjia, H. Awadalla, A. Awadallah, A. A. Awan, N. Bach, A. Bahree, A. Bakhtiari, J. Bao, H. Behl, *et al.*, "Phi-3 technical report: A highly capable language model locally on your phone," *arXiv preprint arXiv:2404.14219*, 2024. **1, 5**
- [18] A. Fan, M. Lewis, and Y. Dauphin, "Hierarchical neural story generation," *arXiv preprint arXiv:1805.04833*, 2018. **1**
- [19] C. Chhun, P. Colombo, C. Clavel, and F. M. Suchanek, "Of human criteria and automatic metrics: A benchmark of the evaluation of story generation," *arXiv preprint arXiv:2208.11646*, 2022. **1**
- [20] S. Arif, S. Farid, A. H. Azeemi, A. Athar, and A. A. Raza, "The fellowship of the llms: Multi-agent workflows for synthetic preference optimization dataset generation," *arXiv preprint arXiv:2408.08688*, 2024. **1**
- [21] S. Ye, Y. Jo, D. Kim, S. Kim, H. Hwang, and M. Seo, "Selfsee: Iterative self-revising llm empowered by self-feedback generation," *Blog post*, 2023. **1**
- [22] M. Musolesi, "Creative beam search: Llm-as-a-judge for improving response generation," *ICCC*, 2024. **1**
- [23] J. Ren, Y. Zhao, T. Vu, P. J. Liu, and B. Lakshminarayanan, "Self-evaluation improves selective generation in large language models," in *Proceedings on*, pp. 49–64, PMLR, 2023. **1**
- [24] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, *et al.*, "Retrieval-augmented generation for knowledge-intensive nlp tasks," *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020. **1, 2**
- [25] Y. Lai, C. Li, Y. Wang, T. Zhang, R. Zhong, L. Zettlemoyer, W.-t. Yih, D. Fried, S. Wang, and T. Yu, "Ds-1000: A natural and reliable benchmark for data science code generation," in *International Conference on Machine Learning*, pp. 18319–18345, PMLR, 2023. **1**
- [26] B. Zhu, E. Frick, T. Wu, H. Zhu, K. Ganesan, W.-L. Chiang, J. Zhang, and J. Jiao, "Starling-7b: Improving helpfulness and harmlessness with rlai," in *First Conference on Language Modeling*, 2024. **1, 5**
- [27] D. Paul, M. Ismayilzada, M. Peyrard, B. Borges, A. Bosselut, R. West, and B. Faltings, "Refiner: Reasoning feedback on intermediate representations," *arXiv preprint arXiv:2304.01904*, 2023. **1**
- [28] Y. Xie, K. Kawaguchi, Y. Zhao, J. X. Zhao, M.-Y. Kan, J. He, and M. Xie, "Self-evaluation guided beam search for reasoning," *Advances in Neural Information Processing Systems*, vol. 36, 2024. **1**
- [29] H. Ye and H. T. Ng, "Self-judge: Selective instruction following with alignment self-evaluation," *arXiv preprint arXiv:2409.00935*, 2024. **1, 20**
- [30] Z. Luo, H. Wu, D. Li, J. Ma, M. Kankanhalli, and J. Li, "Videocutout: An automated arena for evaluating large multimodal models in video analysis through user simulation," *arXiv preprint arXiv:2411.13281*, 2024. **1**
- [31] S. Deng, W. Zhao, Y.-J. Li, K. Wan, D. Miranda, A. Kale, and Y. Tian, "Efficient self-improvement in multimodal large language models: A model-level judge-free approach," *arXiv preprint arXiv:2411.17760*, 2024. **1, 2**
- [32] llm labels," in *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pp. 1–21, 2024. **1**
- [39] R. OpenAI, "Gpt-4 technical report. arxiv 2303.08774," *View in Article*, vol. 2, no. 5, 2023. **1, 4, 11**
- [40] D. Guo, D. Yang, H. Zhang, J. Song, R. Zhang, R. Xu, Q. Zhu, S. Ma, P. Wang, X. Bi, *et al.*, "DeepSeek-R1: Incentivizing reasoning capability in llms via reinforcement learning," *arXiv preprint arXiv:2501.12948*, 2025. **1, 4, 10, 11, 19**
- [41] M. T. Hicks, J. Humphries, and J. Slater, "Chatgpt is bullshit," *Ethics and Information Technology*, vol. 26, no. 2, pp. 1–10, 2024. **1**
- [42] N. Maleki, B. Padmanabhan, and K. Dutta, "Ai hallucinations: A misnomer worth clarifying," 2024. **1**
- [43] A. Bruno, P. L. Mazzeo, A. Chetouani, M. Tliba, and M. A. Kerkouri, "Insights into classifying and mitigating llms' hallucinations," *arXiv preprint arXiv:2311.08117*, 2023. **1**
- [44] S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal, "Detecting hallucinations in large language models using semantic entropy," *Nature*, vol. 630, no. 8017, pp. 625–630, 2024. **1**
- [45] F. Leiser, S. Eckhardt, V. Leuthe, M. Knaeble, A. Maedche, G. Schwabe, and A. Sunyaev, "Hill: A hallucination identifier for large language models," in *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pp. 1–13, 2024. **1**
- [46] A. Gunjal, J. Yin, and E. Bas, "Detecting and preventing hallucinations in large vision language models," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, pp. 18135–18143, 2024. **1**
- [47] S. Hao, Y. Gu, H. Ma, J. J. Hong, Z. Wang, D. Z. Wang, and Z. Hu, "Reasoning with language model is planning with world model," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, EMNLP 2023, Singapore, December 6-10, 2023* (H. Bouamor, J. Pino, and K. Bali, eds.), pp. 8154–8173, Association for Computational Linguistics, 2023. **1**
- [48] Y. Ye, Z. Huang, Y. Xiao, E. Chern, S. Xia, and P. Liu, "Limo: Less is more for reasoning," 2025. **1, 3**
- [49] C. Li, W. Wu, H. Zhang, Y. Xia, S. Mao, L. Dong, I. Vulic, and F. Wei, "Imagine while reasoning in space: Multimodal visualization-of-thought," 2025. **1, 3, 21**
- [50] T. Xia, B. Yu, Y. Wu, Y. Chang, and C. Zhou, "Language models can evaluate themselves via probability discrepancy," *arXiv preprint arXiv:2405.10516*, 2024. **1**
- [51] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023. **1, 2**
- [52] H. He and W. J. Su, "A law of next-token prediction in large language models," 2024. **1**
- [53] Y. Bai, S. Kadavath, S. Kundu, A. Askell, J. Kernion, A. Jones, A. Chen, A. Goldie, A. Mirhoseini, C. McKinnon, *et al.*, "Constitutional ai: Harmlessness from ai feedback," *arXiv preprint arXiv:2212.08073*, 2022. **2, 19**
- [54] interconnects.ai, "blob reinforcement fine-tuning." (Accessed: 2025-12-6). **2**
- [55] E. Lobo, C. Agarwal, and H. Lakkaraju, "On the impact of fine-tuning on chain-of-thought reasoning," *arXiv preprint arXiv:2411.15382*, 2024. **2**
- [56] L. Trung, X. Zhang, Z. Jie, P. Sun, X. Jin, and H. Li, "Reft: Reasoning with reinforced fine-tuning," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 7601–7614, 2024. **2**
- [57] R. Rafailov, A. Sharma, E. Mitchell, C. D. Manning, S. Ermon, and C. Finn, "Direct preference optimization: Your language model is secretly a reward model," *Advances in Neural Information Processing Systems*, vol. 36, 2024. **2, 5, 19**
- [58] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, *et al.*, "Training language models to follow instructions with human feedback," *Advances in neural information processing systems*, vol. 35, pp. 27730–27744, 2022. **2, 4, 8, 12, 14**
- [59] Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. Li, Y. Wu, *et al.*, "Deepseekmath: Pushing the limits of mathematical reasoning in open language models," *arXiv preprint arXiv:2402.03300*, 2024. **2, 5, 7, 9, 10, 11, 21**
- [60] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," *arXiv preprint arXiv:2106.09685*, 2021. **2, 12, 13, 19, 21**
- [61] S. Ebrahimi, S. Ö. Arik, T. Nama, and T. Pfister, "Crome: Cross-modal adapters for efficient multimodal llm," *ArXiv*, vol. abs/2408.06610, 2024. **2, 21**
- [62] O. Tafjord, B. D. Mishra, and P. Clark, "Proofwriter: Generating implications, proofs, and abductive statements over natural language," 2021. **2**
- [63] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-rag: Learning to retrieve, generate, and critique through self-reflection," *arXiv preprint arXiv:2310.11511*, 2023. **2**
- [64] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, M. Wang, and H. Wang, "Retrieval-augmented generation for large language models: A survey," *arXiv preprint arXiv:2312.10997*, 2023. **2**
- [65] Z. Hu, L. Song, J. Zhang, Z. Xiao, J. Wang, Z. Chen, and H. Xiong, "Rethinking llm-based preference evaluation," *arXiv preprint arXiv:2407.01085*, 2024. **2**
- [66] S. Yue, W. Chen, S. Wang, B. Li, C. Shen, S. Liu, Y. Zhou, Y. Xiao, S. Yun, X. Huang, *et al.*, "Disc-lawllm: Fine-tuning large language models for intelligent legal services," *arXiv preprint arXiv:2309.11325*, 2023. **2**
- [67] Y. Luo, Z. Yang, F. Meng, Y. Li, J. Zhou, and Y. Zhang, "An empirical study of catastrophic forgetting in large language models during continual fine-tuning," *arXiv preprint arXiv:2308.08747*, 2023. **2**
- [68] H. Li, J. Chen, W. Su, Q. Ai, and Y. Liu, "Towards better web search performance: Pre-training, fine-tuning and learning to rank," 2023. **2**
- [69] OpenAI, "Reinforcement fine-tuning." (Accessed: 2025-12-6). **2**
- [70] W. Zhang, Y. Deng, B. Liu, S. J. Pan, and L. Bing, "Sentiment analysis in the era of large language models: A reality check," *arXiv preprint arXiv:2305.15005*, 2023. **2**
- [71] A. Maas, R. E. Daly, P. T. Pham, D. Huang, A. Y. Ng, and C. Potts, "Learning word vectors for sentiment analysis," in *Proceedings of the 49th annual meeting of the association for computational linguistics: Human language technologies*, pp. 142–150, 2011. **2**
- [72] Y. Wang, S. Wang, Y. Li, and D. Dou, "Recognizing medical search query intent by few-shot learning," in *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 502–512, 2022. **2**
- [73] R. Luo, L. Sun, Y. Xia, T. Qin, S. Zhang, H. Poon, and T.-Y.

- [32] T. Ashraf and J. Bashir, "Fate: Focal-modulated attention encoder for temperature prediction," 2024.
- [33] D. Chen, R. Chen, S. Zhang, Y. Liu, Y. Wang, H. Zhou, Q. Zhang, Y. Wan, P. Zhou, and L. Sun, "Mllm-as-a-judge: Assessing multimodal llm-as-a-judge with vision-language benchmark," *arXiv preprint arXiv:2402.04788*, 2024. [1](#)
- [34] T. Hagendorf, S. Fabi, and M. Kosinski, "Human-like intuitive behavior and reasoning biases emerged in large language models but disappeared in chatgpt," *Nature Computational Science*, vol. 3, no. 10, pp. 833–838, 2023. [1](#)
- [35] Q. Pan, Z. Ashktorab, M. Desmond, M. S. Cooper, J. Johnson, R. Nair, E. Daly, and W. Geyer, "Human-centered design recommendations for llm-as-a-judge," *arXiv preprint arXiv:2407.03479*, 2024. [1](#), [20](#)
- [36] G. H. Chen, S. Chen, Z. Liu, F. Jiang, and B. Wang, "Humans or llms as the judge? a study on judgement biases," *arXiv preprint arXiv:2402.10669*, 2024. [1](#)
- [37] A. Newell, "Human problem solving," *Upper Saddle River/Prentice Hall*, 1972. [1](#), [16](#)
- [38] X. Wang, H. Kim, S. Rahman, K. Mitra, and Z. Miao, "Human-llm collaborative annotation through effective verification of llm labels," in *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pp. 1–21, 2024. [1](#)
- [39] R. OpenAI, "Gpt-4 technical report, arxiv 2303.08774," *View in Article*, vol. 2, no. 5, 2023. [1](#), [5](#), [12](#)
- [40] D. Guo, D. Yang, H. Zhang, J. Song, R. Zhang, R. Xu, Q. Zhu, S. Ma, P. Wang, X. Bi, *et al.*, "DeepSeek-R1: Incentivizing reasoning capability in llms via reinforcement learning," *arXiv preprint arXiv:2501.12948*, 2025. [1](#), [5](#), [11](#), [12](#), [20](#)
- [41] M. T. Hicks, J. Humphries, and J. Slater, "Chatgpt is bullshit," *Ethics and Information Technology*, vol. 26, no. 2, pp. 1–10, 2024. [1](#)
- [42] N. Maleki, B. Padmanabhan, and K. Dutta, "Ai hallucinations: A misnomer worth clarifying," 2024. [1](#)
- [43] A. Bruno, P. L. Mazzeo, A. Chetouani, M. Tliba, and M. A. Kerkouri, "Insights into classifying and mitigating llms' hallucinations," *arXiv preprint arXiv:2311.08117*, 2023. [1](#)
- [44] S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal, "Detecting hallucinations in large language models using semantic entropy," *Nature*, vol. 630, no. 8017, pp. 625–630, 2024. [1](#)
- [45] F. Leiser, S. Eckhardt, V. Leuthe, M. Knaeble, A. Maedche, G. Schwabe, and A. Sunyaev, "Hill: A hallucination identifier for large language models," in *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pp. 1–13, 2024. [1](#)
- [46] A. Gunjal, J. Yin, and E. Bas, "Detecting and preventing hallucinations in large vision language models," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, pp. 18135–18143, 2024. [1](#)
- [47] S. Hao, Y. Gu, H. Ma, J. J. Hong, Z. Wang, D. Z. Wang, and Z. Hu, "Reasoning with language model is planning with world model," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, EMNLP 2023, Singapore, December 6-10, 2023* (H. Bouamor, J. Pino, and K. Bali, eds.), pp. 8154–8173, Association for Computational Linguistics, 2023. [1](#)
- [48] Y. Ye, Z. Huang, Y. Xiao, E. Chern, S. Xia, and P. Liu, "Limo: Less is more for reasoning," 2025. [1](#), [3](#)
- [49] C. Li, W. Wu, H. Zhang, Y. Xia, S. Mao, L. Dong, I. Vulić, and F. Wei, "Imagine while reasoning in space: Multimodal visualization-of-thought," 2025. [1](#), [3](#), [22](#)
- [50] T. Xia, B. Yu, Y. Wu, Y. Chang, and C. Zhou, "Language models can evaluate themselves via probability discrepancy," *arXiv preprint arXiv:2405.10516*, 2024. [1](#)
- [51] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023. [1](#), [3](#)
- [52] H. He and W. J. Su, "A law of next-token prediction in large language models," 2024. [2](#)
- [53] Y. Bai, S. Kadavath, S. Kundu, A. Askell, J. Kernion, A. Jones, A. Chen, A. Goldie, A. Mirhoseini, C. McKinmon, *et al.*, "Constitutional ai: Harmlessness from ai feedback," *arXiv preprint arXiv:2312.08073*, 2022. [2](#), [20](#), [21](#)
- [54] interconnects.ai, "blob reinforcement fine-tuning." (Accessed: 2025-12-6). [2](#)
- [55] E. Lobo, C. Agarwal, and H. Lakkaraju, "On the impact of fine-tuning on chain-of-thought reasoning," *arXiv preprint arXiv:2411.15382*, 2024. [2](#), [3](#)
- [56] L. Trung, X. Zhang, Z. Jie, P. Sun, X. Jin, and H. Li, "Reft: Reasoning with reinforced fine-tuning," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 7601–7614, 2024. [2](#)
- [57] R. Rafailov, A. Sharma, E. Mitchell, C. D. Manning, S. Ermon, and C. Finn, "Direct preference optimization: Your language model is secretly a reward model," *Advances in Neural Information Processing Systems*, vol. 36, 2024. [2](#), [3](#), [6](#), [20](#)
- [58] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, *et al.*, "Training language models to follow instructions with human feedback," *Advances in neural information processing systems*, vol. 35, pp. 27730–27744, 2022. [2](#), [3](#), [5](#), [9](#), [13](#), [15](#)
- [59] Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. Li, Y. Wu, *et al.*, "Deepseekmath: Pushing the limits of mathematical reasoning in open language models," *arXiv preprint arXiv:2402.03300*, 2024. [2](#), [3](#), [6](#), [8](#), [10](#), [12](#), [22](#)
- [60] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," *arXiv preprint arXiv:2106.09685*, 2021. [2](#), [13](#), [20](#), [22](#)
- [61] S. Ebrahimi, S. Ö. Arik, T. Nama, and T. Pfister, "Crome: Cross-modal adapters for efficient multimodal llm," *arXiv, abs/2408.06610*, 2024. [2](#), [22](#)
- [62] O. Tafjord, B. D. Mishra, and P. Clark, "Proofwriter: Generating implications, proofs, and abductive statements over natural language," 2021.
- [63] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-rag: Learning to retrieve, generate, and critique through self-reflection," *arXiv preprint arXiv:2310.11511*, 2023. [2](#)
- [64] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, M. Wang, and H. Wang, "Retrieval-augmented generation for large language models: A survey," *arXiv preprint arXiv:2312.10997*, 2023. [2](#)
- [65] Z. Hu, L. Song, J. Zhang, Z. Xiao, J. Wang, Z. Chen, and H. Xiong, "Rethinking llm-based preference evaluation," *arXiv preprint arXiv:2407.01085*, 2024. [2](#)
- [66] S. Yue, W. Chen, S. Wang, B. Li, C. Shen, S. Liu, Y. Zhou, Y. Xiao, S. Yun, X. Huang, *et al.*, "Disc-lawllm: Fine-tuning large language models for intelligent legal services," *arXiv preprint arXiv:2309.11325*, 2023. [2](#)
- [67] Y. Luo, Z. Yang, F. Meng, Y. Li, J. Zhou, and Y. Zhang, "An empirical study of catastrophic forgetting in large language models during continual fine-tuning," *arXiv preprint arXiv:2308.08747*, 2023. [2](#)
- [68] H. Li, J. Chen, W. Su, Q. Ai, and Y. Liu, "Towards better web search performance: Pre-training, fine-tuning and learning to rank," 2023. [2](#)
- [69] OpenAI, "Reinforcement fine-tuning." (Accessed: 2025-12-6). [2](#)
- [70] W. Zhang, Y. Deng, B. Liu, S. J. Pan, and L. Bing, "Sentiment analysis in the era of large language models: A reality check," *arXiv preprint arXiv:2305.15005*, 2023. [2](#)
- [71] A. Maas, R. E. Daly, P. T. Pham, D. Huang, A. Y. Ng, and C. Potts, "Learning word vectors for sentiment analysis," in *Proceedings of the 49th annual meeting of the association for computational linguistics: Human language technologies*, pp. 142–150, 2011. [2](#)
- [72] Y. Wang, S. Wang, Y. Li, and D. Dou, "Recognizing medical search query intent by few-shot learning," in *Proceedings of the 43th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 502–512, 2022. [2](#)
- [73] R. Luo, L. Sun, Y. Xia, T. Qin, S. Zhang, H. Poon, and T.-Y. Liu, "Biogpt: generative pre-trained transformer for biomedical text generation and mining," *Briefings in bioinformatics*, vol. 23, no. 6, p. bbac409, 2022. [2](#), [13](#)
- [74] O. Wysocki, M. Wysocka, D. Carvalho, A. T. Bogatu, D. M. Guscuma, M. Delmas, H. Unsworth, and A. Freitas, "An llm-based knowledge synthesis and scientific reasoning framework for biomedical discovery," *arXiv preprint arXiv:2406.18626*, 2024. [2](#)
- [75] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017. [2](#), [5](#), [6](#), [8](#), [10](#)
- [76] Y. Xie, A. Goyal, W. Zheng, M.-Y. Kan, T. P. Lillicrap, K. Kawaguchi, and M. Shieh, "Monte carlo tree search boosts
- Liu, "Biogpt: generative pre-trained transformer for biomedical text generation and mining," *Briefings in bioinformatics*, vol. 23, no. 6, p. bbac409, 2022. [2](#), [12](#)
- [74] O. Wysocki, M. Wysocka, D. Carvalho, A. T. Bogatu, D. M. Guscuma, M. Delmas, H. Unsworth, and A. Freitas, "An llm-based knowledge synthesis and scientific reasoning framework for biomedical discovery," *arXiv preprint arXiv:2406.18626*, 2024. [2](#)
- [75] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017. [2](#), [5](#), [7](#), [8](#), [9](#)
- [76] Y. Xie, A. Goyal, W. Zheng, M.-Y. Kan, T. P. Lillicrap, K. Kawaguchi, and M. Shieh, "Monte carlo tree search boosts reasoning via iterative preference learning," *arXiv preprint arXiv:2405.00451*, 2024. [2](#), [17](#), [21](#)
- [77] M. Shanahan, K. McDonnell, and L. Reynolds, "Role play with large language models," *Nature*, vol. 623, no. 7987, pp. 493–498, 2023. [2](#)
- [78] T. Xu, E. Helenowski, K. A. Sankararaman, D. Jin, K. Peng, E. Han, S. Nie, C. Zhu, H. Zhang, W. Zhou, *et al.*, "The perfect blend: Redefining rlhf with mixture of judges," *arXiv preprint arXiv:2409.20370*, 2024. [2](#)
- [79] M. Cao, L. Shu, L. Yu, Y. Zhu, N. Wichers, Y. Liu, and L. Meng, "Beyond sparse rewards: Enhancing reinforcement learning with language model critique in text generation," 2024. [2](#)
- [80] Y. Dubois, C. X. Li, R. Taori, T. Zhang, I. Gulrajani, J. Ba, C. Guestrin, P. S. Liang, and T. B. Hashimoto, "Alpacafarm: A simulation framework for methods that learn from human feedback," *Advances in Neural Information Processing Systems*, vol. 36, 2024. [2](#), [14](#)
- [81] L. Shani, A. Rosenberg, A. Cassel, O. Lang, D. Calandriello, A. Zipori, H. Noga, O. Keller, B. Piot, I. Szpektor, *et al.*, "Multi-turn reinforcement learning from preference human feedback," *arXiv preprint arXiv:2405.14655*, 2024. [2](#)
- [82] Z. Li, Y. He, L. He, J. Wang, T. Shi, B. Lei, Y. Li, and Q. Chen, "Falcon: Feedback-driven adaptive long/short-term memory reinforced coding optimization system," *arXiv preprint arXiv:2410.21349*, 2024. [2](#)
- [83] R. Nakano, J. Hilton, S. Balaji, J. Wu, L. Ouyang, C. Kim, C. Hesse, S. Jain, V. Kosaraju, W. Saunders, *et al.*, "Webgpt: Browser-assisted question-answering with human feedback," *arXiv preprint arXiv:2112.09332*, 2021. [2](#), [11](#), [14](#)
- [84] L. Gao, J. Schulman, and J. Hilton, "Scaling laws for reward model overoptimization," in *International Conference on Machine Learning*, pp. 10835–10866, PMLR, 2023. [2](#), [14](#), [19](#)
- [85] C. Snell, J. Lee, K. Xu, and A. Kumar, "Scaling llm test-time compute optimally can be more effective than scaling model parameters," *arXiv preprint arXiv:2408.03314*, 2024. [2](#), [14](#), [20](#)
- [86] S. Yao, D. Yu, J. Zhao, I. Shafraan, T. Griffiths, Y. Cao, and K. Narasimhan, "Tree of thoughts: Deliberate problem solving with large language models," *Advances in Neural Information Processing Systems*, vol. 36, 2024. [2](#), [13](#), [15](#), [19](#)
- [87] J. Jiang, D. He, and J. Allan, "Searching, browsing, and clicking in a search session: Changes in user behavior by task and over time," in *Proceedings of the 37th international ACM SIGIR conference on Research & development in information retrieval*, pp. 607–616, 2014. [2](#)
- [88] Y. Xie, K. Kawaguchi, Y. Zhao, J. X. Zhao, M.-Y. Kan, J. He, and M. Xie, "Self-evaluation guided beam search for reasoning,"
- Advances in Neural Information Processing Systems*, vol. 36, 2024. [2](#)
- [89] Y. Tian, B. Peng, L. Song, L. Jin, D. Yu, H. Mi, and D. Yu, "Toward self-improvement of llms via imagination, searching, and criticizing," *arXiv preprint arXiv:2404.12253*, 2024. [2](#), [17](#)
- [90] K. Gandhi, D. H. J. Lee, G. Grand, M. Liu, W. Cheng, A. Sharma, and N. Goodman, "Stream of search (SoS): Learning to search in language," in *First Conference on Language Modeling*, 2024. [2](#)
- [91] K. Yang, A. M. Swope, A. Gu, R. Chalamala, P. Song, S. Yu, S. Godil, R. Prenger, and A. Anandkumar, "Leandjo: Theorem proving with retrieval-augmented language models," 2023. [2](#)
- [92] C. Sun, S. Huang, and D. Pompili, "Retrieval-augmented hierarchical in-context reinforcement learning and hindsight modular reflections for task planning with llms," 2024. [2](#)
- [93] E. Davis, "Testing gpt-4o-1-preview on math and science problems: A follow-up study," *arXiv preprint arXiv:2410.22340*, 2024. [2](#)
- [94] J. Y. Wang, N. Sukiennik, T. Li, W. Su, Q. Hao, J. Xu, Z. Huang, F. Xu, and Y. Li, "A survey on human-centric llms," *arXiv preprint arXiv:2411.14491*, 2024. [2](#)
- [95] J. Wu, S. Yang, R. Zhan, Y. Yuan, L. S. Chao, and D. F. Wong, "A survey on llm-generated text detection: Necessity, methods, and future directions," *Computational Linguistics*, pp. 1–65, 2025. [2](#), [13](#), [21](#)
- [96] Y. Chang, X. Wang, J. Wang, Y. Wu, L. Yang, K. Zhu, H. Chen, X. Yi, C. Wang, Y. Wang, *et al.*, "A survey on evaluation of large language models," *ACM Transactions on Intelligent Systems and Technology*, vol. 15, no. 3, pp. 1–45, 2024. [2](#)
- [97] H. Lee, S. Phatale, H. Mansoor, K. R. Lu, T. Mesnard, J. Ferret, C. Bishop, E. Hall, V. Carbone, and A. Rastogi, "Rlaif: Scaling reinforcement learning from human feedback with ai feedback," 2023. [2](#), [8](#), [19](#), [20](#)
- [98] Q. Dong, L. Li, D. Dai, C. Zheng, Z. Wu, B. Chang, X. Sun, J. Xu, and Z. Sui, "A survey on in-context learning," *arXiv preprint arXiv:2301.00234*, 2022. [2](#)
- [99] W. X. Zhao, K. Zhou, J. Li, T. Tang, X. Wang, Y. Hou, Y. Min, B. Zhang, J. Zhang, Z. Dong, *et al.*, "A survey of large language models," *arXiv preprint arXiv:2303.18223*, 2023. [2](#), [21](#)
- [100] S. Han, H. Schoelkopf, Y. Zhao, Z. Qi, M. Riddell, W. Zhou, J. Coady, D. Peng, Y. Qiao, L. Benson, L. Sun, A. Wardle-Solano, H. Szabo, E. Zubova, M. Burtell, J. Fan, Y. Liu, B. Wong, M. Sailor, A. Ni, L. Nan, J. Kasai, T. Yu, R. Zhang, A. R. Fabbri, W. Kryscinski, S. Yavuz, Y. Liu, X. V. Lin, S. Joty, Y. Zhou, C. Xiong, R. Ying, A. Cohan, and D. Radev, "Folio: Natural language reasoning with first-order logic," 2024. [2](#)
- [101] Z. Xi, S. Jin, Y. Zhou, R. Zheng, S. Gao, T. Gui, Q. Zhang, and X. Huang, "Self-polish: Enhance reasoning in large language models via problem refinement," 2024. [2](#)
- [102] A. Saparov and H. He, "Language models are greedy reasoners: A systematic formal analysis of chain-of-thought," 2023. [2](#)
- [103] J. Liu, C. Wang, C. Y. Liu, L. Zeng, R. Yan, Y. Sun, Y. Liu, and Y. Zhou, "Improving multi-step reasoning abilities of large language models with direct advantage policy optimization," *arXiv preprint arXiv:2412.18279*, 2024. [2](#)
- [104] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," *Advances in neural information processing systems*, vol. 35, pp. 22199–

- reasoning via iterative preference learning,” *arXiv preprint arXiv:2405.00451*, 2024. [2](#), [18](#), [22](#)
- [77] M. Shanahan, K. McDonnell, and L. Reynolds, “Role play with large language models,” *Nature*, vol. 623, no. 7987, pp. 493–498, 2023. [2](#)
- [78] T. Xu, E. Helenowski, K. A. Sankararaman, D. Jin, K. Peng, E. Han, S. Nie, C. Zhu, H. Zhang, W. Zhou, *et al.*, “The perfect blend: Redefining rlhf with mixture of judges,” *arXiv preprint arXiv:2409.20370*, 2024. [2](#)
- [79] M. Cao, L. Shu, L. Yu, Y. Zhu, N. Wichers, Y. Liu, and L. Meng, “Beyond sparse rewards: Enhancing reinforcement learning with language model critique in text generation,” 2024. [2](#)
- [80] Y. Dubois, C. X. Li, R. Taori, T. Zhang, I. Gulrajani, J. Ba, C. Guestrin, P. S. Liang, and T. B. Hashimoto, “AlpacaFarm: A simulation framework for methods that learn from human feedback,” *Advances in Neural Information Processing Systems*, vol. 36, 2024. [2](#), [15](#)
- [81] L. Shani, A. Rosenberg, A. Cassel, O. Lang, D. Calandriello, A. Zipori, H. Noga, O. Keller, B. Piot, I. Szepesktor, *et al.*, “Multi-turn reinforcement learning from preference human,” *arXiv preprint arXiv:2405.14655*, 2024. [2](#)
- [82] Z. Li, Y. He, L. He, J. Wang, T. Shi, B. Lei, Y. Li, and Q. Chen, “Falcon: Feedback-driven adaptive long/short-term memory reinforced coding optimization system,” *arXiv preprint arXiv:2410.21349*, 2024. [2](#)
- [83] R. Nakano, J. Hilton, S. Balaji, J. Wu, L. Ouyang, C. Kim, C. Hesse, S. Jain, V. Kosaraju, W. Saunders, *et al.*, “Webgpt: Browser-assisted question-answering with human feedback,” *arXiv preprint arXiv:2112.09332*, 2021. [2](#), [11](#), [15](#)
- [84] L. Gao, J. Schulman, and J. Hilton, “Scaling laws for reward model overoptimization,” in *International Conference on Machine Learning*, pp. 10835–10866, PMLR, 2023. [2](#), [15](#), [20](#)
- [85] C. Snell, J. Lee, K. Xu, and A. Kumar, “Scaling llm test-time compute optimally can be more effective than scaling model parameters,” *arXiv preprint arXiv:2408.03314*, 2024. [2](#), [15](#), [21](#)
- [86] S. Yao, D. Yu, J. Zhao, I. Shafraan, T. Griffiths, Y. Cao, and K. Narasimhan, “Tree of thoughts: Deliberate problem solving with large language models,” *Advances in Neural Information Processing Systems*, vol. 36, 2024. [2](#), [14](#), [16](#), [20](#)
- [87] J. Jiang, D. He, and J. Allan, “Searching, browsing, and clicking in a search session: Changes in user behavior by task and over time,” in *Proceedings of the 37th international ACM SIGIR conference on Research & development in information retrieval*, pp. 607–616, 2014. [2](#)
- [88] Y. Xie, K. Kawaguchi, Y. Zhao, J. X. Zhao, M.-Y. Kan, J. He, and M. Xie, “Self-evaluation guided beam search for reasoning,” *Advances in Neural Information Processing Systems*, vol. 36, 2024. [2](#)
- [89] Y. Tian, B. Peng, L. Song, L. Jin, D. Yu, H. Mi, and D. Yu, “Toward self-improvement of llms via imagination, searching, and criticizing,” *arXiv preprint arXiv:2404.12253*, 2024. [2](#), [18](#)
- [90] K. Gandhi, D. H. J. Lee, G. Grand, M. Liu, W. Cheng, A. Sharma, and N. Goodman, “Stream of search (SoS): Learning to search in language,” in *First Conference on Language Modeling*, 2024. [2](#)
- [91] K. Yang, A. M. Swope, A. Gu, R. Chalamala, P. Song, S. Yu, S. Godil, R. Prenger, and A. Anandkumar, “Leandjo: Theorem proving with retrieval-augmented language models,” 2023. [2](#)
- [92] C. Sun, S. Huang, and D. Pompili, “Retrieval-augmented hierarchical in-context reinforcement learning and hindsight modular reflections for task planning with llms,” 2024. [2](#)
- [93] E. Davis, “Testing gpt-4o-preview on math and science problems: A follow-up study,” *arXiv preprint arXiv:2410.22340*, 2024. [2](#)
- [94] J. Y. Wang, N. Sukienik, T. Li, W. Su, Q. Hao, J. Xu, Z. Huang, F. Xu, and Y. Li, “A survey on human-centric llms,” *arXiv preprint arXiv:2411.14491*, 2024. [3](#)
- [95] J. Wu, S. Yang, R. Zhan, Y. Yuan, L. S. Chao, and D. F. Wong, “A survey on llm-generated text detection: Necessity, methods, and future directions,” *Computational Linguistics*, pp. 1–65, 2025. [3](#), [14](#), [22](#)
- [96] Y. Chang, X. Wang, J. Wang, Y. Wu, L. Yang, K. Zhu, H. Chen, X. Yi, C. Wang, Y. Wang, *et al.*, “A survey on evaluation of large language models,” *ACM Transactions on Intelligent Systems and Technology*, vol. 15, no. 3, pp. 1–45, 2024. [3](#)
- [97] H. Lee, S. Phatale, H. Mansoor, K. R. Lu, T. Mesnard, J. Ferret, C. Bishop, E. Hall, V. Carbone, and A. Rastogi, “Rlaif: Scaling reinforcement learning from human feedback with ai feedback,” 2023. [3](#), [9](#), [20](#), [21](#)
- [98] Q. Dong, L. Li, D. Dai, C. Zheng, Z. Wu, B. Chang, X. Sun, J. Xu, and Z. Sui, “A survey on in-context learning,” *arXiv preprint arXiv:2301.00231*, 2022. [3](#)
- [99] W. X. Zhao, K. Zhou, J. Li, T. Tang, X. Wang, Y. Hou, Y. Min, B. Zhang, J. Zhang, Z. Dong, *et al.*, “A survey of large language models,” *arXiv preprint arXiv:2303.18223*, 2023. [3](#), [22](#)
- [100] S. Han, H. Schoelkopf, Y. Zhao, Z. Qi, M. Riddell, W. Zhou, J. Coady, D. Peng, Y. Qiao, L. Benson, L. Sun, A. Wardle-Solano, H. Szabo, E. Zubova, M. Burtell, J. Fan, Y. Liu, B. Wong, M. Sailor, A. Ni, L. Nan, J. Kasai, T. Yu, R. Zhang, A. R. Fabbri, W. Kryscinski, S. Yavuz, Y. Liu, X. V. Lin, S. Joty, Y. Zhou, C. Xiong, R. Ying, A. Cohan, and D. Radev, “Folio: Natural language reasoning with first-order logic,” 2024. [3](#)
- [101] Z. Xi, S. Jin, Y. Zhou, R. Zheng, S. Gao, T. Gui, Q. Zhang, and X. Huang, “Self-polish: Enhance reasoning in large language models via problem refinement,” 2024. [3](#)
- [102] A. Saparov and H. He, “Language models are greedy reasoners: A systematic formal analysis of chain-of-thought,” 2023. [3](#)
- [103] J. Liu, C. Wang, C. Y. Liu, L. Zeng, R. Yan, Y. Sun, Y. Liu, and Y. Zhou, “Improving multi-step reasoning abilities of large language models with direct advantage policy optimization,” *arXiv preprint arXiv:2412.18279*, 2024. [3](#)
- [104] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, “Large language models are zero-shot reasoners,” *Advances in neural information processing systems*, vol. 35, pp. 22199–22213, 2022. [3](#)
- [105] X. Deng, Y. Su, A. Lees, Y. Wu, C. Yu, and H. Sun, “Reasonbert: Pre-trained to reason with distant supervision,” *arXiv preprint arXiv:2109.04912*, 2021. [3](#)
- [106] T. Sawada, D. Paleka, A. Havrilla, P. Tadepalli, P. Vidas, A. Kranias, J. J. Nay, K. Gupta, and A. Komatsuzaki, “Arb: Advanced reasoning benchmark for large language models,” 2023. [3](#)
- [107] A. Radford, “Improving language understanding by generative pre-training,” 2018. [3](#)
- [108] I. J. Myung, “Tutorial on maximum likelihood estimation,” *Journal of mathematical Psychology*, vol. 47, no. 1, pp. 90–100, 2003. [3](#)
- [109] Y. Shao, J. Mao, Y. Liu, W. Ma, K. Satoh, M. Zhang, and S. Ma, “Bert-pli: Modeling paragraph-level interactions for legal case retrieval,” in *IJCAI*, pp. 3501–3507, 2020. [3](#)
- [110] J. Wei, Y. Tay, R. Bommasani, C. Raffel, B. Zoph, S. Borgeaud, D. Yogatama, M. Bosma, D. Zhou, D. Metzler, *et al.*, “Emergent abilities of large language models,” *arXiv preprint arXiv:2206.07682*, 2022. [3](#)
- [111] R. Bellman, “A markovian decision process,” *Journal of mathematics and mechanics*, pp. 679–684, 1957. [9](#)
- [112] S. Hao, Y. Gu, H. Luo, T. Liu, X. Shao, X. Wang, S. Xie, H. Ma, A. Samavedhi, Q. Gao, Z. Wang, and Z. Hu, “Llm reasoners: New evaluation, library, and analysis of step-by-step reasoning with large language models,” 2024. [3](#)
- [113] J. Geiping, S. McLeish, N. Jain, J. Kirchenbauer, S. Singh, B. R. Bartoldson, B. Kaikhura, A. Bhatle, and T. Goldstein, “Scaling up test-time compute with latent reasoning: A recurrent depth approach,” 2025. [3](#), [19](#)
- [114] H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, and K. Cobbe, “Let’s verify step by step,” *arXiv preprint arXiv:2305.20050*, 2023. [3](#)
- [115] G. Chen, M. Liao, C. Li, and K. Fan, “Step-level value preference optimization for mathematical reasoning,” *arXiv preprint arXiv:2406.10858*, 2024. [3](#)
- [116] K. Nguyen, H. Daumé III, and J. Boyd-Graber, “Reinforcement learning for bandit neural machine translation with simulated human feedback,” *arXiv preprint arXiv:1707.07402*, 2017. [3](#), [4](#), [5](#)
- [117] G. Williams, “Substantive and adjectival law,” in *Learning the Law*, pp. 19–23, Law Book Company, Limited, 1982. [3](#)
- [118] M. Ranzato, S. Chopra, M. Auli, and W. Zaremba, “Sequence level training with recurrent neural networks,” 2016. [3](#), [5](#)
- [119] S. J. Rennie, E. Marcheret, Y. Mroueh, J. Ross, and V. Goel, “Self-critical sequence training for image captioning,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 7008–7024, 2017. [4](#), [5](#)
- [120] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “Bleu: a method for automatic evaluation of machine translation,” in *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pp. 311–318, 2002. [4](#)
- [121] R. Vedantam, C. L. Zitnick, and D. Parikh, “Cider: Consensus-based image description evaluation,” 2015. [4](#)
- [122] OpenAI, “Openai gpt-4.5 system card,” 2025. Accessed: 2025-02-28. [4](#)
- [123] Anthropic, “Claude 3.7 sonnet,” 2025. Accessed: 2025-02-26. [4](#), [18](#)
- [124] R. Team, A. Ormazabal, C. Zheng, C. d. M. d’Autume, D. Yogatama, D. Fu, D. Ong, E. Chen, E. Lamprecht, H. Pham, *et al.*, “Reka core, flash, and edge: A series of powerful multimodal language models,” *arXiv preprint arXiv:2404.12387*, 2024. [4](#)
- [125] B. Adler, N. Agarwal, A. Aithal, D. H. Anh, P. Bhattacharya, A. Brundyn, J. Casper, B. Catanzaro, S. Clay, J. Cohen, *et al.*, “Nemotron-4 340b technical report,” *arXiv preprint arXiv:2406.11704*, 2024. [4](#)
- [126] E. Almazrouei, H. Alolaidi, A. Alshamsi, A. Cappelli, R. Cojocar, M. Debbah, Étienne Goffinet, D. Hesslow, J. Lau-nay, Q. Malartic, D. Mazzotta, B. Nouné, B. Pannier, and G. Penedo, “The falcon series of open language models,” 2023. [4](#)
- [127] B. Hui, J. Yang, Z. Cui, J. Yang, D. Liu, L. Zhang, T. Liu, J. Zhang, B. Yu, K. Lu, *et al.*, “Qwen2. 5-coder technical report,” *arXiv preprint arXiv:2409.12186*, 2024. [4](#)
- [128] A. Défossez, L. Mazari, M. Orsini, A. Royer, P. Pérez, H. Jégou, E. Grave, and N. Zeghidour, “Moshi: a speech-text foundation model for real-time dialogue,” tech. rep., 2024. [4](#)
- [129] Nexusflow, “Athene: An rlhf-enhanced language model,” 2024. Accessed: 2025-02-26. [4](#)
- [130] R. Teknium, J. Quesnelle, and C. Guang, “Hermes 3 technical report,” *arXiv preprint arXiv:2408.11857*, 2024. [4](#)
- [131] Z. AI, “Zed,” 2025. 500B, Zed AI, RLHF, Multi-modal, Open. [4](#)
- [132] R. Anil, A. M. Dai, O. Firat, M. Johnson, D. Lepikhin, A. Passos, S. Shakeri, E. Taropa, P. Bailey, Z. Chen, *et al.*, “Palm 2 technical report,” *arXiv preprint arXiv:2305.10403*, 2023. [4](#)
- [133] Z. Cai, M. Cao, H. Chen, K. Chen, K. Chen, X. Chen, X. Chen, Z. Chen, Z. Chen, P. Chu, *et al.*, “Internlm2 technical report,” *arXiv preprint arXiv:2403.17297*, 2024. [4](#)
- [134] S. Labs, “Supernova,” 2025. 220B, Supernova Labs, RLHF, Multi-modal, Open. [4](#)
- [135] xAI, “Grok-3: The next generation ai model by xai,” tech. rep., xAI, 2025. Accessed: 2025-02-24. [4](#)
- [136] P. Agrawal, S. Antoniak, *et al.*, “Pixtral 12b,” 2024. [4](#)
- [137] MiniMax, A. Li, *et al.*, “Minimax-01: Scaling foundation models with lightning attention,” 2025. [4](#)
- [138] A. A. G. Intelligence, “The amazon nova family of models: Technical report and model card,” *Amazon Technical Reports*, 2024. [4](#)
- [139] Fujitsu and T. I. of Technology, “Fugaku-llm: The largest cpu-only trained language model,” *Fujitsu Research Press Release*, May 2024. [4](#)
- [140] R. AI, “Nova: A family of ai models by rubik’s ai,” *Rubik’s AI Research*, October 2024. [4](#)
- [141] OpenAI, “Openai o3 system card,” technical report, OpenAI, 2025. [4](#)
- [142] M. R. Team *et al.*, “Introducing dbrx: a new state-of-the-art open llm,” *Mosaic AI Research*, 2024. [4](#)
- [143] A. Köpf, Y. Kilcher, D. von Rütte, S. Anagnostidis, Z.-R. Tam, K. Stevens, A. Barhoum, N. M. Duc, O. Stanley, R. Nagyi, S. ES, S. Suri, D. Glushkov, A. Dantuluri, A. Maguire, C. Schuhmann, H. Nguyen, and A. Mattick, “Openassistant conversations – democratizing large language model alignment,” 2023. [4](#)
- [144] T. GLM, A. Zeng, B. Xu, B. Wang, C. Zhang, D. Yin, D. Zhang, D. Rojas, G. Feng, *et al.*, “Chatglm: A family of large language models from glm-130b to glm-4 all tools,” *arXiv preprint arXiv:2406.12793*, 2024. [4](#)
- [145] A. Bartolome, J. Hong, N. Lee, K. Rasul, and L. Tunstall,

- recognition, pp. 7008–7024, 2017. 4, 5
- [120] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “Bleu: a method for automatic evaluation of machine translation,” in *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pp. 311–318, 2002. 4
- [121] R. Vedantam, C. L. Zitnick, and D. Parikh, “Cider: Consensus-based image description evaluation,” 2015. 4
- [122] OpenAI, “Openai gpt-4.5 system card,” 2025. Accessed: 2025-02-28. 5
- [123] Anthropic, “Claude 3.7 sonnet,” 2025. Accessed: 2025-02-26. 5, 19, 20
- [124] R. Team, A. Ormazabal, C. Zheng, C. d. M. d’Autume, D. Yagatama, D. Fu, D. Ong, E. Chen, E. Lamprecht, H. Pham, et al., “Reka core, flash, and edge: A series of powerful multimodal language models,” *arXiv preprint arXiv:2404.12387*, 2024. 5
- [125] B. Adler, N. Agarwal, A. Aithal, D. H. Anh, P. Bhattacharya, A. Brundyn, J. Casper, B. Catanzaro, S. Clay, J. Cohen, et al., “Nemotron-4 340b technical report,” *arXiv preprint arXiv:2406.11704*, 2024. 5
- [126] E. Almazrouei, H. Alobeidli, A. Alshamsi, A. Cappelli, R. Cojocaru, M. Debbab, Étienne Goffinet, D. Hessel, J. Lau-nay, Q. Malaric, D. Mazzotta, B. Nouné, B. Pannier, and G. Penedo, “The falcon series of open language models,” 2023. 5
- [127] B. Hui, J. Yang, Z. Cui, J. Yang, D. Liu, L. Zhang, T. Liu, J. Zhang, B. Yu, K. Lu, et al., “Qwen2. 5-coder technical report,” *arXiv preprint arXiv:2409.12186*, 2024. 5
- [128] A. Défossez, L. Mazaré, M. Orsini, A. Royer, P. Pérez, H. Jégou, E. Grave, and N. Zeghidour, “Moshi: a speech-text foundation model for real-time dialogue,” tech. rep., 2024. 5
- [129] Nexusflow, “Athene: An rllm-enhanced language model,” 2024. Accessed: 2025-02-26. 5
- [130] R. Teknium, J. Quesnelle, and C. Guang, “Hermes 3 technical report,” *arXiv preprint arXiv:2408.11857*, 2024. 5
- [131] Z. AI, “Zed,” 2025. 500B, Zed AI, RLHF, Multi-modal, Open. 5
- [132] R. Anil, A. M. Dai, O. Firat, M. Johnson, D. Lepikhin, A. Pas-sos, S. Shakeri, E. Taropa, P. Bailey, Z. Chen, et al., “Palm 2 technical report,” *arXiv preprint arXiv:2305.10403*, 2023. 5
- [133] Z. Cai, M. Cao, H. Chen, K. Chen, K. Chen, X. Chen, X. Chen, Z. Chen, Z. Chen, P. Chu, et al., “Internlm2 technical report,” *arXiv preprint arXiv:2403.17297*, 2024. 5
- [134] S. Labs, “Supernova,” 2025. 202B, Supernova Labs, RLHF, Multi-modal, Open. 5
- [135] xAI, “Grok-3: The next generation ai model by xai,” tech. rep., xAI, 2025. Accessed: 2025-02-24. 5
- [136] P. Agrawal, S. Antoniak, et al., “Pixtral 12b,” 2024. 5
- [137] MiniMax, A. Li, et al., “Minimax-01: Scaling foundation models with lightning attention,” 2025. 5
- [138] A. A. G. Intelligence, “The amazon nova family of models: Technical report and model card,” *Amazon Technical Reports*, 2024. 5
- [139] Fujitsu and T. I. of Technology, “Fugaku-llm: The largest cpu-only trained language model,” *Fujitsu Research Press Release*, May 2024. 5
- [140] R. AI, “Nova: A family of ai models by rubik’s ai,” *Rubik’s AI Research*, October 2024. 5
- [141] OpenAI, “Openai o3 system card,” technical report, OpenAI, 2025. 5
- [142] M. R. Team et al., “Introducing dbx: a new state-of-the-art open llm,” *Mosaic AI Research*, 2024. 5
- [143] A. Köpf, Y. Kilcher, D. von Rütte, S. Anagnostidis, Z.-R. Tam, K. Stevens, A. Barhoum, N. Duc, O. Stanley, R. Nagyfi, S. ES, S. Suri, D. Glushkov, A. Dattuluri, A. Maguire, C. Schuhmann, H. Nguyen, and A. Mantlik, “Openassistant conversations – democratizing large language model alignment,” 2023. 5
- [144] T. GLM, A. Zeng, B. Xu, B. Wang, C. Zhang, D. Yin, D. Zhang, D. Rojas, G. Feng, H. Zhao, et al., “Chatglm: A family of large language models from glm-130b to glm-4 all tools,” *arXiv preprint arXiv:2406.12793*, 2024. 5
- [145] A. Bartolome, J. Hong, N. Lee, K. Rasul, and L. Tunstall, “Zephyr 141b a39b,” 2024. 5
- [146] O. Lieber, O. Sharir, B. Lenz, and Y. Shoham, “Jurassic-1: Technical details and evaluation,” *White Paper. A121 Labs*, vol. 1, no. 9, pp. 1–17, 2021. 5
- [147] K. Team, A. Du, B. Gao, B. Xing, C. Jiang, C. Chen, C. Li, C. Xiao, C. Du, C. Liao, et al., “Kimi k1. 5: Scaling reinforcement learning with llms,” *arXiv preprint arXiv:2501.12599*, 2025. 5, 22
- [148] M. Abdin, J. Aneja, H. Behl, S. Bubeck, R. Eldan, S. Gu-nasekar, M. Harrison, R. J. Hewett, M. Javaheripi, P. Kauff-mann, J. R. Lee, Y. T. Lee, Y. Li, W. Liu, C. C. T. Mendes, A. Nguyen, E. Price, G. de Rosa, O. Saarikivi, A. Salim, S. Shah, X. Wang, R. Ward, Y. Wu, D. Yu, C. Zhang, and Y. Zhang, “Phi-4 technical report,” 2024. 5
- [149] C. Team, “Chameleon: Mixed-modal early-fusion foundation models,” *arXiv preprint arXiv:2405.09818*, 2024. 5
- [150] N. Dey, G. Gosal, Zhiming, Chen, H. Khachane, W. Marshall, R. Pathria, M. Tom, and J. Hestness, “Cerebras-gpt: Open compute-optimal language models trained on the cerebras wafer-scale cluster,” 2023. 5
- [151] S. Wu, O. Irsoy, S. Lu, V. Dabravolski, M. Dredze, S. Gehrmann, P. Kambadur, D. Rosenberg, and G. Mann, “Bloomberggpt: A large language model for finance,” 2023. 5
- [152] J. Hoffmann, S. Borgeaud, A. Mensch, E. Buchatskaya, T. Cai, E. Rutherford, D. de Las Casas, L. A. Hendricks, J. Welbl, A. Clark, T. Hennigan, E. Noland, K. Millican, G. van den Driessche, B. Damoc, A. Guy, S. Osindero, K. Simonyan, E. Elsen, J. W. Rae, O. Vinyals, and L. Sifre, “Training compute-optimal large language models,” 2022. 5
- [153] S. Shen, Y. Cheng, Z. He, W. He, H. Wu, M. Sun, and Y. Liu, “Minimum risk training for neural machine translation,” 2016. 4, 5
- [154] V. Konda and J. Tsitsiklis, “Actor-critic algorithms,” *Advances in neural information processing systems*, vol. 12, 1999. 4
- [155] S. Bhatnagar, M. Ghavamzadeh, M. Lee, and R. S. Sutton, “In-cremental natural actor-critic algorithms,” *Advances in neural information processing systems*, vol. 20, 2007. 4
- [156] V. Mnih, “Asynchronous methods for deep reinforcement learning,” *arXiv preprint arXiv:1602.01783*, 2016. 4, 5
- [157] M. Babaeizadeh, I. Frosio, S. Tyree, J. Clemons, and J. Kautz, “Reinforcement learning through asynchronous advantage actor-critic on a gpu,” *arXiv preprint arXiv:1611.06256*, 2016. 4, 5
- [158] S. M. Kakade, “A natural policy gradient,” *Advances in neural information processing systems*, vol. 14, 2001. 4
- [159] A. G. Barto, R. S. Sutton, and C. W. Anderson, “Neuronlike adaptive elements that can solve difficult learning control problems,” *IEEE transactions on systems, man, and cybernetics*, no. 5, pp. 834–846, 1983. 5
- [160] L. Yu, W. Zhang, J. Wang, and Y. Yu, “Seqgan: Sequence generative adversarial nets with policy gradient,” 2017. 5
- [161] A. Yang, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Li, D. Liu, F. Huang, H. Wei, et al., “Qwen2. 5 technical report,” *arXiv preprint arXiv:2412.15115*, 2024. 5, 6
- [162] J. Schulman, S. Levine, P. Moritz, M. I. Jordan, and P. Abbeel, “Trust region policy optimization,” 2017. 6, 8, 9
- [163] N. Vieillard, T. Kozuno, B. Scherrer, O. Pietquin, R. Munos, and M. Geist, “Leverage the average: an analysis of kl regularization in reinforcement learning,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 12163–12174, 2020. 6
- [164] S. Guo, B. Zhang, T. Liu, T. Liu, M. Khalman, F. Llinares, A. Rame, T. Mesnard, Y. Zhao, B. Piot, et al., “Direct language model alignment from online ai feedback,” *arXiv preprint arXiv:2402.04792*, 2024. 6, 9, 10
- [165] C. An, M. Zhong, Z. Wu, Q. Zhu, X. Huang, and X. Qiu, “Colo: A contrastive learning based re-ranking framework for one-stage summarization,” *arXiv preprint arXiv:2209.14569*, 2022. 6
- [166] R. A. Bradley and M. E. Terry, “Rank analysis of incomplete block designs: I. the method of paired comparisons,” *Biometrika*, vol. 39, no. 3/4, pp. 324–345, 1952. 6
- [167] R. L. Plackett, “The analysis of permutations,” *Journal of the Royal Statistical Society Series C: Applied Statistics*, vol. 24, no. 2, pp. 193–202, 1975. 6
- [168] A. Setlur, C. Nagpal, A. Fisch, X. Geng, J. Eisenstein, R. Agarwal, A. Agarwal, J. Berant, and A. Kumar, “Rewarding progress: Scaling automated process verifiers for llm reasoning,” *arXiv preprint arXiv:2410.08146*, 2024. 7, 20
- [169] N. Lambert, V. Pyatkin, J. Morrison, L. Miranda, B. Y. Lin, K. Chandu, N. Dziri, S. Kumar, T. Zick, Y. Choi, et al., “Rewardbench: Evaluating reward models for language modeling,” 2024. 4
- [170] J. Hoffmann, S. Borgeaud, A. Mensch, E. Buchatskaya, T. Cai, E. Rutherford, D. de Las Casas, L. A. Hendricks, J. Welbl, A. Clark, T. Hennigan, E. Noland, K. Millican, G. van den Driessche, B. Damoc, A. Guy, S. Osindero, K. Simonyan, E. Elsen, J. W. Rae, O. Vinyals, and L. Sifre, “Training compute-optimal large language models,” 2022. 4
- [171] S. Shen, Y. Cheng, Z. He, W. He, H. Wu, M. Sun, and Y. Liu, “Minimum risk training for neural machine translation,” 2016. 4, 5
- [172] V. Konda and J. Tsitsiklis, “Actor-critic algorithms,” *Advances in neural information processing systems*, vol. 12, 1999. 4
- [173] S. Bhatnagar, M. Ghavamzadeh, M. Lee, and R. S. Sutton, “In-cremental natural actor-critic algorithms,” *Advances in neural information processing systems*, vol. 20, 2007. 4
- [174] V. Mnih, “Asynchronous methods for deep reinforcement learning,” *arXiv preprint arXiv:1602.01783*, 2016. 4, 5
- [175] M. Babaeizadeh, I. Frosio, S. Tyree, J. Clemons, and J. Kautz, “Reinforcement learning through asynchronous advantage actor-critic on a gpu,” *arXiv preprint arXiv:1611.06256*, 2016. 4, 5
- [176] S. M. Kakade, “A natural policy gradient,” *Advances in neural information processing systems*, vol. 14, 2001. 5
- [177] A. G. Barto, R. S. Sutton, and C. W. Anderson, “Neuronlike adaptive elements that can solve difficult learning control problems,” *IEEE transactions on systems, man, and cybernetics*, no. 5, pp. 834–846, 1983. 5
- [178] L. Yu, W. Zhang, J. Wang, and Y. Yu, “Seqgan: Sequence generative adversarial nets with policy gradient,” 2017. 5
- [179] A. Yang, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Li, D. Liu, F. Huang, H. Wei, et al., “Qwen2. 5 technical report,” *arXiv preprint arXiv:2412.15115*, 2024. 5
- [180] J. Schulman, S. Levine, P. Moritz, M. I. Jordan, and P. Abbeel, “Trust region policy optimization,” 2017. 5, 7, 8
- [181] N. Vieillard, T. Kozuno, B. Scherrer, O. Pietquin, R. Munos, and M. Geist, “Leverage the average: an analysis of kl regularization in reinforcement learning,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 12163–12174, 2020. 6
- [182] S. Guo, B. Zhang, T. Liu, T. Liu, M. Khalman, F. Llinares, A. Rame, T. Mesnard, Y. Zhao, B. Piot, et al., “Direct language model alignment from online ai feedback,” *arXiv preprint arXiv:2402.04792*, 2024. 6, 9, 10
- [183] C. An, M. Zhong, Z. Wu, Q. Zhu, X. Huang, and X. Qiu, “Colo: A contrastive learning based re-ranking framework for one-stage summarization,” *arXiv preprint arXiv:2209.14569*, 2022. 6
- [184] R. A. Bradley and M. E. Terry, “Rank analysis of incomplete block designs: I. the method of paired comparisons,” *Biometrika*, vol. 39, no. 3/4, pp. 324–345, 1952. 6
- [185] R. L. Plackett, “The analysis of permutations,” *Journal of the Royal Statistical Society Series C: Applied Statistics*, vol. 24, no. 2, pp. 193–202, 1975. 6
- [186] A. Setlur, C. Nagpal, A. Fisch, X. Geng, J. Eisenstein, R. Agarwal, A. Agarwal, J. Berant, and A. Kumar, “Rewarding progress: Scaling automated process verifiers for llm reasoning,” *arXiv preprint arXiv:2410.08146*, 2024. 7, 20
- [187] N. Lambert, V. Pyatkin, J. Morrison, L. Miranda, B. Y. Lin, K. Chandu, N. Dziri, S. Kumar, T. Zick, Y. Choi, et al., “Rewardbench: Evaluating reward models for language modeling,” 2024. 4
- [188] O. Lieber, O. Sharir, B. Lenz, and Y. Shoham, “Jurassic-1: Technical details and evaluation,” *White Paper. A121 Labs*, vol. 1, no. 9, pp. 1–17, 2021. 5
- [189] K. Team, A. Du, B. Gao, B. Xing, C. Jiang, C. Chen, C. Li, “Zephyr 141b a39b,” 2024. 4
- [190] O. Lieber, O. Sharir, B. Lenz, and Y. Shoham, “Jurassic-1: Technical details and evaluation,” *White Paper. A121 Labs*, vol. 1, no. 9, pp. 1–17, 2021. 4
- [191] K. Team, A. Du, B. Gao, B. Xing, C. Jiang, C. Chen, C. Li, C. Xiao, C. Du, C. Liao, et al., “Kimi k1. 5: Scaling reinforcement learning with llms,” *arXiv preprint arXiv:2501.12599*, 2025. 4, 21
- [192] M. Abdin, J. Aneja, H. Behl, S. Bubeck, R. Eldan, S. Gu-nasekar, M. Harrison, R. J. Hewett, M. Javaheripi, P. Kauff-mann, J. R. Lee, Y. T. Lee, Y. Li, W. Liu, C. C. T. Mendes, A. Nguyen, E. Price, G. de Rosa, O. Saarikivi, A. Salim, S. Shah, X. Wang, R. Ward, Y. Wu, D. Yu, C. Zhang, and Y. Zhang, “Phi-4 technical report,” 2024. 4
- [193] C. Team, “Chameleon: Mixed-modal early-fusion foundation models,” *arXiv preprint arXiv:2405.09818*, 2024. 4
- [194] N. Dey, G. Gosal, Zhiming, Chen, H. Khachane, W. Marshall, R. Pathria, M. Tom, and J. Hestness, “Cerebras-gpt: Open compute-optimal language models trained on the cerebras wafer-scale cluster,” 2023. 4
- [195] S. Wu, O. Irsoy, S. Lu, V. Dabravolski, M. Dredze, S. Gehrmann, P. Kambadur, D. Rosenberg, and G. Mann, “Bloomberggpt: A large language model for finance,” 2023. 4
- [196] J. Hoffmann, S. Borgeaud, A. Mensch, E. Buchatskaya, T. Cai, E. Rutherford, D. de Las Casas, L. A. Hendricks, J. Welbl, A. Clark, T. Hennigan, E. Noland, K. Millican, G. van den Driessche, B. Damoc, A. Guy, S. Osindero, K. Simonyan, E. Elsen, J. W. Rae, O. Vinyals, and L. Sifre, “Training compute-optimal large language models,” 2022. 4
- [197] S. Shen, Y. Cheng, Z. He, W. He, H. Wu, M. Sun, and Y. Liu, “Minimum risk training for neural machine translation,” 2016. 4, 5
- [198] V. Konda and J. Tsitsiklis, “Actor-critic algorithms,” *Advances in neural information processing systems*, vol. 12, 1999. 4
- [199] S. Bhatnagar, M. Ghavamzadeh, M. Lee, and R. S. Sutton, “In-cremental natural actor-critic algorithms,” *Advances in neural information processing systems*, vol. 20, 2007. 4
- [200] V. Mnih, “Asynchronous methods for deep reinforcement learning,” *arXiv preprint arXiv:1602.01783*, 2016. 4, 5
- [201] M. Babaeizadeh, I. Frosio, S. Tyree, J. Clemons, and J. Kautz, “Reinforcement learning through asynchronous advantage actor-critic on a gpu,” *arXiv preprint arXiv:1611.06256*, 2016. 4, 5
- [202] S. M. Kakade, “A natural policy gradient,” *Advances in neural information processing systems*, vol. 14, 2001. 4
- [203] A. G. Barto, R. S. Sutton, and C. W. Anderson, “Neuronlike adaptive elements that can solve difficult learning control problems,” *IEEE transactions on systems, man, and cybernetics*, no. 5, pp. 834–846, 1983. 5
- [204] L. Yu, W. Zhang, J. Wang, and Y. Yu, “Seqgan: Sequence generative adversarial nets with policy gradient,” 2017. 5
- [205] A. Yang, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Li, D. Liu, F. Huang, H. Wei, et al., “Qwen2. 5 technical report,” *arXiv preprint arXiv:2412.15115*, 2024. 5
- [206] J. Schulman, S. Levine, P. Moritz, M. I. Jordan, and P. Abbeel, “Trust region policy optimization,” 2017. 5, 7, 8
- [207] N. Vieillard, T. Kozuno, B. Scherrer, O. Pietquin, R. Munos, and M. Geist, “Leverage the average: an analysis of kl regularization in reinforcement learning,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 12163–12174, 2020. 6
- [208] S. Guo, B. Zhang, T. Liu, T. Liu, M. Khalman, F. Llinares, A. Rame, T. Mesnard, Y. Zhao, B. Piot, et al., “Direct language model alignment from online ai feedback,” *arXiv preprint arXiv:2402.04792*, 2024. 6, 9
- [209] C. An, M. Zhong, Z. Wu, Q. Zhu, X. Huang, and X. Qiu, “Colo: A contrastive learning based re-ranking framework for one-stage summarization,” *arXiv preprint arXiv:2209.14569*, 2022. 5
- [210] R. A. Bradley and M. E. Terry, “Rank analysis of incomplete block designs: I. the method of paired comparisons,” *Biometrika*, vol. 39, no. 3/4, pp. 324–345, 1952. 6
- [211] R. L. Plackett, “The analysis of permutations,” *Journal of the Royal Statistical Society Series C: Applied Statistics*, vol. 24, no. 2, pp. 193–202, 1975. 6
- [212] A. Setlur, C. Nagpal, A. Fisch, X. Geng, J. Eisenstein, R. Agarwal, A. Agarwal, J. Berant, and A. Kumar, “Rewarding progress: Scaling automated process verifiers for llm reasoning,” *arXiv preprint arXiv:2410.08146*, 2024. 6, 19
- [213] N. Lambert, V. Pyatkin, J. Morrison, L. Miranda, B. Y. Lin, K. Chandu, N. Dziri, S. Kumar, T. Zick, Y. Choi, et al., “Rewardbench: Evaluating reward models for language modeling,” *arXiv preprint arXiv:2403.13787*, 2024. 6, 18, 19
- [214] J. Hong, N. Lee, and J. Thorne, “Orpo: Monolithic preference optimization without reference model,” 2024. 8
- [215] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, “High-dimensional continuous control using generalized advantage estimation,” 2018. 8
- [216] R. Rafailov, Y. Chittipuru, R. Park, H. S. Sikchi, J. Hejna, B. Knox, C. Finn, and S. Niekum, “Scaling laws for reward model abstraction in direct alignment algorithms,” *ArXiv*, vol. abs/2406.02900, 2024. 8
- [217] H. Wang, S. Hao, H. Dong, S. Zhang, Y. Bao, Z. Yang, and Y. Wu, “Offline reinforcement learning for llm multi-step reasoning,” 2024. 9, 20
- [218] G. Mukobi, P. Chatain, S. Fong, R. Windesheim, G. Kutyniok, K. Bhatia, and S. Alberti, “Superhf: Supervised iterative learning from human feedback,” *arXiv preprint arXiv:2310.16763*, 2023. 10
- [219] C. Li, H. Zhou, G. Glavaš, A. Korhonen, and I. Vulić, “On task performance and model calibration with supervised and self-ensembled in-context learning,” *arXiv preprint arXiv:2312.13772*, 2023. 10
- [220] C. Wang, Z. Zhao, C. Zhu, K. A. Sankararaman, M. Valko, X. Cao, Z. Chen, M. Khabsa, Y. Chen, H. Ma, and S. Wang, “Preference optimization with multi-sample comparisons,” 2024. 10
- [221] M. Suzgun, N. Scales, N. Schärli, S. Gehrmann, Y. Tay, H. W. Chung, A. Chowdhery, Q. V. Le, E. H. Chi, D. Zhou, , and J. Wei, “Challenging big-bench tasks and whether chain-of-thought can solve them,” *arXiv preprint arXiv:2210.09261*, 2022. 11, 18
- [222] X. Zhang, C. Du, T. Pang, Q. Liu, W. Gao, and M. Lin, “Chain of preference optimization: Improving chain-of-thought reasoning in LLMs,” in *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. 11
- [223] Z. Zhang, A. Zhang, M. Li, and A. Smola, “Automatic chain of thought prompting in large language models,” *arXiv preprint arXiv:2210.03493*, 2022. 11
- [224] V. Sanh, A. Webson, C. Raffel, S. H. Bach, L. Sutawika,

- arXiv preprint arXiv:2403.13787*, 2024. **7, 19, 20**
- [170] J. Hong, N. Lee, and J. Thorne, “Open: Monolithic preference optimization without reference model,” 2024. **8**
- [171] J. Schuman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, “High-dimensional continuous control using generalized advantage estimation,” 2018. **8**
- [172] R. Rafailov, Y. Chittepuru, R. Park, H. S. Sikchi, J. Hejna, B. Knox, C. Finn, and S. Niekum, “Scaling laws for reward model overoptimization in direct alignment algorithms,” *ArXiv*, vol. abs/2406.02900, 2024. **9**
- [173] H. Wang, S. Hao, H. Dong, S. Zhang, Y. Bao, Z. Yang, and Y. Wu, “Offline reinforcement learning for llm multi-step reasoning,” 2024. **10, 22**
- [174] G. Mukobi, P. Chatain, S. Fong, R. Windesheim, G. Kutyniok, K. Bhatia, and S. Alberti, “SuperH: Supervised iterative learning from human feedback,” *arXiv preprint arXiv:2310.16763*, 2023. **11**
- [175] C. Li, H. Zhou, G. Glavaš, A. Korhonen, and I. Vulić, “On task performance and model calibration with supervised and self-ensembled in-context learning,” *arXiv preprint arXiv:2312.13772*, 2023. **11**
- [176] C. Wang, Z. Zhao, C. Zhu, K. A. Sankaranarayanan, M. Valko, X. Cao, Z. Chen, M. Khabsa, Y. Chen, H. Ma, and S. Wang, “Preference optimization with multi-sample comparisons,” 2024. **11**
- [177] M. Sugun, N. Scales, N. Schärli, S. Gehrmann, Y. Tay, H. W. Chung, A. Chowdhery, Q. V. Le, E. H. Chi, D. Zhou, , and J. Wei, “Challenging big-bench tasks and whether chain-of-thought can solve them,” *arXiv preprint arXiv:2210.09261*, 2022. **11, 19**
- [178] X. Zhang, C. Du, T. Pang, Q. Liu, W. Gao, and M. Lin, “Chain of preference optimization: Improving chain-of-thought reasoning in LLMs,” in *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. **11**
- [179] Z. Zhang, A. Zhang, M. Li, and A. Smola, “Automatic chain of thought prompting in large language models,” *arXiv preprint arXiv:2210.03493*, 2022. **11**
- [180] V. Sanh, A. Webson, C. Raffel, S. H. Bach, L. Sutawika, Z. Alyafeai, A. Chaffin, A. Stiegler, T. L. Scao, A. Raja, *et al.*, “Multitask prompted training enables zero-shot task generalization,” *arXiv preprint arXiv:2110.08207*, 2021. **12, 22**
- [181] J. Wei, M. Bosma, V. Y. Zhao, K. Guu, A. W. Yu, B. Lester, N. Du, A. M. Dai, and Q. V. Le, “Finetuned language models are zero-shot learners,” *arXiv preprint arXiv:2109.01652*, 2021. **12**
- [182] R. Taori, I. Gulrajani, T. Zhang, Y. Dubois, X. Li, C. Guestrin, P. Liang, and T. B. Hashimoto, “Stanford alpaca: An instruction-following llama model,” 2023. **12**
- [183] W.-L. Chiang, Z. Li, Z. Lin, Y. Sheng, Z. Wu, H. Zhang, L. Zheng, S. Zhuang, Y. Zhuang, J. E. Gonzalez, I. Stoica, and E. P. Xing, “Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality,” March 2023. **12**
- [184] M. Conover, M. Hayes, A. Mathur, J. Xie, J. Wan, S. Shah, A. Ghodsi, P. Wendell, M. Zaharia, and R. Xin, “Free dolly: Introducing the world’s first truly open instruction-tuned llm,” 2023. **12**
- [185] R. Thoppilan, D. De Freitas, J. Hall, N. Shazeer, A. Kulshreshtha, H.-T. Cheng, A. Jin, T. Bos, L. Baker, Y. Du, *et al.*, “Llama: Language models for dialog applications,” *arXiv preprint arXiv:2201.08239*, 2022. **12**
- [186] X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. H. Chi, S. Narang, A. Chowdhery, and D. Zhou, “Self-consistency improves chain of thought reasoning in language models,” in *The Eleventh International Conference on Learning Representations*, 2023. **12, 15, 20**
- [187] L. C. Magister, J. Mallinson, J. Adamek, E. Malmi, and A. Severny, “Teaching small language models to reason,” *arXiv preprint arXiv:2212.08410*, 2022. **12, 13**
- [188] G. Xu, P. Jin, H. Li, Y. Song, L. Sun, and L. Yuan, “Llava-cot: Let vision language models reason step-by-step,” 2024. **12**
- [189] O. Thawakar, D. Dissanayake, K. More, R. Thawkar, A. Heakl, N. Ahsan, Y. Li, M. Zumri, J. Lahoud, R. M. Anwer, H. Cholakkal, I. Laptev, M. Shah, F. S. Khan, and S. Khan, “Llamav-o1: Rethinking step-by-step visual reasoning in llms,” 2025. **12, 18**
- [190] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “Qlora: Efficient finetuning of quantized llms,” *arXiv preprint arXiv:2305.14314*, 2023. **12, 13**
- [191] E. Frantar, S. Ashkboos, T. Hoeffer, and D. Alistarh, “GPTQ: Accurate post-training compression for generative pretrained transformers,” *arXiv preprint arXiv:2210.17323*, 2022. **13**
- [192] E. Frantar and D. Alistarh, “SparseGPT: Massive language models can be accurately pruned in one-shot,” *arXiv preprint arXiv:2301.00774*, 2023. **13, 14**
- [193] S. Mangrulkar, S. Gugger, L. Debut, Y. Belkada, S. Paul, and B. Bossan, “Pft: State-of-the-art parameter-efficient finetuning methods,” 2022. **13**
- [194] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer, “Llm.int8(): 8-bit matrix multiplication for transformers at scale,” *arXiv preprint arXiv:2208.07339*, 2022. **13**
- [195] Q. Zhang, M. Chen, A. Bukharin, P. He, Y. Cheng, W. Chen, and T. Zhao, “Adaptive budget allocation for parameter-efficient fine-tuning,” in *The Eleventh International Conference on Learning Representations*, 2023. **13, 20**
- [196] X. Liu, K. Ji, Y. Fu, Z. Du, Z. Yang, and J. Tang, “P-tuning v2: Prompt tuning can be comparable to fine-tuning universally across scales and tasks,” *CoRR*, vol. abs/2110.07602, 2021. **13**
- [197] Q. Lhoest, A. Villanova del Moral, Y. Jernite, A. Thakur, P. von Platen, S. Patil, J. Chaumond, M. Drame, J. Plu, L. Tunstall, J. Davison, M. Šaško, G. Chhablani, B. Malik, S. Brandeis, T. Le Scao, V. Sanh, C. Xu, N. Patry, A. McMillan-Major, P. Schmid, S. Gugger, C. Delangue, T. Matysiński, L. Debut, S. Bekman, P. Cistac, T. Goehringer, V. Mustar, F. Lagunas, A. Rush, and T. Wolf, “Datasets: A community library for natural language processing,” in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, (Online and Punta Cana, Dominican Republic), pp. 175–184, Association for Computational Linguistics, Nov. 2021. **13**
- [198] A. Torralba and Others, “Webdataset: A format for petascale deep learning,” Efficient tar-based sharding format for petascale distributed training. **13**
- [199] I. Iterative, “Dvc: Data version control,” Git-like version control for datasets and machine learning pipelines. **13**
- [200] N. Richardson, I. Cook, N. Crane, D. Dunnington, R. François, J. Keane, D. Moldovan-Grünfeld, J. Ooms, J. Wujciak-Jens, and Apache Arrow, *arrow: Integration to 'Apache' 'Arrow'*, 2025. R package version 19.0.0, <https://arrow.apache.org/docs/r/>. **13**
- [201] I. Facebook, “Zstandard: High-speed compression algorithm,” High-speed compression algorithm for training data storage/-transfer. **13**
- [202] C. Team, “Cleanlab: The standard data-centric ai package for machine learning with noisy labels,” Automatic detection of label errors and outliers in training datasets. **13**
- [203] R. Y. Aminabadi, S. Rajbhandari, M. Zhang, A. A. Awan, C. Li, D. Li, E. Zheng, J. Rasley, S. Smith, O. Ruwase, and Y. He, “DeepSpeed inference: Enabling efficient inference of transformer models at unprecedented scale,” 2022. **13**
- [204] M. Shoyebi, M. Patwary, R. Puri, P. LeGresley, J. Casper, and B. Catanzaro, “Megatron-lm: Training multi-billion parameter language models using model parallelism,” 2020. **13**
- [205] S. Li, H. Liu, Z. Bian, J. Fang, H. Huang, Y. Liu, B. Wang, and Y. You, “Colossal-ai: A unified deep learning system for large-scale parallel training,” 2023. **13**
- [206] A. Sergeev and M. D. Balso, “Horovod: fast and easy distributed deep learning in tensorflow,” 2018. **13**
- [207] P. Moritz, R. Nishihara, S. Wang, A. Tumanov, R. Liaw, E. Liang, M. Elibol, Z. Yang, W. Paul, M. I. Jordan, and I. Stoica, “Ray: A distributed framework for emerging ai applications,” 2018. **13**
- [208] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, and I. Stoica, “Efficient memory management for large language model serving with pagedattention,” 2023. **13**
- [209] Y. Zhou and K. Yang, “Exploring tensorrt to improve real-time inference for deep learning,” in *2022 IEEE 24th Int Conf on High Performance Computing & Communications; 8th Int Conf on Data Science & Systems; 20th Int Conf on Smart City; 8th Int Conf on Dependability in Sensor, Cloud & Big Data Systems & Application (HPCC/DSS/SmartCity/DependSys)*, pp. 2011–2018, 2022. **13**
- [210] P. Tillet, H.-T. Kung, and D. Cox, “Triton: an intermediate language and compiler for tiled neural network computations,” in *Proceedings of the 3rd ACM SIGPLAN International Workshop on Machine Learning and Programming Languages*, pp. 10–19, 2019. **13**
- [211] O. Community, “Onnx: Open neural network exchange.” Uni-
- Z. Alyafeai, A. Chaffin, A. Stiegler, T. L. Scao, A. Raja, *et al.*, “Multitask prompted training enables zero-shot task generalization,” *arXiv preprint arXiv:2110.08207*, 2021. **11, 21**
- [181] J. Wei, M. Bosma, V. Y. Zhao, K. Guu, A. W. Yu, B. Lester, N. Du, A. M. Dai, and Q. V. Le, “Finetuned language models are zero-shot learners,” *arXiv preprint arXiv:2109.01652*, 2021. **11**
- [182] R. Taori, I. Gulrajani, T. Zhang, Y. Dubois, X. Li, C. Guestrin, P. Liang, and T. B. Hashimoto, “Stanford alpaca: An instruction-following llama model,” 2023. **11**
- [183] W.-L. Chiang, Z. Li, Z. Lin, Y. Sheng, Z. Wu, H. Zhang, L. Zheng, S. Zhuang, Y. Zhuang, J. E. Gonzalez, I. Stoica, and E. P. Xing, “Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality,” March 2023. **11**
- [184] M. Conover, M. Hayes, A. Mathur, J. Xie, J. Wan, S. Shah, A. Ghodsi, P. Wendell, M. Zaharia, and R. Xin, “Free dolly: Introducing the world’s first truly open instruction-tuned llm,” 2023. **11**
- [185] R. Thoppilan, D. De Freitas, J. Hall, N. Shazeer, A. Kulshreshtha, H.-T. Cheng, A. Jin, T. Bos, L. Baker, Y. Du, *et al.*, “Llama: Language models for dialog applications,” *arXiv preprint arXiv:2201.08239*, 2022. **11**
- [186] X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. H. Chi, S. Narang, A. Chowdhery, and D. Zhou, “Self-consistency improves chain of thought reasoning in language models,” in *The Eleventh International Conference on Learning Representations*, 2023. **11, 14, 19**
- [187] L. C. Magister, J. Mallinson, J. Adamek, E. Malmi, and A. Severny, “Teaching small language models to reason,” *arXiv preprint arXiv:2212.08410*, 2022. **12**
- [188] G. Xu, P. Jin, H. Li, Y. Song, L. Sun, and L. Yuan, “Llava-cot: Let vision language models reason step-by-step,” 2024. **12**
- [189] O. Thawakar, D. Dissanayake, K. More, R. Thawkar, A. Heakl, N. Ahsan, Y. Li, M. Zumri, J. Lahoud, R. M. Anwer, H. Cholakkal, I. Laptev, M. Shah, F. S. Khan, and S. Khan, “Llamav-o1: Rethinking step-by-step visual reasoning in llms,” 2025. **12, 18**
- [190] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “Qlora: Efficient finetuning of quantized llms,” *arXiv preprint arXiv:2305.14314*, 2023. **12, 13**
- [191] E. Frantar, S. Ashkboos, T. Hoeffer, and D. Alistarh, “GPTQ: Accurate post-training compression for generative pretrained transformers,” *arXiv preprint arXiv:2210.17323*, 2022. **13**
- [192] E. Frantar and D. Alistarh, “SparseGPT: Massive language models can be accurately pruned in one-shot,” *arXiv preprint arXiv:2301.00774*, 2023. **12, 13**
- [193] S. Mangrulkar, S. Gugger, L. Debut, Y. Belkada, S. Paul, and B. Bossan, “Pft: State-of-the-art parameter-efficient finetuning methods,” 2022. **13**
- [194] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer, “Llm.int8(): 8-bit matrix multiplication for transformers at scale,” *arXiv preprint arXiv:2208.07339*, 2022. **13**
- [195] Q. Zhang, M. Chen, A. Bukharin, P. He, Y. Cheng, W. Chen, and T. Zhao, “Adaptive budget allocation for parameter-efficient fine-tuning,” in *The Eleventh International Conference on Learning Representations*, 2023. **13, 19**
- [196] X. Liu, K. Ji, Y. Fu, Z. Du, Z. Yang, and J. Tang, “P-tuning v2: Prompt tuning can be comparable to fine-tuning universally across scales and tasks,” *CoRR*, vol. abs/2110.07602, 2021. **13**
- [197] Q. Lhoest, A. Villanova del Moral, Y. Jernite, A. Thakur, P. von

- Proceedings of the 3rd ACM SIGPLAN International Workshop on Machine Learning and Programming Languages*, pp. 10–19, 2019. [13](#)
- [211] Q. Community, “Onnx: Open neural network exchange,” Unified inference engine with hardware-specific optimizations. [13](#)
- [212] I. Corporation, “Openvino: Intel optimization toolkit,” 2025. Runtime for Intel CPUs/iGPUs with pruning/quantization support. [13](#)
- [213] M. Dukhan, “The indirect convolution algorithm,” 2019. [13](#)
- [214] I. Groq, “Groq: Ai accelerator,” 2025. Deterministic low-latency inference via custom tensor streaming processor. [13](#)
- [215] J. Castaño, S. Martínez-Fernández, X. Franch, and J. Bogner, “Analyzing the evolution and maintenance of ml models on hugging face,” 2024. [13](#)
- [216] A. Paszke, S. Gross, S. Chintala, G. Chanan, E. Yang, Z. DeVito, Z. Lin, A. Desmaison, L. Antiga, and A. Lerer, “Automatic differentiation in pytorch,” in *NIPS-W*, 2017. [13](#)
- [217] S. Hao, Y. Gu, H. Luo, T. Liu, X. Shao, X. Wang, S. Xie, H. Ma, A. Samavedhi, Q. Gao, et al., “Llm reasoners: New evaluation, library, and analysis of step-by-step reasoning with large language models,” *arXiv preprint arXiv:2404.05221*, 2024. [13](#)
- [218] S. Pieri, S. S. Mullapilly, F. S. Khan, R. M. Anwer, S. Khan, T. Baldwin, and H. Cholakkal, “Bimedix: Bilingual medical mixture of experts llm,” *arXiv preprint arXiv:2402.13253*, 2024. [13](#)
- [219] Y. Yang, M. C. S. Uy, and A. Huang, “Finbert: A pretrained language model for financial communications,” *arXiv preprint arXiv:2006.08097*, 2020. [13](#)
- [220] D. Thulke, Y. Gao, P. Pelsner, R. Brune, R. Jalota, F. Fok, M. Ramos, I. van Wyk, A. Nasir, H. Goldstein, et al., “Climategpt: Towards ai synthesizing interdisciplinary research on climate change,” *arXiv preprint arXiv:2401.09646*, 2024. [13](#)
- [221] S. S. Mullapilly, A. Shaker, O. Thawakar, H. Cholakkal, R. M. Anwer, S. Khan, and F. S. Khan, “Arabic mini-climategpt: A climate change and sustainability tailored arabic llm,” *arXiv preprint arXiv:2312.09366*, 2023. [13](#)
- [222] Y. Wang, W. Wang, S. Joty, and S. C. Hoi, “Codet5: Identifier-aware unified pre-trained encoder-decoder models for code understanding and generation,” *arXiv preprint arXiv:2109.00859*, 2021. [13](#)
- [223] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. S. Khan, “Geochat: Grounded large vision-language model for remote sensing,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 27831–27840, 2024. [13](#)
- [224] S. S. Mullapilly, M. I. Kurpath, S. Pieri, S. Y. Alseieri, S. Cholakkal, K. Aldahmani, F. Khan, R. Anwer, S. Khan, T. Baldwin, et al., “Bimedix2: Bio-medical expert llm for diverse medical modalities,” *arXiv preprint arXiv:2412.07769*, 2024. [13](#)
- [225] M. Maaz, H. Rasheed, S. Khan, and F. S. Khan, “Videochatgpt: Towards detailed video understanding via large vision and language models,” *arXiv preprint arXiv:2306.05424*, 2023. [13](#)
- [226] B. Lin, Y. Ye, B. Zhu, J. Cui, M. Ning, P. Jin, and L. Yuan, “Video-llava: Learning united visual representation by alignment before projection,” *arXiv preprint arXiv:2311.10122*, 2023. [13](#)
- [227] H. Zhang, X. Li, and L. Bing, “Video-llama: An instruction-tuned audio-visual language model for video understanding,” *arXiv preprint arXiv:2306.02858*, 2023. [13](#)
- [228] Y. Han, C. Zhang, X. Chen, X. Yang, Z. Wang, G. Yu, B. Fu, and H. Zhang, “Chartllama: A multimodal llm for chart understanding and generation,” *arXiv preprint arXiv:2311.16483*, 2023. [13](#)
- [229] X. Zhu, J. Li, Y. Liu, C. Ma, and W. Wang, “A survey on model compression for large language models,” *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 1556–1577, 2024. [13](#)
- [230] Z. Wan, X. Wang, C. Liu, S. Alam, Y. Zheng, J. Liu, Z. Qu, S. Yan, Y. Zhu, Q. Zhang, et al., “Efficient large language models: A survey,” *arXiv preprint arXiv:2312.03863*, 2023. [13](#)
- [231] C.-Y. Hsieh, C.-L. Li, C.-K. Yeh, H. Nakhost, Y. Fujii, A. Ratner, R. Krishna, C.-Y. Lee, and T. Pfister, “Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes,” *arXiv preprint arXiv:2305.02301*, 2023. [13](#)
- [232] Y. Gu, L. Dong, F. Wei, and M. Huang, “Minillm: Knowledge distillation of large language models,” *arXiv preprint arXiv:2306.08543*, 2023. [13](#)
- [233] X. L. Li and P. Liang, “Prefix-tuning: Optimizing continuous prompts for generation,” *arXiv preprint arXiv:2101.00190*, 2021. [13](#)
- [234] N. Housby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. de Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, “Parameter-efficient transfer learning for nlp,” 2019. [13](#)
- [235] B. P. Lowerre and B. R. Reddy, “Harpy, a connected speech recognition system,” *The Journal of the Acoustical Society of America*, vol. 59, no. S1, pp. S97–S97, 1976. [14](#)
- [236] A. Graves, “Sequence transduction with recurrent neural networks,” *arXiv preprint arXiv:1211.3711*, 2012. [14](#)
- [237] H. Sun, M. Haider, R. Zhang, H. Yang, J. Qiu, M. Yin, M. Wang, P. Bartlett, and A. Zanette, “Fast best-of-n decoding via speculative rejection,” 2024. [14](#)
- [238] A. Askell, Y. Bai, A. Chen, D. Drain, D. Ganguli, T. Henighan, A. Jones, N. Joseph, B. Mann, N. DasSarma, et al., “A general language assistant as a laboratory for alignment,” *arXiv preprint arXiv:2112.00861*, 2021. [14](#)
- [239] A. Glaese, N. McAleese, M. Trębacz, J. Aslanides, V. Firoiu, T. Ewalds, M. Rauh, L. Weidinger, M. Chadwick, P. Thacker, et al., “Improving alignment of dialogue agents via targeted human judgements,” *arXiv preprint arXiv:2209.14375*, 2022. [14](#)
- [240] N. Stiennon, L. Ouyang, J. Wu, D. Ziegler, R. Lowe, C. Voss, A. Radford, D. Amodei, and P. F. Christiano, “Learning to summarize with human feedback,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 3008–3021, 2020. [14](#)
- [241] J. Q. Yang, S. Salamatian, Z. Sun, A. T. Suresh, and A. Beirami, “Asymptotics of language model alignment,” *arXiv preprint arXiv:2404.01730*, 2024. [15](#)
- [242] H. Sun, M. Haider, R. Zhang, H. Yang, J. Qiu, M. Yin, M. Wang, P. Bartlett, and A. Zanette, “Fast best-of-n decoding via speculative rejection,” *arXiv preprint arXiv:2410.20290*, 2024. [15](#)
- [243] J. Hilton and L. Gao, “Measuring goodhart’s law,” *OpenAI Research Blog*, 2022. [15](#)
- [244] L. Wang, W. Xu, Y. Lan, Z. Hu, Y. Lan, R. K.-W. Lee, and E.-P. Lim, “Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models,” *arXiv preprint arXiv:2305.04091*, 2023. [15](#)
- [245] Y. Wang, Y. Kordi, S. Mishra, A. Liu, N. A. Smith, D. Khashabi, and H. Hajishirzi, “Self-instruct: Aligning language models with self-generated instructions,” *arXiv preprint arXiv:2212.10560*, 2022. [15](#)
- [246] X. Chen, R. Aksitov, U. Alon, J. Ren, K. Xiao, P. Yin, S. Prakash, C. Sutton, X. Wang, and D. Zhou, “Universal self-consistency for large language models,” in *ICML 2024 Workshop on In-Context Learning*. [16](#)
- [247] A. Newell, “On the analysis of human problem solving protocols,” 1966. [16](#)
- [248] F. Haji, M. Bethany, M. Tabar, J. Chiang, A. Rios, and P. Najafirad, “Improving llm reasoning with multi-agent tree-of-thought validator agent,” *arXiv preprint arXiv:2409.11527*, 2024. [16](#)
- [249] M. Besta, N. Blach, A. Kubicek, R. Gerstenberger, M. Podstawski, L. Gianinazzi, J. Gajda, T. Lehmann, H. Niewiadomski, P. Nyczyk, and T. Hoefler, “Graph of thoughts: Solving elaborate problems with large language models,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, p. 17682–17690, Mar. 2024. [16](#)
- [250] D. Wilson, “Llm tree search,” *arXiv preprint arXiv:2410.19117*, 2024. [17](#)
- [251] D. Hendrycks and K. Gimpel, “A baseline for detecting misclassified and out-of-distribution examples in neural networks,” *arXiv preprint arXiv:1610.02136*, 2016. [17](#)
- [252] Z. Xie, S. Thiem, J. Martin, E. Wainwright, S. Marmorstein, and P. Jansen, “WorldTree v2: A corpus of science-domain structured explanations and inference patterns supporting multi-hop inference,” in *Proceedings of the Twelfth Language Resources and Evaluation Conference* (N. Calzolari, F. Béchet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, and S. Piperidis, eds.), (Marseille, France), pp. 5456–5473, European Language Resources Association, May 2020. [17, 19](#)
- ified inference engine with hardware-specific optimizations. [13](#)
- [212] I. Corporation, “Openvino: Intel optimization toolkit,” 2025. Runtime for Intel CPUs/iGPUs with pruning/quantization support. [13](#)
- [213] M. Dukhan, “The indirect convolution algorithm,” 2019. [13](#)
- [214] I. Groq, “Groq: Ai accelerator,” 2025. Deterministic low-latency inference via custom tensor streaming processor. [13](#)
- [215] J. Castaño, S. Martínez-Fernández, X. Franch, and J. Bogner, “Analyzing the evolution and maintenance of ml models on hugging face,” 2024. [13](#)
- [216] A. Paszke, S. Gross, S. Chintala, G. Chanan, E. Yang, Z. DeVito, Z. Lin, A. Desmaison, L. Antiga, and A. Lerer, “Automatic differentiation in pytorch,” in *NIPS-W*, 2017. [13](#)
- [217] S. Hao, Y. Gu, H. Luo, T. Liu, X. Shao, X. Wang, S. Xie, H. Ma, A. Samavedhi, Q. Gao, et al., “Llm reasoners: New evaluation, library, and analysis of step-by-step reasoning with large language models,” *arXiv preprint arXiv:2404.05221*, 2024. [13](#)
- [218] S. Pieri, S. S. Mullapilly, F. S. Khan, R. M. Anwer, S. Khan, T. Baldwin, and H. Cholakkal, “Bimedix: Bilingual medical mixture of experts llm,” *arXiv preprint arXiv:2402.13253*, 2024. [13](#)
- [219] Y. Yang, M. C. S. Uy, and A. Huang, “Finbert: A pretrained language model for financial communications,” *arXiv preprint arXiv:2006.08097*, 2020. [13](#)
- [220] D. Thulke, Y. Gao, P. Pelsner, R. Brune, R. Jalota, F. Fok, M. Ramos, I. van Wyk, A. Nasir, H. Goldstein, et al., “Climategpt: Towards ai synthesizing interdisciplinary research on climate change,” *arXiv preprint arXiv:2401.09646*, 2024. [13](#)
- [221] S. S. Mullapilly, A. Shaker, O. Thawakar, H. Cholakkal, R. M. Anwer, S. Khan, and F. S. Khan, “Arabic mini-climategpt: A climate change and sustainability tailored arabic llm,” *arXiv preprint arXiv:2312.09366*, 2023. [13](#)
- [222] Y. Wang, W. Wang, S. Joty, and S. C. Hoi, “Codet5: Identifier-aware unified pre-trained encoder-decoder models for code understanding and generation,” *arXiv preprint arXiv:2109.00859*, 2021. [13](#)
- [223] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. S. Khan, “Geochat: Grounded large vision-language model for remote sensing,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 27831–27840, 2024. [13](#)
- [224] S. S. Mullapilly, M. I. Kurpath, S. Pieri, S. Y. Alseieri, S. Cholakkal, K. Aldahmani, F. Khan, R. Anwer, S. Khan, T. Baldwin, et al., “Bimedix2: Bio-medical expert llm for diverse medical modalities,” *arXiv preprint arXiv:2412.07769*, 2024. [13](#)
- [225] M. Maaz, H. Rasheed, S. Khan, and F. S. Khan, “Videochatgpt: Towards detailed video understanding via large vision and language models,” *arXiv preprint arXiv:2306.05424*, 2023. [13](#)
- [226] B. Lin, Y. Ye, B. Zhu, J. Cui, M. Ning, P. Jin, and L. Yuan, “Video-llava: Learning united visual representation by alignment before projection,” *arXiv preprint arXiv:2311.10122*, 2023. [13](#)
- [227] H. Zhang, X. Li, and L. Bing, “Video-llama: An instruction-tuned audio-visual language model for video understanding,” *arXiv preprint arXiv:2306.02858*, 2023. [13](#)
- [228] Y. Han, C. Zhang, X. Chen, X. Yang, Z. Wang, G. Yu, B. Fu, and H. Zhang, “Chartllama: A multimodal llm for chart understanding and generation,” *arXiv preprint arXiv:2311.16483*, 2023. [13](#)
- [229] X. Zhu, J. Li, Y. Liu, C. Ma, and W. Wang, “A survey on model compression for large language models,” *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 1556–1577, 2024. [13](#)
- [230] Z. Wan, X. Wang, C. Liu, S. Alam, Y. Zheng, J. Liu, Z. Qu, S. Yan, Y. Zhu, Q. Zhang, et al., “Efficient large language models: A survey,” *arXiv preprint arXiv:2312.03863*, 2023. [13](#)
- [231] C.-Y. Hsieh, C.-L. Li, C.-K. Yeh, H. Nakhost, Y. Fujii, A. Ratner, R. Krishna, C.-Y. Lee, and T. Pfister, “Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes,” *arXiv preprint arXiv:2305.02301*, 2023. [13](#)
- [232] Y. Gu, L. Dong, F. Wei, and M. Huang, “Minillm: Knowledge distillation of large language models,” *arXiv preprint arXiv:2306.08543*, 2023. [13](#)
- [233] X. L. Li and P. Liang, “Prefix-tuning: Optimizing continuous prompts for generation,” *arXiv preprint arXiv:2101.00190*, 2021. [13](#)
- [234] N. Housby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. de Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, “Parameter-efficient transfer learning for nlp,” 2019. [12](#)
- [235] B. P. Lowerre and B. R. Reddy, “Harpy, a connected speech recognition system,” *The Journal of the Acoustical Society of America*, vol. 59, no. S1, pp. S97–S97, 1976. [12, 14](#)
- [236] A. Graves, “Sequence transduction with recurrent neural networks,” *arXiv preprint arXiv:1211.3711*, 2012. [12, 14](#)
- [237] H. Sun, M. Haider, R. Zhang, H. Yang, J. Qiu, M. Yin, M. Wang, P. Bartlett, and A. Zanette, “Fast best-of-n decoding via speculative rejection,” 2024. [13](#)
- [238] A. Askell, Y. Bai, A. Chen, D. Drain, D. Ganguli, T. Henighan, A. Jones, N. Joseph, B. Mann, N. DasSarma, et al., “A general language assistant as a laboratory for alignment,” *arXiv preprint arXiv:2112.00861*, 2021. [13](#)
- [239] A. Glaese, N. McAleese, M. Trębacz, J. Aslanides, V. Firoiu, T. Ewalds, M. Rauh, L. Weidinger, M. Chadwick, P. Thacker, et al., “Improving alignment of dialogue agents via targeted human judgements,” *arXiv preprint arXiv:2209.14375*, 2022. [13](#)
- [240] N. Stiennon, L. Ouyang, J. Wu, D. Ziegler, R. Lowe, C. Voss, A. Radford, D. Amodei, and P. F. Christiano, “Learning to summarize with human feedback,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 3008–3021, 2020. [13](#)
- [241] J. Q. Yang, S. Salamatian, Z. Sun, A. T. Suresh, and A. Beirami, “Asymptotics of language model alignment,” *arXiv preprint arXiv:2404.01730*, 2024. [14](#)
- [242] H. Sun, M. Haider, R. Zhang, H. Yang, J. Qiu, M. Yin, M. Wang, P. Bartlett, and A. Zanette, “Fast best-of-n decoding via speculative rejection,” *arXiv preprint arXiv:2410.20290*, 2024. [14](#)
- [243] J. Hilton and L. Gao, “Measuring goodhart’s law,” *OpenAI Research Blog*, 2022. [14](#)
- [244] L. Wang, W. Xu, Y. Lan, Z. Hu, Y. Lan, R. K.-W. Lee, and E.-P. Lim, “Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models,” *arXiv preprint arXiv:2305.04091*, 2023. [14](#)
- [245] Y. Wang, Y. Kordi, S. Mishra, A. Liu, N. A. Smith, D. Khashabi, and H. Hajishirzi, “Self-instruct: Aligning language models with self-generated instructions,” *arXiv preprint arXiv:2212.10560*, 2022. [14](#)

- [253] G. Portillo Wightman, A. DeLucia, and M. Dredze, "Strength in numbers: Estimating confidence of large language models by prompt agreement," in *Proceedings of the 3rd Workshop on Trustworthy Natural Language Processing (TrustNLP 2023)*, pp. 326–362, 2023. [17](#)
- [254] J. Qi, H. Tang, and Z. Zhu, "Verifierq: Enhancing llm test time compute with q-learning-based verifiers," *arXiv preprint arXiv:2410.08048*, 2024. [17](#)
- [255] A. Madaan, N. Tandon, P. Gupta, S. Hallinan, L. Gao, S. Wiegrefe, U. Alon, N. Dziri, S. Prabhunoye, Y. Yang, S. Gupta, B. P. Majumder, K. Hermann, S. Welleck, A. Yazdanbakhsh, and P. Clark, "Self-refine: Iterative refinement with self-feedback," 2023. [17](#)
- [256] Z. Xi, W. Chen, X. Guo, W. He, Y. Ding, B. Hong, M. Zhang, J. Wang, S. Jin, E. Zhou, *et al.*, "The rise and potential of large language model based agents: A survey," *arXiv preprint arXiv:2309.07864*, 2023. [17](#)
- [257] R. Coulom, "Efficient selectivity and backup operators in monte-carlo tree search," in *International conference on computers and games*, pp. 72–83, Springer, 2006. [18](#)
- [258] J. X. Chen, "The evolution of computing: AlphaGo," *Computing in Science & Engineering*, vol. 18, no. 4, pp. 4–7, 2016. [18](#)
- [259] L. Kocsis and C. Szepesvári, "Bandit based monte-carlo planning," in *European conference on machine learning*, pp. 282–293, Springer, 2006. [18](#)
- [260] H. W. Spruiell, C. Edwards, M. V. Olarte, U. Sanyal, H. Ji, and S. Choudhury, "Monte carlo thought search: Large language model querying for complex scientific reasoning in catalyst design," *arXiv preprint arXiv:2310.14420*, 2023. [18](#), [20](#)
- [261] M. DeLorenzo, A. B. Chowdhury, V. Gohil, S. Thakur, R. Karri, S. Garg, and J. Rajendran, "Make every move count: Llm-based high-quality rtl code generation using mcts," *arXiv preprint arXiv:2402.03289*, 2024. [18](#)
- [262] S. Park, X. Liu, Y. Gong, and E. Choi, "Ensembling large language models with process reward-guided tree search for better complex reasoning," *arXiv preprint arXiv:2412.15797*, 2024. [18](#)
- [263] M. Shen, G. Zeng, Z. Qi, Z.-W. Hong, Z. Chen, W. Lu, G. Wornell, S. Das, D. Cox, and C. Gan, "Satori: Reinforcement learning with chain-of-action-thought enhances llm reasoning via autoregressive search," *arXiv preprint arXiv:2502.02508*, 2025. [18](#)
- [264] S. R. Motwani, C. Smith, R. J. Das, R. Rafailov, I. Laptev, P. H. S. Torr, F. Pizzati, R. Clark, and C. S. de Witt, "Malt: Improving reasoning with multi-agent llm training," 2025. [18](#)
- [265] U. Anwar, A. Saparov, J. Rando, D. Paleka, M. Turpin, P. Hase, E. S. Lubana, E. Jenner, S. Casper, O. Sourbut, *et al.*, "Foundational challenges in assuring alignment and safety of large language models," *arXiv preprint arXiv:2404.09932*, 2024. [18](#)
- [266] W. Zhang, P. H. Torr, M. Elhoseiny, and A. Bibi, "Bi-factorial preference optimization: Balancing safety-helpfulness in language models," *arXiv preprint arXiv:2408.15313*, 2024. [18](#)
- [267] C. Li, W. Wu, H. Zhang, Y. Xia, S. Mao, L. Dong, I. Vulić, and F. Wei, "Imagine while reasoning in space: Multimodal visualization-of-thought," *arXiv preprint arXiv:2501.07542*, 2025. [19](#)
- [268] F. Nowak, A. Svete, A. Butoi, and R. Cotterell, "On the representational capacity of neural language models with chain-of-thought reasoning," *arXiv preprint arXiv:2406.14197*, 2024. [19](#)
- [269] Z. Li, H. Liu, D. Zhou, and T. Ma, "Chain of thought empowers transformers to solve inherently serial problems," *arXiv preprint arXiv:2402.12875*, vol. 1, 2024. [19](#)
- [270] W. Merrill and A. Sabharwal, "The expressive power of transformers with chain of thought," *arXiv preprint arXiv:2310.07923*, 2023. [19](#)
- [271] C. V. Snell, J. Lee, K. Xu, and A. Kumar, "Scaling test-time compute optimally can be more effective than scaling llm parameters," in *The Thirteenth International Conference on Learning Representations*. [19](#)
- [272] Saxton *et al.*, "Analysing mathematical reasoning abilities of neural models," *arXiv:1904.01557*, 2019. [19](#)
- [273] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, and J. Schulman, "Training verifiers to solve math word problems," *arXiv preprint arXiv:2110.14168*, 2021. [19](#)
- [274] L. Yu, W. Jiang, H. Shi, J. Yu, Z. Liu, Y. Zhang, J. T. Kwok, Z. Li, A. Weller, and W. Liu, "Metamath: Bootstrap your own mathematical questions for large language models," *arXiv preprint arXiv:2309.12284*, 2023. [19](#)
- [275] F. Liu, E. Bugliarello, E. M. Ponti, S. Reddy, N. Collier, and D. Elliott, "Visually grounded reasoning across languages and cultures," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, (Online and Punta Cana, Dominican Republic), pp. 10467–10485, Association for Computational Linguistics, Nov. 2021. [19](#)
- [276] X. Yue, Y. Ni, K. Zhang, T. Zheng, R. Liu, G. Zhang, S. Stevens, D. Jiang, W. Ren, Y. Sun, C. Wei, B. Yu, R. Yuan, R. Sun, M. Yin, B. Zheng, Z. Yang, Y. Liu, W. Huang, H. Sun, Y. Su, and W. Chen, "Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi," in *Proceedings of CVPR*, 2024. [19](#)
- [277] S. Lin, J. Hilton, and O. Evans, "Truthfulqa: Measuring how models mimic human falsehoods," *arXiv preprint arXiv:2109.07958*, 2021. [19](#)
- [278] X. Yue *et al.*, "Mammoth: Building math generalist models through hybrid instruction tuning," *arXiv preprint arXiv:2309.05653*, 2023. [19](#)
- [279] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt, "Measuring massive multitask language understanding," *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021. [19](#), [20](#)
- [280] D. Hendrycks, C. Burns, S. Basart, A. Critch, J. Li, D. Song, and J. Steinhardt, "Aligning ai with shared human values," *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021. [19](#), [20](#)
- [281] D. Dua, Y. Wang, P. Dasigi, G. Stanovsky, S. Singh, and M. Gardner, "DROP: A reading comprehension benchmark requiring discrete reasoning over paragraphs," in *Proc. of NAACL*, 2019. [19](#)
- [282] Z. Wang, Y. Dong, J. Zeng, V. Adams, M. N. Sreedhar, D. Egert, O. Delalleau, J. P. Scowcroft, N. Kant, A. Swopce, *et al.*, "Helpsteer: Multi-attribute helpfulness dataset for steerlm," *arXiv preprint arXiv:2311.09528*, 2023. [19](#), [20](#)
- [283] G. Cui, L. Yuan, N. Ding, G. Yao, W. Zhu, Y. Ni, G. Xie, Z. Liu, and M. Sun, "Ultrafeedback: Boosting language models with high-quality feedback," 2023. [19](#), [20](#)
- [284] J. Fu, A. Kumar, O. Nachum, G. Tucker, and S. Levine, "D4rl: Datasets for deep data-driven reinforcement learning," 2020. [19](#), [20](#)
- [285] T. Schmied, M. Hofmarcher, F. Paischer, R. Pascanu, and S. Hochreiter, "Learning to modulate pre-trained models in rl," *Advances in Neural Information Processing Systems*, vol. 36, 2024. [19](#), [20](#)
- [286] W. H. Guss, B. Houghton, N. Topin, P. Wang, C. Codel, M. Veloso, and R. Salakhutdinov, "Minerl: A large-scale dataset of minecraft demonstrations," 2019. [19](#), [20](#)
- [287] T. Nguyen, C. V. Nguyen, V. D. Lai, H. Man, N. T. Ngo, F. Dernoncourt, R. A. Rossi, and T. H. Nguyen, "CulturaX: A cleaned, enormous, and multilingual dataset for large language models in 167 languages," in *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)* (N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, and N. Xue, eds.), (Torino, Italia), pp. 4226–4237, ELRA and ICCL, May 2024. [19](#), [20](#)
- [288] X. Yue, Y. Song, A. Asai, S. Kim, J. de Dieu Nyandwi, S. Khanuja, A. Kantharuban, L. Sutawika, S. Ramamoorthy, and G. Neubig, "Pangea: A fully open multilingual multimodal llm for 39 languages," *arXiv preprint arXiv:2410.16153*, 2024. [19](#), [20](#)
- [289] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: pre-training of deep bidirectional transformers for language understanding," *CoRR*, vol. abs/1810.04805, 2018. [19](#), [20](#)
- [290] Y. Liang, N. Duan, Y. Gong, N. Wu, F. Guo, W. Qi, M. Gong, L. Shou, D. Jiang, G. Cao, X. Fan, R. Zhang, R. Agrawal, E. Cui, S. Wei, T. Bhatti, Y. Qiao, J.-H. Chen, W. Wu, S. Liu, F. Yang, D. Campos, R. Majumder, and M. Zhou, "Xglue: A new benchmark dataset for cross-lingual pre-training, understanding and generation," *arXiv*, vol. abs/2004.01401, 2020. [19](#)
- [291] G. Son, D. Yoon, J. Suk, J. Anla-Blasco, M. Aslan, V. T. Kim, S. B. Islam, J. Prats-Cristià, L. Tormo-Bañuelos, and S. Kim, "Mm-eval: A multilingual meta-evaluation benchmark for llm-as-a-judge and reward models," 2024. [19](#), [20](#)
- [246] X. Chen, R. Aksitov, U. Alon, J. Ren, K. Xiao, P. Yin, S. Prakash, C. Sutton, X. Wang, and D. Zhou, "Universal self-consistency for large language models," in *ICML 2024 Workshop on In-Context Learning*. [15](#)
- [247] A. Newell, "On the analysis of human problem solving protocols," 1966. [15](#)
- [248] F. Haji, M. Bethany, M. Tabar, J. Chiang, A. Rios, and P. Najafirad, "Improving llm reasoning with multi-agent tree-of-thought validator agent," *arXiv preprint arXiv:2409.11527*, 2024. [15](#)
- [249] M. Besta, N. Blach, A. Kubicek, R. Gerstenberger, M. Podstawski, L. Gianinazzi, J. Gajda, T. Lehmann, H. Niewiadomski, P. Nycz, and T. Hoefler, "Graph of thoughts: Solving elaborate problems with large language models," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, p. 17682–17690, Mar. 2024. [15](#)
- [250] D. Wilson, "Llm tree search," *arXiv preprint arXiv:2410.19117*, 2024. [16](#)
- [251] D. Hendrycks and K. Gimpel, "A baseline for detecting misclassified and out-of-distribution examples in neural networks," *arXiv preprint arXiv:1610.02136*, 2016. [16](#)
- [252] Z. Xie, S. Thiem, J. Martin, E. Wainwright, S. Marmorstein, and P. Jansen, "WorldTree v2: A corpus of science-domain structured explanations and inference patterns supporting multi-hop inference," in *Proceedings of the Twelfth Language Resources and Evaluation Conference* (N. Calzolari, F. Béchet, P. Blache, K. Choukri, C. Cieri, T. Declercq, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, and S. Piperidis, eds.), (Marseille, France), pp. 5456–5473, European Language Resources Association, May 2020. [16](#), [18](#)
- [253] G. Portillo Wightman, A. DeLucia, and M. Dredze, "Strength in numbers: Estimating confidence of large language models by prompt agreement," in *Proceedings of the 3rd Workshop on Trustworthy Natural Language Processing (TrustNLP 2023)*, pp. 326–362, 2023. [16](#)
- [254] J. Qi, H. Tang, and Z. Zhu, "Verifierq: Enhancing llm test time compute with q-learning-based verifiers," *arXiv preprint arXiv:2410.08048*, 2024. [16](#)
- [255] A. Madaan, N. Tandon, P. Gupta, S. Hallinan, L. Gao, S. Wiegrefe, U. Alon, N. Dziri, S. Prabhunoye, Y. Yang, S. Gupta, B. P. Majumder, K. Hermann, S. Welleck, A. Yazdanbakhsh, and P. Clark, "Self-refine: Iterative refinement with self-feedback," 2023. [16](#)
- [256] Z. Xi, W. Chen, X. Guo, W. He, Y. Ding, B. Hong, M. Zhang, J. Wang, S. Jin, E. Zhou, *et al.*, "The rise and potential of large language model based agents: A survey," *arXiv preprint arXiv:2309.07864*, 2023. [16](#)
- [257] R. Coulom, "Efficient selectivity and backup operators in monte-carlo tree search," in *International conference on computers and games*, pp. 72–83, Springer, 2006. [16](#)
- [258] J. X. Chen, "The evolution of computing: AlphaGo," *Computing in Science & Engineering*, vol. 18, no. 4, pp. 4–7, 2016. [17](#)
- [259] L. Kocsis and C. Szepesvári, "Bandit based monte-carlo planning," in *European conference on machine learning*, pp. 282–293, Springer, 2006. [17](#)
- [260] H. W. Spruiell, C. Edwards, M. V. Olarte, U. Sanyal, H. Ji, and S. Choudhury, "Monte carlo thought search: Large language model querying for complex scientific reasoning in catalyst design," *arXiv preprint arXiv:2310.14420*, 2023. [17](#), [19](#)
- [261] M. DeLorenzo, A. B. Chowdhury, V. Gohil, S. Thakur, R. Karri, S. Garg, and J. Rajendran, "Make every move count: Llm-based high-quality rtl code generation using mcts," *arXiv preprint arXiv:2402.03289*, 2024. [17](#)
- [262] S. Park, X. Liu, Y. Gong, and E. Choi, "Ensembling large language models with process reward-guided tree search for better complex reasoning," *arXiv preprint arXiv:2412.15797*, 2024. [17](#)
- [263] M. Shen, G. Zeng, Z. Qi, Z.-W. Hong, Z. Chen, W. Lu, G. Wornell, S. Das, D. Cox, and C. Gan, "Satori: Reinforcement learning with chain-of-action-thought enhances llm reasoning via autoregressive search," *arXiv preprint arXiv:2502.02508*, 2025. [17](#)
- [264] S. R. Motwani, C. Smith, R. J. Das, R. Rafailov, I. Laptev, P. H. S. Torr, F. Pizzati, R. Clark, and C. S. de Witt, "Malt: Improving reasoning with multi-agent llm training," 2025. [17](#)
- [265] U. Anwar, A. Saparov, J. Rando, D. Paleka, M. Turpin, P. Hase, E. S. Lubana, E. Jenner, S. Casper, O. Sourbut, *et al.*, "Foundational challenges in assuring alignment and safety of large language models," *arXiv preprint arXiv:2404.09932*, 2024. [17](#)
- [266] W. Zhang, P. H. Torr, M. Elhoseiny, and A. Bibi, "Bi-factorial preference optimization: Balancing safety-helpfulness in language models," *arXiv preprint arXiv:2408.15313*, 2024. [17](#)
- [267] C. Li, W. Wu, H. Zhang, Y. Xia, S. Mao, L. Dong, I. Vulić, and F. Wei, "Imagine while reasoning in space: Multimodal visualization-of-thought," *arXiv preprint arXiv:2501.07542*, 2025. [17](#)
- [268] F. Nowak, A. Svete, A. Butoi, and R. Cotterell, "On the representational capacity of neural language models with chain-of-thought reasoning," *arXiv preprint arXiv:2406.14197*, 2024. [17](#)
- [269] Z. Li, H. Liu, D. Zhou, and T. Ma, "Chain of thought empowers transformers to solve inherently serial problems," *arXiv preprint arXiv:2402.12875*, vol. 1, 2024. [17](#)
- [270] W. Merrill and A. Sabharwal, "The expressive power of transformers with chain of thought," *arXiv preprint arXiv:2310.07923*, 2023. [17](#)
- [271] C. V. Snell, J. Lee, K. Xu, and A. Kumar, "Scaling test-time compute optimally can be more effective than scaling llm parameters," in *The Thirteenth International Conference on Learning Representations*. [17](#)
- [272] Saxton *et al.*, "Analysing mathematical reasoning abilities of neural models," *arXiv:1904.01557*, 2019. [18](#)
- [273] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, and J. Schulman, "Training verifiers to solve math word problems," *arXiv preprint arXiv:2110.14168*, 2021. [18](#)
- [274] L. Yu, W. Jiang, H. Shi, J. Yu, Z. Liu, Y. Zhang, J. T. Kwok, Z. Li, A. Weller, and W. Liu, "Metamath: Bootstrap your own mathematical questions for large language models," *arXiv preprint arXiv:2309.12284*, 2023. [18](#)
- [275] F. Liu, E. Bugliarello, E. M. Ponti, S. Reddy, N. Collier, and D. Elliott, "Visually grounded reasoning across languages and cultures," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, (Online and Punta Cana, Dominican Republic), pp. 10467–10485, Association for Computational Linguistics, Nov. 2021. [18](#)
- [276] X. Yue, Y. Ni, K. Zhang, T. Zheng, R. Liu, G. Zhang, S. Stevens, D. Jiang, W. Ren, Y. Sun, C. Wei, B. Yu, R. Yuan, R. Sun, M. Yin, B. Zheng, Z. Yang, Y. Liu, W. Huang, H. Sun, Y. Su, and W. Chen, "Mmmu: A massive multi-discipline

- [292] S. et.al, "Beyond the imitation game: Quantifying and extrapolating the capabilities of language models," 2022. [19](#), [20](#)
- [293] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. Xing, et al., "Judging llm-as-a-judge with mt-bench and chatbot arena," *Advances in Neural Information Processing Systems*, vol. 36, pp. 46595–46623, 2023. [19](#), [20](#)
- [294] E. Dinan, V. Logacheva, V. Malykh, A. H. Miller, K. Shuster, J. Urbanek, D. Kiela, A. Szlam, I. Serban, R. Lowe, S. Prabh-moye, A. W. Black, A. I. Rudnicky, J. Williams, J. Pineau, M. S. Burtsev, and J. Weston, "The second conversational intelligence challenge (conval2)," *CoRR*, vol. abs/1902.00098, 2019. [19](#), [20](#)
- [295] M. Eric, R. Goel, S. Paul, A. Sethi, S. Agarwal, S. Gao, and D. Hakkani-Tur, "MultiWOZ 2.1: A consolidated multi-domain dialogue dataset with state corrections and state tracking baselines," in *Proceedings of the 12th Language Resources and Evaluation Conference*, (Marseille, France), pp. 422–428, European Language Resources Association, May 2020. [19](#), [20](#)
- [296] X. Li and D. Roth, "Learning question classifiers," in *COLING 2002: The 19th International Conference on Computational Linguistics*, 2002. [19](#), [20](#)
- [297] E. Hovy, L. Gerber, U. Hermjakob, C.-Y. Lin, and D. Ravichandran, "Toward semantics-based answer pinpointing," in *Proceedings of the First International Conference on Human Language Technology Research*, 2001. [19](#), [20](#)
- [298] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, and I. Gurevych, "BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models," in *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*, 2021. [19](#), [20](#)
- [299] C. Chhun, F. M. Suchanek, and C. Clavel, "Do language models enjoy their own stories? Prompting large language models for automatic story evaluation," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 1122–1142, 2024. [19](#)
- [300] H. Chen, D. M. Vo, H. Takamura, Y. Miyao, and H. Nakayama, "Storyer: Automatic story evaluation via ranking, rating and reasoning," 2022. [19](#)
- [301] J. Ji, M. Liu, J. Dai, X. Pan, C. Zhang, C. Bian, B. Chen, R. Sun, Y. Wang, and Y. Yang, "Beavertails: Towards improved safety alignment of llm via a human-preference dataset," *Advances in Neural Information Processing Systems*, vol. 36, 2024. [19](#), [20](#)
- [302] G. Xu, J. Liu, M. Yan, H. Xu, J. Si, Z. Zhou, P. Yi, X. Gao, J. Sang, R. Zhang, et al., "Cvalues: Measuring the values of chinese large language models from safety to responsibility," *arXiv preprint arXiv:2307.09705*, 2023. [19](#)
- [303] S. Mishra, D. Khashabi, C. Baral, and H. Hajishirzi, "Cross-task generalization via natural language crowdsourcing instructions," in *ACL*, 2022. [19](#)
- [304] Y. Wang, S. Mishra, P. Alipoormolabashi, Y. Kordi, A. Mirzaei, A. Arunkumar, A. Ashok, A. S. Dhanasekaran, A. Naik, D. Stap, et al., "Super-naturalinstructions:generalization via declarative instructions on 1600+ tasks," in *EMNLP*, 2022. [19](#)
- [305] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Askell, P. Welinder, P. F. Christiano, J. Leike, and R. Lowe, "Training language models to follow instructions with human feedback," in *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022* (S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, eds.), 2022. [20](#)
- [306] W. Saunders, C. Yeh, J. Wu, S. Bills, L. Ouyang, J. Ward, and J. Leike, "Self-critiquing models for assisting human evaluators," *arXiv preprint arXiv:2306.05802*, 2022. [20](#)
- [307] J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, and D. Amodei, "Scaling laws for neural language models," *arXiv preprint arXiv:2001.08361*, 2020. [20](#)
- [308] S. Roy and D. Roth, "Solving general arithmetic word problems," 2016. [20](#)
- [309] Y. Jinmai, T. Morimura, K. Ariu, and K. Abe, "Regularized best-of-n sampling to mitigate reward hacking for language model alignment," *arXiv preprint arXiv:2404.01054*, 2024. [20](#)
- [310] L. Chen, C. Zhu, D. Soselia, J. Chen, T. Zhou, T. Goldstein, H. Huang, M. Shoeibi, and B. Catanzaro, "Odin: Disentangled reward mitigates hacking in rlhf," *ArXiv*, vol. abs/2402.07319, 2024. [20](#)
- [311] T. Liu, W. Xiong, J. Ren, L. Chen, J. Wu, R. Joshi, Y. Gao, J. Shen, Z. Qin, T. Yu, D. Sohn, A. Makarova, J. Liu, Y. Liu, B. Piot, A. Ittycheriah, A. Kumar, and M. Saleh, "Rrm: Robust reward model training mitigates reward hacking," *ArXiv*, vol. abs/2409.13156, 2024. [20](#)
- [312] C. Wang, Z. Zhao, Y. Jiang, Z. Chen, C. Zhu, Y. Chen, J. Liu, L. Zhang, X. Fan, H. Ma, and S.-Y. Wang, "Beyond reward hacking: Causal rewards for large language model alignment," 2025. [20](#)
- [313] C. B. Browne, E. Powley, D. Whitehouse, S. M. Lucas, P. I. Cowling, P. Rohlfshagen, S. Tavener, D. Perez, S. Samothrakis, and S. Colton, "A survey of monte carlo tree search methods," *IEEE Transactions on Computational Intelligence and AI in games*, vol. 4, no. 1, pp. 1–43, 2012. [20](#)
- [314] A. Madaan, N. Tandon, P. Gupta, S. Hallinan, L. Gao, S. Wiegrefe, U. Alon, N. Dzir, S. Prabh-moye, Y. Yang, et al., "Self-refine: Iterative refinement with self-feedback," *Advances in Neural Information Processing Systems*, vol. 36, 2024. [20](#)
- [315] Y. Fu, H. Peng, A. Sabharwal, P. Clark, and T. Khot, "Complexity-based prompting for multi-step reasoning," in *The Eleventh International Conference on Learning Representations*, 2022. [20](#)
- [316] B. Johnson, "Metacognition for artificial intelligence system safety—an approach to safe and desired behavior," *Safety Science*, vol. 151, p. 105743, 2022. [20](#), [21](#)
- [317] OpenAI, "Early access for safety testing," 2024. [20](#)
- [318] Y. Yan, X. Lou, J. Li, Y. Zhang, J. Xie, C. Yu, Y. Wang, D. Yan, and Y. Shen, "Reward-robust rlhf in llms," *ArXiv*, vol. abs/2409.15360, 2024. [20](#)
- [319] W. Yu, Z. Sun, J. Xu, Z. Dong, X. Chen, H. Xu, and J.-R. Wen, "Explainable legal case matching via inverse optimal transport-based rationale extraction," in *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 657–668, 2022. [20](#)
- [320] A. Amini, S. Gabriel, P. Lin, R. Koncel-Kedziorski, Y. Choi, and H. Hajishirzi, "Mathqa: Towards interpretable math word problem solving with operation-based formalisms," 2019. [20](#)
- [321] L. Chen, O. Sinavski, J. Hünermann, A. Karnsund, A. J. Willmott, D. Birch, D. Maund, and J. Shotton, "Driving with llms: Fusing object-level vector modality for explainable autonomous driving," *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 14093–14100, 2023. [20](#)
- [322] R. Poulain, H. Fayyaz, and R. Beheshti, "Bias patterns in the application of llms for clinical decision support: A comprehensive study," *arXiv preprint arXiv:2404.15149*, 2024. [20](#)
- [323] Z. Fan, R. Chen, R. Xu, and Z. Liu, "Biasalert: A plug-and-play tool for social bias detection in llms," *arXiv preprint arXiv:2407.10241*, 2024. [20](#)
- [324] M. Li, T. Shi, C. Ziem, M.-Y. Kan, N. F. Chen, Z. Liu, and D. Yang, "Coannotating: Uncertainty-guided work allocation between human and large language models for data annotation," *arXiv preprint arXiv:2310.15638*, 2023. [20](#)
- [325] A. Elangovan, J. Ko, L. Xu, M. Elyasi, L. Liu, S. Bodapati, and D. Roth, "Beyond correlation: The impact of human uncertainty in measuring the effectiveness of automatic evaluation and llm-as-a-judge," *arXiv preprint arXiv:2410.03775*, 2024. [20](#)
- [326] Y. R. Dong, T. Hu, and N. Collier, "Can llm be a personalized judge?," *arXiv preprint arXiv:2406.11657*, 2024. [20](#)
- [327] D. Wang, K. Yang, H. Zhu, X. Yang, A. Cohen, L. Li, and Y. Tian, "Learning personalized story evaluation," *arXiv preprint arXiv:2310.03304*, 2023. [20](#)
- [328] H. Du, S. Liu, L. Zheng, Y. Cao, A. Nakamura, and L. Chen, "Privacy in fine-tuning large language models: Attacks, defenses, and future directions," 2024. [20](#), [21](#), [22](#)
- [329] L. Yuan, W. Li, H. Chen, G. Cui, N. Ding, K. Zhang, B. Zhou, Z. Liu, and H. Peng, "Free process rewards without process labels," *arXiv preprint arXiv:2412.01981*, 2024. [20](#)
- [330] Y. J. Ma, W. Liang, G. Wang, D.-A. Huang, O. Bastani, D. Jayaraman, Y. Zhu, L. Fan, and A. Anandkumar, "Eureka: Human-level reward design via coding large language models," *ArXiv*, vol. abs/2310.12931, 2023. [20](#)
- [331] A. Havrilla, S. C. Raparthy, C. Nalmpantis, J. Dweivedi-Yu, M. Zhuravinskiy, E. Hambro, and R. Raulineau, "Glore: When, where, and how to improve llm reasoning via global and local multimodal understanding and reasoning benchmark for expert agi," in *Proceedings of CVPR*, 2024. [18](#)
- [277] S. Lin, J. Hilton, and O. Evans, "Truthfulqa: Measuring how models mimic human falsehoods," *arXiv preprint arXiv:2109.07958*, 2021. [18](#)
- [278] X. Yue et al., "Mammoth: Building math generalist models through hybrid instruction tuning," *arXiv preprint arXiv:2309.05653*, 2023. [18](#)
- [279] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt, "Measuring massive multitask language understanding," *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021. [18](#), [19](#)
- [280] D. Hendrycks, C. Burns, S. Basart, A. Critch, J. Li, D. Song, and J. Steinhardt, "Aligning ai with shared human values," *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021. [18](#), [19](#)
- [281] D. Dua, Y. Wang, P. Dasigi, G. Stanovsky, S. Singh, and M. Gardner, "DROP: A reading comprehension benchmark requiring discrete reasoning over paragraphs," in *Proc. of NAACL*, 2019. [18](#)
- [282] Z. Wang, Y. Dong, J. Zeng, V. Adams, M. N. Sreedhar, D. Egert, O. Delalleau, J. P. Scowcroft, N. Kant, A. Swope, et al., "Helpsteer: Multi-attribute helpfulness dataset for steerlm," *arXiv preprint arXiv:2311.09528*, 2023. [18](#)
- [283] G. Cui, L. Yuan, N. Ding, G. Yao, W. Zhu, Y. Ni, G. Xie, Z. Liu, and M. Sun, "Ultrafeedback: Boosting language models with high-quality feedback," 2023. [18](#)
- [284] J. Fu, A. Kumar, O. Nachum, G. Tucker, and S. Levine, "D4rl: Datasets for deep data-driven reinforcement learning," 2020. [18](#)
- [285] T. Schmied, M. Hofmarcher, F. Paischer, R. Pascanu, and S. Hochreiter, "Learning to modulate pre-trained models in rl," *Advances in Neural Information Processing Systems*, vol. 36, 2024. [18](#)
- [286] W. H. Guss, B. Houghton, N. Topin, P. Wang, C. Codel, M. Veloso, and R. Salakhutdinov, "Miner: A large-scale dataset of minecraft demonstrations," 2019. [18](#)
- [287] T. Nguyen, C. V. Nguyen, V. D. Lai, H. Man, N. T. Ngo, F. Démoncourt, R. A. Rossi, and T. H. Nguyen, "CulturaX: A cleaned, enormous, and multilingual dataset for large language models in 167 languages," in *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)* (N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, and N. Xue, eds.), (Torino, Italia), pp. 4226–4237, ELRA and ICCL, May 2024. [18](#)
- [288] X. Yue, Y. Song, A. Asai, S. Kim, J. de Dieu Nyandwi, S. Khanuja, A. Kantharuban, L. Sutawika, S. Ramamoorthy, and G. Neubig, "Pangea: A fully open multilingual multimodal llm for 39 languages," *arXiv preprint arXiv:2410.16153*, 2024. [18](#)
- [289] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: pre-training of deep bidirectional transformers for language understanding," *CoRR*, vol. abs/1810.04805, 2018. [18](#)
- [290] Y. Liang, N. Duan, Y. Gong, N. Wu, F. Guo, W. Qi, M. Gong, L. Shou, D. Jiang, G. Cao, X. Fan, R. Zhang, R. Agrawal, E. Cui, S. Wei, T. Bhatti, Y. Qiao, J.-H. Chen, W. Wu, S. Liu, F. Yang, D. Campos, R. Majumder, and M. Zhou, "Xglue: A new benchmark dataset for cross-lingual pre-training, understanding and generation," *arXiv*, vol. abs/2004.01401, 2020. [18](#)
- [291] G. Son, D. Yoon, J. Suk, J. Aulá-Blasco, M. Aslan, V. T. Kim, S. B. Islam, J. Prats-Cristià, L. Tormo-Bañuelos, and S. Kim, "Mm-eval: A multilingual meta-evaluation benchmark for llm-as-a-judge and reward models," 2024. [18](#)
- [292] S. et.al, "Beyond the imitation game: Quantifying and extrapolating the capabilities of language models," 2022. [18](#), [19](#)
- [293] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. Xing, et al., "Judging llm-as-a-judge with mt-bench and chatbot arena," *Advances in Neural Information Processing Systems*, vol. 36, pp. 46595–46623, 2023. [18](#), [19](#)
- [294] E. Dinan, V. Logacheva, V. Malykh, A. H. Miller, K. Shuster, J. Urbanek, D. Kiela, A. Szlam, I. Serban, R. Lowe, S. Prabh-moye, A. W. Black, A. I. Rudnicky, J. Williams, J. Pineau, M. S. Burtsev, and J. Weston, "The second conversational intelligence challenge (conval2)," *CoRR*, vol. abs/1902.00098, 2019. [18](#), [19](#)
- [295] M. Eric, R. Goel, S. Paul, A. Sethi, S. Agarwal, S. Gao, and D. Hakkani-Tur, "MultiWOZ 2.1: A consolidated multi-domain dialogue dataset with state corrections and state tracking baselines," in *Proceedings of the 12th Language Resources and Evaluation Conference*, (Marseille, France), pp. 422–428, European Language Resources Association, May 2020. [18](#), [19](#)
- [296] X. Li and D. Roth, "Learning question classifiers," in *COLING 2002: The 19th International Conference on Computational Linguistics*, 2002. [18](#), [19](#)
- [297] E. Hovy, L. Gerber, U. Hermjakob, C.-Y. Lin, and D. Ravichandran, "Toward semantics-based answer pinpointing," in *Proceedings of the First International Conference on Human Language Technology Research*, 2001. [18](#), [19](#)
- [298] N. Thakur, N. Reimers, A. Rücklé, A. Srivastava, and I. Gurevych, "BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models," in *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*, 2021. [18](#), [19](#)
- [299] C. Chhun, F. M. Suchanek, and C. Clavel, "Do language models enjoy their own stories? Prompting large language models for automatic story evaluation," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 1122–1142, 2024. [18](#)
- [300] H. Chen, D. M. Vo, H. Takamura, Y. Miyao, and H. Nakayama, "Storyer: Automatic story evaluation via ranking, rating and reasoning," 2022. [18](#)
- [301] J. Ji, M. Liu, J. Dai, X. Pan, C. Zhang, C. Bian, B. Chen, R. Sun, Y. Wang, and Y. Yang, "Beavertails: Towards improved safety alignment of llm via a human-preference dataset," *Advances in Neural Information Processing Systems*, vol. 36, 2024. [18](#), [19](#)
- [302] G. Xu, J. Liu, M. Yan, H. Xu, J. Si, Z. Zhou, P. Yi, X. Gao, J. Sang, R. Zhang, et al., "Cvalues: Measuring the values of chinese large language models from safety to responsibility," *arXiv preprint arXiv:2307.09705*, 2023. [18](#)
- [303] S. Mishra, D. Khashabi, C. Baral, and H. Hajishirzi, "Cross-task generalization via natural language crowdsourcing instructions," in *ACL*, 2022. [18](#)
- [304] Y. Wang, S. Mishra, P. Alipoormolabashi, Y. Kordi, A. Mirzaei, A. Arunkumar, A. Ashok, A. S. Dhanasekaran, A. Naik, D. Stap, et al., "Super-naturalinstructions:generalization via declarative instructions on 1600+ tasks," in *EMNLP*, 2022. [18](#)
- [305] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Askell,

- refinements,” *ArXiv*, vol. abs/2402.10963, 2024. [20](#)
- [332] M. Fawi, “Curlora: Stable llm continual fine-tuning and catastrophic forgetting mitigation,” 2024. [20](#)
- [333] C. Fu, P. Chen, Y. Shen, Y. Qin, M. Zhang, X. Lin, Z. Qiu, W. Lin, J. Yang, X. Zheng, K. Li, X. Sun, and R. Ji, “Mme: A comprehensive evaluation benchmark for multimodal large language models,” *ArXiv*, vol. abs/2306.13394, 2023. [20](#)
- [334] Y. Wang, Z. Yu, Z. Zeng, L. Yang, C. Wang, H. Chen, C. Jiang, R. Xie, J. Wang, X. Xie, W. Ye, S.-B. Zhang, and Y. Zhang, “Pandalm: An automatic evaluation benchmark for llm instruction tuning optimization,” *ArXiv*, vol. abs/2306.05087, 2023. [20](#)
- [335] Y. Sun, Z. Li, Y. Li, and B. Ding, “Improving lora in privacy-preserving federated learning,” *ArXiv*, vol. abs/2403.12313, 2024. [20](#)
- [336] Y. He, Y. Kang, L. Fan, and Q. Yang, “Fedeval-llm: Federated evaluation of large language models on downstream tasks with collective wisdom,” *arXiv preprint arXiv:2404.12273*, 2024. [20](#)
- [337] J. Park, S. Jwa, M. Ren, D. Kim, and S. Choi, “Offsetbias: Leveraging debiased data for tuning evaluators,” *arXiv preprint arXiv:2407.06551*, 2024. [20](#)
- [338] P. Lu, L. Qiu, K.-W. Chang, Y. N. Wu, S.-C. Zhu, T. Rajpurohit, P. Clark, and A. Kalyan, “Dynamic prompt learning via policy gradient for semi-structured mathematical reasoning,” 2023. [20](#)
- [339] L. Zhang, A. Hosseini, H. Bansal, M. Kazemi, A. Kumar, and R. Agarwal, “Generative verifiers: Reward modeling as next-token prediction,” in *The 4th Workshop on Mathematical Reasoning and AI at NeurIPS’24*, 2024. [20](#)
- [340] S. Yang and D. Song, “FPC: Fine-tuning with prompt curriculum for relation extraction,” in *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)* (Y. He, H. Ji, S. Li, Y. Liu, and C.-H. Chang, eds.), (Online only), pp. 1065–1077, Association for Computational Linguistics, Nov. 2022. [20](#)
- [341] Y. Yu, W. Ping, Z. Liu, B. Wang, J. You, C. Zhang, M. Shoenybi, and B. Catanzaro, “Rankrag: Unifying context ranking with retrieval-augmented generation in llms,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 121156–121184, 2025. [20](#)
- [342] X. V. Lin, X. Chen, M. Chen, W. Shi, M. Lomeli, R. James, P. Rodriguez, J. Kahn, G. Szilvasy, M. Lewis, et al., “Rad-it: Retrieval-augmented dual instruction tuning,” in *The Twelfth International Conference on Learning Representations*, 2023. [20](#)
- [343] T. Zhang, S. G. Patil, N. Jain, S. Shen, M. Zaharia, I. Stoica, and J. E. Gonzalez, “Raft: Adapting language model to domain specific rag,” in *First Conference on Language Modeling*, 2024. [20](#)
- [344] T. Huang, S. Hu, F. Ilhan, S. F. Tekin, and L. Liu, “Harmful fine-tuning attacks and defenses for large language models: A survey,” *arXiv preprint arXiv:2409.18169*, 2024. [21](#)
- [345] S. R. Bowman, J. Hyun, E. Perez, E. Chen, C. Pettit, S. Heiner, K. Lukosiute, A. Askell, A. Jones, A. Chen, et al., “Measuring progress on scalable oversight for large language models,” *arXiv preprint arXiv:2211.03540*, 2022. [21](#)
- [346] N. Hollmann, S. Müller, and F. Hutter, “Llms for semi-automated data science: Introducing caafe for context-aware automated feature engineering,” *CoRR*, 2023. [21](#)
- [347] S. R. Motwani, C. Smith, R. J. Das, M. Rybchuk, P. H. Torr, I. Laptev, F. Pizzati, R. Clark, and C. S. de Witt, “Malt: Improving reasoning with multi-agent llm training,” *arXiv preprint arXiv:2412.01928*, 2024. [21](#)
- [348] A. Estornell, J.-F. Ton, Y. Yao, and Y. Liu, “Acc-debate: An actor-critic approach to multi-agent debate,” *arXiv preprint arXiv:2411.00053*, 2024. [21](#)
- [349] I. Puri, S. Sudalairaj, G. Xu, K. Xu, and A. Srivastava, “A probabilistic inference approach to inference-time scaling of llms using particle-based monte carlo methods,” *arXiv preprint arXiv:2502.01618*, 2025.
- [350] L. Luo, Y. Liu, R. Liu, S. Phatale, H. Lara, Y. Li, L. Shu, Y. Zhu, L. Meng, J. Sun, et al., “Improve mathematical reasoning in language models by automated process supervision,” *arXiv preprint arXiv:2406.06592*, 2024. [22](#)
- [351] W. Shen, X. Zhang, Y. Yao, R. Zheng, H. Guo, and Y. Liu, “Improving reinforcement learning from human feedback using contrastive rewards,” *arXiv preprint arXiv:2403.07708*, 2024. [22](#)
- [352] M. Ma, P. D’Oro, Y. Bengio, and P.-L. Bacon, “Long-term credit assignment via model-based temporal shortcuts,” in *Deep RL Workshop NeurIPS 2021*, 2021. [22](#)
- [353] E. Pignatelli, J. Ferret, M. Geist, T. Mesnard, H. van Hasselt, O. Pietquin, and L. Toni, “A survey of temporal credit assignment in deep reinforcement learning,” *arXiv preprint arXiv:2312.01072*, 2023. [22](#)
- [354] H. Zhang and Y. Guo, “Generalization of reinforcement learning with policy-aware adversarial data augmentation,” 2021. [22](#)
- [355] A. Ahmadian, C. Cremer, M. Gallé, M. Fadaee, J. Kreutzer, O. Pietquin, A. Üstün, and S. Hooker, “Back to basics: Revisiting reinforce style optimization for learning from human feedback in llms,” *arXiv preprint arXiv:2402.14740*, 2024. [22](#)
- [356] S. Lee, G. Lee, J. W. Kim, J. Shin, and M.-K. Lee, “Hetal: Efficient privacy-preserving transfer learning with homomorphic encryption,” 2024. [22](#)
- [357] Y. Wei, J. Jia, Y. Wu, C. Hu, C. Dong, Z. Liu, X. Chen, Y. Peng, and S. Wang, “Distributed differential privacy via shuffling versus aggregation: A curious study,” *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 2501–2516, 2024. [22](#)
- [358] W. Zhang, K. Tang, H. Wu, M. Wang, Y. Shen, G. Hou, Z. Tan, P. Li, Y. Zhuang, and W. Lu, “Agent-pro: Learning to evolve via policy-level reflection and optimization,” *arXiv preprint arXiv:2402.17574*, 2024. [22](#)
- [359] C. Ma, J. Zhang, Z. Zhu, C. Yang, Y. Yang, Y. Jin, Z. Lan, L. Kong, and J. He, “Agentboard: An analytical evaluation board of multi-turn llm agents,” *arXiv preprint arXiv:2401.13178*, 2024. [22](#)
- [360] J. Zhang, J. Xiang, Z. Yu, F. Teng, X. Chen, J. Chen, M. Zhuge, X. Cheng, S. Hong, J. Wang, et al., “Aflow: Automating agentic workflow generation,” *arXiv preprint arXiv:2410.10762*, 2024. [22](#)
- [361] H. D. Le, X. Xia, and Z. Chen, “Multi-agent causal discovery using large language models,” *arXiv preprint arXiv:2407.15073*, 2024. [22](#)
- [362] D. M. Owens, R. A. Rossi, S. Kim, T. Yu, F. Dernoncourt, X. Chen, R. Zhang, J. Gu, H. Deilamsalehy, and N. Lipka, “A multi-llm debiasing framework,” *arXiv preprint arXiv:2409.13884*, 2024. [22](#)
- [363] H. Zou, Q. Zhao, L. Bariah, Y. Tian, M. Bennis, S. Lasaulce, M. Debbah, and F. Bader, “Genaint: Enabling wireless collective intelligence via knowledge transfer and reasoning,” *ArXiv*, vol. abs/2402.16631, 2024. [22](#)
- [364] R. Lee, O. J. Mengshoel, A. Saksena, R. Gardner, D. Genin, J. Silbermann, M. Owen, and M. J. Kochenderfer, “Adaptive stress testing: Finding likely failure events with reinforcement learning,” 2020. [22](#)
- [365] A. G. Baydin, B. A. Pearlmutter, D. Syme, F. Wood, and P. Torr, “Gradients without backpropagation,” 2022. [22](#)
- [366] Y. Liu, C. Cai, X. Zhang, X. Yuan, and C. Wang, “Arondight: Red teaming large vision language models with auto-generated multi-modal jailbreak prompts,” in *ACM Multimedia*, 2024. [22](#)
- [367] D. Kim, K. Lee, J. Shin, and J. Kim, “Aligning large language models with self-generated preference data,” *arXiv preprint arXiv:2406.04412*, 2024. [22](#)
- [368] H. Xia, Y. Li, C. T. Leong, W. Wang, and W. Li, “Tokenskip: Controllable chain-of-thought compression in llms,” 2025. [22](#)
- [369] Z. Ma, W. Wu, Z. Zheng, Y. Guo, Q. Chen, S. Zhang, and X. Chen, “Leveraging speech ptm, text llm, and emotional tts for speech emotion recognition,” *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 11146–11150, 2023. [22](#)
- [370] Z. Xi, W. Chen, B. Hong, S. Jin, R. Zheng, W. He, Y. Ding, S. Liu, X. Guo, J. Wang, H. Guo, W. Shen, X. Fan, Y. Zhou, S. Dou, X. Wang, X. Zhang, P. Sun, T. Gui, Q. Zhang, and X. Huang, “Training large language models for reasoning through reverse curriculum reinforcement learning,” *ArXiv*, vol. abs/2402.05808, 2024. [22](#)
- [371] O. Y. Lee, A. Xie, K. Fang, K. Pertsch, and C. Finn, “Affordance-guided reinforcement learning via visual prompting,” *ArXiv*, vol. abs/2407.10341, 2024. [22](#)
- [372] H. Xu, Z. Zhu, D. Ma, S. Zhang, S. Fan, L. Chen, and K. Yu, “Rejection improves reliability: Training llms to refuse unknown questions using rl from knowledge feedback,” *ArXiv*, 2019. [22](#)
- [373] P. Welinder, P. F. Christiano, J. Leike, and R. Lowe, “Training language models to follow instructions with human feedback,” in *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022* (S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, eds.), 2022. [19](#)
- [374] W. Saunders, C. Yeh, J. Wu, S. Bills, L. Ouyang, J. Ward, and J. Leike, “Self-critiquing models for assisting human evaluators,” *arXiv preprint arXiv:2206.05802*, 2022. [19](#)
- [375] J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, and D. Amodei, “Scaling laws for neural language models,” *arXiv preprint arXiv:2001.08361*, 2020. [19](#)
- [376] S. Roy and D. Roth, “Solving general arithmetic word problems,” 2016. [19](#)
- [377] Y. Jinmai, T. Morimura, K. Ariu, and K. Abe, “Regularized best-of-n sampling to mitigate reward hacking for language model alignment,” *arXiv preprint arXiv:2404.01054*, 2024. [19](#)
- [378] L. Chen, C. Zhu, D. Soesela, J. Chen, T. Zhou, T. Goldstein, H. Huang, M. Shoenybi, and B. Catanzaro, “Odin: Disentangled reward mitigates hacking in rlhf,” *ArXiv*, vol. abs/2402.07319, 2024. [19](#)
- [379] T. Liu, W. Xiong, J. Ren, L. Chen, J. Wu, R. Joshi, Y. Gao, J. Shen, Z. Qin, T. Yu, D. Sohn, A. Makarova, J. Liu, Y. Liu, B. Piot, A. Ittycheriah, A. Kumar, and M. Saleh, “Rrm: Robust reward model training mitigates reward hacking,” *ArXiv*, vol. abs/2409.13156, 2024. [19](#)
- [380] C. Wang, Z. Zhao, Y. Jiang, Z. Chen, C. Zhu, Y. Chen, J. Liu, L. Zhang, X. Fan, H. Ma, and S.-Y. Wang, “Beyond reward hacking: Causal rewards for large language model alignment,” 2025. [19](#)
- [381] C. B. Browne, E. Powley, D. Whitehouse, S. M. Lucas, P. I. Cowling, P. Rohlfshagen, S. Tavener, D. Perez, S. Samothrakis, and S. Colton, “A survey of monte carlo tree search methods,” *IEEE Transactions on Computational Intelligence and AI in games*, vol. 4, no. 1, pp. 1–43, 2012. [19](#)
- [382] A. Madaan, N. Tandon, P. Gupta, S. Hallinan, L. Gao, S. Wiegrefe, U. Alon, N. Dziri, S. Prabhunoye, Y. Yang, et al., “Self-refine: Iterative refinement with self-feedback,” *Advances in Neural Information Processing Systems*, vol. 36, 2024. [19](#)
- [383] Y. Fu, H. Peng, A. Sabharwal, P. Clark, and T. Khot, “Complexity-based prompting for multi-step reasoning,” in *The Eleventh International Conference on Learning Representations*, 2022. [19](#)
- [384] B. Johnson, “Metacognition for artificial intelligence system safety—an approach to safe and desired behavior,” *Safety Science*, vol. 151, p. 105743, 2022. [19](#)
- [385] OpenAI, “Early access for safety testing,” 2024. [19](#)
- [386] Y. Yan, X. Lou, J. Li, Y. Zhang, J. Xie, C. Yu, Y. Wang, D. Yan, and Y. Shen, “Reward-robust rlhf in llms,” *ArXiv*, vol. abs/2409.15360, 2024. [19](#)
- [387] W. Yu, Z. Sun, J. Xu, Z. Dong, X. Chen, H. Xu, and J.-R. Wen, “Explainable legal case matching via inverse optimal transport-based rationale extraction,” in *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 657–668, 2022. [19](#)
- [388] A. Amini, S. Gabriel, P. Lin, R. Koncel-Kedziorski, Y. Choi, and H. Hajishirzi, “Mathqa: Towards interpretable math word problem solving with operation-based formalisms,” 2019. [19](#)
- [389] L. Chen, O. Sinavski, J. Hünemann, A. Karnsund, A. J. Willmott, D. Birch, D. Maund, and J. Shotton, “Driving with llms: Fusing object-level vector modality for explainable autonomous driving,” 2024 *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 14093–14100, 2023. [19](#)
- [390] R. Poulain, H. Fayyaz, and R. Beheshti, “Bias patterns in the application of llms for clinical decision support: A comprehensive study,” *arXiv preprint arXiv:2404.15149*, 2024. [19](#)
- [391] Z. Fan, R. Chen, R. Xu, and Z. Liu, “Biasalert: A plug-and-play tool for social bias detection in llms,” *arXiv preprint arXiv:2407.10241*, 2024. [19](#)
- [392] M. Li, T. Shi, C. Ziemis, M.-Y. Kan, N. F. Chen, Z. Liu, and D. Yang, “Coannotating: Uncertainty-guided work allocation between human and large language models for data annotation,” *arXiv preprint arXiv:2310.15638*, 2023. [19](#)
- [393] A. Elangovan, J. Ko, L. Xu, M. Elyasi, L. Liu, S. Bodapati, and D. Roth, “Beyond correlation: The impact of human uncertainty in measuring the effectiveness of automatic evaluation and llm-as-a-judge,” *arXiv preprint arXiv:2410.03775*, 2024. [19](#)
- [394] Y. R. Dong, T. Hu, and N. Collier, “Can llm be a personalized judge?,” *arXiv preprint arXiv:2406.11657*, 2024. [19](#)
- [395] D. Wang, K. Yang, H. Zhu, X. Yang, A. Cohen, L. Li, and Y. Tian, “Learning personalized story evaluation,” *arXiv preprint arXiv:2401.03304*, 2023. [19](#)
- [396] H. Du, S. Liu, L. Zheng, Y. Cao, A. Nakamura, and L. Chen, “Privacy in fine-tuning large language models: Attacks, defenses, and future directions,” 2024. [19](#)
- [397] L. Yuan, W. Li, H. Chen, G. Cui, N. Ding, K. Zhang, B. Zhou, Z. Liu, and H. Peng, “Free process rewards without process labels,” *arXiv preprint arXiv:2412.01981*, 2024. [19](#)
- [398] Y. J. Ma, W. Liang, G. Wang, D.-A. Huang, O. Bastani, D. Jayaraman, Y. Zhu, L. Fan, and A. Anandkumar, “Eureka: Human-level reward design via coding large language models,” *ArXiv*, vol. abs/2310.12931, 2023. [19](#)
- [399] A. Havrilla, S. C. Raparthy, C. Nalmpantis, J. Dwivedi-Yu, M. Zhuravinskiy, E. Hambro, and R. Railneau, “Glore: When, where, and how to improve llm reasoning via global and local refinements,” *ArXiv*, vol. abs/2402.10963, 2024. [19](#)
- [400] M. Fawi, “Curlora: Stable llm continual fine-tuning and catastrophic forgetting mitigation,” 2024. [19](#)
- [401] C. Fu, P. Chen, Y. Shen, Y. Qin, M. Zhang, X. Lin, Z. Qiu, W. Lin, J. Yang, X. Zheng, K. Li, X. Sun, and R. Ji, “Mme: A comprehensive evaluation benchmark for multimodal large language models,” *ArXiv*, vol. abs/2306.13394, 2023. [19](#)
- [402] Y. Wang, Z. Yu, Z. Zeng, L. Yang, C. Wang, H. Chen, C. Jiang, R. Xie, J. Wang, X. Xie, W. Ye, S.-B. Zhang, and Y. Zhang, “Pandalm: An automatic evaluation benchmark for llm instruction tuning optimization,” *ArXiv*, vol. abs/2306.05087, 2023. [19](#)
- [403] Y. Sun, Z. Li, Y. Li, and B. Ding, “Improving lora in privacy-preserving federated learning,” *ArXiv*, vol. abs/2403.12313, 2024. [19](#)
- [404] Y. He, Y. Kang, L. Fan, and Q. Yang, “Fedeval-llm: Federated evaluation of large language models on downstream tasks with collective wisdom,” *arXiv preprint arXiv:2404.12273*, 2024. [19](#)
- [405] J. Park, S. Jwa, M. Ren, D. Kim, and S. Choi, “Offsetbias: Leveraging debiased data for tuning evaluators,” *arXiv preprint arXiv:2407.06551*, 2024. [19](#)
- [406] P. Lu, L. Qiu, K.-W. Chang, Y. N. Wu, S.-C. Zhu, T. Rajpurohit, P. Clark, and A. Kalyan, “Dynamic prompt learning via policy gradient for semi-structured mathematical reasoning,”

- vol. abs/2403.18349, 2024. [22](#)
- [373] X. Chen, J. Xu, T. Liang, Z. He, J. Pang, D. Yu, L. Song, Q. Liu, M. Zhou, Z. Zhang, R. Wang, Z. Tu, H. Mi, and D. Yu, “Do not think that much for 2+3=? on the overthinking of ol-like llms,” *ArXiv*, vol. abs/2412.21187, 2024. [22](#)
- [374] M. Kemmerling, D. Lütticke, and R. H. Schmitt, “Beyond games: a systematic review of neural monte carlo tree search applications,” *Applied Intelligence*, vol. 54, no. 1, pp. 1020–1046, 2024. [22](#)
- [375] Y. Li, H. Wen, W. Wang, X. Li, Y. Yuan, G. Liu, J. Liu, W. Xu, X. Wang, Y. Sun, R. Kong, Y. Wang, H. Geng, J. Luan, X. Jin, Z.-L. Ye, G. Xiong, F. Zhang, X. Li, M. Xu, Z. Li, P. Li, Y. Liu, Y. Zhang, and Y. Liu, “Personal llm agents: Insights and survey about the capability, efficiency and security,” *ArXiv*, vol. abs/2401.05459, 2024. [22](#)
- [376] H. Li, L. Ding, M. Fang, and D. Tao, “Revisiting catastrophic forgetting in large language model tuning,” 2024. [23](#)
- [377] N. Alzahrani, H. A. Alyahya, Y. Alnumay, S. Alrashed, S. Alsubaie, Y. Almushaykeh, F. Mirza, N. Alotaibi, N. Altwairesh, A. Alowisheq, M. S. Bari, and H. Khan, “When benchmarks are targets: Revealing the sensitivity of large language model leaderboards,” 2024. [23](#)
- [339] L. Zhang, A. Hosseini, H. Bansal, M. Kazemi, A. Kumar, and R. Agarwal, “Generative verifiers: Reward modeling as next-token prediction,” in *The 4th Workshop on Mathematical Reasoning and AI at NeurIPS’24*, 2024. [19](#)
- [340] S. Yang and D. Song, “FPC: Fine-tuning with prompt curriculum for relation extraction,” in *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)* (Y. He, H. Ji, S. Li, Y. Liu, and C.-H. Chang, eds.), (Online only), pp. 1065–1077, Association for Computational Linguistics, Nov. 2022. [19](#)
- [341] Y. Yu, W. Ping, Z. Liu, B. Wang, J. You, C. Zhang, M. Shoyebi, and B. Catanzaro, “Rankrag: Unifying context ranking with retrieval-augmented generation in llms,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 121156–121184, 2025. [19](#)
- [342] X. V. Lin, X. Chen, M. Chen, W. Shi, M. Lomeli, R. James, P. Rodriguez, J. Kahn, G. Szilvasy, M. Lewis, *et al.*, “Ra-dit: Retrieval-augmented dual instruction tuning,” in *The Twelfth International Conference on Learning Representations*, 2023. [19](#)
- [343] T. Zhang, S. G. Patil, N. Jain, S. Shen, M. Zaharia, I. Stoica, and J. E. Gonzalez, “Raft: Adapting language model to domain specific rag,” in *First Conference on Language Modeling*, 2024. [19](#)
- [344] T. Huang, S. Hu, F. Ilhan, S. F. Tekin, and L. Liu, “Harmful fine-tuning attacks and defenses for large language models: A survey,” *arXiv preprint arXiv:2409.18169*, 2024. [19](#)
- [345] S. R. Bowman, J. Hyun, E. Perez, E. Chen, C. Pettit, S. Heiner, K. Lukosiuć, A. Askell, A. Jones, A. Chen, *et al.*, “Measuring progress on scalable oversight for large language models,” *arXiv preprint arXiv:2211.03540*, 2022. [19](#)
- [346] N. Hollmann, S. Müller, and F. Hutter, “Llms for semi-automated data science: Introducing caafe for context-aware automated feature engineering,” *CoRR*, 2023. [20](#)
- [347] S. R. Motwani, C. Smith, R. J. Das, M. Rybchuk, P. H. Torr, I. Laptev, F. Pizzati, R. Clark, and C. S. de Witt, “Malt: Improving reasoning with multi-agent llm training,” *arXiv preprint arXiv:2412.01928*, 2024. [20](#)
- [348] A. Estornell, J.-F. Ton, Y. Yao, and Y. Liu, “Acc-debate: An actor-critic approach to multi-agent debate,” *arXiv preprint arXiv:2411.00053*, 2024. [20](#)
- [349] I. Puri, S. Sudalairaj, G. Xu, K. Xu, and A. Srivastava, “A probabilistic inference approach to inference-time scaling of llms using particle-based monte carlo methods,” *arXiv preprint arXiv:2502.01618*, 2025. [20](#)
- [350] L. Luo, Y. Liu, R. Liu, S. Phatale, H. Lara, Y. Li, L. Shu, Y. Zhu, L. Meng, J. Sun, *et al.*, “Improve mathematical reasoning in language models by automated process supervision,” *arXiv preprint arXiv:2406.06592*, 2024. [20](#)
- [351] W. Shen, X. Zhang, Y. Yao, R. Zheng, H. Guo, and Y. Liu, “Improving reinforcement learning from human feedback using contrastive rewards,” *arXiv preprint arXiv:2403.07708*, 2024. [20](#)
- [352] M. Ma, P. D’Oro, Y. Bengio, and P.-L. Bacon, “Long-term credit assignment via model-based temporal shortcuts,” in *Deep RL Workshop NeurIPS 2021*, 2021. [20](#)
- [353] E. Pignatelli, J. Ferret, M. Geist, T. Mesnard, H. van Has-
self, O. Pietquin, and L. Toni, “A survey of temporal credit assignment in deep reinforcement learning,” *arXiv preprint arXiv:2312.01072*, 2023. [20](#)
- [354] H. Zhang and Y. Guo, “Generalization of reinforcement learning with policy-aware adversarial data augmentation,” 2021. [20](#)
- [355] A. Ahmadian, C. Cremer, M. Gallé, M. Fadaee, J. Kreutzer, O. Pietquin, A. Ustün, and S. Hooker, “Back to basics: Revisiting reinforce style optimization for learning from human feedback in llms,” *arXiv preprint arXiv:2402.14740*, 2024. [21](#)
- [356] S. Lee, G. Lee, J. W. Kim, J. Shin, and M.-K. Lee, “Hetal: Efficient privacy-preserving transfer learning with homomorphic encryption,” 2024. [21](#)
- [357] Y. Wei, J. Jia, Y. Wu, C. Hu, C. Dong, Z. Liu, X. Chen, Y. Peng, and S. Wang, “Distributed differential privacy via shuffling versus aggregation: A curious study,” *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 2501–2516, 2024. [21](#)
- [358] W. Zhang, K. Tang, H. Wu, M. Wang, Y. Shen, G. Hou, Z. Tan, P. Li, Y. Zhuang, and W. Lu, “Agent-pro: Learning to evolve via policy-level reflection and optimization,” *arXiv preprint arXiv:2402.17574*, 2024. [21](#)
- [359] C. Ma, J. Zhang, Z. Zhu, C. Yang, Y. Yang, Y. Jin, Z. Lan, L. Kong, and J. He, “Agentboard: An analytical evaluation board of multi-turn llm agents,” *arXiv preprint arXiv:2401.13178*, 2024. [21](#)
- [360] J. Zhang, J. Xiang, Z. Yu, F. Teng, X. Chen, J. Chen, M. Zhuge, X. Cheng, S. Hong, J. Wang, *et al.*, “Aflow: Automating agentic workflow generation,” *arXiv preprint arXiv:2410.10762*, 2024. [21](#)
- [361] H. D. Le, X. Xia, and Z. Chen, “Multi-agent causal discovery using large language models,” *arXiv preprint arXiv:2407.15073*, 2024. [21](#)
- [362] D. M. Owens, R. A. Rossi, S. Kim, T. Yu, F. Dernoncourt, X. Chen, R. Zhang, J. Gu, H. Deilamsalehy, and N. Lipka, “A multi-llm debiasing framework,” *arXiv preprint arXiv:2409.13884*, 2024. [21](#)
- [363] H. Zou, Q. Zhao, L. Bariah, Y. Tian, M. Bennis, S. Lasaulce, M. Debbah, and F. Bader, “Genainet: Enabling wireless collective intelligence via knowledge transfer and reasoning,” *ArXiv*, vol. abs/2402.16631, 2024. [21](#)
- [364] R. Lee, O. J. Mengshoel, A. Sakkena, R. Gardner, D. Genin, J. Silbermann, M. Owen, and M. J. Kochenderfer, “Adaptive stress testing: Finding likely failure events with reinforcement learning,” 2020. [21](#)
- [365] A. G. Baydin, B. A. Pearlmutter, D. Syme, F. Wood, and P. Torr, “Gradients without backpropagation,” 2022. [21](#)
- [366] Y. Liu, C. Cai, X. Zhang, X. Yuan, and C. Wang, “Arondight: Red teaming large vision language models with auto-generated multi-modal jailbreak prompts,” in *ACM Multimedia*, 2024. [21](#)
- [367] D. Kim, K. Lee, J. Shin, and J. Kim, “Aligning large language models with self-generated preference data,” *arXiv preprint arXiv:2406.04412*, 2024. [21](#)
- [368] H. Xia, Y. Li, C. T. Leong, W. Wang, and W. Li, “Tokenskip: Controllable chain-of-thought compression in llms,” 2025. [21](#)
- [369] Z. Ma, W. Wu, Z. Zheng, Y. Guo, Q. Chen, S. Zhang, and X. Chen, “Leveraging speech ptm, text llm, and emotional tts for speech emotion recognition,” *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 11146–11150, 2023. [21](#)
- [370] Z. Xi, W. Chen, B. Hong, S. Jin, R. Zheng, W. He, Y. Ding,

- S. Liu, X. Guo, J. Wang, H. Guo, W. Shen, X. Fan, Y. Zhou, S. Dou, X. Wang, X. Zhang, P. Sun, T. Gui, Q. Zhang, and X. Huang, "Training large language models for reasoning through reverse curriculum reinforcement learning," *ArXiv*, vol. abs/2402.05808, 2024. [21](#)
- [371] O. Y. Lee, A. Xie, K. Fang, K. Pertsch, and C. Finn, "Affordance-guided reinforcement learning via visual prompting," *ArXiv*, vol. abs/2407.10341, 2024. [21](#)
- [372] H. Xu, Z. Zhu, D. Ma, S. Zhang, S. Fan, L. Chen, and K. Yu, "Rejection improves reliability: Training llms to refuse unknown questions using rl from knowledge feedback," *ArXiv*, vol. abs/2403.18349, 2024. [21](#)
- [373] X. Chen, J. Xu, T. Liang, Z. He, J. Pang, D. Yu, L. Song, Q. Liu, M. Zhou, Z. Zhang, R. Wang, Z. Tu, H. Mi, and D. Yu, "Do not think that much for $2+3=?$ on the overthinking of o1-like llms," *ArXiv*, vol. abs/2412.21187, 2024. [21](#)
- [374] M. Kemmerling, D. Lütticke, and R. H. Schmitt, "Beyond games: a systematic review of neural monte carlo tree search applications," *Applied Intelligence*, vol. 54, no. 1, pp. 1020–1046, 2024. [21](#)
- [375] Y. Li, H. Wen, W. Wang, X. Li, Y. Yuan, G. Liu, J. Liu, W. Xu, X. Wang, Y. Sun, R. Kong, Y. Wang, H. Geng, J. Luan, X. Jin, Z.-L. Ye, G. Xiong, F. Zhang, X. Li, M. Xu, Z. Li, P. Li, Y. Liu, Y. Zhang, and Y. Liu, "Personal llm agents: Insights and survey about the capability, efficiency and security," *ArXiv*, vol. abs/2401.05459, 2024. [21](#)
- [376] H. Li, L. Ding, M. Fang, and D. Tao, "Revisiting catastrophic forgetting in large language model tuning," 2024. [21](#)
- [377] N. Alzahrani, H. A. Alyahya, Y. Alnumay, S. Alrashed, S. Alsubaie, Y. Almushaykeh, F. Mirza, N. Alotaibi, N. Altwairesh, A. Alowisheq, M. S. Bari, and H. Khan, "When benchmarks are targets: Revealing the sensitivity of large language model leaderboards," 2024. [21](#)